TYPE II REGULARITY FOR NAVIER–STOKES

GUILLEM DURAN BALLESTER

Abstract. We establish a local Type II exclusion criterion for the three-dimensional Navier–Stokes equations near a possible singular point. From a positive Caffarelli–Kohn–Nirenberg concentration sequence we select renormalized local windows around a retained core and analyze the resulting alternatives by compactness, pressure reconstruction, and local energy estimates. The argument combines a compact single-core exclusion criterion, a repaired-gauge representation for nondegenerate concentration cores, local Calderon–Zygmund pressure control, a Caccioppoli estimate on compact cylinders, multibubble and cascade reductions, and scale-collapse cost estimates. The conclusion is conditional on the analytic inputs and rigidity hypotheses stated in the paper; no additional stationary or self-similar classification theorem is used implicitly.

Contents

1. Further Local Reductions for Rough Cores, Scale Collapse, and Cascades

This section records the remaining local reductions used to separate the rough-core, scale-collapse, multibubble, and same-point cascade alternatives. All statements are conditional analytic statements for represented Type II branches.

1.1. Vorticity-Controlled Rough-Core Alternative.

Let (V, P, a, b) be a repaired-gauge renormalized Navier–Stokes branch,

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0,$$

and set $\Omega = \nabla \times V$. Then, in distributions,

$$\partial_\tau \Omega + (V \cdot \nabla)\Omega - (\Omega \cdot \nabla)V = \nu\Delta\Omega + a(\tau)(2\Omega + y \cdot \nabla\Omega) + b(\tau) \cdot \nabla\Omega.$$

At the smooth approximation level, a localized enstrophy estimate contains

$$\int (\Omega \cdot \nabla V) \cdot \Omega\, \chi_R^2.$$

The standard bound is

$$\left| \int (\Omega \cdot \nabla V) \cdot \Omega\, \chi_R^2 \right| \le C\|\Omega\chi_R\|_{L^2}^{1/2} \|\nabla(\Omega\chi_R)\|_{L^2}^{3/2} \|\nabla V\|_{L^2(B_{2R})}.$$

After Young's inequality the differential inequality has a coefficient

$$C_\nu \|\nabla V\|_{L^2(B_{2R})}^4$$

in front of the localized enstrophy. Thus the vorticity equation does not by itself close a uniform enstrophy-window estimate without an independent gradient-window estimate. The vorticity argument below is therefore a conditional rough-core exclusion, while the primary rough-core exclusion remains the Caccioppoli estimate in Theorem 7.15.

Lemma 1.1 (Local div-curl estimate)**.** *For each $R > 0$ there is $C_R < \infty$ such that every $U \in L^2(B_{2R}; \mathbb{R}^3)$ satisfying*

$$\nabla \cdot U = 0 \quad \text{in } \mathcal{D}'(B_{2R})$$

and

$$\nabla \times U \in L^2(B_{2R}; \mathbb{R}^3)$$

belongs to $H^1(B_R; \mathbb{R}^3)$, with

$$\|\nabla U\|_{L^2(B_R)} \le C_R \left(\|\nabla \times U\|_{L^2(B_{2R})} + \|U\|_{L^2(B_{2R})} \right).$$

Proof. Choose $\chi \in C_c^\infty(B_{2R})$ with $\chi \equiv 1$ on B_R, and set

$$W = \chi U.$$

After extension by zero, $W \in L^2(\mathbb{R}^3)$ is compactly supported. Moreover,

$$\nabla \times W = \chi \nabla \times U + \nabla\chi \times U \in L^2(\mathbb{R}^3),$$

and, using $\nabla \cdot U = 0$,

$$\nabla \cdot W = \nabla\chi \cdot U \in L^2(\mathbb{R}^3).$$

For compactly supported distributions with W, $\nabla \times W$, and $\nabla \cdot W$ in L^2, the Fourier div-curl identity gives

$$\|\nabla W\|_{L^2(\mathbb{R}^3)}^2 = \|\nabla \times W\|_{L^2(\mathbb{R}^3)}^2 + \|\nabla \cdot W\|_{L^2(\mathbb{R}^3)}^2.$$

Indeed, Plancherel and

$$|\xi|^2 |\widehat{W}(\xi)|^2 = |\xi \times \widehat{W}(\xi)|^2 + |\xi \cdot \widehat{W}(\xi)|^2$$

yield the identity after integration in ξ. Hence $W \in H^1(\mathbb{R}^3)$. Since $W = U$ on B_R,

$$\|\nabla U\|_{L^2(B_R)} \le \|\nabla W\|_{L^2(\mathbb{R}^3)} \le C_R \left(\|\nabla \times U\|_{L^2(B_{2R})} + \|U\|_{L^2(B_{2R})}\right).$$

□

Lemma 1.2 (Critical L^3 control gives local L^2 control)**.** *If*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M,$$

then, for every $R > 0$*,*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|^2_{L^2(B_R)} \le CRM^2.$$

Proof. Hölder's inequality gives

$$\int_{B_R} |V(y,\tau)|^2 \, dy \le |B_R|^{1/3} \left(\int_{B_R} |V(y,\tau)|^3 \, dy\right)^{2/3} \le CRM^2.$$

Taking the supremum in τ proves the estimate. □

Proposition 1.3 (Vorticity-window control implies local windowed H^1)**.** *Assume V is a represented divergence-free branch,*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M,$$

$\Omega = \nabla \times V$ *in distributions, and*

$$\forall R > 0 : \qquad \sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|\Omega(\tau)\|^2_{L^2(B_{2R})} \, d\tau < \infty.$$

Then

$$\forall R > 0 : \qquad \sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_R)} \, d\tau < \infty.$$

Proof. Fix $R > 0$. By Theorem 1.1, for almost every τ,

$$\|\nabla V(\tau)\|^2_{L^2(B_R)} \le C_R \left(\|\Omega(\tau)\|^2_{L^2(B_{2R})} + \|V(\tau)\|^2_{L^2(B_{2R})}\right).$$

Integrating over $[T, T+1]$, the vorticity term is uniformly bounded by the assumed vorticity-window control, while the local L^2 term is uniformly bounded by Theorem 1.2:

$$\int_T^{T+1} \|V(\tau)\|^2_{L^2(B_{2R})} \, d\tau \le C_R M^2.$$

Therefore

$$\sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|\nabla V(\tau)\|^2_{L^2(B_R)} \, d\tau < \infty.$$

The same local L^2 bound gives

$$\sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|^2_{L^2(B_R)} \, d\tau < \infty,$$

and the asserted windowed H^1 bound follows. □

Corollary 1.4 (Vorticity-controlled rough-core exclusion)**.** *Let a represented Type II branch have positive finite critical mass and local vorticity-window control as in Theorem 1.3. Then the rough-core alternative, namely failure of local windowed H^1 control, cannot hold.*

Proof. The hypotheses give the local windowed H^1 bound by Theorem 1.3. This is the negation of the rough-core alternative. □

Consequently, a represented bounded-critical-norm rough-core branch must fail at least one of the following analytic hypotheses: local vorticity-window control, the representation or critical-mass hypotheses used before the vorticity argument, bounded modulation, Caccioppoli regularity, or one of their integrated refinements.

1.2. **Scale-Collapse Autonomous Reduction.** Let a represented repaired-gauge Type II branch be in the remaining scale-collapse class when it satisfies the representation hypotheses, positive finite critical mass, a nonzero localized L^2 core floor, bounded modulation, local windowed H^1 control, $\lambda(\tau) \to 0$, and the scale-collapse drift obstruction

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty, \qquad M_R(\tau) = \int |V(y,\tau)|^2 \phi_{R(\tau)}(y)\, dy.$$

The core floor supplies constants $m_0 > 0$, τ_1, and moving cutoffs such that

$$M_R(\tau) \geq m_0 \qquad \text{for a.e. } \tau \geq \tau_1.$$

This is the remaining scale-drift class; it is distinct from the radiative and rough-core alternatives.

Definition 1.5 (Thick negative-drift windows)**.** The branch has thick negative-drift windows if there exist $T > 0$, $c > 0$, and $\tau_n \to \infty$ such that

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) M_R(\tau)\, d\tau \geq c \qquad \text{for every } n.$$

If $M_R \leq M_1$ on these windows, then

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau)\, d\tau \geq \frac{c}{M_1}.$$

The divergence of $a_- M_R$ alone does not imply the existence of Theorem 1.5; a bounded nonnegative function can have infinite integral while its unit-window averages tend to zero. Without thick windows, the branch remains in the thin-drift alternative rather than entering a self-similar regime.

Definition 1.6 (Autonomous modulation limit)**.** On fixed windows $[\tau_n, \tau_n + T]$, the branch has an autonomous modulation limit (a_∞, b_∞) if, after translation,

$$a(\tau_n + \cdot) \to a_\infty, \qquad b(\tau_n + \cdot) \to b_\infty$$

in a coefficient topology strong enough to pass the products

$$a(V + y \cdot \nabla V), \qquad b \cdot \nabla V$$

to the distributional limit. Strong $L^1(0,T)$ convergence is sufficient. The parameter a_∞ may be negative, zero, or any value allowed by the bounded modulation range. The case $a_\infty < 0$ is a generalized self-similar regime, while $a_\infty = 0$ gives a stationary Navier–Stokes regime, possibly with constant translation drift b_∞.

Lemma 1.7 (Thick autonomous drift gives a negative scale parameter)**.** *Assume Theorem 1.5 holds on windows $[\tau_n, \tau_n + T]$, $M_R \leq M_1$ on those windows, and*

$$a(\tau_n + \cdot) \to a_\infty \quad \text{in } L^1(0,T).$$

Then

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. The upper mass bound and thick weighted drift give

$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)\,d\tau \geq \frac{c}{M_1}.$$

After translation and L^1 convergence of a to the constant a_∞, the left-hand side converges to $(-a_\infty)_+$. Hence

$$(-a_\infty)_+ \geq \frac{c}{M_1},$$

and $a_\infty \leq -c/M_1 < 0$. □

Definition 1.8 (Compactness on autonomous windows)**.** The branch has compactness on the windows (T, τ_n) if, for every ball B_R, the translated sequence

$$V_n(y,s) = V(y, \tau_n + s), \qquad 0 < s < T,$$

is precompact strongly in $L^2(0,T;L^q(B_R))$ for every $q < 6$, weakly in $L^2(0,T;H^1(B_R))$, and weak-* in

$$L^\infty(0,T;L^3(\mathbb{R}^3)).$$

The pressures are normalized modulo constants and converge in a topology sufficient to pass

$$-\Delta P = \partial_i\partial_j(V_iV_j)$$

to the local limit. The compactness, modulation convergence, and any thick drift condition are always imposed on the same window sequence τ_n.

Theorem 1.9 (Autonomous reduced-limit theorem)**.** *Assume the represented scale-collapse hypotheses above, compactness on autonomous windows, and an autonomous modulation limit* (a_∞, b_∞)*. Then, after passing to a subsequence, the translated branch converges to a nonzero weak solution* (U,Π) *on* $\mathbb{R}^3 \times (0,T)$ *of*

$$\partial_s U + (U\cdot\nabla)U + \nabla\Pi = \nu\Delta U + a_\infty(U + y\cdot\nabla U) + b_\infty\cdot\nabla U, \qquad \nabla\cdot U = 0.$$

Moreover,

$$U \in L^\infty(0,T;L^3(\mathbb{R}^3)),$$

the pressure satisfies the Navier–Stokes pressure reconstruction, and the localized core floor passes to the limit on a smaller compact cutoff. In particular $U \not\equiv 0$.

Proof. The weak-* $L^\infty L^3$ bound follows from the positive finite critical annulus. The local L^2H^1 bound and time-derivative compactness are part of the compactness hypothesis, so Aubin–Lions gives strong local convergence in L^2L^q for $q < 6$. This strong convergence, together with weak L^2H^1 convergence, passes the nonlinear term to $(U\cdot\nabla)U$ in distributions. The pressure convergence passes the Poisson reconstruction to Π.

The modulation-limit hypothesis passes the coefficient terms to

$$a_\infty(U + y\cdot\nabla U), \qquad b_\infty\cdot\nabla U.$$

Therefore the limit solves the autonomous reduced equation. Finally, the localized L^2 core floor gives positive localized mass on the selected branch. The compactness hypothesis includes convergence strong enough for this localized pairing, or for a compact subcutoff below it, to pass to the limit. Hence $U \not\equiv 0$. □

Definition 1.10 (Stationary omega-limit hypothesis)**.** For a parameter pair (a_∞, b_∞), the autonomous limit of Theorem 1.9 has a stationary omega-limit if there is a nonzero stationary profile (W,Q) satisfying

$$(W\cdot\nabla)W + \nabla Q = \nu\Delta W + a_\infty(W + y\cdot\nabla W) + b_\infty\cdot\nabla W, \qquad \nabla\cdot W = 0,$$

with

$$W \in L^3(\mathbb{R}^3),$$

and with the admissibility hypotheses required by the rigidity theorem to be invoked.

The stationary omega-limit hypothesis is an additional analytic hypothesis: an autonomous bounded trajectory need not be stationary merely by time translation. A Lyapunov or asymptotic-compactness argument is required to produce W.

Definition 1.11 (Rigidity-covered stationary class)**.** For (a_∞, b_∞), the stationary class is rigidity-covered if every admissible stationary solution of

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty(W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \qquad W \in L^3(\mathbb{R}^3),$$

is zero, or at least is not a terminal Type II blow-up branch in the class under consideration. In the classical backward self-similar normalization, after rescaling and constant translations, this is the role of the Nečas–Růžička–Šverák rigidity theorem. For forward self-similar classes, the required hypothesis is not nonexistence of profiles, but the narrower statement that the admissible forward profile is not a terminal Type II blow-up branch.

Theorem 1.12 (Conditional scale-collapse exclusion)**.** *Assume the represented scale-collapse hypotheses, compactness on autonomous windows, an autonomous modulation limit, a stationary omega-limit, and rigidity coverage for the corresponding parameter pair. Then the scale-collapse alternative is excluded for that branch.*

Proof. Theorem 1.9 gives a nonzero autonomous limit. The stationary omega-limit hypothesis yields a nonzero stationary profile (W, Q) in the admissible class for (a_∞, b_∞). Rigidity coverage says that no such nonzero profile can represent a terminal Type II branch in the class under consideration. This contradicts the assumption that the original branch remains in the remaining scale-collapse class. □

If, in addition, the branch is in the supercritical Type II scale alternative and the analytic implication from scale-collapse exclusion to that Type II alternative is assumed, then Theorem 1.12 excludes the corresponding Type II scale alternative.

The scale-collapse drift obstruction implies

$$\int_{\tau_0}^{\infty} a_-(\tau)\, d\tau = \infty$$

only when M_R is bounded above on the same moving cutoffs, because then

$$a_-(\tau) M_R(\tau) \leq M_1 a_-(\tau).$$

The local mass floor is used for nontriviality of limits, not for this unweighted-drift implication. Thus the drift obstruction proves sustained weighted contraction, but by itself does not prove convergence of $a(\tau)$ to a negative constant, nor uniformly negative fixed-window averages.

The scale-collapse alternatives are therefore:

(i) thick autonomous drift, compactness, modulation convergence, and a stationary omega-limit reduce the branch to a generalized self-similar stationary profile, which is excluded precisely in parameter ranges covered by rigidity;
(ii) thin or oscillatory drift prevents the self-similar rigidity reduction;
(iii) an autonomous limit with $a_\infty = 0$ gives the stationary Navier–Stokes equation with possible constant advection, hence a stationary $L^3(\mathbb{R}^3)$ Liouville problem unless an appropriate theorem is available.

The remaining analytic alternatives are thin drift, compactness failure, failure of modulation convergence, failure to obtain a stationary omega-limit, and absence of the relevant rigidity theorem.

1.3. **Local L^3-Compactness from L^2-Compactness.** We record a standard interpolation argument that converts local compactness in L^2, under uniform local L^6 control, into local compactness in L^3 for translated renormalized Navier–Stokes trajectories. Let $R > 0$, $I = (0,1)$, $\overline{I} = [0,1]$, and let the shifted trajectories be defined on $\overline{I} \times B_{2R}$ by

$$W_n(s,y) := V(\tau_n + s, y), \qquad (s,y) \in \overline{I} \times B_{2R},$$

where $\tau_n \to \infty$. All spacetime norms below are taken over I; the notation $W_n(0,\cdot)$ denotes the endpoint trace whenever that trace is used.

Lemma 1.13 (Spacetime compactness in $L^2_t L^2_x$)**.** *Assume*

$$\sup_n \|W_n\|_{L^2(I;H^1(B_{2R}))} < \infty$$

and

$$\sup_n \|\partial_s W_n\|_{L^2(I;H^{-1}(B_R))} < \infty.$$

Then there are a subsequence, not relabeled, and a limit

$$W \in L^2(I; L^2(B_R))$$

such that

$$W_n \to W \qquad \textit{strongly in } L^2(I; L^2(B_R)).$$

Proof. Restrict W_n to $I \times B_R$. The first hypothesis gives (W_n) bounded in $L^2(I; H^1(B_R))$, and the second gives $(\partial_s W_n)$ bounded in $L^2(I; H^{-1}(B_R))$. The embeddings

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-1}(B_R)$$

are, respectively, compact and continuous. The Aubin–Lions–Simon compactness theorem therefore implies that (W_n) is relatively compact in

$$L^2(I; L^2(B_R)).$$

After passing to a subsequence, there is

$$W \in L^2(I; L^2(B_R))$$

such that $W_n \to W$ strongly in this space. □

Lemma 1.14 (Spacetime compactness in $L^2_t L^3_x$)**.** *Assume the hypotheses of Lemma 1.13. Assume also*

$$\sup_n \|W_n\|_{L^2(I;L^6(B_{2R}))} < \infty.$$

Then, after passing to a subsequence if necessary,

$$W_n \to W \qquad \textit{strongly in } L^2(I; L^3(B_R)).$$

Proof. By the $L^2(I; L^6(B_{2R}))$ bound, the restricted sequence is bounded in the reflexive space $L^2(I; L^6(B_R))$. Hence, after passing to a subsequence,

$$W_n \rightharpoonup \widetilde{W} \qquad \text{weakly in } L^2(I; L^6(B_R)).$$

The same subsequence converges strongly to W in $L^2(I; L^2(B_R))$ by Lemma 1.13. To identify the weak $L^2_t L^6_x$ limit, let

$$\varphi \in C_c^\infty(I \times B_R; \mathbb{R}^3).$$

Strong $L^2_tL^2_x$ convergence gives

$$\int_I\int_{B_R} W_n\cdot\varphi\,dy\,ds\to\int_I\int_{B_R} W\cdot\varphi\,dy\,ds,$$

while weak convergence in $L^2_tL^6_x$ gives the same limit with $\widetilde{W}$ in place of W. Hence $W=\widetilde{W}$ in $\mathcal{D}'(I\times B_R)$, and therefore

$$W\in L^2(I;L^6(B_R)),$$

with

$$\|W_n-W\|_{L^2(I;L^6(B_R))}$$

uniformly bounded in n. Indeed,

$$\|W_n-W\|_{L^2(I;L^6(B_R))}\le\sup_k\|W_k\|_{L^2(I;L^6(B_R))}+\|W\|_{L^2(I;L^6(B_R))}.$$

The Gagliardo–Nirenberg interpolation inequality on the bounded domain B_R gives, for almost every $s\in I$,

$$\|W_n(s)-W(s)\|_{L^3(B_R)}\le C_R\|W_n(s)-W(s)\|^{1/2}_{L^2(B_R)}\|W_n(s)-W(s)\|^{1/2}_{L^6(B_R)}$$

because $W_n(s)-W(s)\in L^2(B_R)\cap L^6(B_R)$ for a.e. s. Squaring and integrating in s gives

$$\int_0^1\|W_n(s)-W(s)\|^2_{L^3(B_R)}\,ds\le C_R\int_0^1\|W_n(s)-W(s)\|_{L^2(B_R)}\|W_n(s)-W(s)\|_{L^6(B_R)}\,ds.$$

Hölder's inequality in time, applied to the product of the $L^2(B_R)$- and $L^6(B_R)$-norms, yields

$$\|W_n-W\|^2_{L^2(I;L^3(B_R))}\le C_R\|W_n-W\|_{L^2(I;L^2(B_R))}\|W_n-W\|_{L^2(I;L^6(B_R))}.$$

The first factor tends to zero by Lemma 1.13, while the second factor is uniformly bounded. Therefore the left-hand side tends to zero. □

Lemma 1.15 (Fixed-time L^3 convergence from fixed-time L^2 convergence)**.** *Assume that, after passing to a subsequence, the time slices satisfy*

$$W_n(0,\cdot)\to f\qquad\textit{strongly in } L^2(B_R)$$

and

$$\sup_n\|W_n(0,\cdot)\|_{L^6(B_R)}<\infty.$$

Then

$$W_n(0,\cdot)\to f\qquad\textit{strongly in } L^3(B_R).$$

Proof. First we show that the strong L^2 limit belongs to $L^6(B_R)$. Since $L^6(B_R)$ is reflexive, the uniform $L^6(B_R)$ bound gives a subsequence and a function $g\in L^6(B_R)$ such that

$$W_n(0,\cdot)\rightharpoonup g\qquad\text{weakly in } L^6(B_R).$$

For every $\phi\in C_c^\infty(B_R)$, the strong $L^2(B_R)$ convergence also gives

$$\int_{B_R} W_n(0,y)\cdot\phi(y)\,dy\to\int_{B_R} f(y)\cdot\phi(y)\,dy.$$

The weak L^6 limit gives the same distributional limit with g in place of f, so $g=f$. Thus $f\in L^6(B_R)$. Returning to the original sequence, the triangle inequality gives

$$\sup_n\|W_n(0,\cdot)-f\|_{L^6(B_R)}\le\sup_n\|W_n(0,\cdot)\|_{L^6(B_R)}+\|f\|_{L^6(B_R)}<\infty.$$

Interpolating between $L^2(B_R)$ and $L^6(B_R)$ at the fixed time $s=0$,

$$\|W_n(0,\cdot)-f\|_{L^3(B_R)}\le C_R\|W_n(0,\cdot)-f\|^{1/2}_{L^2(B_R)}\|W_n(0,\cdot)-f\|^{1/2}_{L^6(B_R)}.$$

The L^6 factor is uniformly bounded and the L^2 factor tends to zero by hypothesis. Hence the $L^3(B_R)$ norm tends to zero. □

Remark 1.16 (Limitation at a fixed time). The preceding two spacetime compactness lemmas do not by themselves imply convergence at the fixed time $s = 0$. The fixed-time conclusion in Lemma 1.15 requires a separate L^2 compactness hypothesis for the traces. One sufficient condition is

$$\sup_n \|W_n\|_{L^\infty(I;H^1(B_R))} < \infty$$

and, for some $q > 1$,

$$\sup_n \|\partial_s W_n\|_{L^q(I;H^{-1}(B_R))} < \infty.$$

Simon's compactness theorem then gives precompactness of (W_n) in

$$C([0,1];L^2(B_R)),$$

and hence the fixed-time L^2 convergence after subsequence extraction. Without such an additional condition, fixed-time convergence at $s = 0$ is not a consequence of Aubin–Lions compactness in $L^2(I;L^2(B_R))$.

Lemma 1.17 (Time selection from averaged localized dissipation and local H^1 control)**.** *Let $J_n = [s_n, s_n + \ell]$ be time intervals with fixed length $\ell > 0$, and let $\delta_n \downarrow 0$. Assume that $\widetilde{\mathfrak{D}}_{R_0}$ is a nonnegative measurable function and, for a fixed radius R, that*

$$\int_{J_n} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau \le \delta_n$$

and

$$\int_{J_n} \|V(\tau)\|^2_{H^1(B_R)}\, d\tau \le C_R.$$

Then there exist times $\sigma_n \in J_n$ such that

$$\widetilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0, \qquad \|V(\sigma_n)\|_{H^1(B_R)} \le \left(\frac{2C_R}{\ell}\right)^{1/2}.$$

Consequently, after passing to a subsequence,

$$V(\sigma_n) \to V_* \qquad \text{strongly in } L^2(B_R)$$

and

$$V(\sigma_n) \to V_* \qquad \text{strongly in } L^3(B_R).$$

Proof. Define

$$E_n := \left\{\tau \in J_n : \widetilde{\mathfrak{D}}_{R_0}(\tau) \le \delta_n^{1/2}\right\}.$$

By Chebyshev's inequality,

$$|J_n \setminus E_n| \le \delta_n^{1/2}.$$

Indeed,

$$\delta_n^{1/2}|J_n \setminus E_n| \le \int_{J_n\setminus E_n} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau \le \delta_n.$$

For all sufficiently large n, $\delta_n^{1/2} \le \ell/2$, and hence $|E_n| \ge \ell/2$. Since

$$\int_{E_n} \|V(\tau)\|^2_{H^1(B_R)}\, d\tau \le C_R,$$

there exists $\sigma_n \in E_n$ such that

$$\|V(\sigma_n)\|^2_{H^1(B_R)} \le \frac{2C_R}{\ell}.$$

For otherwise the integral over E_n would be strictly larger than $(2C_R/\ell)|E_n| \geq C_R$, contradicting the displayed bound. Because $\sigma_n \in E_n$,

$$\widetilde{\mathfrak{D}}_{R_0}(\sigma_n) \leq \delta_n^{1/2} \to 0.$$

The estimate on $\|V(\sigma_n)\|_{H^1(B_R)}$ gives a bounded sequence in $H^1(B_R)$. By weak compactness in $H^1(B_R)$, after passing to a subsequence there is $V_* \in H^1(B_R)$ such that

$$V(\sigma_n) \rightharpoonup V_* \qquad \text{weakly in } H^1(B_R).$$

The Rellich compactness theorem, after passing to a further subsequence if necessary, gives the corresponding strong convergence

$$V(\sigma_n) \to V_* \qquad \text{strongly in } L^2(B_R).$$

The same $H^1(B_R)$ bound and the Sobolev embedding

$$H^1(B_R) \hookrightarrow L^6(B_R)$$

give a uniform $L^6(B_R)$ bound for $V(\sigma_n)$. Since $V_* \in H^1(B_R)$, the same embedding gives $V_* \in L^6(B_R)$, and $\|V(\sigma_n) - V_*\|_{L^6(B_R)}$ is uniformly bounded by the triangle inequality. Interpolation gives

$$\|V(\sigma_n) - V_*\|_{L^3(B_R)} \leq C_R \|V(\sigma_n) - V_*\|_{L^2(B_R)}^{1/2} \|V(\sigma_n) - V_*\|_{L^6(B_R)}^{1/2}.$$

The L^6 factor is uniformly bounded and the L^2 factor tends to zero. This proves strong convergence in $L^3(B_R)$. □

Remark 1.18 (Role of the time selection). Lemma 1.17 does not require endpoint trace compactness. Instead of first fixing times τ_n and then proving compactness of $V(\tau_n)$, one selects times of small localized dissipation inside intervals where the same intervals also carry local H^1 control. This supplies the fixed-time L^2-compactness hypothesis in Lemma 1.15 whenever the argument provides common intervals with these two properties.

Corollary 1.19 (Compactness at translated renormalized times)**.** *Assume that, for each fixed radius R, the shifted sequence satisfies the hypotheses of Lemma 1.13 and the time slices satisfy the two hypotheses of Lemma 1.15. Then, after passing to a subsequence,*

$$V(\tau_n) \to V_* \qquad \textit{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3),$$

where V_* is the local L^2 limit of the time slices $V(\tau_n)$.

Proof. Apply Lemma 1.15 first with $R = 1$ and pass to a subsequence along which $V(\tau_n)$ converges strongly in $L^3(B_1)$. Apply the same argument to that subsequence with $R = 2$, and continue inductively. The diagonal subsequence converges strongly in $L^3(B_m)$ for every integer $m \geq 1$. The limits on nested balls agree because the convergence in $L^3(B_m)$ restricts to convergence in $L^3(B_k)$ whenever $k < m$. Hence these local limits define a single function $V_* \in L^3_{\mathrm{loc}}(\mathbb{R}^3)$, and

$$V(\tau_n) \to V_* \qquad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3)$$

along the diagonal subsequence. □

Remark 1.20 (Role in the compactness argument). The compactness argument uses the preceding statements as follows:

(i) a Caccioppoli-type estimate gives the $L^2(I; H^1)$ bound in Lemma 1.13;
(ii) a time-regularity estimate gives the $\partial_s W_n$ bound;
(iii) a local L^6-type estimate gives the hypothesis of Lemma 1.14 and fixed-time L^6 boundedness;
(iv) Lemmas 1.13 and 1.14 give spacetime L^3 compactness;
(v) an additional fixed-time L^2-compactness hypothesis gives Lemma 1.15;

(vi) Lemma 1.15 gives the fixed-time strong L^3-compactness used in vanishing-dissipation contradiction arguments.

This is the local compactness step needed to pass from renormalized L^2_{loc} compactness to the critical local L^3 topology at the selected times.

Corollary 1.21 (Local compactness from a common sequence of selected times)**.** *Assume there is a single sequence of selected times σ_n such that, for every fixed radius R,*

$$\sup_n \|V(\sigma_n)\|_{H^1(B_R)} < \infty$$

and

$$\widetilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0.$$

Then, after diagonal subsequence extraction,

$$V(\sigma_n) \to V_* \qquad \textit{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3).$$

For each fixed radius, Lemma 1.17 supplies a radius-dependent sequence from intervals with local H^1 control. If the time selection is made on common intervals for an exhaustion of balls, then Lemma 1.17 replaces the separate fixed-time compactness hypothesis in Lemma 1.15.

Proof. Fix an integer $m \geq 1$. The uniform $H^1(B_m)$ bound gives a subsequence and a limit $V_*^{(m)} \in H^1(B_m)$ such that

$$V(\sigma_n) \rightharpoonup V_*^{(m)} \qquad \text{weakly in } H^1(B_m).$$

Rellich compactness gives, after passing to a further subsequence,

$$V(\sigma_n) \to V_*^{(m)} \qquad \text{strongly in } L^2(B_m),$$

after relabelling the subsequence. The Sobolev embedding $H^1(B_m) \hookrightarrow L^6(B_m)$ gives a uniform $L^6(B_m)$ bound, and the limit $V_*^{(m)}$ also belongs to $L^6(B_m)$. Therefore

$$\|V(\sigma_n) - V_*^{(m)}\|_{L^6(B_m)}$$

is uniformly bounded by the triangle inequality. The interpolation inequality

$$\|V(\sigma_n) - V_*^{(m)}\|_{L^3(B_m)} \leq C_m \|V(\sigma_n) - V_*^{(m)}\|_{L^2(B_m)}^{1/2} \|V(\sigma_n) - V_*^{(m)}\|_{L^6(B_m)}^{1/2}$$

then gives strong convergence in $L^3(B_m)$. Repeating this construction for $m = 1, 2, \dots$ and taking a diagonal subsequence gives strong convergence on every B_m. The nested limits agree by restriction, so they define a single $V_* \in L^3_{\mathrm{loc}}(\mathbb{R}^3)$, and the diagonal subsequence converges to V_* strongly in $L^3_{\mathrm{loc}}(\mathbb{R}^3)$. □

1.4. **Reduction to Multibubble Alternatives.** Let the structural Type II hypotheses consist of exhaustion of the local Type II alternatives, repaired-gauge representation, positive finite critical mass, pressure reconstruction, bounded repaired-gauge modulation, Caccioppoli regularity, and the cost comparison used for the compact Type II criterion. If one of these hypotheses fails, the branch is classified by the corresponding analytic failure rather than by a multibubble alternative. If, in addition, the Navier–Stokes implication from compact scale exclusion to the relevant Type II scale alternative is included, the compact conclusion excludes that alternative.

Under these structural hypotheses, the following non-multibubble strata are excluded under separate analytic hypotheses.

(i) Far radiation consists of radiative critical mass whose physical radius stays away from the blow-up point:

$$\lambda(\tau_n)R_n \geq r_* > 0$$

along a radiative annular sequence $R_n \to \infty$. Under a single-point blow-up localization hypothesis, this mass lies in a physically regular exterior region. An exterior-regularity hypothesis asserts that such mass can be separated from the concentrating Type II core and does not count as a core-radiation alternative.

(ii) The rough-core stratum is excluded by the Caccioppoli implication

$$\text{representation}, \quad \text{bounded critical norm}, \quad \text{pressure reconstruction},$$
$$\text{bounded modulation}, \quad \text{Caccioppoli regularity}$$

which yields local windowed H^1 control.

(iii) The scale-collapse drift stratum is excluded whenever the scale-collapse argument extracts a nonzero stationary L^3 profile in a Nečas–Růžička–Šverák rigidity-covered self-similar class. The nonzero conclusion uses the same positive critical-mass and tightness mechanism as the compact Type II criterion.

Definition 1.22 (Same-point secondary concentration)**.** Same-point secondary concentration means that at least two concentration scales occur at the same physical blow-up point:

$$\lambda_1(t) \ll \lambda_2(t) \to 0.$$

In the primary renormalized coordinates, the secondary bubble appears as radiation at radius

$$R(t) = \frac{\lambda_2(t)}{\lambda_1(t)} \to \infty.$$

Regular strict cascades of this form are excluded by Theorem 1.32, and all same-point cascades are excluded under the nonlinear decoupling hypothesis of Theorem 1.35. Before that hypothesis is imposed, the remaining same-point alternative is precisely the failure of nonlinear no-splitting/profile compatibility for scale-separated profiles.

Definition 1.23 (Multi-point concentration)**.** Multi-point concentration means that at least two physical concentration centers are active. In a renormalization centered at one bubble, the other bubble appears as an escaping profile. This case is excluded by a separated-point decoupling hypothesis reducing each active physical point to its own single-center branch. Before that hypothesis is imposed, the remaining multi-point alternative is failure of separated-point decoupling.

Theorem 1.24 (Reduction to multibubble alternatives)**.** *Assume an admissible Navier–Stokes Type II branch satisfies the structural Type II hypotheses above, the single-point blow-up localization and exterior-regular removal hypotheses for far radiation, and the scale-collapse rigidity hypothesis for scale-collapse drift. If the branch is not a multibubble branch in either the same-point scale-separated form of Theorem 1.22 or the multi-point form of Theorem 1.23, then the compact Type II scale-exclusion conclusion holds. If the Navier–Stokes implication from this scale exclusion to the relevant Type II scale alternative is also included, then that Type II scale alternative is excluded.*

Proof. The structural hypotheses supply representation, critical-mass control, pressure reconstruction, bounded modulation, Caccioppoli regularity, and cost comparison. If the branch is globally L^3-tight, the Caccioppoli estimate gives local windowed H^1 control. Together with positive finite critical mass and tightness, the compact Type II criterion gives infinite localized renormalization cost, and the cost comparison gives the Type II scale-exclusion condition.

If uniform compact-cylinder tightness fails, the radiative branch is present. Far radiation is excluded by the single-point blow-up localization and exterior-regular removal hypotheses, because it lies in a physically smooth exterior region and is separated from the concentrating Type II core. Regular strict same-point cascades are excluded by Theorem 1.32, and all same-point cascades are excluded under the nonlinear same-point decoupling hypothesis. Separated-point multibubbles are excluded under the separated-point decoupling hypothesis. If the branch is assumed not to be in either multibubble form, no radiative or noncompact alternative remains.

If scale-collapse drift occurs, the scale-collapse rigidity hypothesis extracts a nonzero L^3 stationary profile in a rigidity-covered self-similar class. Rigidity forces this profile to vanish, contradicting nonzero extraction. The scale-collapse drift alternative is therefore excluded.

The remaining rough-core branch is excluded by the Caccioppoli implication, because the structural hypotheses include the pressure, modulation, and regularity hypotheses required for local windowed H^1 control. Thus all non-multibubble alternatives are exhausted, and the compact Type II scale-exclusion conclusion holds. The final Type II scale conclusion follows from the corresponding Navier–Stokes implication when that hypothesis is included. □

Thus, after the structural hypotheses, same-point regular cascade estimates, and separated-point reductions are supplied, the only remaining multibubble alternatives are failure of same-point nonlinear decoupling and failure of separated-point decoupling. Once nonlinear profile evolution and terminal active-frame compactness provide those decouplings, a multibubble failure is an earlier failure of profile completeness, representation, Caccioppoli regularity, or scale-collapse rigidity, not an additional singularity class.

1.5. **Same-Point Multiscale Cascade Exclusion.** For $\lambda > 0$, $x_0 \in \mathbb{R}^3$, and $\phi \in L^3(\mathbb{R}^3)$, define

$$(\Lambda_{\lambda,x_0}\phi)(x) = \lambda^{-1}\phi\left(\frac{x-x_0}{\lambda}\right).$$

Then

$$\|\Lambda_{\lambda,x_0}\phi\|_{L^3(\mathbb{R}^3)} = \|\phi\|_{L^3(\mathbb{R}^3)}.$$

Two parameter sequences (λ_i^n, x_i^n) and (λ_j^n, x_j^n) are asymptotically orthogonal if

$$\frac{\lambda_i^n}{\lambda_j^n} + \frac{\lambda_j^n}{\lambda_i^n} + \frac{|x_i^n - x_j^n|}{\max(\lambda_i^n, \lambda_j^n)} \longrightarrow \infty.$$

Lemma 1.25 (Finite L^3 decoupling). *Let $K < \infty$, let $\phi^1, \dots, \phi^K \in L^3(\mathbb{R}^3)$, and set*

$$g_n^j = \Lambda_{\lambda_j^n, x_j^n}\phi^j.$$

If the parameter sequences are pairwise asymptotically orthogonal, then

$$\left\|\sum_{j=1}^K g_n^j\right\|_{L^3}^3 = \sum_{j=1}^K \|\phi^j\|_{L^3}^3 + o_n(1).$$

Proof. It suffices first to prove the statement for $\phi^j \in C_c^\infty(\mathbb{R}^3)$. The general L^3 case follows by density. For each $\varepsilon > 0$, choose $\psi^j \in C_c^\infty$ with

$$\|\phi^j - \psi^j\|_3 < \varepsilon.$$

Since each $\Lambda_{\lambda_j^n, x_j^n}$ is an L^3 isometry,

$$\left\|\sum_{j=1}^K \Lambda_{\lambda_j^n, x_j^n}(\phi^j - \psi^j)\right\|_3 \le K\varepsilon.$$

On bounded L^3 sets, $f \mapsto \|f\|_3^3$ is locally Lipschitz:

$$\big|\|f\|_3^3 - \|h\|_3^3\big| \le C(\|f\|_3 + \|h\|_3)^2 \|f-h\|_3.$$

Thus the smooth compactly supported case implies the general case after $\varepsilon \downarrow 0$.

Assume now $\phi^j \in C_c^\infty$. The proof is by induction on K. The case $K = 1$ is the scaling invariance of the L^3 norm. Assume the result for $K-1$, and write

$$G_n = \sum_{j=1}^{K-1} \Lambda_{\lambda_j^n, x_j^n} \phi^j, \qquad g_n^K = \Lambda_{\lambda_K^n, x_K^n} \phi^K.$$

Applying the inverse scaling of the K-th profile gives

$$(\Lambda^{-1}_{\lambda_K^n, x_K^n} G_n)(y) = \sum_{j=1}^{K-1} \frac{\lambda_K^n}{\lambda_j^n} \phi^j \left(\frac{\lambda_K^n y + x_K^n - x_j^n}{\lambda_j^n} \right).$$

For each $j < K$, asymptotic orthogonality gives, along a subsequence, one of the following alternatives:

$$\lambda_K^n / \lambda_j^n \to 0, \qquad \lambda_j^n / \lambda_K^n \to 0,$$

or the scale ratio stays bounded above and below while

$$|x_K^n - x_j^n| / \lambda_j^n \to \infty.$$

In the first case the displayed term is bounded pointwise by

$$\frac{\lambda_K^n}{\lambda_j^n} \|\phi^j\|_\infty \to 0.$$

In the second case, if $\operatorname{supp} \phi^j \subset B_A$, the set on which the term is nonzero is

$$E_n^j = \left\{ y : \left| \frac{\lambda_K^n y + x_K^n - x_j^n}{\lambda_j^n} \right| \le A \right\}.$$

Since $\lambda_j^n / \lambda_K^n \to 0$, the diameter of E_n^j is

$$O(\lambda_j^n / \lambda_K^n) \to 0.$$

Thus $\mathbf{1}_{E_n^j} \to 0$ pointwise outside at most one limiting point after passing to a subsequence, and the term converges to zero a.e. In the third case, the spatial translation tends to infinity in the rescaled variables; hence for each fixed compact B_R, the argument of ϕ^j lies outside $\operatorname{supp} \phi^j$ for all large n. Therefore

$$\Lambda^{-1}_{\lambda_K^n, x_K^n} G_n \to 0 \quad \text{a.e. on } \mathbb{R}^3.$$

The sequence is bounded in L^3, because the inverse scaling is an L^3 isometry and the sum is finite. The Brézis–Lieb lemma, applied to

$$\phi^K + \Lambda^{-1}_{\lambda_K^n, x_K^n} G_n,$$

gives

$$\left\| \phi^K + \Lambda^{-1}_{\lambda_K^n, x_K^n} G_n \right\|_3^3 = \|\phi^K\|_3^3 + \left\| \Lambda^{-1}_{\lambda_K^n, x_K^n} G_n \right\|_3^3 + o_n(1).$$

Using the L^3 isometry again,

$$\|G_n + g_n^K\|_3^3 = \|\phi^K\|_3^3 + \|G_n\|_3^3 + o_n(1).$$

The induction hypothesis gives

$$\|G_n\|_3^3 = \sum_{j=1}^{K-1} \|\phi^j\|_3^3 + o_n(1),$$

and the result follows for K. □

Definition 1.26 (Admissible same-point critical cascade)**.** Let $u_n = u(\cdot, t_n)$, $t_n \uparrow T^*$. A same-point critical cascade decomposition at x_* consists of profiles $\{\phi^j\}_{j\in\mathcal{J}} \subset L^3(\mathbb{R}^3)$, scales $\lambda_j^n \downarrow 0$, centers $x_j^n \to x_*$, and remainders r_n^J such that, for every finite $J \subset \mathcal{J}$,

$$u_n = \sum_{j\in J} \Lambda_{\lambda_j^n, x_j^n} \phi^j + r_n^J.$$

The decomposition is admissible if:

(i) the parameters are pairwise asymptotically orthogonal:

$$\frac{\lambda_i^n}{\lambda_j^n} + \frac{\lambda_j^n}{\lambda_i^n} + \frac{|x_i^n - x_j^n|}{\max(\lambda_i^n, \lambda_j^n)} \to \infty \qquad (i \neq j);$$

(ii) critical L^3 decoupling with remainder holds:

$$\|u_n\|_3^3 = \sum_{j\in J} \|\phi^j\|_3^3 + \|r_n^J\|_3^3 + o_n(1);$$

(iii) there is $\eta > 0$ such that

$$\|\phi^j\|_3 \geq \eta \qquad \text{for every } j \in \mathcal{J}.$$

Theorem 1.27 (Uniform bound on cascade length)**.** *Assume*

$$\sup_n \|u_n\|_{L^3(\mathbb{R}^3)} \leq M < \infty.$$

If u_n admits an admissible same-point critical cascade decomposition and every active profile satisfies

$$\|\phi^j\|_3 \geq \eta > 0,$$

then the number of active profiles is finite and

$$\#\mathcal{J} \leq \left\lfloor \frac{M^3}{\eta^3} \right\rfloor.$$

Proof. Let $J \subset \mathcal{J}$ be finite. By L^3 decoupling with remainder,

$$\|u_n\|_3^3 = \sum_{j\in J} \|\phi^j\|_3^3 + \|r_n^J\|_3^3 + o_n(1).$$

The remainder contribution is nonnegative, hence

$$\limsup_{n\to\infty} \|u_n\|_3^3 \geq \sum_{j\in J} \|\phi^j\|_3^3.$$

Since $\|u_n\|_3 \leq M$,

$$M^3 \geq \sum_{j\in J} \|\phi^j\|_3^3 \geq (\#J)\eta^3.$$

Thus

$$\#J \leq \frac{M^3}{\eta^3}$$

for every finite $J \subset \mathcal{J}$. If $\mathcal{J}$ had more than $\lfloor M^3/\eta^3 \rfloor$ elements, choosing

$$\#J = \left\lfloor \frac{M^3}{\eta^3} \right\rfloor + 1$$

would contradict the preceding inequality. $\square$

Definition 1.28 (Regular strict same-point cascade)**.** Assume there are $J \geq 2$ active profiles ordered by scale:

$$\lambda_1^n \ll \lambda_2^n \ll \cdots \ll \lambda_J^n, \qquad \lambda_j^n \to 0, \qquad x_j^n \to x_*.$$

Set

$$\mu_j^n = \frac{\lambda_1^n}{\lambda_j^n} \to 0 \qquad (j \geq 2).$$

The cascade is regular in the innermost coordinates if, after passing to a subsequence:

(i)
$$\xi_j^n = \frac{x_* - x_j^n}{\lambda_j^n} \longrightarrow \xi_j \in \mathbb{R}^3 \qquad (j \geq 2);$$

(ii) each outer profile is C^2 near ξ_j;

(iii) for every $R < \infty$,
$$\sup_{|z-\xi_j| \leq R\mu_j^n} \left(|\phi^j(z)| + |\nabla \phi^j(z)| + |\nabla^2 \phi^j(z)|\right) < \infty$$
uniformly for all large n;

(iv) the innermost repaired-gauge representation is available:
$$V_1^n(y,\tau) = \lambda_1(t) u(x_*(t) + \lambda_1(t) y, t)$$
and solves the repaired-gauge Navier–Stokes equation on compact (y,τ)-cylinders, modulo the outer-bubble perturbation.

If $|\xi_j^n| \to \infty$, the corresponding outer bubble is locally invisible in the innermost variables by Theorem 1.33, provided the pressure interaction satisfies the decoupling hypothesis stated below.

Lemma 1.29 (Inner expansion of regular outer bubbles)**.** *For $j \geq 2$, define*

$$W_j^n(y) = \lambda_1^n (\lambda_j^n)^{-1} \phi^j \left(\frac{\lambda_1^n y + x_* - x_j^n}{\lambda_j^n} \right) = \mu_j^n \phi^j (\xi_j^n + \mu_j^n y).$$

For every fixed $R < \infty$,

$$W_j^n(y) = \mu_j^n \phi^j(\xi_j) + (\mu_j^n)^2 D\phi^j(\xi_j) y + \mathcal{R}_j^n(y) \qquad \text{on } B_R,$$

where

$$\|\mathcal{R}_j^n\|_{L^\infty(B_R)} \leq C_{j,R} \left((\mu_j^n)^3 + \mu_j^n |\xi_j^n - \xi_j|^2 + (\mu_j^n)^2 |\xi_j^n - \xi_j| \right) = o(\mu_j^n) + O((\mu_j^n)^3).$$

In particular,

$$W_j^n = c_j^n + A_j^n y + o_{L^\infty(B_R)}(\mu_j^n),$$

with

$$c_j^n = \mu_j^n \phi^j(\xi_j), \qquad A_j^n = (\mu_j^n)^2 D\phi^j(\xi_j), \qquad \|A_j^n\| = O((\mu_j^n)^2).$$

Proof. Fix R. For $y \in B_R$, Taylor's theorem at ξ_j gives

$$\phi^j(\xi_j^n + \mu_j^n y) = \phi^j(\xi_j) + D\phi^j(\xi_j)(\xi_j^n - \xi_j + \mu_j^n y) + \frac{1}{2}(\xi_j^n - \xi_j + \mu_j^n y)^T D^2\phi^j(\zeta_{j,n,y})(\xi_j^n - \xi_j + \mu_j^n y),$$

where $\zeta_{j,n,y}$ lies on the segment between ξ_j and $\xi_j^n + \mu_j^n y$. Multiplying by μ_j^n yields

$$W_j^n(y) = \mu_j^n \phi^j(\xi_j) + \mu_j^n D\phi^j(\xi_j)(\xi_j^n - \xi_j) + (\mu_j^n)^2 D\phi^j(\xi_j)y + O\left(\mu_j^n |\xi_j^n - \xi_j + \mu_j^n y|^2\right).$$

The term $\mu_j^n D\phi^j(\xi_j)(\xi_j^n - \xi_j)$ is independent of y and may be absorbed into the constant part if one uses

$$c_j^n = \mu_j^n \phi^j(\xi_j^n).$$

With the displayed choice $c_j^n = \mu_j^n \phi^j(\xi_j)$, it is part of the remainder. Since $\xi_j^n \to \xi_j$ and $\mu_j^n \to 0$, the stated estimate follows. □

For the dynamical reduction it is useful to use the exact constant

$$c_j^n = \mu_j^n \phi^j(\xi_j^n),$$

so that

$$W_j^n(y) = c_j^n + (\mu_j^n)^2 D\phi^j(\xi_j^n)y + O_{L^\infty(B_R)}((\mu_j^n)^3 R^2).$$

Since Theorem 1.27 gives $J < \infty$,

$$c_{\mathrm{out}}^n = \sum_{j=2}^{J} c_j^n = O\left(\sum_{j=2}^{J} \mu_j^n\right) \to 0$$

on every fixed cascade sequence, provided the profiles are locally bounded at their offsets.

Lemma 1.30 (Absorption of constant ambient velocity)**.** *Suppose on a compact cylinder*

$$V = V_{\mathrm{in}} + c(\tau) + S,$$

where $c(\tau)$ is independent of y and S is a perturbation. Ignoring S, the equation for V_{in} is

$$\partial_\tau V_{\mathrm{in}} + (V_{\mathrm{in}} \cdot \nabla)V_{\mathrm{in}} + \nabla P_{\mathrm{in}} = \nu\Delta V_{\mathrm{in}} + a(V_{\mathrm{in}} + y \cdot \nabla V_{\mathrm{in}}) + (b - c) \cdot \nabla V_{\mathrm{in}},$$

modulo spatially constant forcing terms, which are absorbed by the moving translation gauge. Thus $b_{\mathrm{eff}} = b - c$.

Proof. Since c is independent of y,

$$(V_{\mathrm{in}} + c) \cdot \nabla(V_{\mathrm{in}} + c) = (V_{\mathrm{in}} \cdot \nabla)V_{\mathrm{in}} + c \cdot \nabla V_{\mathrm{in}}.$$

Terms depending only on τ, such as $\partial_\tau c - ac$, are spatially constant forces and are absorbed in the equivalent translation-gauge description. With the stated sign convention, the effective translation parameter is $b - c$. □

The leading constant part of every outer bubble in the inner coordinates is absorbed into b_{eff}. Since

$$|c_{\mathrm{out}}^n| \le \sum_{j=2}^{J} \mu_j^n |\phi^j(\xi_j^n)|,$$

finite cascade length and local boundedness of the profiles imply

$$\sup_n |c_{\mathrm{out}}^n| < \infty, \qquad c_{\mathrm{out}}^n \to 0.$$

Thus bounded modulation is preserved whenever the unperturbed repaired-gauge b-parameter is bounded.

Definition 1.31 (Perturbative scale-collapse robustness)**.** Let E_n be the remaining inner-coordinate error after subtracting the innermost bubble and absorbing c^n_{out} into b_{eff}. It contains the linear shear terms

$$\sum_{j=2}^{J} (\mu_j^n)^2 D\phi^j(\xi_j^n)y,$$

Taylor remainders of size $O((\mu_j^n)^3)$ on fixed balls, the rescaled critical remainder, and pressure cross-terms generated by the interaction of the innermost profile with the outer perturbation. Perturbative scale-collapse robustness means that on every compact cylinder $B_R \times [\tau_0, \tau_0 + L]$:

(i) $E_n \to 0$ in $L^1_\tau H^{-1}_y(B_R)$, or in a stronger topology sufficient for the scale-collapse compactness passage;
(ii) the pressure generated by cross-terms converges to zero in the pressure topology used by the repaired-gauge and scale-collapse arguments;
(iii) local compactness, autonomous modulation, and stationary-limit extraction in the scale-collapse limit are stable under adding $E_n \to 0$;
(iv) the limiting stationary equation is the Nečas–Růžička–Šverák-covered self-similar profile equation, not a perturbed equation.

Under Theorem 1.31, the innermost bubble gives a single-bubble scale-collapse branch with bounded modulation and positive critical mass.

Theorem 1.32 (Regular same-point cascade exclusion)**.** *Let u be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T^*)$ with finite initial energy and*

$$\sup_{t<T^*} \|u(t)\|_{L^3(\mathbb{R}^3)} \le M < \infty.$$

Let $t_n \uparrow T^*$ be a concentrating sequence at x_*. Assume:

(i) *$u_n = u(\cdot, t_n)$ admits an admissible same-point critical cascade decomposition;*
(ii) *every active bubble satisfies $\|\phi^j\|_3 \ge \eta > 0$;*
(iii) *the active bubbles form a regular strict same-point cascade;*
(iv) *the innermost scale admits the repaired-gauge representation with bounded modulation, pressure reconstruction, Caccioppoli regularity, and positive finite critical mass;*
(v) *perturbative scale-collapse robustness holds;*
(vi) *the scale-collapse extraction of a nonzero limiting profile and Nečas–Růžička–Šverák rigidity hold for the innermost branch.*

Then the cascade cannot occur.

Proof. By Theorem 1.27, the number of active bubbles is finite:

$$J \le \left\lfloor \frac{M^3}{\eta^3} \right\rfloor.$$

Let λ_1^n be the smallest scale, and for $j \ge 2$ set

$$\mu_j^n = \lambda_1^n / \lambda_j^n \to 0.$$

Theorem 1.29 gives, on every fixed B_R,

$$\lambda_1^n (\lambda_j^n)^{-1} \phi^j \left(\frac{\lambda_1^n y + x_* - x_j^n}{\lambda_j^n} \right) = c_j^n + (\mu_j^n)^2 D\phi^j(\xi_j^n) y + O_{L^\infty(B_R)}((\mu_j^n)^3 R^2).$$

Summing over $2 \le j \le J$, the constant part

$$c^n_{\mathrm{out}} = \sum_{j=2}^{J} c^n_j$$

is bounded and tends to zero. By Theorem 1.30, it is absorbed into the repaired-gauge translation parameter:

$$b^n_{\mathrm{eff}} = b^n - c^n_{\mathrm{out}}.$$

The bounded-modulation hypothesis for b^n, together with boundedness of c^n_{out}, gives

$$\sup_n |b^n_{\mathrm{eff}}| < \infty.$$

The nonconstant outer contribution is

$$\sum_{j=2}^{J} (\mu^n_j)^2 D\phi^j(\xi^n_j) y + O_{L^\infty(B_R)}\left(\sum_{j=2}^{J} (\mu^n_j)^3 R^2\right),$$

which converges to zero uniformly on each fixed B_R. The rescaled profile-decomposition remainder and pressure cross-terms are included in E_n, and perturbative scale-collapse robustness gives $E_n \to 0$ in the topology required by the scale-collapse compactness and rigidity passage.

Consequently, the innermost branch satisfies the repaired-gauge Navier–Stokes equation with bounded effective modulation, positive finite critical mass, pressure reconstruction, Caccioppoli regularity, and a perturbation that vanishes in the scale-collapse limit. The scale $\lambda_1(t) \to 0$ gives the scale-collapse hypothesis. The scale-collapse rigidity hypotheses therefore produce a nonzero stationary $L^3(\mathbb{R}^3)$ profile in the Nečas–Růžička–Šverák-covered self-similar class.

The Nečas–Růžička–Šverák rigidity theorem rules out every nonzero $L^3(\mathbb{R}^3)$ stationary profile in that class. This contradicts the nonzero extraction, which ultimately comes from $\|\phi^1\|_3 \ge \eta > 0$. Hence the regular strict same-point cascade cannot occur. □

The preceding theorem uses the following analytic hypotheses: strong L^3 profile decomposition with Brézis–Lieb decoupling and remainder decoupling; a uniform active-bubble mass floor $\|\phi^j\|_3 \ge \eta$; finite-offset C^2 nested-cascade regularity; repaired-gauge representation of the innermost branch with pressure, Caccioppoli regularity, bounded modulation, and positive finite critical mass; perturbative scale-collapse robustness; and the scale-collapse extraction of a nonzero limiting profile together with Nečas–Růžička–Šverák rigidity. Bounded critical mass alone proves only uniformly bounded cascade length. Full exclusion of the regular same-point cascade requires the inner-decoupling and perturbative scale-collapse hypotheses.

Lemma 1.33 (Escaping offsets vanish locally)**.** *Let*

$$W^n_j(y) = \mu^n_j \phi^j(\xi^n_j + \mu^n_j y), \qquad \mu^n_j = \frac{\lambda^n_1}{\lambda^n_j} \to 0.$$

If $\phi^j \in L^3(\mathbb{R}^3)$ and $|\xi^n_j| \to \infty$, then, for every fixed $R < \infty$,

$$\|W^n_j\|_{L^3(B_R)} \to 0.$$

If additionally

$$\|\nabla W^n_j\|_{L^3(B_R)} + \|D^2 W^n_j\|_{L^3(B_R)} \to 0$$

for every fixed R, then $W^n_j \to 0$ in $W^{2,3}_{\mathrm{loc}}$.

Proof. With $z = \xi_j^n + \mu_j^n y$ and $dy = (\mu_j^n)^{-3}\,dz$,

$$\|W_j^n\|_{L^3(B_R)}^3 = \int_{B_R} (\mu_j^n)^3 |\phi^j(\xi_j^n + \mu_j^n y)|^3\,dy = \int_{B(\xi_j^n,\mu_j^n R)} |\phi^j(z)|^3\,dz.$$

Since $|\xi_j^n| \to \infty$ and $\mu_j^n R \to 0$, the balls

$$B(\xi_j^n, \mu_j^n R)$$

eventually lie in $\{|z| > A\}$ for every fixed $A < \infty$. Since $\phi^j \in L^3(\mathbb{R}^3)$,

$$\int_{|z|>A} |\phi^j(z)|^3\,dz \to 0 \quad \text{as } A \to \infty.$$

Thus, for all sufficiently large n,

$$\|W_j^n\|_{L^3(B_R)}^3 \le \int_{|z|>A} |\phi^j(z)|^3\,dz,$$

and the right-hand side is arbitrarily small for large A. This proves local L^3 vanishing. The $W^{2,3}_{\mathrm{loc}}$ conclusion is precisely the additional derivative-tail hypothesis. □

The pressure contribution of an escaping-offset bubble is not automatic from local velocity vanishing, because pressure is nonlocal. The needed escaping-offset pressure-decoupling hypothesis is that, for every compact B_R and compact time window, the pressure generated by the cross source

$$U_n \odot W_j^n + W_j^n \odot U_n + W_j^n \otimes W_j^n$$

converges to zero modulo constants in the pressure topology used for the repaired-gauge and scale-collapse passages. Under this hypothesis, escaping-offset outer profiles are perturbative for the innermost scale-collapse branch.

Proposition 1.34 (Perturbative estimates for the same-point cascade)**.** *Let U_n be the innermost renormalized branch after absorbing c^n_{out} into b^n, and let S_n be the remaining outer perturbation on $B_R \times I$. Assume*

$$\|S_n\|_{L^\infty_\tau W^{2,\infty}_y(B_R)} + \|\partial_\tau S_n\|_{L^1_\tau H^{-1}_y(B_R)} \longrightarrow 0,$$
$$\|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R,$$

and assume the pressure reconstruction satisfies the local Calderon–Zygmund estimate for cross terms. Then every perturbative term produced by replacing $U_n + S_n$ with U_n vanishes in

$$L^1_\tau H^{-1}_y(B_R).$$

Proof. The time and Laplacian perturbations satisfy

$$\|\partial_\tau S_n\|_{L^1 H^{-1}} \to 0, \qquad \|\Delta S_n\|_{L^1 H^{-1}} \le C_R \|S_n\|_{L^1 W^{2,\infty}} \to 0.$$

The linear transport terms obey

$$\|(U_n \cdot \nabla) S_n\|_{L^1 H^{-1}(B_R)} \le C_R \|U_n\|_{L^2 L^2(B_R)} \|\nabla S_n\|_{L^2 L^\infty(B_R)} \to 0,$$

and

$$\|(S_n \cdot \nabla) U_n\|_{L^1 H^{-1}(B_R)} \le C_R \|S_n\|_{L^\infty L^\infty(B_R)} \|\nabla U_n\|_{L^1 L^2(B_R)} \to 0.$$

The quadratic perturbation satisfies

$$\|(S_n \cdot \nabla) S_n\|_{L^1 H^{-1}(B_R)} \le C_R \|S_n\|_{L^\infty L^\infty} \|\nabla S_n\|_{L^1 L^\infty} \to 0.$$

The modulation terms satisfy

$$\|a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n\|_{L^1 H^{-1}(B_R)} \le C_R(M_{ab}) \|S_n\|_{L^1 W^{1,\infty}(B_R)} \to 0.$$

For the pressure, the cross source is

$$2U_n \odot S_n + S_n \otimes S_n.$$

On B_{2R},

$$\|U_n \odot S_n\|_{L^1_\tau L^{3/2}_y} \le \|S_n\|_{L^\infty_{\tau,y}} \|U_n\|_{L^1_\tau L^3_y} \to 0,$$

and

$$\|S_n \otimes S_n\|_{L^1_\tau L^{3/2}_y} \to 0.$$

The local pressure estimate then gives convergence of the cross pressure modulo constants in the required pressure topology, hence its gradient vanishes in $L^1_\tau H^{-1}_y(B_R)$. □

For a regular finite strict cascade, Theorem 1.29 and bounded relative modulation give the required $S_n \to 0$ estimates. Therefore regular nested-cascade structure, bounded modulation, pressure reconstruction, and local windowed H^1 control imply perturbative scale-collapse robustness.

Definition 1.35 (Same-point nonlinear decoupling)**.** Same-point nonlinear decoupling means that every finite same-point strict scale-separated profile family satisfies:

(i) strong critical decomposition,

$$u_n = \sum_{j=1}^{J} \Lambda_{\lambda_j^n, x_j^n} \phi^j + r_n, \qquad \|u_n\|_3^3 = \sum_{j=1}^{J} \|\phi^j\|_3^3 + \|r_n\|_3^3 + o_n(1);$$

(ii) nonlinear profile stability: profiles below the critical small-data threshold are perturbative, and every active singular profile has L^3-mass at least $\varepsilon_{\mathrm{sd}}$;

(iii) innermost velocity decoupling: after rescaling by the smallest scale and absorbing all constant ambient velocities into the translation gauge,

$$V_n = U_n + S_n, \qquad S_n \to 0$$

in the local velocity topology required by perturbative scale-collapse robustness;

(iv) pressure decoupling: all pressure cross-terms between the innermost profile, outer profiles, and the remainder vanish modulo constants in the repaired-gauge scale-collapse pressure topology;

(v) modulation compatibility: the effective repaired-gauge parameters of the innermost branch remain bounded after ambient constants are absorbed;

(vi) scale-collapse topology compatibility: the full interaction error vanishes in the compactness and stationarity passage.

Theorem 1.36 (Conditional no same-point multibubble theorem)**.** *Assume an admissible Navier–Stokes Type II branch has the structural hypotheses needed to enter the repaired-gauge Type II class, and assume same-point nonlinear decoupling, the innermost repaired-gauge representation, and the scale-collapse extraction of a nonzero limiting profile together with Nečas–Růžička–Šverák rigidity. Then no same-point multibubble cascade of active singular profiles can occur.*

Proof. Let a same-point strict scale-separated cascade be given. By Theorem 1.35, the decomposition is strongly decoupled in critical L^3, and every active singular bubble carries at least $\varepsilon_{\mathrm{sd}}$ critical mass. Since the ambient Type II branch has bounded critical mass,

$$\sup_n \|u_n\|_3 \le M,$$

Theorem 1.27 gives

$$J \le \left\lfloor \frac{M^3}{\varepsilon_{\mathrm{sd}}^3} \right\rfloor.$$

Choose the innermost scale λ_1^n. The nonlinear decoupling hypothesis says that, in innermost variables and after translation-gauge absorption of constant ambient velocities,

$$V_n = U_n + S_n, \qquad S_n \to 0$$

in the perturbative topology required for scale-collapse compactness, including pressure and modulation compatibility. Hence the innermost branch U_n satisfies the repaired-gauge Navier–Stokes equation with bounded effective modulation, positive finite critical mass, pressure reconstruction, and Caccioppoli regularity, up to an error that vanishes in the scale-collapse limiting passage.

The scale $\lambda_1^n \to 0$ gives the scale-collapse hypothesis. Therefore the scale-collapse rigidity hypotheses produce a nonzero stationary $L^3(\mathbb{R}^3)$ profile in the Nečas–Růžička–Šverák-covered self-similar class. Nonzero extraction follows from the active bubble mass floor:

$$\|\phi^1\|_3 \geq \varepsilon_{\mathrm{sd}} > 0.$$

The Nečas–Růžička–Šverák rigidity theorem rules out such a nonzero L^3 stationary self-similar profile, contradicting the extraction. □

Thus same-point cascade exclusion leaves only the failure of same-point nonlinear decoupling, namely the no-splitting/profile-compatibility multibubble alternative. If a regular strict same-point cascade of active singular profiles is not excluded, then one of the following has failed: profile decomposition or remainder decoupling, active-bubble small-data separation, escaping-offset pressure decoupling, local profile regularity or innermost representation, or the scale-collapse extraction of a nonzero limiting profile and rigidity hypotheses. These are analytic profile, representation, interaction, or rigidity failures; a below-threshold profile is regular and is not a singular cascade bubble.

2. Local Single-Core Criterion

2.1. **Introduction.** We first isolate a local compactness mechanism for Type II concentration in the three-dimensional Navier–Stokes equations, in the standard class of suitable weak solutions introduced by Leray and the partial-regularity theory of Scheffer and Caffarelli–Kohn–Nirenberg [Ler34, Sch76, CKN82, Lin98]. The setting is a sequence of renormalized solutions on fixed parabolic cylinders, centered at a selected concentration core and written in scale-translation variables. The result excludes the case in which one compact core carries a fixed positive Caffarelli–Kohn–Nirenberg density while the localized dissipation along selected windows vanishes. The theorem is conditional on the local representation and pressure decomposition/stability inputs stated below; the compact-cylinder A, E, C, D bounds are part of the branch hypothesis.

The notation and estimates follow the usual local energy, critical-space, and monograph conventions for Navier–Stokes [Lad69, Tem77, CF88, Kat84, ESS03, LR02]. The argument is local. Uniform Caffarelli–Kohn–Nirenberg bounds and bounded modulation coefficients give compactness on smaller cylinders. Vanishing localized dissipation forces the compactness limit to be spatially constant. If the constant is zero, strong velocity convergence and pressure stability contradict the positive core density. If the constant is nonzero, the sequence has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition.

2.2. **Setup and definitions.** For $R > 0$, write

$$Q_R = B_R \times (-R^2, 0).$$

For a suitable pair (v, q) on Q_R, define

$$A_v(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |v(y,s)|^2 \, dy,$$

$$E_v(R) = R^{-1} \int_{Q_R} |\nabla v|^2 \, dy \, ds,$$

$$C_v(R) = R^{-2} \int_{Q_R} |v|^3 \, dy \, ds,$$

$$D_q(R) = R^{-2} \int_{-R^2}^{0} \int_{B_R} |q(y,s) - (q)_{B_R}(s)|^{3/2} \, dy \, ds.$$

All pressure quantities are understood modulo functions of time.

Definition 2.1 (Bounded selected-window limit)**.** Let (V_n, P_n) be a sequence of represented local rescalings on a compact cylinder Q_r. A selected-window subsequence has a bounded selected-window limit on Q_r if, after passing to a subsequence,

$$V_n \to V_* \qquad \text{strongly in } L^3(Q_r),$$

where $V_* \in L^\infty(Q_r)$. The limit is nonzero if $V_* \not\equiv 0$ on Q_r.

Definition 2.2 (Local Type II sequence)**.** A represented sequence is called a local Type II sequence if it has positive local Caffarelli–Kohn–Nirenberg concentration on some fixed compact cylinder and no selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder in the sense of Theorem 2.1.

Definition 2.3 (Compact single-core vanishing-dissipation branch)**.** Fix radii $0 < r_* < R_*$, constants $M < \infty$, $\eta_0 > 0$, and a sequence of represented local rescalings (V_n, P_n, a_n, b_n) on Q_{R_*}. The sequence is a compact single-core vanishing-dissipation branch if:

(i) each (V_n, P_n) is suitable on Q_{R_*} and solves the represented Navier–Stokes equation

$$\partial_s V_n + (V_n \cdot \nabla) V_n + \nabla P_n = \nu \Delta V_n + a_n(s)(V_n + y \cdot \nabla V_n) + b_n(s) \cdot \nabla V_n, \qquad \nabla \cdot V_n = 0; \tag{1}$$

(ii) the compact-cylinder suitable estimates are uniformly bounded:

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M;$$

(iii) the selected core has positive CKN concentration on Q_{r_*}:

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0 \qquad \text{for all } n;$$

(iv) the localized scale and center are nondegenerate for the selected core, so the representation theorem gives the variables in (1);

(v) the modulation coefficients are uniformly bounded on the selected cylinder:

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M;$$

(vi) the branch has vanishing localized dissipation in the sense of Theorem 2.4 below.

Definition 2.4 (Vanishing localized dissipation)**.** Let $r_* < \rho < R_*$. Selected windows are first translated and rescaled to the fixed cylinders $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. The localized dissipation on Q_ρ is

$$\mathcal{D}_n(\rho) = \int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds.$$

The branch has vanishing localized dissipation if, after passing to a selected-window subsequence, there is $\rho \in (r_*, R_*)$ such that

$$\mathcal{D}_n(\rho) \to 0.$$

2.3. **Local analytic inputs.** The compact single-core argument is based on the following local analytic statements.

Theorem 2.5 (Local concentration dichotomy)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then, using the pressure-normalized quantities C, D, there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0,T), r_n) + D((x_0,T), r_n) \ge \eta \qquad \textit{for all } n.$$

Moreover, at the concentration point either a local Type I scale-invariant bound holds, or the point lies in the local Type II alternative, meaning that no such local Type I bound holds.

Proof. This is the local concentration dichotomy used throughout the paper. □

Proposition 2.6 (Passage to local Type II sequences)**.** *Let (x_0, T) be a concentration point in the local Type II alternative of Theorem 2.5. After choosing the positive concentration scales given there and applying the local scale-translation selection used in the present section to a nondegenerate compact single core, the represented selected windows form a sequence with positive local CKN concentration on a fixed compact cylinder. In the selected-window Type II branch, no subsequence has a nonzero bounded selected-window limit on any compact cylinder. Hence the represented selected windows are a local Type II sequence in the sense of Theorem 2.2.*

Proof. The local concentration dichotomy supplies the positive CKN concentration sequence and the exclusion of the local Type I alternative at the chosen concentration point. The scale-translation selection rewrites those shrinking cylinders as fixed selected windows. The last condition is the selected-window formulation of the Type II branch considered here: any branch with a nonzero bounded selected-window limit belongs to the bounded-window alternative, not to Theorem 2.2. □

Proposition 2.7 (Local representation input)**.** *For a nondegenerate selected single core, the local scale-translation construction gives represented variables (V_n, P_n, a_n, b_n) satisfying (1) on compact cylinders. Failure of the localized nondegeneracy condition is a multibubble, cascade, or gauge-degenerate alternative, not a compact single-core branch.*

Proposition 2.8 (Local pressure decomposition and stability). *For every $B_r \Subset B_R$, the represented pressure decomposes as*

$$P_n = P_{\mathrm{loc},n} + H_n, \qquad -\Delta P_{\mathrm{loc},n} = \partial_i\partial_j(\zeta V_{n,i}V_{n,j}),$$

where $\zeta \in C_c^\infty(B_R)$ equals one on B_r, and H_n is harmonic in B_r. The local part satisfies the Calderon–Zygmund bound [CZ52, Ste70]

$$\|P_{\mathrm{loc},n}\|_{L^{3/2}(B_R)} \le C\|V_n\|^2_{L^3(B_R)}$$

for a.e. time, and the harmonic part is controlled modulo spatial means by the local pressure norm.

Moreover, if $V_n \to 0$ strongly in $L^3(Q_R)$, then for every $r < R$,

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

Lemma 2.9 (Localization of pressure oscillation bounds). *Let $0 < r < R$, and let $q \in L^{3/2}(Q_R)$. Then*

$$D_q(r) \le C\left(\frac{R}{r}\right)^2 D_q(R),$$

where C is universal.

Proof. For a.e. $s \in (-r^2, 0)$, Jensen's inequality gives

$$|(q)_{B_r}(s) - (q)_{B_R}(s)|^{3/2} \le \frac{1}{|B_r|}\int_{B_r} |q(y,s) - (q)_{B_R}(s)|^{3/2}\,dy.$$

Hence

$$\int_{B_r} |q - (q)_{B_r}(s)|^{3/2}\,dy \le C\int_{B_r} |q - (q)_{B_R}(s)|^{3/2}\,dy.$$

Integrating over $(-r^2, 0)$ and using $Q_r \subset Q_R$,

$$\begin{aligned} D_q(r) &\le Cr^{-2}\int_{-r^2}^0\int_{B_r} |q - (q)_{B_R}(s)|^{3/2}\,dy\,ds \\ &\le Cr^{-2}\int_{-R^2}^0\int_{B_R} |q - (q)_{B_R}(s)|^{3/2}\,dy\,ds = C\left(\frac{R}{r}\right)^2 D_q(R). \end{aligned}$$

□

2.4. **Compactness and pressure stability.**

Proposition 2.10 (Compactness on selected windows). *Let (V_n, P_n) be suitable on Q_{R_*}, solve* (1)*, and satisfy*

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M,$$

with

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M.$$

Then for every $r < R_*$, after passing to a subsequence there are (V, P) such that

$$\begin{aligned} V_n &\overset{*}{\rightharpoonup} V \quad \textit{in } L^\infty(-r^2, 0; L^2(B_r)), \\ V_n &\rightharpoonup V \quad \textit{in } L^2(-r^2, 0; H^1(B_r)), \\ V_n &\to V \quad \textit{strongly in } L^3(Q_r), \end{aligned}$$

and

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \textit{in } L^{3/2}(Q_r).$$

After passing to a further subsequence, a_n and b_n converge weak- in L^∞ to bounded coefficients, and the limit satisfies the corresponding represented equation distributionally on smaller cylinders.*

Proof. The bounds on A and E give weak-* compactness in $L^\infty L^2$ and weak compactness in $L^2 H^1$ on smaller cylinders. Fix $r < r_1 < R_*$, and choose m so large that

$$H_0^m(B_{r_1}) \hookrightarrow W_0^{1,\infty}(B_{r_1}).$$

We claim that $\partial_s V_n$ is uniformly bounded in

$$L^1(-r_1^2, 0; H^{-m}(B_{r_1})).$$

Let $\varphi \in H_0^m(B_{r_1})$. Pairing (1) with φ, integrating by parts in space where needed, and using the embedding above gives

$$|\langle (V_n \cdot \nabla)V_n, \varphi\rangle| = \left|\int_{B_{r_1}} V_{n,i}V_{n,j}\partial_j\varphi_i \, dy\right| \le C\|V_n\|^2_{L^3(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle \nu\Delta V_n, \varphi\rangle| \le C\|\nabla V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle \nabla P_n, \varphi\rangle| = \left|\int_{B_{r_1}} (P_n - (P_n)_{B_{r_1}})\nabla\cdot\varphi \, dy\right| \le C\|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(B_{r_1})}\|\varphi\|_{H^m}.$$

The modulation terms obey

$$|\langle a_n V_n, \varphi\rangle| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle a_n\, y\cdot\nabla V_n, \varphi\rangle| = \left|\int_{B_{r_1}} a_n V_{n,i}\partial_j(y_j\varphi_i)\, dy\right| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

and

$$|\langle b_n\cdot\nabla V_n, \varphi\rangle| = \left|\int_{B_{r_1}} V_{n,i}b_{n,j}\partial_j\varphi_i\, dy\right| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

where the constant uses r_1 and the uniform L^∞ bounds for a_n, b_n. Integrating these estimates in time gives a uniform bound because

$$\int_{-r_1^2}^0 \|V_n(s)\|^2_{L^3(B_{r_1})}\, ds \le r_1^{2/3}\|V_n\|^2_{L^3(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|\nabla V_n(s)\|_{L^2(B_{r_1})}\, ds \le r_1\|\nabla V_n\|_{L^2(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|P_n(s) - (P_n)_{B_{r_1}}(s)\|_{L^{3/2}(B_{r_1})}\, ds \le r_1^{2/3}\|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(Q_{r_1})},$$

and

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^2(B_{r_1})}\, ds \le r_1^2\|V_n\|_{L^\infty(-r_1^2,0;L^2(B_{r_1}))}.$$

The uniform C, E, D, and A bounds therefore prove the claimed time-derivative bound; the D-bound on Q_{r_1} follows from the bound on Q_{R_*} by Theorem 2.9.

The compact embedding $H^1(B_{r_1}) \Subset L^2(B_r)$, together with Aubin–Lions compactness [Aub63, LM72, Sim87], gives

$$V_n \to V \qquad \text{strongly in } L^2(Q_r)$$

after passing to a subsequence. The same $L^\infty L^2 \cap L^2 H^1$ bounds give the parabolic interpolation estimate

$$\|V_n\|_{L^{10/3}(Q_r)} \le C(r, M).$$

Indeed, for a.e. s, the Sobolev and Gagliardo–Nirenberg inequalities give

$$\|V_n(s)\|_{L^{10/3}(B_r)} \le C\|V_n(s)\|^{2/5}_{L^2(B_r)}\|\nabla V_n(s)\|^{3/5}_{L^2(B_r)}.$$

Raising to the power 10/3, integrating in time, and using the A and E bounds yields

$$\int_{-r^2}^{0} \|V_n(s)\|_{L^{10/3}(B_r)}^{10/3} \, ds \le C \left(\operatorname*{ess\,sup}_{-r^2<s<0} \|V_n(s)\|_{L^2(B_r)}^2 \right)^{2/3} \int_{Q_r} |\nabla V_n|^2 \, dy \, ds \le C(r, M).$$

The same estimate applies to V by weak lower semicontinuity, and therefore to $V_n - V$ by the triangle inequality. Interpolating between L^2 and $L^{10/3}$, one obtains

$$\|V_n - V\|_{L^3(Q_r)} \le \|V_n - V\|_{L^2(Q_r)}^{1/6} \|V_n - V\|_{L^{10/3}(Q_r)}^{5/6}.$$

The second factor is uniformly bounded and the first tends to zero, so

$$V_n \to V \qquad \text{strongly in } L^3(Q_r).$$

The pressure compactness follows from the uniform D-bound: after subtracting spatial means,

$$P_n - (P_n)_{B_r} \rightharpoonup P \qquad \text{in } L^{3/2}(Q_r).$$

We pass to the limit distributionally using the preceding convergences. It remains to justify the modulation terms. For every compactly supported test field φ,

$$\int a_n(s) \, y \cdot \nabla V_n \cdot \varphi = -\int a_n(s) \, V_{n,i} \, \partial_j (y_j \varphi_i),$$

where summation over repeated spatial indices is understood. Equivalently,

$$\partial_j (y_j \varphi_i) = 3\varphi_i + y_j \partial_j \varphi_i.$$

Hence this term passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of a_n: if $F_n \to F$ in L^1 and $a_n \rightharpoonup^* a$ in L^∞, then

$$\int a_n F_n - \int a F = \int a_n (F_n - F) + \int (a_n - a) F \to 0.$$

The term $a_n V_n$ is immediate from the same argument, while

$$\int b_n(s) \cdot \nabla V_n \cdot \varphi = -\int V_{n,i} \, b_{n,j}(s) \, \partial_j \varphi_i$$

passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of b_n. These passages give the limiting represented equation in the sense of distributions. ☐

Remark 2.11 (Use of bounded modulation). The uniform L^∞ bounds on a_n and b_n enter the compactness argument in exactly two places. First, they give the negative Sobolev time-derivative bound for the three lower-order modulation terms in (1). Second, after subsequential weak-* convergence of a_n, b_n, they allow the modulation terms to pass to the distributional limit once $V_n \to V$ strongly in L^2_{loc}.

Lemma 2.12 (Stability of local pressure decomposition)**.** *Let $V_n \to V$ strongly in $L^3(Q_R)$, and use the pressure decomposition of Theorem 2.8. Then for every $r < R$,*

$$P_{\mathrm{loc},n} \to P_{\mathrm{loc}} \quad \textit{strongly in } L^{3/2}(Q_r),$$

where $-\Delta P_{\mathrm{loc}} = \partial_i \partial_j (\zeta V_i V_j)$. If $V \equiv 0$ on Q_R, then

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

More locally in time, if $V_n \to 0$ strongly in

$$L^3(B_R \times (-r^2, 0)),$$

then

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

Proof. Strong convergence in L^3 gives

$$V_{n,i}V_{n,j} \to V_iV_j \quad \text{strongly in } L^{3/2}(Q_R).$$

Indeed,

$$\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)} \le \big(\|V_n\|_{L^3(Q_R)} + \|V\|_{L^3(Q_R)}\big)\|V_n - V\|_{L^3(Q_R)}.$$

Calderon–Zygmund estimates for the compactly supported localized source yield strong convergence of the localized pressures in $L^{3/2}$ on smaller balls:

$$\|P_{\mathrm{loc},n} - P_{\mathrm{loc}}\|_{L^{3/2}(Q_r)} \le C_{r,R}\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)}.$$

Assume now that $V \equiv 0$ on Q_R. Then $P_{\mathrm{loc}} = 0$. By the zero-limit stability part of Theorem 2.8,

$$P_n - (P_n)_{B_r} \to 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

The same proof is local in time. If $V_n \to 0$ strongly in $L^3(B_R \times (-r^2, 0))$, the Calderon–Zygmund estimate and harmonic pressure control are applied only for times $s \in (-r^2, 0)$, giving the strong convergence on Q_r. □

Lemma 2.13 (Zero-dissipation limit)**.** *If*

$$V_n \rightharpoonup V \quad \text{in } L^2(I; H^1(B_\rho))$$

and

$$\int_I \int_{B_\rho} |\nabla V_n|^2 \, dy \, ds \to 0,$$

then $\nabla V = 0$ a.e. on $B_\rho \times I$. Hence for a.e. time, $V(\cdot, s)$ is spatially constant on B_ρ.

Proof. Weak lower semicontinuity gives

$$\int_I \int_{B_\rho} |\nabla V|^2 \, dy \, ds \le \liminf_{n\to\infty} \int_I \int_{B_\rho} |\nabla V_n|^2 \, dy \, ds = 0.$$

Thus $\nabla V = 0$ a.e. on $B_\rho \times I$. By Fubini, for a.e. $s \in I$ one has $\nabla V(\cdot, s) = 0$ in $L^2(B_\rho)$. Since B_ρ is connected, every $H^1(B_\rho)$ function with zero weak gradient is a.e. equal to a constant. Hence $V(\cdot, s)$ is spatially constant for a.e. s. □

Lemma 2.14 (Spatially constant limits)**.** *Let V be a limit on Q_ρ and suppose $\nabla V = 0$ a.e. there. Then $V(y, s) = c(s)$ on Q_ρ, with $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the approximating selected-window subsequence has a nonzero bounded selected-window limit.*

Proof. The spatial constancy follows from Theorem 2.13. Since $V(\cdot, s) = c(s)$ on B_ρ, the local $L^\infty_s L^2_y$ bound gives

$$|c(s)|^2 |B_\rho| \le \operatorname*{ess\,sup}_{-\rho^2 < s < 0} \int_{B_\rho} |V(y, s)|^2 \, dy,$$

so $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the strong L^3 convergence of Theorem 2.10 gives a subsequential limit to a nonzero bounded function on Q_ρ, which is precisely a nonzero bounded selected-window limit in the sense of Theorem 2.1. □

2.5. **Contradiction lemmas.**

Lemma 2.15 (Persistence of positive core concentration)**.** *For a compact single-core branch in the sense of Theorem 2.3, there are fixed $r_* > 0$ and $\eta_0 > 0$ such that along the selected-window subsequence,*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0 \qquad \textit{for all } n.$$

Proof. The lower bound is part of the definition of the selected compact core and is preserved by passing to subsequences of the selected windows. □

Lemma 2.16 (Zero constant contradicts positive CKN concentration)**.** *Let $r_* < \rho$. Suppose the compactness limit in Theorem 2.10 is spatially constant on Q_ρ and is identically zero on Q_{r_*}. Then*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0.$$

Consequently this case contradicts Theorem 2.15.

Proof. Strong $L^3(Q_{r_*})$ convergence to zero gives

$$C_{V_n}(r_*) = r_*^{-2} \int_{Q_{r_*}} |V_n|^3 \, dy \, ds = r_*^{-2} \|V_n\|^3_{L^3(Q_{r_*})} \to 0.$$

It remains to control the pressure term. Since V is spatially constant on Q_ρ, the identity $V = 0$ on Q_{r_*} implies

$$V = 0 \qquad \text{on } B_\rho \times (-r_*^2, 0).$$

Indeed, for a.e. $s \in (-r_*^2, 0)$, the spatially constant value on B_ρ vanishes on the nonempty subball B_{r_*}, hence vanishes on all of B_ρ. The strong L^3_{loc} convergence therefore gives

$$V_n \to 0 \qquad \text{strongly in } L^3(B_\rho \times (-r_*^2, 0)).$$

Applying Theorem 2.12 with $R = \rho$ and $r = r_*$, in its local-in-time form, gives

$$P_n - (P_n)_{B_{r_*}} \to 0 \qquad \text{strongly in } L^{3/2}(Q_{r_*}).$$

Therefore

$$D_{P_n}(r_*) = r_*^{-2} \|P_n - (P_n)_{B_{r_*}}\|^{3/2}_{L^{3/2}(Q_{r_*})} \to 0.$$

Thus

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting the uniform lower bound

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0.$$

□

Lemma 2.17 (Nonzero constant contradicts the Type II regime)**.** *If the compactness limit is spatially constant and nonzero on a compact subcylinder, then the original sequence is not a local Type II sequence.*

Proof. By Theorem 2.14, the selected-window subsequence has a nonzero bounded selected-window limit. This contradicts Theorem 2.2. □

2.6. Main theorem.

Theorem 2.18 (Local single-core Type II criterion)**.** *Under the analytic inputs Theorems 2.6 to 2.8, a compact single-core vanishing-dissipation branch in the sense of Theorem 2.3 cannot be a local Type II sequence.*

Proof. By the definition of vanishing localized dissipation, after passing to a selected-window subsequence there is a fixed radius $\rho \in (r_*, R_*)$ such that

$$\int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds \to 0$$

and the positive core bound on Q_{r_*} remains valid along this subsequence. Apply compactness on the fixed compact cylinder Q_ρ and pass to a subsequence. The compactness proposition gives

$$V_n \rightharpoonup V \quad \text{in } L^2(-\rho^2, 0; H^1(B_\rho))$$

and strong convergence in L^3 on every smaller cylinder. Since the localized dissipation tends to zero, the zero-dissipation lemma gives

$$\nabla V = 0 \quad \text{a.e. on } Q_\rho.$$

Thus $V(y, s) = c(s)$ on Q_ρ, and in particular on the selected core cylinder Q_{r_*}.

If this spatially constant limit is zero on the core cylinder, the zero-limit contradiction lemma gives

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting persistence of positive core concentration.

If the spatially constant limit is not identically zero on Q_{r_*}, the nonzero-limit lemma produces a nonzero bounded selected-window limit, contradicting the definition of a local Type II sequence. The two cases exhaust all spatially constant limits, so the branch cannot occur. □

Remark 2.19 (Minimal local inputs)*.* The proof uses two local analytic inputs from the Type II argument: the localized representation for a nondegenerate selected core, the compact-ball pressure reconstruction with stability on smaller balls. The compact-cylinder A, E, C, D bounds and vanishing localized dissipation are part of the branch hypothesis, not separate analytic inputs. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are outside this single-core theorem; they are treated by the local multibubble and cascade exclusion theorem.

3. Repaired-Gauge Construction

3.1. **Assumptions and notation.** Let $u : \mathbb{R}^3 \times (T-\delta, T) \to \mathbb{R}^3$, $p : \mathbb{R}^3 \times (T-\delta, T) \to \mathbb{R}$ be a suitable weak Navier–Stokes solution with viscosity $\nu > 0$. For $z = (x_0, t_0)$ and $R > 0$,

$$Q_R(z) := B_R(x_0) \times (t_0 - R^2, t_0), \qquad Q_R^* := B_R \times I_*.$$

Fix a reference physical center-time $(x_\bullet, t_\star)$ around which $B_{R_*}(x_\bullet)$ is taken. For a fixed spatial center x_0 we also use

$$Q_R(t) := Q_R(x_0, t).$$

Fix a compact window $I_* := [\tau_0 - \ell, \tau_0 + \ell]$, $\ell > 0$, and write $I_{\mathrm{ph}} := [t_\star - \Lambda_{\max}^2 \ell,\ t_\star + \Lambda_{\max}^2 \ell]$.

For a cylinder $Q_R(x_0, t_0) = B_R(x_0) \times (t_0 - R^2, t_0)$, set

$$\begin{aligned}
A(u; Q_R(x_0,t_0)) &:= R^{-1} \sup_{s \in (t_0 - R^2, t_0)} \int_{B_R(x_0)} |u(x,s)|^2\,dx,\\
E(u; Q_R(x_0,t_0)) &:= R^{-1} \int_{Q_R(x_0,t_0)} |\nabla u|^2\,dx\,ds,\\
C(u; Q_R(x_0,t_0)) &:= R^{-2} \int_{Q_R(x_0,t_0)} |u|^3\,dx\,ds,\\
D(p; Q_R(x_0,t_0)) &:= R^{-2} \int_{t_0 - R^2}^{t_0} \int_{B_R(x_0)} \left|p(x,s) - \langle p(s)\rangle_{B_R(x_0)}\right|^{3/2}\,dx\,ds.
\end{aligned}$$

Here $\langle p(s)\rangle_{B_R(x_0)} := |B_R|^{-1} \int_{B_R(x_0)} p(x,s)\,dx$; pressure is always understood modulo time-dependent constants.

Definition 3.1 (Single-core hypotheses)**.** Let $(R_*, r_*, M, \eta, \kappa, \Lambda_{\min}, \Lambda_{\max})$ satisfy $\Lambda_{\min} \le \Lambda_{\max} < \infty$. We say (u,p) satisfies $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$ on $Q_{R_*}(x_\bullet, t_\star)$ if the following hold:

(SC1) $A(u; Q_{R_*}(x_\bullet, t)) + E(u; Q_{R_*}(x_\bullet, t)) + C(u; Q_{R_*}(x_\bullet, t)) + D(p; Q_{R_*}(x_\bullet, t)) \le M$ for every $t \in [t_\star - R_*^2, t_\star)$.

(SC2) There exists a physically selected core map $(x_{\mathrm{c}}(t), \rho_{\mathrm{c}}(t))$ on I_{ph} such that $\rho_{\mathrm{c}}(t) \in (0, R_*]$, $x_{\mathrm{c}}(t) \in B_{R_*/2}(x_\bullet)$, and for some $\Theta_0 > 0$,

$$\int_{B_{r_*}(0)} |\rho_{\mathrm{c}}(t)^{-1} u(x_{\mathrm{c}}(t) + \rho_{\mathrm{c}}(t) y, t)|^3 \mathrm{d}y \ge \eta \quad \text{for a.e. } t \in I_{\mathrm{ph}}.$$

(SC3) There exists $0 < r_{\mathrm{sep}} \le r_*/2$ such that for every $0 < r \le r_{\mathrm{sep}}$, every competitor pair

$$\mathcal{C}_{\mathrm{core}}(t) := \left\{ (\bar{x}, \bar{\rho}) : |\bar{x} - x_\bullet| \le \frac{R_*}{2},\ 0 < \bar{\rho} \le R_* \right\},$$

$(\bar{x}, \bar{\rho}) \in \mathcal{C}_{\mathrm{core}}(t) \setminus \{(x_{\mathrm{c}}(t), \rho_{\mathrm{c}}(t))\}$ with

$$\rho_{\mathrm{c}}(t)^{-1} |\bar{x} - x_{\mathrm{c}}(t)| + \left| \frac{\bar{\rho}}{\rho_{\mathrm{c}}(t)} - 1 \right| \ge r_{\mathrm{sep}}$$

has separated local weighted mass in the same core-scale variables:

$$\left| \int_{B_r} |\rho_{\mathrm{c}}(t)^{-1} u(x_{\mathrm{c}}(t) + \rho_{\mathrm{c}}(t) y, t)|^3\,\mathrm{d}y - \int_{B_r} |\bar{\rho}^{-1} u(\bar{x} + \bar{\rho} y, t)|^3\,\mathrm{d}y \right| \ge \eta.$$

This quantitative non-competing core condition is imposed on the selected physical core directly; it is not a consequence of the repaired-gauge construction.

(SC4) For the selected core,

$$\sup_{t\in I_{\rm ph}} \int_{B_{2R_*}} |y|^{-p} |\rho_{\rm c}(t)^{-1} u(x_{\rm c}(t)+\rho_{\rm c}(t)y,t)|^3\,dy \le M,$$

for some $0<p<3$.

(SC5) The preliminary frame associated with the selected core is the absolutely continuous representative

$$0<\Lambda_{\min}\le\Lambda_{\rm pre}(\tau)\le\Lambda_{\max},\quad \Lambda_{\rm pre}\in AC(I_*),\quad X_{\rm pre}\in AC(I_*;\mathbb{R}^3).$$

It is tied to the selected physical core by the time change $dt_{\rm c}/d\tau=\Lambda_{\rm pre}(\tau)^2$, $t_{\rm c}(\tau_0)=t_\star$: one has

$$X_{\rm pre}(\tau)=x_{\rm c}(t_{\rm c}(\tau)),\qquad \Lambda_{\rm pre}(\tau)=\rho_{\rm c}(t_{\rm c}(\tau))$$

for a.e. $\tau\in I_*$.

The parameter κ denotes the quantitative constants obtained in the local gauge estimates and is not used in items (SC1)–(SC5).

Remark 3.2. The selected core $(x_{\rm c},\rho_{\rm c})$ and its separation/nondegeneracy conditions are specified in physical variables; no repaired-gauge constraint is used in their definition. This avoids the circularity of defining the core via the chart to be constructed.

3.2. **Preliminary frame map from the selected core.** The transport map is constructed from the selected core; the core selection itself is part of the hypotheses.

Define

$$\frac{dt_{\rm c}}{d\tau}=\Lambda_{\rm pre}(\tau)^2,\qquad t_{\rm c}(\tau_0)=t_\star,$$

and use the compatibility in (SC5) to identify $\Lambda_{\rm pre}(\tau)=\rho_{\rm c}(t_{\rm c}(\tau))$ and $X_{\rm pre}(\tau)=x_{\rm c}(t_{\rm c}(\tau))$ a.e.

Proposition 3.3 (Preliminary frame map regularity)**.** *Under* $\mathrm{SC}(R_*,r_*,M,\eta,\kappa)$*, the map*

$$\Phi_{\rm pre}(\tau,Y):=(x,t)=(X_{\rm pre}(\tau)+\Lambda_{\rm pre}(\tau)Y,\ t_{\rm c}(\tau))$$

is well-defined and absolutely continuous in τ*. Moreover,* $1/\Lambda_{\rm pre}$*,* $\Lambda_{\rm pre}\in L^\infty(I_*)$*,* $X_{\rm pre}\in W^{1,1}_{\rm loc}(I_*;\mathbb{R}^3)$*, and for every* $R\le R_*$*,*

$$\Phi_{\rm pre}:Q^*_R\to\Phi_{\rm pre}(Q^*_R)\subset Q_{2R_*}(x_\bullet,t_\star)$$

is bi-Lipschitz with $\det\nabla_Y\Phi_{\rm pre}=\Lambda_{\rm pre}^3$*, hence Jacobian and inverse-Jacobian bounds depend only on* $\Lambda_{\min},\Lambda_{\max}$.

Proof. From (SC5), $\Lambda_{\rm pre}\in AC(I_*)$, $0<\Lambda_{\min}\le\Lambda_{\rm pre}\le\Lambda_{\max}<\infty$, and $X_{\rm pre}\in AC(I_*;\mathbb{R}^3)$. The scalar ODE

$$\frac{dt_{\rm c}}{d\tau}=\Lambda_{\rm pre}(\tau)^2,\qquad t_{\rm c}(\tau_0)=t_\star$$

therefore has an absolutely continuous strictly increasing solution on I_*, and

$$\frac{dt_{\rm c}}{d\tau}=\Lambda_{\rm pre}(\tau)^2\quad\text{a.e. on } I_*.$$

The compatibility condition (SC5) identifies this absolutely continuous frame with the selected physical core a.e.; no differentiability of $x_{\rm c}$ or $\rho_{\rm c}$ beyond the frame regularity in (SC5) is used. For fixed τ, $\nabla_Y\Phi_{\rm pre}(\tau,\cdot)=\Lambda_{\rm pre}(\tau)I$, so

$$\det\nabla_Y\Phi_{\rm pre}(\tau,\cdot)=\Lambda_{\rm pre}(\tau)^3,\qquad \Lambda_{\min}^3\le\det\nabla_Y\Phi_{\rm pre}\le\Lambda_{\max}^3.$$

Hence, if $Y_1,Y_2\in B_R$,

$$\Lambda_{\min}|Y_1-Y_2|\le|\Phi_{\rm pre}(\tau,Y_1)-\Phi_{\rm pre}(\tau,Y_2)|\le\Lambda_{\max}|Y_1-Y_2|,$$

and therefore $\Phi_{\rm pre}(\tau,\cdot)$ is bi-Lipschitz on B_R. Its inverse has Lipschitz constant $\Lambda_{\min}^{-1}$, so Jacobian and inverse-Jacobian bounds are controlled by $\Lambda_{\min}, \Lambda_{\max}$.

For image inclusions, if $x = X_{\rm pre}(\tau) + \Lambda_{\rm pre}(\tau)Y$ with $|Y| \le R$, then

$$|x - X_{\rm pre}(\tau)| \le \Lambda_{\max}R, \qquad X_{\rm pre}(\tau) \in B_{R_*/2}(x_\bullet),$$

so $x \in B_{R_*/2+\Lambda_{\max}R}(x_\bullet)$. As $\tau \in I_*$, we also have $t = t_{\rm c}(\tau) \in I_{\rm ph}$. Therefore

$$\Phi_{\rm pre}(Q_R^*) \subset B_{R_*/2+\Lambda_{\max}R}(x_\bullet) \times I_{\rm ph}.$$

□

Definition 3.4 (Preliminary profile)**.** For $(x,t) = \Phi_{\rm pre}(\tau, Y)$, define

$$V_{\rm pre}(Y,\tau) := \Lambda_{\rm pre}(\tau)\, u(x,t), \qquad P_{\rm pre}(Y,\tau) := \Lambda_{\rm pre}(\tau)^2\, p(x,t).$$

Hence $u(x,t) = \Lambda_{\rm pre}(\tau)^{-1}V_{\rm pre}(Y,\tau)$ and $p(x,t) = \Lambda_{\rm pre}(\tau)^{-2}P_{\rm pre}(Y,\tau)$ under $\Phi_{\rm pre}$.

3.3. **Exact renormalized change of variables.** Define renormalized fields on Q_{2R}^* by

$$Y = \frac{x - X(\tau)}{\Lambda(\tau)}, \qquad u(x,t) = \Lambda(\tau)^{-1}V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2}P(Y,\tau),$$

where $dt/d\tau = \Lambda(\tau)^2$, $X : I_* \to \mathbb{R}^3$, $\Lambda : I_* \to (0,\infty)$, and $\Lambda, X \in AC(I_*)$. Write

$$a(\tau) := -\Lambda'(\tau), \qquad b(\tau) := -X'(\tau).$$

Lemma 3.5 (Exact differential identities)**.** *For smooth data and fixed (x,t), the chain rule gives*

$$\partial_t u = \Lambda^{-3}\left(\partial_\tau V - (X' + \Lambda' Y)\cdot\nabla_Y V - \Lambda' V\right),$$

$$\nabla_x u = \Lambda^{-2}\nabla_Y V, \qquad \Delta_x u = \Lambda^{-3}\Delta_Y V, \qquad \nabla_x p = \Lambda^{-3}\nabla_Y P.$$

In particular

$$\nabla_x \cdot u = \Lambda^{-2}\nabla_Y\cdot V.$$

Proof. For fixed (x,t), $\tau = \tau(t)$ and $Y = (x - X(\tau))/\Lambda(\tau)$. Differentiate

$$u(x,t) = \Lambda(\tau)^{-1}V\big(Y,\tau\big)$$

in t:

$$\partial_t u = -\frac{\Lambda'}{\Lambda^2}V + \frac{1}{\Lambda}\partial_\tau V\,\frac{d\tau}{dt} + \frac{1}{\Lambda}\nabla_Y V\cdot\frac{dY}{dt}.$$

Since $d\tau/dt = \Lambda^{-2}$ and

$$Y_\tau = -\Lambda^{-1}X' - \frac{\Lambda'}{\Lambda}Y, \qquad \frac{dY}{dt} = Y_\tau\,\frac{d\tau}{dt},$$

therefore

$$\partial_t u = \Lambda^{-3}\left(\partial_\tau V - (X' + \Lambda' Y)\cdot\nabla_Y V - \Lambda' V\right).$$

For spatial derivatives, $Y = (x - X)/\Lambda$ gives $\nabla_x Y = \Lambda^{-1}I$, hence

$$\nabla_x u = \Lambda^{-2}\nabla_Y V, \qquad \Delta_x u = \Lambda^{-3}\Delta_Y V.$$

Similarly $p(x,t) = \Lambda^{-2}P(Y,\tau)$ yields

$$\nabla_x p = \Lambda^{-3}\nabla_Y P.$$

Taking divergence of $u = \Lambda^{-1}V$ gives

$$\nabla_x\cdot u = \Lambda^{-2}\nabla_Y\cdot V.$$

□

Theorem 3.6 (Renormalized Navier–Stokes equation)**.** *Assume u, p solve*

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu\Delta_x u = 0, \qquad \nabla_x \cdot u = 0$$

in distributions. Under the transformation above, V, P satisfy, on the image renormalized cylinder, the exact equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu\Delta V + a(\tau)\big(V + Y \cdot \nabla V\big) + b(\tau) \cdot \nabla V = 0, \qquad \nabla \cdot V = 0,$$

with the sign convention fixed by $a = -\Lambda'$, $b = -X'$. Conversely, any V, P satisfying this equation pull back to a distributional solution u, p of Navier–Stokes on the physical image.

Proof. To avoid hidden regularity loss, first write the physical equation tested against $\Phi \in C_c^\infty$ in the physical cylinder and then use the smooth pull-back formula (an approximation argument gives distributions). By Lemma 3.5,

$$\partial_t u + \big(u \cdot \nabla_x\big)u + \nabla_x p - \nu\Delta_x u = 0$$

becomes

$$\Lambda^{-3}\big(\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla V - \Lambda' V\big) + \Lambda^{-3}(V \cdot \nabla)V + \Lambda^{-3}\nabla P - \Lambda^{-3}\nu\Delta V = 0.$$

Multiplying by Λ^3 and using $a = -\Lambda'$, $b = -X'$ gives

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu\Delta V + a\big(V + Y \cdot \nabla V\big) + b \cdot \nabla V = 0.$$

The divergence condition is

$$\nabla_x \cdot u = \Lambda^{-2}\nabla \cdot V = 0 \quad \Longrightarrow \quad \nabla \cdot V = 0.$$

Conversely, assume (V, P) solve this equation on the renormalized cylinder. Set u, p by the same change-of-variables map. Then the preceding identities run in reverse and give

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu\Delta_x u = 0, \qquad \nabla_x \cdot u = 0$$

in $\mathcal{D}'$ on the physical image. □

3.4. **Domain localization and support.** Let $R > 0$ be fixed and $0 < \ell \le \ell_0$. Set $Q_R^* = B_R \times I_*$. With Φ as above define physical support maps

$$\mathcal{Q}_R^{\mathrm{ph}} := \Phi(Q_R^*), \qquad \mathcal{Q}_{2R}^{\mathrm{ph}} := \Phi(Q_{2R}^*).$$

Proposition 3.7 (Compact support transport)**.** *Assume $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$ on I_*. Then*

$$X(\tau) \in B_{R_*/2}(x_\bullet), \qquad |x - X(\tau)| \le \Lambda_{\max} R,$$
$$\mathcal{Q}_R^{\mathrm{ph}} \subset B_{R_*/2 + \Lambda_{\max} R}(x_\bullet) \times I_{\mathrm{ph}},$$
$$\mathcal{Q}_{2R}^{\mathrm{ph}} \subset B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet) \times I_{\mathrm{ph}},$$

for every $\tau \in I_*$ and $x = X(\tau) + \Lambda(\tau)Y$ with $Y \in B_{2R}$. In particular, if $|c(\tau)| \le \delta$ from Proposition 3.20, then $X(\tau) \in B_{R_*/2 + \Lambda_{\max}\delta}(x_\bullet)$ on I_*. Hence, $\chi_R \in C_c^\infty(B_{2R})$ pulls back to $\chi_R^\tau := \chi_R \circ \Phi^{-1} \in C_c^\infty(\mathcal{Q}_{2R}^{\mathrm{ph}})$ with support depending only on $R, \Lambda_{\min}, \Lambda_{\max}$.

Proof. Fix $\tau \in I_*$ and $Y \in B_R$. Since $x = X(\tau) + \Lambda(\tau)Y$,

$$|x - X(\tau)| \le |Y|\,\Lambda_{\max} \le \Lambda_{\max} R, \qquad \Lambda_{\max}^{-1}|x - X(\tau)| \le |Y|.$$

Because $X(\tau) \in B_{R_*/2}(x_\bullet)$, this gives

$$x \in B_{R_*/2 + \Lambda_{\max} R}(x_\bullet).$$

Similarly, if $Y \in B_{2R}$, then $x \in B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet)$. Together with $t = t_c(\tau) \in I_{\mathrm{ph}}$, these give the claimed inclusions for $\mathcal{Q}_R^{\mathrm{ph}}$ and $\mathcal{Q}_{2R}^{\mathrm{ph}}$.

For pull-backs of cutoffs, Φ is bi-Lipschitz at every τ, so composition

$$\chi_R^\tau(x,t) := \chi_R\left(\frac{x - X(\tau)}{\Lambda(\tau)}\right)$$

is smooth on $\mathcal{Q}_{2R}^{\mathrm{ph}}$, supported where $Y \in B_{2R}$, hence supported where $(x,t) \in \mathcal{Q}_{2R}^{\mathrm{ph}}$. The support size depends only on $R, \Lambda_{\min}, \Lambda_{\max}$ and the fixed C_c^∞ norm of χ_R, not on the particular branch. □

3.5. **Transport of local quantities under the chart.** Let $t = t(\tau) = t_{\mathrm{c}}(\tau)$ and, for $R > 0$, define

$$J_\tau^-(R) := \big(t(\tau) - \Lambda_{\max}^2 R^2,\ t(\tau)\big].$$

Define renormalized local quantities on $Q_R^*(\tau)$ by

$$A_V(\tau,R) := \sup_{s\in(\tau-R^2,\tau)} \int_{B_R} |V(\cdot,s)|^2\,dY, \qquad E_V(\tau,R) := \int_{\tau-R^2}^{\tau}\!\int_{B_R} |\nabla V|^2\,dY\,ds,$$

$$C_V(\tau,R) := \int_{\tau-R^2}^{\tau}\!\int_{B_R} |V|^3\,dY\,ds, \qquad D_V(\tau,R) := \int_{\tau-R^2}^{\tau}\!\int_{B_R} \Big|P - \langle P\rangle_{B_R}\Big|^{3/2}\,dY\,ds.$$

For interval notation and a center X, define

$$A_u(J,R;X) := \sup_{s\in J} \int_{B_R(X)} |u(\cdot,s)|^2\,dx, \quad E_u(J,R;X) := \int_J \int_{B_R(X)} |\nabla u|^2\,dx\,ds,$$

$$C_u(J,R;X) := \int_J \int_{B_R(X)} |u|^3\,dx\,ds, \quad D_u(J,R;X) := \int_J \int_{B_R(X)} |p - \langle p\rangle_{B_R(X)}|^{3/2}\,dx\,ds.$$

In the inequalities below we abbreviate

$$A_u(J,R) := A_u(J,R;X(\tau)), \quad E_u(J,R) := E_u(J,R;X(\tau)),$$
$$C_u(J,R) := C_u(J,R;X(\tau)), \quad D_u(J,R) := D_u(J,R;X(\tau)).$$

Proposition 3.8 (Cylinder comparison on compact windows)**.** *Assume $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$ and formulas of Lemma 3.5. Then for every $\tau \in I_*$, $R > 0$,*

$$A_V(\tau,R) \le \Lambda_{\min}^{-1} A_u(J_\tau^-(R), \Lambda_{\max} R),$$

$$E_V(\tau,R) \le \Lambda_{\min}^{-3} E_u(J_\tau^-(R), \Lambda_{\max} R),$$

$$C_V(\tau,R) \le \Lambda_{\min}^{-2} C_u(J_\tau^-(R), \Lambda_{\max} R),$$

$$D_V(\tau,R) \le \Lambda_{\min}^{-2} D_u(J_\tau^-(R), \Lambda_{\max} R).$$

In particular, all four renormalized quantities are bounded whenever the physical quantities are bounded on the corresponding compact physical cylinders.

Proof. These are direct changes of variables under Φ_{pre}:

$$V(\cdot,\tau) = \Lambda(\tau) u(\cdot, t(\tau)), \qquad P(\cdot,\tau) = \Lambda(\tau)^2 p(\cdot, t(\tau)),$$

$$\nabla_Y V = \Lambda \nabla_x u, \qquad dY\,d\tau = \Lambda^{-5}\,dx\,dt, \qquad d\tau = \Lambda^2\,dt.$$

Insert these formulas into each definition and use $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$, plus

$$\tau - s \in (0, R^2) \implies t(\tau) - \Lambda_{\max}^2 R^2 \le t(s) \le t(\tau) - \Lambda_{\min}^2 R^2.$$

□

3.6. **Local repaired gauge map.** Fix $R > 0$, $0 < p < 3$, and $\chi_R \in C_c^\infty(B_{2R})$, $\chi_R \equiv 1$ on B_R. Let the local energy profile space be

$$\mathcal{V}_R := \left\{ V \in L^3(B_{2R}) \cap W^{1,3/2}(B_{2R};\mathbb{R}^3) : |Y|^{-p/3} V \in L^3(B_{2R}) \right\},$$

with norm

$$\|V\|_{\mathcal{V}_R} := \|V\|_{L^3(B_{2R})} + \||Y|^{-p/3}V\|_{L^3(B_{2R})} + \|\nabla V\|_{L^{3/2}(B_{2R})}.$$

For differentiating the repaired scale and centering gauges we use the admissible gauge class

$$\mathcal{A}_{p,R} := \left\{ V : \begin{array}{l} V \in L^3(B_{4R}) \cap W^{1,3}_{\mathrm{loc}}(B_{4R};\mathbb{R}^3), \\ \displaystyle\int_{B_{4R}} |Y|^{-p}|V|^3\,dY < \infty, \quad \int_{B_{4R}} |Y|^{-p}|V|\,|Y\cdot\nabla V|\,dY < \infty, \\ \displaystyle\int_{B_{4R}} |Y|^{-p-1}|V|^3\,dY < \infty \end{array} \right\}.$$

The larger ball is only a domain buffer: all gauge integrals are supported in B_{2R}, while (λ, c) is restricted near $(1, 0)$ so that $\lambda^{-1}(B_{2R} - c) \subset B_{4R}$.

Definition 3.9 (Gauge actions)**.** For $(\lambda, c) \in (0,\infty) \times \mathbb{R}^3$, define

$$V_{\lambda,c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

For pressure profiles use

$$P_{\lambda,c}(Y) := \lambda^{-2} P(\lambda^{-1}(Y - c)).$$

We also write $V^{\lambda,c} := V_{\lambda,c}$ and $P^{\lambda,c} := P_{\lambda,c}$.

Definition 3.10 (Finite-dimensional gauge map)**.** For $V \in \mathcal{V}_R$ and $(\lambda, c) \in (0,\infty) \times \mathbb{R}^3$, set

$$\mathcal{F}(\lambda, c; V) := \big(\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3\big),$$

with

$$\mathcal{F}_0(\lambda, c; V) := \int_{\mathbb{R}^3} \chi_R(Y)\,|Y|^{-p}|V_{\lambda,c}(Y)|^3 \mathrm{d}Y - \Theta_0, \qquad \mathcal{F}_j(\lambda, c; V) := \int_{\mathbb{R}^3} Y_j\,\chi_R(Y)\,|V_{\lambda,c}(Y)|^2 \mathrm{d}Y$$

for $j = 1, 2, 3$. The gauge conditions are $\mathcal{F}(\lambda, c; V) = 0$. Codomain is $\mathbb{R}^4$, domain $(0,\infty) \times \mathbb{R}^3 \times \mathcal{V}_R$.

Definition 3.11 (Localized modulation matrix)**.** For $J \subset I_*$ and a profile path $W : J \to \mathcal{V}_R$, define

$$J_W(\tau) := D_{(\lambda,c)}\mathcal{F}\big(1, 0; W(\tau)\big) \in \mathbb{R}^{4\times4}.$$

The rows are ordered as the scale constraint followed by the three centering constraints.

Lemma 3.12 (Weighted integrability and differentiation under integrals)**.** *Assume $V, W \in \mathcal{A}_{p,R}$. Then $G_0(V) := \int \chi_R |Y|^{-p}|V|^3$ is finite and*

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big|_{\epsilon=0} \int \chi_R |Y|^{-p}|V + \epsilon W|^3\,\mathrm{d}Y = \int 3\chi_R |Y|^{-p}|V|\,V\cdot W\,\mathrm{d}Y.$$

Proof. For $W \in \mathcal{A}_{p,R}$, define

$$F_\varepsilon(Y) := \chi_R(Y)|Y|^{-p}|V(Y) + \varepsilon W(Y)|^3.$$

For each Y, $\varepsilon \mapsto F_\varepsilon(Y)$ is C^1 and

$$\partial_\varepsilon F_\varepsilon = 3\chi_R |Y|^{-p}|V + \varepsilon W|(V + \varepsilon W)\cdot W.$$

For $|\varepsilon| \le 1$,

$$|F_\varepsilon(Y)| + |\partial_\varepsilon F_\varepsilon| \le C\chi_R |Y|^{-p}\big(|V|^3 + |W|^3\big) \in L^1(B_{2R})$$

by the defining weighted L^3 condition of $\mathcal{A}_{p,R}$. Hence dominated convergence allows differentiation under the integral, and

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0}\int_{B_{2R}}\chi_R|Y|^{-p}|V+\epsilon W|^3\,\mathrm{d}Y=\int_{B_{2R}}3\chi_R|Y|^{-p}|V|\,V\cdot W\,\mathrm{d}Y.$$

□

Lemma 3.13 (Gauge covariance)**.** *For $V\in\mathcal{V}_R$, $(\lambda,c)\in(0,\infty)\times\mathbb{R}^3$, and $\alpha=0,\dots,3$,*

$$\mathcal{F}_\alpha(1,0;V_{\lambda,c})=\mathcal{F}_\alpha(\lambda,c;V).$$

Proof. For each component this is a direct substitution:

$$\mathcal{F}_0(1,0;V_{\lambda,c})=\int\chi_R(Y)|Y|^{-p}|V_{\lambda,c}(Y)|^3-\Theta_0\,\mathrm{d}Y=\mathcal{F}_0(\lambda,c;V),$$

$$\mathcal{F}_j(1,0;V_{\lambda,c})=\int Y_j\chi_R(Y)|V_{\lambda,c}(Y)|^2\,\mathrm{d}Y=\mathcal{F}_j(\lambda,c;V),\quad j=1,2,3.$$

□

Proposition 3.14 (Differentiability of $\mathcal{F}$)**.** *For each fixed $V\in\mathcal{A}_{p,R}$, the map $(\lambda,c)\mapsto\mathcal{F}(\lambda,c;V)$ is C^1 in a neighborhood of $(1,0)$. In each component $\mathcal{F}_\alpha$ all derivatives under the integral sign are legitimate.*

Proof. Fix (λ,c) near $(1,0)$, and abbreviate

$$V_{\lambda,c}(Y):=\lambda^{-1}V\big(\lambda^{-1}(Y-c)\big).$$

By direct differentiation,

$$\partial_\lambda V_{\lambda,c}=-\lambda^{-1}V_{\lambda,c}-\lambda^{-1}(Y-c)\cdot\nabla V_{\lambda,c},\qquad\partial_{c_k}V_{\lambda,c}=-\lambda^{-2}\partial_kV(\lambda^{-1}(Y-c))=-\partial_kV_{\lambda,c}.$$

The neighborhood of $(1,0)$ is chosen so that the arguments of V remain inside B_{4R} for $Y\in\operatorname{supp}\chi_R$. The scale derivative is controlled by the two weighted integrability conditions

$$\int|Y|^{-p}|V|^3<\infty,\qquad\int|Y|^{-p}|V|\,|Y\cdot\nabla V|<\infty,$$

transported by the smooth change of variables. The translation derivative in the scale component is treated by the source identity

$$3\int|Y|^{-p}|V|\,V\cdot\partial_kV=p\int Y_k|Y|^{-p-2}|V|^3,$$

obtained first for smooth compactly supported profiles and then by approximation in $\mathcal{A}_{p,R}$; the right-hand side is finite because $\int|Y|^{-p-1}|V|^3<\infty$. The translation components are supported in B_{2R} and are controlled by $V\in L^2_{\mathrm{loc}}$ and $\nabla V\in L^3_{\mathrm{loc}}$. These bounds give an integrable majorant for the difference quotients, so dominated convergence applies.

$$\mathcal{F}_0(\lambda,c;V)=\int_{\mathbb{R}^3}\chi_R(Y)\,|Y|^{-p}|V_{\lambda,c}|^3\,\mathrm{d}Y-\Theta_0,$$

for which dominated convergence gives

$$\partial_\lambda\mathcal{F}_0=\int_{\mathbb{R}^3}\chi_R|Y|^{-p}\,3|V_{\lambda,c}|\,V_{\lambda,c}\cdot\partial_\lambda V_{\lambda,c}\,dY,$$

$$\partial_{c_k}\mathcal{F}_0=\int_{\mathbb{R}^3}\chi_R|Y|^{-p}\,3|V_{\lambda,c}|\,V_{\lambda,c}\cdot\partial_{c_k}V_{\lambda,c}\,dY.$$

Similarly

$$\mathcal{F}_j(\lambda,c;V)=\int_{\mathbb{R}^3}Y_j\chi_R(Y)\,|V_{\lambda,c}|^2\,\mathrm{d}Y,\quad j=1,2,3,$$

hence

$$\partial_\lambda \mathcal{F}_j = \int_{\mathbb{R}^3} 2Y_j \chi_R \, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} \, dY,$$

$$\partial_{c_k} \mathcal{F}_j = \int_{\mathbb{R}^3} 2Y_j \chi_R \, V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c} \, dY.$$

Thus each component admits pointwise C^1 formulas and these are continuous in (λ, c) near $(1,0)$. Therefore $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V) \in C^1$ near $(1,0)$. □

Proposition 3.15 (Full local Jacobian at $(1,0)$)**.** *Let $V \in \mathcal{A}_{p,R}$. Set $V_0(Y) = V(Y)$. At $(\lambda, c) = (1,0)$, the matrix*

$$J(V) := D_{(\lambda,c)}\mathcal{F}(1,0;V) \in \mathbb{R}^{4\times 4}$$

has rows

$$J_\alpha = \big(\partial_\lambda \mathcal{F}_\alpha,\ \partial_{c_1}\mathcal{F}_\alpha,\ \partial_{c_2}\mathcal{F}_\alpha,\ \partial_{c_3}\mathcal{F}_\alpha\big), \quad \alpha = 0, \dots, 3.$$

Writing $W_{\mathrm{scl}} = V_0 + Y \cdot \nabla V_0$, one has

$$\partial_\lambda \mathcal{F}_0(1,0;V) = -3 \int \chi_R |Y|^{-p} |V_0| \, V_0 \cdot W_{\mathrm{scl}} \, \mathrm{d}Y, \tag{2}$$

$$\partial_{c_k} \mathcal{F}_0(1,0;V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| \, V_0 \cdot \partial_k V_0 \, \mathrm{d}Y, \tag{3}$$

$$\partial_\lambda \mathcal{F}_j(1,0;V) = -2 \int_{\mathbb{R}^3} Y_j \chi_R \, V_0 \cdot W_{\mathrm{scl}} \, \mathrm{d}Y, \tag{4}$$

$$\partial_{c_k} \mathcal{F}_j(1,0;V) = -2 \int_{\mathbb{R}^3} \chi_R Y_j V_0 \cdot \partial_k V_0 \, \mathrm{d}Y, \tag{5}$$

for $j, k = 1, 2, 3$.

Proof. Differentiate the formulas in Definition 3.10 at $(\lambda, c) = (1,0)$.

Since $V_{\lambda,c}(Y) = \lambda^{-1} V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c} = -\lambda^{-1} V_{\lambda,c} - \lambda^{-1} Y \cdot \nabla V_{\lambda,c}, \qquad \partial_{c_k} V_{\lambda,c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)).$$

At $(1,0)$, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V)$, $\partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V$.

For $\mathcal{F}_0$, write

$$\mathcal{F}_0(\lambda, c; V) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 \, dY - \Theta_0.$$

Derivative in λ:

$$\partial_\lambda \mathcal{F}_0 = \int \chi_R |Y|^{-p} 3 |V_{\lambda,c}| \, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} \, dY$$

Evaluating at $(1,0)$ yields (2).

For $k = 1, 2, 3$, c_k-derivative of $\mathcal{F}_0$ has no cutoff term:

$$\partial_{c_k} \mathcal{F}_0(1,0;V) = \int \chi_R |Y|^{-p} \, 3|V| \, V \cdot (-\partial_k V) \, dY,$$

which is (3).

For $\mathcal{F}_j$,

$$\mathcal{F}_j(\lambda, c; V) = \int Y_j \chi_R(Y) \, |V_{\lambda,c}|^2 \, dY.$$

Hence

$$\partial_\lambda \mathcal{F}_j = \int 2Y_j \chi_R \, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} \, dY,$$

and at $(1,0)$:

$$\partial_\lambda \mathcal{F}_j(1,0;V) = -2\int Y_j \chi_R\, V\cdot W_{\rm scl}\, dY,$$

which is (4). Similarly,

$$\partial_{c_k}\mathcal{F}_j(1,0;V) = \int 2Y_j\chi_R\, V\cdot \partial_{c_k}V_{\lambda,c}\big|_{(1,0)}\, dY = -2\int \chi_R Y_j V\cdot \partial_k V\, dY,$$

giving (5). $\square$

Proposition 3.16 (Translation block and Schur-complement invertibility)**.** *Let $V : J\to \mathcal{A}_{p,R}$ satisfy the repaired gauge constraints*

$$\mathcal{F}(1,0;V(\tau)) = 0 \qquad \textit{for a.e. } \tau\in J.$$

Write the modulation matrix in block form

$$J(V(\tau)) = \begin{pmatrix} \alpha(\tau) & u(\tau)^T \\ v(\tau) & A(\tau)\end{pmatrix},$$

where $\alpha = \partial_\lambda \mathcal{F}_0$, $u_\ell = \partial_{c_\ell}\mathcal{F}_0$, $v_j = \partial_\lambda \mathcal{F}_j$, and $A_{j\ell} = \partial_{c_\ell}\mathcal{F}_j$. Assume that for a.e. $\tau\in J$

$$m_\chi(\tau) := \int |V(Y,\tau)|^2\chi_R(Y)\, dY \ge m_0 > 0,$$

$$\int_{A_R} |V(Y,\tau)|^2\, dY \le \varepsilon_0, \qquad 3C_\chi\varepsilon_0 \le \frac{m_0}{2},$$

where $A_R = \{R\le |Y|\le 2R\}$ and

$$C_\chi := \max_{j,\ell}\|Y_j\partial_\ell \chi_R\|_{L^\infty}.$$

Assume also

$$C_u := \operatorname*{ess\,sup}_{\tau\in J}\int |Y|^{-p-1}|V(Y,\tau)|^3\, dY < \infty$$

and, with $C_\nabla := \||Y|\nabla\chi_R\|_{L^\infty}$,

$$6p\left(\frac{2}{m_0}\right) C_u\, RC_\nabla \varepsilon_0 \le \frac{p\Theta_0}{2}.$$

Then $J(V(\tau))$ is invertible for a.e. $\tau\in J$, and

$$\operatorname*{ess\,sup}_{\tau\in J}\|J(V(\tau))^{-1}\|_{\infty\to\infty} \le K(R,p,\Theta_0,m_0,C_u,\varepsilon_0,\chi_R).$$

Proof. The entries are finite by $V(\tau)\in\mathcal{A}_{p,R}$ and the compact support of χ_R. With the present parameter convention, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y\cdot\nabla V)$. Hence the scale entry satisfies the signed transversality identity

$$\alpha(\tau) = -p\int |Y|^{-p}|V(Y,\tau)|^3\, dY = -p\Theta_0$$

on the gauge surface. Thus $|\alpha|\ge p\Theta_0$.

For the centering block, integration by parts gives

$$A_{j\ell} = -\delta_{j\ell}\int |V|^2\chi_R - \int |V|^2 Y_j\partial_\ell\chi_R.$$

Writing $A = -m_\chi I_3 + E$, the annular support of $\nabla\chi_R$ gives

$$|E_{j\ell}| \le C_\chi \int_{A_R}|V|^2 \le C_\chi\varepsilon_0, \qquad \|E\|_{\infty\to\infty}\le 3C_\chi\varepsilon_0 \le m_0/2.$$

Thus $A = -m_\chi(I_3 - m_\chi^{-1}E)$ and $\|m_\chi^{-1}E\|_{\infty\to\infty}\le 1/2$, so A^{-1} exists and $\|A^{-1}\|_{\infty\to\infty}\le 2/m_0$.

For the scale-to-translation block, the scale-constraint formula gives

$$|u_\ell| \le p \int |Y|^{-p-1}|V|^3\,dY \le p \int |Y|^{-p-1}|V|^3\,dY,$$

so $\|u\|_\infty \le pC_u$. For the translation-to-scale block,

$$v_j = -\int |V|^2 Y_j (Y\cdot\nabla\chi_R)\,dY$$

because the centering constraint cancels the interior term. Therefore

$$\|v\|_\infty \le 2R\,C_\nabla\varepsilon_0.$$

The Schur complement $S = \alpha - u^T A^{-1} v$ satisfies

$$|u^T A^{-1} v| \le 3\|u\|_\infty \|A^{-1}\|_{\infty\to\infty}\|v\|_\infty \le 6p\left(\frac{2}{m_0}\right) C_u R C_\nabla \varepsilon_0 \le \frac{p\Theta_0}{2}.$$

Since $|\alpha| = p\Theta_0$, this gives $|S| \ge p\Theta_0/2$. The block inverse formula

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}u^T A^{-1} \\ -A^{-1}vS^{-1} & A^{-1} + A^{-1}vS^{-1}u^T A^{-1} \end{pmatrix}$$

then gives the displayed bound. □

3.7. **Repaired gauge by implicit inversion in time.**

Lemma 3.17 (Time chain rule for preliminary profile)**.** *For each $\alpha = 0,\dots,3$, set $G_\alpha(\tau) := \mathcal{F}_\alpha(1,0;V_{\rm pre}(\tau))$. Then $G_\alpha \in W^{1,1}_{\rm loc}(I_*)$ and for a.e. τ,*

$$G'_\alpha(\tau) = D_V\mathcal{F}_\alpha(1,0;V_{\rm pre}(\tau))[\partial_\tau V_{\rm pre}(\tau)].$$

The right-hand side is obtained from the profile formulas of Proposition 3.14 and is in $L^1_{\rm loc}(I_*)$.

Proof. Let $V^\varepsilon_{\rm pre}$ be a spatial and temporal Steklov mollification inside a slightly larger compact cylinder. For the smoothed profiles the ordinary chain rule gives

$$\frac{d}{d\tau}\mathcal{F}_\alpha(1,0;V^\varepsilon_{\rm pre}(\tau)) = D_V\mathcal{F}_\alpha(1,0;V^\varepsilon_{\rm pre}(\tau))[\partial_\tau V^\varepsilon_{\rm pre}(\tau)].$$

The transformed equation expresses $\partial_\tau V^\varepsilon_{\rm pre}$ as the mollification of $\nu\Delta V_{\rm pre} - (V_{\rm pre}\cdot\nabla)V_{\rm pre} - \nabla P_{\rm pre}$ plus the chart transport terms. Pairing this distribution with the compactly supported Gateaux gradients below is legitimate by the local suitable bounds, the pressure bound, and the weighted L^3 control in (SC4). Passing $\varepsilon \downarrow 0$ gives the asserted $W^{1,1}_{\rm loc}$ identity. For $W \in \mathcal{V}_R$, these derivative formulas hold:

$$D_V\mathcal{F}_0(1,0;V)[W] = \int_{\mathbb{R}^3} 3\chi_R|Y|^{-p}|V|\,V\cdot W\,\mathrm{d}Y,$$

$$D_V\mathcal{F}_j(1,0;V)[W] = \int_{\mathbb{R}^3} 2Y_j\chi_R\,V\cdot W\,\mathrm{d}Y, \qquad j=1,2,3,$$

where both right-hand sides are finite on $\mathcal{V}_R$ by compact support, $p<3$, and $V,W \in L^3\cap W^{1,3/2}_{\rm loc}$. The right-hand side is finite by the explicit formulas above and by the distributional equation paired with compactly supported test gradients; the local energy, pressure, and weighted L^3 bounds give an $L^1_{\rm loc}(I_*)$ majorant. □

Lemma 3.18 (AC implicit inversion with parameters)**.** *Let $U\subset\mathbb{R}^4$ be open and $G: I_*\times U \to \mathbb{R}^4$. Assume:*

- *(i) for each $s\in U$, $\tau\mapsto G(\tau,s)\in W^{1,1}_{\rm loc}(I_*;\mathbb{R}^4)$, and $D_\tau G \in L^1_{\rm loc}(I_*;L^\infty(U;\mathbb{R}^4))$;*
- *(ii) for a.e. $\tau\in I_*$, $s\mapsto G(\tau,s)$ is C^1 on U, with D_sG continuous in (τ,s);*

(iii) for some $(\tau_0, s_0) \in I_* \times U$, $G(\tau_0, s_0) = 0$, and there is $\rho, C > 0$ such that

$$\sup_{(\tau,s)\in I_*\times B_\rho(s_0)} \|D_s G(\tau, s)^{-1}\|_{\mathcal{L}(\mathbb{R}^4,\mathbb{R}^4)} \le C.$$

Then after possibly shrinking I_* to a subinterval $J \ni \tau_0$, there is a unique map $s \in AC(J; U)$, $s(\tau_0) = s_0$, with

$$G(\tau, s(\tau)) = 0 \quad \text{a.e. in } J.$$

Moreover

$$s'(\tau) = -D_s G(\tau, s(\tau))^{-1}\, D_\tau G(\tau, s(\tau)) \quad \text{a.e. on } J.$$

Proof. By (ii), the matrix field $D_s G$ is continuous in s, so there is $r_0 \in (0, \rho)$ such that $\inf_{(\tau,s)\in I_*\times B_{r_0}(s_0)} |\det D_s G(\tau, s)| > 0$ after possibly shrinking I_*; this keeps $D_s G^{-1}$ bounded and continuous on $B_{r_0}(s_0)$.

Set

$$F(\tau, s) := -D_s G(\tau, s)^{-1} D_\tau G(\tau, s).$$

Then $F(\tau, \cdot)$ is continuous on $B_{r_0}(s_0)$ for a.e. τ and $F(\cdot, s) \in L^1_{\mathrm{loc}}(I_*; \mathbb{R}^4)$ by (i), hence F is a Carathéodory right-hand side. Since the gauge map is C^1 in the finite variables, after shrinking r_0 there is $L \in L^1(J)$ such that

$$|F(\tau, s_1) - F(\tau, s_2)| \le L(\tau)|s_1 - s_2| \quad \text{for } s_1, s_2 \in B_{r_0}(s_0).$$

Choose $\delta > 0$ so small that

$$\int_{\tau_0-\delta}^{\tau_0+\delta} \sup_{s\in B_{r_0}(s_0)} |F(\sigma, s)|\, d\sigma < r_0.$$

On $J = [\tau_0 - \delta, \tau_0 + \delta] \cap I_*$ define the Picard operator

$$(\mathcal{T} s)(\tau) := s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma))\, d\sigma$$

on $C(J; B_{r_0}(s_0))$. The preceding bound keeps $\mathcal{T}$ in the same closed ball, and the Lipschitz estimate gives

$$\|\mathcal{T} s_1 - \mathcal{T} s_2\|_{C(J)} \le \left(\int_J L(\sigma)\, d\sigma\right) \|s_1 - s_2\|_{C(J)}.$$

After decreasing δ once more, the coefficient is < 1, so Banach's fixed-point theorem gives a unique solution

$$s(\tau) = s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma))\, d\sigma, \qquad \tau \in J,$$

which stays in $B_{r_0}(s_0) \subset U$. This solution is absolutely continuous.

Define $H(\tau) := G(\tau, s(\tau))$. Since $G(\cdot, s(\cdot)) \in W^{1,1}_{\mathrm{loc}}$ from (i), the chain rule gives for a.e. $\tau \in J$

$$H'(\tau) = D_\tau G(\tau, s(\tau)) + D_s G(\tau, s(\tau)) s'(\tau) = D_\tau G(\tau, s(\tau)) + D_s G(\tau, s(\tau)) F(\tau, s(\tau)) = 0.$$

Hence $H \equiv G(\tau_0, s_0) = 0$ on J, which is the claimed constraint. Uniqueness follows from Grönwall on the difference of two AC solutions on $B_{r_0}(s_0)$, using Lipschitz continuity of $F(\tau, \cdot)$ on that ball. $\square$

Definition 3.19 (Retained nondegenerate repaired-gauge regime)**.** Let $J \subset I_*$ be compact. The retained nondegenerate repaired-gauge regime over the preliminary profile V_{pre} consists of branches for which there are

$$\lambda \in AC(J; (0, \infty)), \qquad c \in AC(J; \mathbb{R}^3),$$

such that, for a.e. $\tau \in J$,

$$V(\tau) := V_{\rm pre}^{\lambda(\tau),c(\tau)} \in \mathcal{A}_{p,R}, \qquad \mathcal{F}(1,0;V(\tau)) = 0.$$

The path is quantitatively nondegenerate if the translation-block, annular, weighted first-moment, and Schur-complement bounds in Proposition 3.16 hold on J. If no such local AC selection exists, or if the quantitative nondegeneracy hypotheses fail persistently, the branch lies in the gauge-degenerate, multibubble, or cascade alternatives rather than in the retained regime.

Proposition 3.20 (Gauge regularity on the retained regime)**.** *Let (λ, c) be the repaired-gauge selection of the retained nondegenerate regime on J. Then*

$$\mathcal{F}\big(\lambda(\tau), c(\tau); V_{\rm pre}(\tau)\big) = 0 \quad \textit{for a.e. } \tau \in J,$$

and $(\lambda, c) \in AC(J; \mathbb{R}^4)$ by definition. If the path is quantitatively nondegenerate, then

$$\operatorname*{ess\,sup}_{\tau \in J} \|J(V(\tau))^{-1}\| < \infty.$$

Proof. The constraint follows from Lemma 3.13 applied to $V(\tau) = V_{\rm pre}^{\lambda(\tau),c(\tau)}$. The absolute continuity is part of Definition 3.19. The uniform inverse bound is Proposition 3.16. □

Definition 3.21 (Repaired chart)**.** Define

$$\Lambda(\tau) := \Lambda_{\rm pre}(\tau)\lambda(\tau), \qquad X(\tau) := X_{\rm pre}(\tau) + \Lambda_{\rm pre}(\tau)c(\tau),$$

$$V(\tau) := V_{\rm pre}^{\lambda(\tau),c(\tau)}, \qquad P(\tau) := P_{\rm pre}^{\lambda(\tau),c(\tau)} \text{ (same physical pressure pull-back under } \Phi).$$

Theorem 3.22 (AC repaired parameters)**.** *Assume* $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$*, and suppose the branch lies in the retained nondegenerate repaired-gauge regime on J, with selection (λ, c). Then the corrected chart (Λ, X) is absolutely continuous on J, satisfies*

$$a(\tau) = -\Lambda'(\tau) \in L^1_{\rm loc}(J), \qquad b(\tau) = -X'(\tau) \in L^1_{\rm loc}(J),$$

and the gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$ hold a.e. Equivalently,

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\rm pre}(\tau)) = 0$$

for the repaired-gauge selection (λ, c).

Proof. By Definition 3.19, $(\lambda, c) \in AC(J)$ and the repaired gauge constraints hold for $V(\tau)$ a.e. By definition $V(\tau) = V_{\rm pre}^{\lambda(\tau),c(\tau)}$, so Lemma 3.13 applied at $(\lambda, c) = (\lambda(\tau), c(\tau))$ gives

$$\mathcal{F}(1, 0; V_{\rm pre}^{\lambda,c}) = \mathcal{F}(\lambda, c; V_{\rm pre})$$

implies $\mathcal{F}(1, 0; V(\tau)) = 0$ a.e.

Since $\Lambda = \Lambda_{\rm pre}\lambda$ and $X = X_{\rm pre} + \Lambda_{\rm pre}c$, and $\Lambda_{\rm pre}, X_{\rm pre}$ are AC by (SC5), the products and sums preserve AC, hence $(\Lambda, X) \in AC(I_*)$. Differentiating,

$$a = -\Lambda' = -(\Lambda'_{\rm pre}\lambda + \Lambda_{\rm pre}\lambda') \in L^1_{\rm loc}, \qquad b = -X' = -(X'_{\rm pre} + \Lambda'_{\rm pre}c + \Lambda_{\rm pre}c') \in L^1_{\rm loc}.$$

This gives the claimed regularity. □

Remark 3.23 (Gauge degeneracy)*.* Failure of invertibility of J is the gauge-degenerate alternative to the retained single-core regime for this fixed renormalization mechanism. In the Type II decomposition of alternatives this case belongs to the multibubble/cascade and scale-rigidity alternatives, rather than to the representation theorem.

3.8. **Pressure transport and decomposition.**

Lemma 3.24 (Renormalized pressure equation)**.** *If V, P satisfy the renormalized Navier–Stokes equation in Theorem 3.6, then*

$$-\Delta P = \partial_i\partial_j(V_iV_j) \qquad \text{in } \mathcal{D}'(Q^*_{2R}).$$

Proof. Take test $\varphi \in C_c^\infty(Q^*_{2R})$. Since $V, P \in L^2_{\mathrm{loc}} \times L^{3/2}_{\mathrm{loc}}$ on Q^*_{2R}, all pairings below are legitimate. Expand

$$\nabla \cdot \big((V\cdot\nabla)V\big) = \sum_{i,j} \partial_i\partial_j(V_iV_j) - \sum_{i,j} \partial_iV_i\,\partial_jV_j.$$

Because $\nabla\cdot V = 0$, this reduces to $\partial_i\partial_j(V_iV_j)$. Taking divergence of

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P - \nu\Delta V + a(V + Y\cdot\nabla V) + b\cdot\nabla V = 0$$

and using $\nabla\cdot V = 0$, $\nabla\cdot(V+Y\cdot\nabla V) = 0$, $\nabla\cdot(b\cdot\nabla V) = b\cdot\nabla(\nabla\cdot V) = 0$, and $\nabla\cdot\Delta V = \Delta(\nabla\cdot V) = 0$, the following identity holds in $\mathcal{D}'$:

$$\Delta P + \partial_i\partial_j(V_iV_j) = 0.$$

Hence $-\Delta P = \partial_i\partial_j(V_iV_j)$ on Q^*_{2R}. □

Theorem 3.25 (Localized pressure decomposition)**.** *Let $\zeta \in C_c^\infty(B_{2R})$, $\zeta \equiv 1$ on B_R, and assume $V \in L^1_{\mathrm{loc}}(I_*; L^3(B_{2R}))$, $P \in L^1_{\mathrm{loc}}(I_*; L^{3/2}(B_{2R}))$. For a.e. $\tau \in I_*$ there exists*

$$P(\cdot,\tau) = P_{\mathrm{loc}}(\cdot,\tau) + H(\cdot,\tau)$$

on B_R such that

$$\begin{cases} -\Delta P_{\mathrm{loc}}(\cdot,\tau) = \partial_i\partial_j\big(\zeta V_iV_j\big), \\ -\Delta H(\cdot,\tau) = 0 \ \text{ on } B_R, \quad \fint_{B_R} H(\cdot,\tau)\,dY = 0. \end{cases}$$

Moreover

$$\|P_{\mathrm{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|^2_{L^3(B_{2R})}, \qquad \|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|P(\tau)\|_{L^{3/2}(B_{2R})} + \|V(\tau)\|^2_{L^3(B_{2R})}$$

for each $1 \le q < \infty$ and each $0 < r < R$, with constants depending only on R, r, and q.

Proof. For fixed τ, set

$$P_{\mathrm{loc}} = \mathcal{R}_i\mathcal{R}_j(\zeta V_iV_j), \qquad H := P - P_{\mathrm{loc}}.$$

$\zeta V_iV_j \in L^1_{\mathrm{loc}}(I_*; L^{3/2}(B_{2R}))$, so the Calderón–Zygmund operator $f \mapsto \mathcal{R}_i\mathcal{R}_jf$ gives

$$P_{\mathrm{loc}}(\cdot,\tau) = \mathcal{R}_i\mathcal{R}_j(\zeta V_iV_j)(\cdot,\tau), \qquad \|P_{\mathrm{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|^2_{L^3(B_{2R})}.$$

Because $\mathcal{R}_i\mathcal{R}_j$ is linear bounded on $L^{3/2}$,

$$P_{\mathrm{loc}} \in L^1_{\mathrm{loc}}(I_*; L^{3/2}(B_R)).$$

Since $-\Delta P_{\mathrm{loc}} = \partial_i\partial_j(\zeta V_iV_j)$ and $\zeta \equiv 1$ on B_R,

$$-\Delta H(\cdot,\tau) = \partial_i\partial_j\big((1-\zeta)V_iV_j\big) = 0 \quad \text{in } B_R,$$

so $H(\cdot,\tau)$ is harmonic on B_R. Imposing $\fint_{B_R} H = 0$ fixes the time-dependent additive constant uniquely in each time slice. For $q \in [1,\infty)$ and $0 < r < R$, harmonicity and local L^p-to-L^q interior estimates imply

$$\|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|H(\tau)\|_{L^{3/2}(B_R)}.$$

Finally, $\|H(\tau)\|_{L^{3/2}(B_R)} \le \|P(\tau)\|_{L^{3/2}(B_R)} + \|P_{\mathrm{loc}}(\tau)\|_{L^{3/2}(B_R)}$, giving the announced bound in terms of P and V. □

Remark 3.26 (Measurability and normalization). If V, P are measurable in τ, then $\tau \mapsto P_{\mathrm{loc}}(\cdot, \tau)$ is measurable in $L^{3/2}(B_R)$ by boundedness of $\mathcal{R}_i\mathcal{R}_j$, hence $\tau \mapsto H(\cdot, \tau)$ is measurable in each $L^q(B_r)$, $0 < r < R$. The normalization $⨍_{B_R} H(\cdot, \tau)\, dY = 0$ fixes the time-dependent additive constant in H uniquely for each τ, and therefore the decomposition is a legitimate renormalized pressure splitting.

3.9. **Modulation from differentiated constraints.**

Lemma 3.27 (Differentiation of constraints)**.** *For the repaired profile V constructed above, with $|Y|^{-p}|V|^3 \in L^1_{\mathrm{loc}}(I_*; L^1)$, Then $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$ are absolutely continuous and*

$$\frac{\mathrm{d}}{\mathrm{d}\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R)\, \partial_\tau V \,\mathrm{d}Y$$

for a.e. τ. Here $\mathbf{D}_\alpha$ is the L^1-integrable Gateaux gradient associated to $\mathcal{F}_\alpha$.

Proof.

$$G_\alpha(\tau) = \mathcal{F}_\alpha(1, 0; V(\tau))$$

is first evaluated on spatial-temporal mollifications V^ε. For those smooth profiles the ordinary chain rule gives

$$\frac{d}{d\tau} G^\varepsilon_\alpha(\tau) = D_V\mathcal{F}_\alpha(1, 0; V^\varepsilon(\tau))\, [\partial_\tau V^\varepsilon(\tau)].$$

The local suitable bounds, pressure estimates, and compact support of the Gateaux gradients below allow passage to the limit $\varepsilon \downarrow 0$, which gives the displayed derivative formula for V. For $\mathcal{F}_0$,

$$\mathbf{D}_0(V, \nabla V, \nabla \chi_R) = 3\chi_R |Y|^{-p}|V|\, V;$$

for $\alpha = 1, 2, 3$,

$$\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) = 2\chi_R\, Y_\alpha V.$$

Each coefficient is $L^1(B_{2R})$: $3\chi_R|Y|^{-p}|V|\, V \in L^{3/2}(B_{2R}) \subset L^1(B_{2R})$ from $p < 3$ and $|Y|^{-p}|V|^3 \in L^1$, and $2\chi_R Y_\alpha V \in L^2(B_{2R}) \subset L^1(B_{2R})$ by compact support.

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot \partial_\tau V \,\mathrm{d}Y,$$

Since $\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \in L^1(B_{2R})$ and $\partial_\tau V \in L^1_{\mathrm{loc}}(I_*; L^2(B_{2R}))$, the right-hand side is in $L^1_{\mathrm{loc}}(I_*)$, so $G_\alpha \in W^{1,1}_{\mathrm{loc}}(I_*) \subset AC_{\mathrm{loc}}(I_*)$. ☐

Lemma 3.28 (Directional derivatives in the profile variable)**.** *For $\mathcal{A}_\tau = V + Y \cdot \nabla V$ and $\alpha = 0, 1, 2, 3$,*

$$D_V\mathcal{F}_\alpha(1, 0; V)[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \qquad D_V\mathcal{F}_\alpha(1, 0; V)[\partial_k V] = -\partial_{c_k}\mathcal{F}_\alpha(1, 0; V).$$

Indeed, from $V_{\lambda,c}(Y) = \lambda^{-1}V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c}\big|_{(1,0)} = -(V + Y \cdot \nabla V) = -\mathcal{A}_\tau, \qquad \partial_{c_k} V_{\lambda,c}\big|_{(1,0)} = -\partial_k V.$$

Substituting these into the Gateaux formulas from Proposition 3.14 gives the two identities.

Theorem 3.29 (Modulation system)**.** *Let (V, P) be the repaired pair on the compact window under consideration. Writing $\theta := (a, b)$, define*

$$\mathcal{A}_\tau(Y) := V(\tau, Y) + Y \cdot \nabla V(\tau, Y).$$

For $\alpha = 0, 1, 2, 3$, define

$$\mathcal{R}_\alpha(\tau) := -\int_{\mathbb{R}^3} \mathbf{D}_\alpha\big(V, \nabla V, \nabla \chi_R\big) \cdot \big((V \cdot \nabla)V + \nabla P - \nu\Delta V\big)\,\mathrm{d}Y.$$

Then for a.e. $\tau \in I_*$,

$$\partial_\lambda \mathcal{F}_\alpha(1,0;V)\,a(\tau) + \sum_{k=1}^{3} \partial_{c_k}\mathcal{F}_\alpha(1,0;V)\,b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

Equivalently,

$$J(\tau)\theta(\tau) = \mathcal{R}(\tau).$$

Hence

$$\theta(\tau) = J(\tau)^{-1}\mathcal{R}(\tau) \in L^1_{\mathrm{loc}}(I_*)^4, \qquad |\theta(\tau)| \le \kappa\,|\mathcal{R}(\tau)|.$$

Proof. Set $G_\alpha(\tau) := \mathcal{F}_\alpha(1,0;V(\tau))$. By Theorem 3.22, $G_\alpha(\tau) = 0$ a.e.; by Lemma 3.27, $G_\alpha \in W^{1,1}_{\mathrm{loc}}$. Hence for a.e. τ,

$$0 = \frac{d}{d\tau} G_\alpha(\tau) = D_V\mathcal{F}_\alpha(1,0;V(\tau))\,[\partial_\tau V(\tau)].$$

Insert the equation from Theorem 3.6,

$$\partial_\tau V = -(V\cdot\nabla)V - \nabla P + \nu\Delta V - a\mathcal{A}_\tau - b\cdot\nabla V.$$

This gives

$$0 = -D_V\mathcal{F}_\alpha\big[(V\cdot\nabla)V + \nabla P - \nu\Delta V\big] - a\,D_V\mathcal{F}_\alpha[\mathcal{A}_\tau] - \sum_{k=1}^{3} b_k\,D_V\mathcal{F}_\alpha[\partial_k V].$$

By Lemma 3.27, for any profile direction W,

$$D_V\mathcal{F}_\alpha(1,0;V)[W] = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V,\nabla V,\nabla\chi_R)\cdot W\,\mathrm{d}Y.$$

By Lemma 3.28,

$$D_V\mathcal{F}_\alpha[\mathcal{A}_\tau] = -\partial_\lambda\mathcal{F}_\alpha(1,0;V), \qquad D_V\mathcal{F}_\alpha[\partial_k V] = -\partial_{c_k}\mathcal{F}_\alpha(1,0;V).$$

Substituting into the previous line yields the linear system

$$\partial_\lambda \mathcal{F}_\alpha(1,0;V)\,a(\tau) + \sum_{k=1}^{3} \partial_{c_k}\mathcal{F}_\alpha(1,0;V)\,b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

which is equivalent to $J(\tau)\theta(\tau) = \mathcal{R}(\tau)$. By definition of $\mathbf{D}_\alpha$, each $\mathcal{R}_\alpha \in L^1_{\mathrm{loc}}(I_*)$, and Proposition 3.16 gives $|J^{-1}(\tau)| \le K$ a.e., hence

$$\theta(\tau) = J^{-1}(\tau)\mathcal{R}(\tau), \quad \theta \in L^1_{\mathrm{loc}}(I_*), \quad |\theta| \le K|\mathcal{R}| \text{ a.e..}$$

□

Corollary 3.30 (Regularity of a, b)**.** *For the repaired pair of Theorem 3.29,* $a, b \in L^1_{\mathrm{loc}}(I_*)$*. If in addition* $\mathcal{R} \in L^\infty_{\mathrm{loc}}(I_*)$*, then* $a, b \in L^\infty_{\mathrm{loc}}(I_*)$*.*

Proof. By Theorem 3.29, $\theta = (a,b) = J^{-1}\mathcal{R}$ a.e., and $J^{-1} \in L^\infty_{\mathrm{loc}}(I_*)$ by Theorem 3.16. The first claim is therefore immediate from $\mathcal{R} \in L^1_{\mathrm{loc}}$, while the second follows if $\mathcal{R} \in L^\infty_{\mathrm{loc}}$. □

3.10. Repaired suitable structure in renormalized variables.

Proposition 3.31 (Transport of suitable local energy inequality)**.** *Suppose (u,p) is suitable on $\mathcal{Q}^{\mathrm{ph}}_{2R}$, and (V,P) is obtained by the repaired map Φ. Set*

$$\tilde a(\tau) := -\Lambda(\tau)^{-1}\Lambda'(\tau), \qquad \tilde b(\tau) := -\Lambda(\tau)^{-1}X'(\tau).$$

Then for every nonnegative $\phi \in C_c^\infty(Q_R^*)$ and $\tau_1 < \tau_2$,

$$\begin{aligned}\mathscr{E}_\phi(V,P) := \frac12|V|^2\partial_\tau(\phi^2) + \frac{\nu}{2}|V|^2\Delta_Y\phi^2 + \frac12|V|^2\, V\cdot\nabla(\phi^2)\\ + (P_{\mathrm{loc}}+H)\,V\cdot\nabla(\phi^2) + \frac{\tilde a}{2}|V|^2\phi^2 + \frac{\tilde a}{2}|V|^2\,Y\cdot\nabla(\phi^2) + \frac12|V|^2\,\tilde b\cdot\nabla(\phi^2).\end{aligned}$$

$$\begin{aligned}\frac12\int_{B_R}|V(\tau_2)|^2\phi(\cdot,\tau_2)^2 + \nu\int_{\tau_1}^{\tau_2}\int_{B_R}|\nabla V|^2\phi^2\,dY\,d\tau \le \frac12\int_{B_R}|V(\tau_1)|^2\phi(\cdot,\tau_1)^2\\ + \int_{\tau_1}^{\tau_2}\int_{B_R}\mathscr{E}_\phi(V,P)\,dY\,d\tau.\end{aligned}$$

The right-hand side is controlled on compact windows by $\|\tilde a\|_{L^1}, \|\tilde b\|_{L^1}, M$, and the local bounds from Theorem 3.25, and support bounds from Section 3.4. Hence (V,P) is locally suitable on Q_R^*.

Proof. Apply the physical local energy inequality on $\mathcal{Q}^{\mathrm{ph}}_{2R}$: for nonnegative $\psi \in C_c^\infty(\mathcal{Q}^{\mathrm{ph}}_{2R})$,

$$\frac12\int|u|^2\psi\Big|_{t_1}^{t_2} + \nu\int_{t_1}^{t_2}\int|\nabla_x u|^2\psi\,dxdt \le \int_{t_1}^{t_2}\int\Big(\frac12|u|^2(\partial_t\psi+\nu\Delta_x\psi) + \Big(\frac{|u|^2}{2}u + p\,u\Big)\cdot\nabla_x\psi\Big)\,dxdt.$$

Choose

$$\psi(x,t) := \Lambda(t)^{-1}\phi\Big(\frac{x-X(t)}{\Lambda(t)},\tau(t)\Big)^2, \qquad \tau=\tau(t),\ d\tau/dt = \Lambda^{-2},$$

let $t_i = t(\tau_i)$, $i=1,2$. By Proposition 3.3 and Section 3.4, $\psi \in C_c^\infty(\mathcal{Q}^{\mathrm{ph}}_{2R})$.

$$\nabla_x\psi = \Lambda^{-2}\nabla_Y\phi^2, \quad \Delta_x\psi = \Lambda^{-3}\Delta_Y\phi^2, \quad \partial_t\psi = -\Lambda^{-4}\Lambda'\phi^2 + \Lambda^{-3}\partial_\tau(\phi^2) - \Lambda^{-4}(X'+\Lambda'Y)\cdot\nabla_Y\phi^2.$$

Using $u=\Lambda^{-1}V$, $p=\Lambda^{-2}P$, $\nabla_x u = \Lambda^{-2}\nabla_Y V$, $dx\,dt = \Lambda^5\,dY\,d\tau$, each term in the physical inequality becomes

$$\frac12\int|u|^2\psi\Big|_{t_1}^{t_2} = \frac12\int|V|^2\phi^2\Big|_{\tau_1}^{\tau_2},$$

$$\nu\int|\nabla_x u|^2\psi\,dxdt = \nu\int|\nabla V|^2\phi^2\,dY\,d\tau,$$

$$\frac12\int|u|^2\partial_t\psi\,dxdt = \frac12\int|V|^2\left(\partial_\tau(\phi^2) + \tilde a\phi^2 + \tilde a\,Y\cdot\nabla_Y\phi^2 + \tilde b\cdot\nabla_Y\phi^2\right)\,dY\,d\tau,$$

$$\frac{\nu}{2}\int|u|^2\Delta_x\psi\,dxdt = \frac{\nu}{2}\int|V|^2\Delta_Y\phi^2\,dY\,d\tau,$$

$$\int\Big(\frac{|u|^2}{2}u + p\,u\Big)\cdot\nabla_x\psi\,dxdt = \int\Big(\frac12|V|^2V + (P_{\mathrm{loc}}+H)V\Big)\cdot\nabla_Y\phi^2\,dY\,d\tau.$$

Substituting these identities and grouping terms gives the proposition's right-hand side and coefficients $\tilde a, \tilde b$. $\square$

3.11. **Stability under subsequences.**

Proposition 3.32. *Let $(u^{(n)}, p^{(n)})$ satisfy* $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$ *with the same data parameters and let $(\Lambda^{(n)}, X^{(n)}, V^{(n)}, P^{(n)}, a^{(n)}, b^{(n)})$ be obtained from retained quantitatively nondegenerate repaired-gauge selections on the same I_*, with the constants in Proposition 3.16 uniform in n. After extraction $n_j \to \infty$, there exist subsequences and limits (Λ, X, V, P, a, b) such that*

$$\Lambda^{(n_j)} \to \Lambda \text{ in } L^1_{\mathrm{loc}}, \quad X^{(n_j)} \to X \text{ in } L^1_{\mathrm{loc}},$$
$$\Lambda^{(n_j)\prime} \rightharpoonup \Lambda' \text{ in } L^1_{\mathrm{loc}}, \quad X^{(n_j)\prime} \rightharpoonup X' \text{ in } L^1_{\mathrm{loc}},$$
$$V^{(n_j)} \to V \text{ in } L^2_{\mathrm{loc}}(I_*; L^2_{\mathrm{loc}}), \quad V^{(n_j)} \rightharpoonup V \text{ in } L^2_{\mathrm{loc}}(I_*; H^1_{\mathrm{loc}}),$$
$$P^{(n_j)} \rightharpoonup P \text{ in } L^{3/2}_{\mathrm{loc}}(Q^*_R), \quad (a^{(n_j)}, b^{(n_j)}) \rightharpoonup (a, b) \text{ in } L^1_{\mathrm{loc}}(I_*)^4.$$

Moreover $a = -\Lambda'$, $b = -X'$ a.e. on I_*, and the limit obeys all conclusions below.

Proof. From SC and the retained repaired-gauge selections, $(V^{(n)}, P^{(n)})$ are uniformly bounded in $L^\infty_\tau L^2(B_{2R}) \cap L^2_\tau H^1(B_{2R})$ and $L^{3/2}_{\mathrm{loc}}(Q^*_R)$. Theorem 3.16 gives uniform invertibility bounds for $J^{(n)}$, and Theorem 3.29 gives $(a^{(n)}, b^{(n)})$ uniformly in L^1_{loc} (since $\mathcal{R}^{(n)} \in L^1_{\mathrm{loc}}$). Hence $\Lambda^{(n)}$ and $X^{(n)}$ are uniformly bounded in $W^{1,1}_{\mathrm{loc}}$, so by BV compactness and 1D subsequences,

$$\Lambda^{(n_j)} \to \Lambda \text{ in } L^1_{\mathrm{loc}}, \qquad X^{(n_j)} \to X \text{ in } L^1_{\mathrm{loc}},$$

and $a^{(n_j)} \rightharpoonup -\Lambda'$, $b^{(n_j)} \rightharpoonup -X'$ in L^1_{loc}. From the renormalized equation,

$$\partial_\tau V^{(n)} \in L^1_\tau(H^{-1}_{\mathrm{loc}}) + L^{4/3}_\tau(L^{12/11}_{\mathrm{loc}}),$$

uniformly on compact windows. Aubin–Lions compactness on bounded balls gives

$$V^{(n_j)} \to V \text{ in } L^2_{\mathrm{loc}}(I_*; L^2_{\mathrm{loc}}), \quad V^{(n_j)} \rightharpoonup V \text{ in } L^2_{\mathrm{loc}}(I_*; H^1_{\mathrm{loc}}).$$

By Calderón–Zygmund and Theorem 3.25, $P^{(n_j)} \rightharpoonup P$ in $L^{3/2}_{\mathrm{loc}}(Q^*_R)$, and with normalized pressure split $P = P_{\mathrm{loc}} + H$, the same for P_{loc} and H. Passing to limits in $\mathcal{F}(1, 0; V^{(n)}(\tau)) = 0$, Theorem 3.6, and the linear modulation system yields the corresponding limit constraints and equations. □

3.12. **Final representation theorem.**

Theorem 3.33 (Chart and repaired-gauge representation)**.** *Assume* $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$*, and suppose the branch lies in the retained repaired-gauge regime with selection (λ, c) on $J = [\tau_0 - \ell, \tau_0 + \ell] \subset I_*$. Then there are fields (V, P) on Q^J_{2R}, with $(\Lambda, X) \in AC(J)$, $a = -\Lambda'$, $b = -X' \in L^1_{\mathrm{loc}}(J)$, such that*

$$u(x,t) = \Lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right), \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

and $\mathcal{F}(1, 0; V(\tau)) = 0$ for $\tau \in J$.

Proof. From (SC5) and Proposition 3.3, $(\Lambda_{\mathrm{pre}}, X_{\mathrm{pre}})$ are AC on I_*, so V_{pre} is defined on Q^*_{2R}. The retained repaired-gauge regime supplies $(\lambda, c) \in AC(J; (0,\infty) \times \mathbb{R}^3)$ and $\mathcal{F}(1, 0; V^{\lambda,c}_{\mathrm{pre}}) = 0$ a.e. By Definition 3.21 and Theorem 3.22, setting

$$\Lambda(\tau) = \Lambda_{\mathrm{pre}}(\tau)\lambda(\tau), \qquad X(\tau) = X_{\mathrm{pre}}(\tau) + \Lambda_{\mathrm{pre}}(\tau)c(\tau), \qquad V(\tau) = V^{\lambda(\tau), c(\tau)}_{\mathrm{pre}},$$

and $P(\tau) = P^{\lambda(\tau), c(\tau)}_{\mathrm{pre}}$, gives $\Lambda, X \in AC(J)$ and $a = -\Lambda'$, $b = -X' \in L^1_{\mathrm{loc}}(J)$, and

$$\mathcal{F}(1, 0; V(\tau)) = 0 \text{ for a.e. } \tau \in J.$$

Finally $u = \Lambda^{-1} V(\cdot, \tau) \circ \Phi^{-1}$ by definition of the repaired map, hence the physical identity with $\frac{dt}{d\tau} = \Lambda^2$ follows. □

Theorem 3.34 (Pressure and modulation)**.** *Assume, in addition to Theorem 3.33, that the retained repaired-gauge selection is quantitatively nondegenerate in the sense of Proposition 3.16. Then (V, P) satisfies the renormalized Navier–Stokes equation with the sign convention of Theorem 3.6; there is a decomposition $P = P_{\mathrm{loc}} + H$ on Q_R^* as in Theorem 3.25; and (a, b) is the unique solution of*

$$J(\tau)\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = \mathcal{R}(\tau), \qquad \begin{pmatrix} a \\ b \end{pmatrix} \in L^1_{\mathrm{loc}}(J)^4,$$

with bounds $|(a, b)| \le K|\mathcal{R}|$, where K depends on $R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R$.

Proof. Equation and sign convention are inherited directly from Theorem 3.6 applied to (Λ, X) from Theorem 3.33 and the definition of V. By Theorem 3.25 applied at each time slice, the decomposition on Q_R^J is defined and unique once $⨍_{B_R} H = 0$ is imposed. The modulation identity is Theorem 3.29; uniqueness follows from invertibility of J (Theorem 3.16) and the pointwise identity $(a, b) = J^{-1}\mathcal{R}$. Uniform bounds on (a, b) are obtained by the bound on J^{-1} there and the stated bound on $\mathcal{R}$. □

Theorem 3.35 (Compact-window representation theorem)**.** *Under the assumptions of Theorems 3.33 and 3.34, for every compact window $J \subset I_*$ and each fixed $R > 0$, setting $Q_R^J := B_R \times J$, one has:*

1. *the exact renormalized equation on Q_{2R}^J,*
2. *the four repaired-gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$,*
3. *bounds on $a, b \in L^1_{\mathrm{loc}}(J)$ (and L^∞_{loc} under $\mathcal{R} \in L^\infty_{\mathrm{loc}}$),*
4. *local pressure decomposition $P = P_{\mathrm{loc}} + H$ with $L^{3/2}$ local bounds and harmonic remainder controls,*
5. *renormalized local suitable energy inequality on Q_R^J,*
6. *stability under subsequences in the topologies above.*

The later reductions use only these conclusions.

Proof. From Theorem 3.33 and the choice of $J \subset I_*$, all quantities are defined on Q_{2R}^J. Items (1) and (2) are Theorems 3.6 and 3.22; item (3) is Theorem 3.29 together with the inverse bound from Proposition 3.16; item (4) is Theorem 3.25; item (5) is Proposition 3.31; item (6) is Proposition 3.32. Combining these results gives the stated compact-window representation. □

Remark 3.36. The preceding results separate two alternatives:

1. the nondegenerate single-core repaired-gauge regime, where the representation theorems apply;
2. gauge degeneracy ($\det J = 0$) or compact-structure failure, which belongs to the multi-bubble, cascade, or scale-rigidity alternatives.

4. Multibubble and Cascade Exclusion

Analytic inputs.

The local reduction theorem uses the following analytic inputs from the preceding sections, with profile decompositions and decoupling estimates in their standard critical-space sense [Gal01, BL83]. The remaining work is the multibubble, cascade, and terminal-frame decoupling.

(i) From Section 2, the compact single-core criterion Theorem 2.18: a compact single-core finite-cost Type II branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded.

(ii) From Section 3, the renormalized equation Theorem 3.6, pressure decomposition Theorem 3.25, modulation system Theorem 3.29, and repaired-gauge representation Theorems 3.33 to 3.35: the renormalized equation, pressure decomposition modulo functions of time, bounded modulation coefficients, renormalized suitable structure, and stability of these structures under subsequences.

(iii) From Section 5, the compact-cylinder Caccioppoli estimate Theorem 5.4 and rough-core exclusion Theorem 5.7.

(iv) From Section 6, the collapse cost exclusion Theorems 6.6 and 6.7 and autonomous reduced-limit theorem Theorem 6.19 for scale-rigid or scale-collapsing branches.

4.1. **Local setting and active frames.** *Normalized local Navier–Stokes setting.* All quantities are written in rescaled local variables $(x,t) \in Q_R = B_R \times (-R^2, 0)$ with viscosity $\nu = 1$ (critical scaling has absorbed ν). A local pair (u_n, p_n) on such a cylinder satisfies

$$\partial_t u_n - \Delta u_n + (u_n \cdot \nabla) u_n + \nabla p_n = 0, \qquad \nabla \cdot u_n = 0,$$

and pressure is normalised through spatial averaging

$$(p_n)_{B_\rho}(t) := |B_\rho|^{-1} \int_{B_\rho} p_n(x,t)\, dx$$

to factor out time-dependent constants.

For $r > 0$, $x_0 \in \mathbb{R}^3$, and $t_0 \in \mathbb{R}$, let

$$Q_r(x_0, t_0) := B_r(x_0) \times (t_0 - r^2, t_0), \qquad Q_r := Q_r(0,0) = B_r \times (-r^2, 0).$$

Let (u_n, p_n) be parabolic rescalings around a potential singular point with positive local Caffarelli–Kohn–Nirenberg concentration on compact cylinders. Concretely, for a base triple (x_n, t_n, λ_n),

$$u_n(y,s) = \lambda_n u(x_n + \lambda_n y,\ t_n + \lambda_n^2 s), \qquad p_n(y,s) = \lambda_n^2 p(x_n + \lambda_n y,\ t_n + \lambda_n^2 s).$$

For each n and $R > 0$, define the scaling-invariant local quantities

$$A_n(R) := R^{-1} \operatorname*{ess\,sup}_{-R^2<t<0} \int_{B_R} |u_n(x,t)|^2\, dx,$$

$$E_n(R) := R^{-1} \int_{Q_R} |\nabla u_n(x,t)|^2\, dx\, dt,$$

$$C_n(R) := R^{-2} \int_{Q_R} |u_n(x,t)|^3\, dx\, dt, \qquad D_n(R) := R^{-2} \int_{-R^2}^0 \int_{B_R} |p_n - (p_n)_{B_R}(t)|^{3/2}\, dx\, dt,$$

where

$$(p_n)_{B_R}(t) := |B_R|^{-1} \int_{B_R} p_n(x,t)\, dx.$$

For an active frame (x_n, λ_n), $\lambda_n > 0$, define the critical rescaling

$$\mathcal{T}_{x_n,\lambda_n} f(y) := \lambda_n f(x_n + \lambda_n y).$$

If $\phi \in L^3_\sigma(\mathbb{R}^3)$ is observed in this frame, its contribution is

$$\phi_n(y) = \mathcal{T}_{x_n,\lambda_n}\phi(y) = \lambda_n \phi(x_n + \lambda_n y).$$

The critical Jacobian identity is exact:

$$\|\mathcal{T}_{x_n,\lambda_n} f\|^3_{L^3(E)} = \int_{x_n+\lambda_n E} |f(x)|^3 \, dx$$

for every measurable $E \subset \mathbb{R}^3$. In particular

$$\|\mathcal{T}_{x_n,\lambda_n} f\|^3_{L^3(\mathbb{R}^3)} = \|f\|^3_{L^3(\mathbb{R}^3)},$$

so $\mathcal{T}_{x_n,\lambda_n}$ is an isometry on $L^3(\mathbb{R}^3)$.

Definition 4.1 (Bounded selected-window limit)**.** A selected-window subsequence has a bounded selected-window limit on a compact cylinder Q_R if there exists $u_\infty \in L^\infty(Q_R)$ such that, after extracting a subsequence,

$$\forall \varepsilon > 0 \ \exists N : \ \forall n \geq N, \ \|u_n - u_\infty\|_{L^3(Q_R)} < \varepsilon.$$

The limit is nonzero if $u_\infty \not\equiv 0$ on Q_R.

Definition 4.2 (Local Type II sequence)**.** A positive local concentration sequence is called a local Type II sequence if every selected-window rescaling has zero bounded selected-window limit on every compact cylinder Q_R:

$$\forall R > 0, \qquad \lim_{n\to\infty} \|u_n\|_{L^3(Q_R)} = 0.$$

Definition 4.3 (Active frame)**.** Fix a concentrating time sequence $t_n \uparrow T$ and a compact time interval $I \Subset (-\infty, 0]$. An active frame is a tuple

$$\mathfrak{F}^j = \big(x_n^j, \lambda_n^j, I, U_n^j, P_n^j\big), \qquad \lambda_n^j \downarrow 0,$$

where $x_n^j \in \mathbb{R}^3$ and

$$U_n^j(y,s) := \lambda_n^j u(x_n^j + \lambda_n^j y, t_n + (\lambda_n^j)^2 s), \qquad P_n^j(y,s) := (\lambda_n^j)^2 p(x_n^j + \lambda_n^j y, t_n + (\lambda_n^j)^2 s)$$

on $B_{2R} \times I$ for every fixed compact cylinder under consideration. The frame is active at threshold $\eta_* > 0$ if for some fixed $R_0 < \infty$

$$\liminf_{n\to\infty} \int_{Q_{R_0}} |U_n^j|^3 \, dy \, ds \geq \eta_*^3.$$

Definition 4.4 (Relations between active frames)**.** Let $\mathfrak{F}^i$ and $\mathfrak{F}^j$ be two active frames. After passing to a subsequence exactly one of the following parameter regimes holds.

(i) They are *same-point comparable-scale* if

$$0 < c \leq \frac{\lambda_n^i}{\lambda_n^j} \leq C < \infty, \qquad \frac{|x_n^i - x_n^j|}{\max(\lambda_n^i, \lambda_n^j)} \leq C$$

for all large n.

(ii) They form a *same-point cascade* if, after relabeling,

$$\frac{\lambda_n^i}{\lambda_n^j} \to 0, \qquad \frac{|x_n^i - x_n^j|}{\lambda_n^j} = O(1).$$

The frame i is then the inner frame and j is an outer frame.

(iii) They are *separated-point* if

$$\frac{|x_n^i - x_n^j|}{\lambda_n^i} \to \infty \quad \text{and} \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} \to \infty.$$

Definition 4.5 (Terminal frame and perturbative remainder)**.** Given a finite active family, a terminal frame is chosen by selecting an active scale with minimal order:

$$\lambda_n^{j_*} \le C_j \lambda_n^j \qquad \text{for every active } j$$

after passing to a subsequence. Comparable frames at the same physical point are grouped before this choice; if several grouped frames remain tied, one is chosen by a fixed lexicographic ordering of their labels. A remainder r_n in a terminal frame is perturbative on $B_{2R} \times I$ if

$$\|r_n\|_{L_I^\infty L^3(B_{2R})} \to 0$$

and its equation residual and pressure satisfy

$$\|f_n\|_{L_I^1 H^{-1}(B_R)} + \|\nabla \pi_n^r\|_{L_I^1 H^{-1}(B_R)} \to 0$$

for every compact $B_R \times I$.

Definition 4.6 (Local multibubble/cascade branch)**.** A local multibubble/cascade branch is a positive-concentration local Type II branch for which at least one of the following alternatives occurs:

(i) the compact-cylinder suitable estimates $A_n + E_n + C_n + D_n$ are unbounded on some fixed rescaled cylinder;
(ii) positive CKN mass splits into two or more comparable active cores;
(iii) a strict same-point scale cascade appears inside bounded rescaled cylinders;
(iv) separated physical points remain active in different local rescaled frames;
(v) the localized scale or centering selection degenerates (no unique core selected).

Definition 4.7 (Active profile family)**.** An active profile family is a family

$$\{(\phi^j, x_n^j, \lambda_n^j)\}_{j \in J}, \qquad \phi^j \in L^3_\sigma(\mathbb{R}^3),$$

with finite or countable index set J, extracted along a positive local concentration sequence. In physical variables write

$$\Lambda_{\lambda, x_0} \phi(x) := \lambda^{-1} \phi\left(\frac{x - x_0}{\lambda}\right).$$

For every finite $J_0 \subset J$ the family gives an expansion

$$u(\cdot, t_n) = \sum_{j \in J_0} \Lambda_{\lambda_n^j, x_n^j} \phi^j + r_n^{J_0}$$

on the compact terminal windows under consideration. After the finite active subfamily has been selected, the corresponding remainder satisfies the perturbative bounds of Theorem 4.5. The parameters are considered only after grouping same-point comparable-scale profiles into compound profiles as in Theorem 4.8. A profile is active if

$$\|\phi^j\|_{L^3(\mathbb{R}^3)} \ge \eta_* > 0.$$

The family has finite critical mass when

$$\sum_{j \in J} \|\phi^j\|^3_{L^3(\mathbb{R}^3)} \le M_* < \infty.$$

It captures positive local concentration if every compact terminal-window sequence with

$$\liminf_{n\to\infty} \int_{Q_R} |u_n|^3 > 0$$

either contains an active profile observed in that window or is absorbed into the selected terminal frame; equivalently, no positive terminal-window concentration remains in the perturbative remainder.

Definition 4.8 (Compound profile)**.** Let $J_k \subset J$ be a same-point comparable-scale class. Choose one representative $(x_n^{j_k}, \lambda_n^{j_k})$. After passing to a subsequence,

$$\rho_j := \lim_{n\to\infty} \frac{\lambda_n^j}{\lambda_n^{j_k}} \in (0,\infty), \qquad z_j := \lim_{n\to\infty} \frac{x_n^j - x_n^{j_k}}{\lambda_n^{j_k}} \in \mathbb{R}^3.$$

The compound profile in the representative frame is the L^3_σ-sum

$$\Phi^k(y) := \sum_{j\in J_k} \rho_j^{-1} \phi^j \left(\frac{y - z_j}{\rho_j} \right),$$

whenever J_k is finite; finiteness follows for active classes from Theorem 4.9. If $\|\Phi^k\|_{L^3} < \varepsilon_\star$, where $\varepsilon_\star$ is the critical small-data threshold, the class is perturbative. Otherwise Φ^k is treated as one active object in its representative frame.

Lemma 4.9 (Finite active profile count)**.** *Every active profile family with finite critical mass contains only finitely many active profiles. More precisely,*

$$\#J \le M_* \eta_*^{-3}.$$

Proof. For each active profile $j \in J$,

$$\|\phi^j\|^3_{L^3(\mathbb{R}^3)} \ge \eta_*^3.$$

Let $J_0 \subset J$ be finite. Summing over J_0 gives

$$\#J_0\, \eta_*^3 \le \sum_{j\in J_0} \|\phi^j\|^3_{L^3(\mathbb{R}^3)} \le \sum_{j\in J} \|\phi^j\|^3_{L^3(\mathbb{R}^3)} \le M_*.$$

Hence $\#J_0 \le M_* \eta_*^{-3}$. Taking the supremum over all finite subsets J_0 yields the same bound for the cardinality of J, so J is finite and

$$\#J \le M_* \eta_*^{-3}.$$

$\square$

Lemma 4.10 (Exhaustive parameter partition)**.** *Let $\mathfrak{F}^i, \mathfrak{F}^j$ be two active frames. After passing to a subsequence, either the pair is same-point comparable-scale, same-point cascade, or separated-point in the sense of Theorem 4.4; if none of these relations holds in a selected compact frame, then one of the two contributions is locally invisible there in L^3.*

Proof. Pass to a subsequence so that

$$\frac{\lambda_n^i}{\lambda_n^j} \to \ell \in [0,\infty]$$

in the extended half-line. If $0 < \ell < \infty$, then either

$$\frac{|x_n^i - x_n^j|}{\max(\lambda_n^i, \lambda_n^j)}$$

is bounded along a further subsequence, giving the same-point comparable-scale case, or it tends to infinity, giving separated points because the two scales are comparable.

If $\ell = 0$, then either

$$\frac{|x_n^i - x_n^j|}{\lambda_n^j}$$

is bounded, giving a same-point cascade with i inner and j outer, or it tends to infinity. In the latter case the j-profile is locally invisible in the i-frame by Theorem 4.13. The case $\ell = \infty$ is the same after interchanging i and j. These alternatives exhaust all subsequential limits of the scale ratio and normalized center distance. □

Theorem 4.11 (Minimal bad-configuration extraction)**.** *Let a positive local Type II branch admit an active profile family with finite critical mass, perturbative remainder in the sense of Theorem 4.5, and concentration capture in the sense of Theorem 4.7. If the branch is not already a single active terminal frame satisfying the single-core hypotheses, then after passing to a subsequence one of the following occurs:*

(i) *compact-cylinder suitable control fails, and Theorem 4.15 produces a bounded-cylinder concentration core;*
(ii) *there are at least two active frames in a same-point comparable-scale class, hence a compound profile in the sense of Theorem 4.8;*
(iii) *there is a same-point cascade;*
(iv) *there are separated active physical points;*
(v) *the scale-center selection of the selected compound or terminal frame is degenerate.*

In every case the positive local concentration is carried by an active frame or compound profile, not by the perturbative remainder.

Proof. If compact-cylinder suitable control fails, item (i) follows from Theorem 4.15. Assume therefore that the compact-cylinder family is bounded on the windows under consideration.

By concentration capture, positive local concentration cannot remain solely in the perturbative remainder. Hence either one selected active frame carries the terminal-window concentration, or another active frame remains non-negligible. If there is one nondegenerate selected active frame and no competing active object, the branch is in the single-core regime. Since this case is excluded by hypothesis, either the selected scale-center map is degenerate, giving item (v), or at least two active frames remain.

For two active frames, apply Theorem 4.10. Same-point comparable-scale frames give item (ii), same-point scale separation gives item (iii), and separated points give item (iv). The remaining alternative in Theorem 4.10 is local invisibility of one contribution in the selected compact frame; such a contribution is absorbed into the perturbative remainder and cannot carry the retained positive local concentration by the capture property. This gives the extraction statement. □

4.2. **Elementary local decoupling estimates.**

Lemma 4.12 (Critical rescalings that escape are locally invisible)**.** *Let $\phi \in L^3(\mathbb{R}^3)$, let $R < \infty$, and define*

$$W_n(y) = \rho_n \phi(z_n + \rho_n y),$$

where $\rho_n > 0$. Assume that the sets

$$z_n + \rho_n B_R$$

either have measure tending to zero, or escape to spatial infinity in the sense that for every compact set $K \subset \mathbb{R}^3$,

$$(z_n + \rho_n B_R) \cap K = \varnothing$$

for all sufficiently large n. Then

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. By the critical change of variables,

$$\|W_n\|^3_{L^3(B_R)} = \int_{B_R} \rho_n^3 |\phi(z_n + \rho_n y)|^3 \, dy = \int_{z_n+\rho_n B_R} |\phi(x)|^3 \, dx.$$

Set

$$m_n := \int_{z_n+\rho_n B_R} |\phi(x)|^3 \, dx.$$

Since $\|W_n\|^3_{L^3(B_R)} = m_n$, it is enough to show $m_n \to 0$.

Fix $\varepsilon > 0$. We split into the two structural alternatives.

Case 1: shrinking measure. If $|z_n + \rho_n B_R| \to 0$, then by absolute continuity of the integral $|\phi|^3 \in L^1(\mathbb{R}^3)$ there exists $\delta > 0$ such that

$$|E| < \delta \Longrightarrow \int_E |\phi|^3 \, dx < \varepsilon.$$

Choose N with $|z_n + \rho_n B_R| < \delta$ for all $n \geq N$. Then

$$m_n = \int_{z_n+\rho_n B_R} |\phi|^3 \, dx < \varepsilon, \qquad n \geq N.$$

Case 2: escape to infinity. Assume now that for every compact set $K \subset \mathbb{R}^3$, $(z_n + \rho_n B_R) \cap K = \varnothing$ for all sufficiently large n. By absolute continuity at infinity, choose compact $K \subset \mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

For every such compact K, there is $N(K)$ with $(z_n + \rho_n B_R) \cap K = \varnothing$ for all $n \geq N(K)$. For these indices,

$$z_n + \rho_n B_R \subset \mathbb{R}^3 \setminus K,$$

and therefore, for $n \geq N(K)$,

$$m_n = \int_{z_n+\rho_n B_R} |\phi|^3 \, dx \leq \int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

In either case, for every $\varepsilon > 0$, $m_n < \varepsilon$ for n sufficiently large. Hence $m_n \to 0$, and therefore

$$\|W_n\|_{L^3(B_R)} \to 0.$$

□

Lemma 4.13 (Separated active profiles vanish in the selected frame)**.** *Let (x_n^p, λ_n^p) and (x_n^q, λ_n^q) be two active frames with separated physical centers in the sense that*

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then the q-profile observed in the p-frame converges to zero locally in L^3: for every $R < \infty$,

$$\left\| \frac{\lambda_n^p}{\lambda_n^q} \phi^q \left(\frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} y \right) \right\|_{L^3(B_R)} \to 0.$$

Proof. The displayed function has the form in Theorem 4.12 with

$$z_n = \frac{x_n^p - x_n^q}{\lambda_n^q}, \qquad \rho_n = \frac{\lambda_n^p}{\lambda_n^q}.$$

The corresponding physical set in the q-profile variables is

$$z_n + \rho_n B_R = \frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} B_R.$$

For every $y \in B_R$,

$$|z_n + \rho_n y| \geq |z_n| - \rho_n R.$$

Set $E_n := z_n + \rho_n B_R$. Then

$$\operatorname{dist}(E_n, 0) = |z_n| - \rho_n R = \frac{\lambda_n^p}{\lambda_n^q}\left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R\right).$$

Since $\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty$, there exists $N_0(R)$ such that

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \geq 1, \quad n \geq N_0(R),$$

and hence for such n, $\operatorname{dist}(E_n, 0) \geq \rho_n$.

There are two cases.

(a) If $\rho_n \to 0$, then

$$|z_n + \rho_n B_R| = \rho_n^3 |B_R| \to 0.$$

Then the first alternative in Theorem 4.12 applies.

(b) Otherwise, after passing to a subsequence, $\rho_n \geq \rho_* > 0$. Then

$$\operatorname{dist}(z_n + \rho_n B_R, 0) \geq \rho_* \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R\right) \to \infty,$$

so the sets escape every fixed compact subset of $\mathbb{R}^3$. The second alternative in Theorem 4.12 applies.

In both subcases the $L^3(B_R)$-norm of the displayed expression converges to zero. □

Lemma 4.14 (Strict outer scales vanish in the innermost frame)**.** *Let several active profiles have the same physical center and scales $\lambda_n^1, \dots, \lambda_n^J$. Suppose λ_n^1 is an innermost scale and*

$$\frac{\lambda_n^1}{\lambda_n^j} \to 0 \qquad (j \neq 1).$$

Then every outer profile $\phi^j \in L^3(\mathbb{R}^3)$, viewed in the λ_n^1-frame, converges to zero in $L^3(B_R)$ for each fixed $R < \infty$. Any constant ambient velocity arising from the regular part of the outer flow is absorbed into the translation modulation.

Proof. In the innermost variables, the j-th outer contribution has the form

$$W_n^j(y) = \frac{\lambda_n^1}{\lambda_n^j} \phi^j \left(\frac{x_n^1 - x_n^j}{\lambda_n^j} + \frac{\lambda_n^1}{\lambda_n^j} y\right).$$

Set

$$\rho_n := \frac{\lambda_n^1}{\lambda_n^j} \to 0, \qquad z_n^j := \frac{x_n^1 - x_n^j}{\lambda_n^j}.$$

Then

$$W_n^j(y) = \rho_n \phi^j \left(z_n^j + \rho_n y\right),$$

and by the critical change of variables,

$$\|W_n^j\|_{L^3(B_R)}^3 = \int_{z_n^j+\rho_n B_R} |\phi^j(x)|^3\,dx.$$

Since $\rho_n \to 0$, the translated domains have

$$|z_n^j + \rho_n B_R| = \rho_n^3 |B_R| \to 0.$$

By absolute continuity of the integral for $\phi^j \in L^3(\mathbb{R}^3)$,

$$\|W_n^j\|_{L^3(B_R)} \to 0.$$

To treat a general ϕ^j, let $\phi_\varepsilon^j \in C_c^\infty(\mathbb{R}^3)$ with $\|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon$. By critical isometry,

$$\|W_n^j - (W_n^j)_\varepsilon\|_{L^3(B_R)} \le \|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon,$$

where $(W_n^j)_\varepsilon$ is the same scaling of ϕ_ε^j. The compactly supported term has vanishing local norm by the shrinking-domain argument above, and then letting first $n \to \infty$, then $\varepsilon \downarrow 0$ implies the claimed convergence. A constant vector field from the regular part of an outer flow contributes only a Galilean drift in the selected frame; it is therefore incorporated into the translation modulation and does not remain as an exterior perturbation. □

Lemma 4.15 (Compactness failure produces another core)**.** *If the compact-cylinder suitable estimates are unbounded on a fixed rescaled cylinder, then after passing to a subsequence the local $C_n + D_n$ quantity is unbounded on a comparable cylinder. Consequently the branch contains an additional local CKN concentration core.*

Proof. Fix $r > 0$. Apply the local Caccioppoli inequality with outer radius $2r$. For each index n, suitable weak solutions satisfy, with C_r independent of n,

$$A_n(r) + E_n(r) \le C_r\big(1 + C_n(2r) + C_n(2r)^{2/3} + D_n(2r)\big),$$

where the pressure is normalized by subtracting spatial means $(p_n)_{B_{2r}}(t)$.

Suppose that $A_n(r) + E_n(r) + C_n(r) + D_n(r)$ is unbounded along a subsequence. If $C_n(2r) + D_n(2r)$ were bounded on this subsequence, the preceding estimate would give uniform boundedness of $A_n(r)+E_n(r)$. Monotonicity of the local integrals also gives $C_n(r)+D_n(r) \le C_n(2r)+D_n(2r)$, contradicting unboundedness of the full family on Q_r. Therefore, after extracting a subsequence,

$$C_n(2r) + D_n(2r) \to \infty.$$

Define the localized CKN concentration at variable center and radius

$$G_n(x,t,\rho) := \rho^{-2}\int_{Q_\rho(x,t)} |u_n|^3\,dx\,ds + \rho^{-2}\int_{Q_\rho(x,t)} |p_n - (p_n)_{B_\rho}(s)|^{3/2}\,dx\,ds.$$

Set

$$\mathbf{M}_n := \sup\{G_n(x,t,\rho) : 0 < \rho \le 2r,\ Q_\rho(x,t) \subset Q_{4r}\}.$$

The choice $(x,t,\rho) = (0,0,2r)$ is admissible because $Q_{2r}(0,0) \subset Q_{4r}$, so

$$\mathbf{M}_n \ge C_n(2r) + D_n(2r) \to \infty.$$

After extracting a further subsequence, we may assume $\mathbf{M}_n \ge 1$ for all n. Choose (x_n,t_n,ρ_n) with $0 < \rho_n \le 2r$, $Q_{\rho_n}(x_n,t_n) \subset Q_{4r}$, such that

$$G_n(x_n,t_n,\rho_n) \ge \tfrac12\,\mathbf{M}_n \ge \tfrac12.$$

Rescale around (x_n,t_n,ρ_n):

$$v_n(y,s) := \rho_n u_n(x_n + \rho_n y, t_n + \rho_n^2 s), \qquad q_n(y,s) := \rho_n^2 p_n(x_n + \rho_n y, t_n + \rho_n^2 s).$$

Using the critical Jacobian identity,

$$\int_{Q_1} |v_n|^3 \, dy \, ds + \int_{Q_1} |q_n - (q_n)_{B_1}(s)|^{3/2} \, dy \, ds = G_n(x_n, t_n, \rho_n) \geq \tfrac{1}{2}.$$

Hence this new frame carries positive local critical mass, i.e. a new local CKN concentration core, as required. □

4.3. **Terminal active-frame decoupling.**

Definition 4.16 (Admissible terminal active family)**.** Fix a compact cylinder $B_{2R} \times I$ and a selected active frame. A terminal active family is admissible on this cylinder if the following hold.

- (i) The active profiles satisfy the finite-mass and active-mass conditions of Theorem 4.7.
- (ii) Every nonselected active profile is either separated from the selected physical point in the sense of Theorem 4.13, or has the same physical point and a strictly outer scale in the sense of Theorem 4.14, after comparable same-point scales have been grouped.
- (iii) The rescaled remainder r_n satisfies

  $$\|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0,$$

  and there exist fields (π^r_n, f_n) such that on $B_{2R} \times I$,

  $$\partial_t r_n - \Delta r_n + a_n\big(r_n + y \cdot \nabla r_n\big) + b_n \cdot \nabla r_n + \nabla \pi^r_n = f_n, \qquad \nabla \cdot r_n = 0,$$

  with

  $$\|f_n\|_{L^1_I H^{-1}(B_R)} \to 0,$$
  $$\|\nabla \pi^r_n\|_{L^1_I H^{-1}(B_R)} \to 0.$$

- (iv) The selected velocity satisfies

  $$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

- (v) The selected modulation coefficients satisfy

  $$\sup_n \big(\|a_n\|_{L^\infty(I)} + \|b_n\|_{L^\infty(I)}\big) < \infty.$$

- (vi) Local pressure decomposition is stable under the splitting: if tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$, then the corresponding pressure gradients tend to zero in $L^1_I H^{-1}(B_R)$, after subtracting spatial means.
- (vii) Cutoff commutators generated by localizing to B_R are controlled by the $L^\infty_I L^3$, $L^1_I L^{3/2}$, and pressure-gradient norms appearing above. In particular, for any compact $\chi_R \in C^\infty_c(B_{2R})$ with $\chi_R \equiv 1$ on B_R, the commutators satisfy

  $$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla \chi_R, \nabla \cdot](u \otimes u)\|_{H^{-1}(B_R)} \leq C_R \|u\|^2_{L^3(B_{2R})}.$$

  uniformly for $u \in L^3(B_{2R})$ and the same type of estimate for S_n-quadratic commutator terms.

Theorem 4.17 (Terminal profile-evolution decoupling)**.** *Consider an admissible terminal active family on $B_{2R} \times I$. Let U_n be the selected-frame velocity and let S_n be the sum, in that frame, of all nonselected active profiles together with the rescaled remainder. Then on $B_R \times I$:*

- (i) $\nabla \cdot S_n = 0$ *distributionally;*
- (ii) $S_n \to 0$ *in* $L^\infty_I L^3(B_{2R})$*;*
- (iii) $U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0$ *in* $L^1_I L^{3/2}(B_R)$*;*

(iv)
$$\begin{gathered}\exists\,\pi_n,\ F_n:\quad \partial_t S_n-\Delta S_n+a_n\big(y\cdot\nabla S_n+S_n\big)+b_n\cdot\nabla S_n+\nabla\pi_n=F_n,\\ \|F_n\|_{L^1_IH^{-1}(B_R)}\to 0,\qquad \nabla\cdot S_n=0\quad \textit{in } \mathcal{D}'(B_R\times I).\\ \|\nabla\pi_n\|_{L^1_IH^{-1}(B_R)}\to 0.\end{gathered}$$

(v) *every pressure source containing at least one factor of S_n has vanishing pressure gradient in $L^1_IH^{-1}(B_R)$;*

(vi) *localization errors introduced by compact cutoffs vanish in $L^1_IH^{-1}(B_R)$;*

(vii) *the effective modulation parameters of the selected frame remain bounded;*

(viii) *after subtracting S_n, the selected frame remains a positive finite-mass local Type II branch with pressure reconstruction and the compact-cylinder bounds required by the single-core analysis.*

Proof. Each profile in the family is divergence-free, and the rescaled remainder is divergence-free. Since divergence commutes with the Navier–Stokes critical rescalings, finite sums remain divergence-free in distributions. This establishes (i).

We prove (ii). By admissibility, every nonselected active profile is either separated from the selected physical point or is a same-point strictly outer scale. The separated profiles vanish locally by Theorem 4.13, and the same-point outer profiles vanish locally by Theorem 4.14. The remainder tends to zero in $L^\infty_IL^3(B_{2R})$ by Theorem 4.16. The number of active profiles is finite by Theorem 4.9. Since every summand in S_n tends to zero in $L^\infty_IL^3(B_{2R})$, the triangle inequality gives

$$\|S_n\|_{L^\infty_IL^3(B_{2R})}\le\sum_{j\neq j_*}\|S^j_n\|_{L^\infty_IL^3(B_{2R})}+\|r_n\|_{L^\infty_IL^3(B_{2R})}\to 0.$$

This establishes (ii).

For (iii), admissibility gives

$$\sup_n\|U_n\|_{L^\infty_IL^3(B_{2R})}<\infty.$$

Hence, by Hölder's inequality,

$$\|U_n\otimes S_n\|_{L^1_IL^{3/2}(B_R)}\le|I|\|U_n\|_{L^\infty_IL^3(B_R)}\|S_n\|_{L^\infty_IL^3(B_R)}\to 0,$$

and the same estimate applies to $S_n\otimes U_n$. Finally,

$$\|S_n\otimes S_n\|_{L^1_IL^{3/2}(B_R)}\le|I|\|S_n\|^2_{L^\infty_IL^3(B_R)}\to 0.$$

For (iv), write the represented Navier–Stokes equation for the full profile sum U_n+S_n and subtract the represented equation for the selected component U_n. The remaining equation for S_n has the displayed linear left-hand side

$$\partial_tS_n-\Delta S_n+a_n\big(y\cdot\nabla S_n+S_n\big)+b_n\cdot\nabla S_n.$$

Collect all right-hand terms into

$$F_n:=-\nabla\cdot(U_n\otimes S_n+S_n\otimes U_n+S_n\otimes S_n)+f_n+E^{\mathrm{cut}}_n+E^{\mathrm{mod}}_n,$$

where f_n is the remainder forcing from item (iii) of admissibility, and $E^{\mathrm{cut}}_n,E^{\mathrm{mod}}_n$ are localization and modulation errors. The nonlinear stresses tend to zero in $L^1_IL^{3/2}(B_R)$, hence in $L^1_IH^{-1}(B_R)$, because $L^{3/2}(B_R)\hookrightarrow H^{-1}(B_R)$ on a bounded three-dimensional ball. The residual term vanishes by Theorem 4.16. The cutoff terms vanish by the localization part of admissibility, recorded explicitly in (vi). This gives the displayed convergence in $L^1_IH^{-1}(B_R)$.

For (v), decompose the pressure source of U_n+S_n into the pure selected source, the pure exterior source, and the two mixed sources. The mixed sources are

$$\partial_i\partial_j(U_{n,i}S_{n,j}+S_{n,i}U_{n,j})\quad\text{and}\quad\partial_i\partial_j(S_{n,i}S_{n,j}).$$

By (iii), their tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$. The pressure-stability part of admissibility, equivalently local Calderon–Zygmund estimates plus harmonic pressure control modulo spatial means, gives convergence of the corresponding pressure gradients to zero in $L^1_I H^{-1}(B_R)$. Pure exterior sources are generated by profiles that are locally invisible in the selected frame; applying the same pressure stability after Theorems 4.13 and 4.14 gives their vanishing pressure gradients.

For (vi), let χ_R be a cutoff equal to one on B_R and supported in B_{2R}. The commutators are finite sums of terms of the following types:

$$(\nabla\chi_R)\cdot\nabla S_n, \qquad (\Delta\chi_R)S_n, \qquad (\nabla\chi_R)\cdot(U_n\otimes S_n+S_n\otimes U_n+S_n\otimes S_n),$$

plus analogous pressure cutoff terms. Their supports lie in the fixed annulus $B_{2R}\setminus B_R$. By the localization part of admissibility, these commutators are controlled by the norms already shown to vanish in (ii), (iii), and (v). Hence all localization errors vanish in $L^1_I H^{-1}(B_R)$.

For (vii), the only exterior contribution that can remain nonzero at first order in a compact selected frame is a constant ambient velocity. This has already been absorbed into the translation modulation. The remaining exterior profiles vanish locally, so their contribution to the modulation equations is small in the same local L^3 topology used above. The selected modulation coefficients are bounded by admissibility; hence the effective parameters remain bounded.

For $(viii)$, the positive CKN mass of the selected profile is unchanged by subtracting a term that tends to zero in local L^3 and whose mixed pressure sources vanish by (v). The pressure reconstruction is stable under the same decomposition by the pressure-stability condition in Theorem 4.16. The compact-cylinder bounds pass to the selected component because the full family is bounded and the nonretained terms vanish on compact cylinders. Thus the selected frame remains a positive finite-mass local Type II branch with the analytic inputs needed for the single-core analysis. ☐

4.4. **Same-point and separated-point reductions.**

Theorem 4.18 (Same-point cascade reduction)**.** *Every same-point multibubble or strict same-point cascade reduces, in the selected innermost or compound frame, to one of the following: the single-core criterion, scale-rigidity, or a perturbative terminal branch covered by Theorem 4.17, provided the terminal family generated in the selected innermost frame is admissible in the sense of Theorem 4.16.*

Proof. Step 1: scale-comparability classes. Split active same-point profiles into equivalence classes by

$$i\sim j \iff \exists c,C\in(0,\infty): c\le\frac{\lambda_n^i}{\lambda_n^j}\le C \quad \text{for all } n$$

after passing to a subsequence. Thus within each class there are constants c_{ij}, C_{ij} with $c_{ij}\le \lambda_n^i/\lambda_n^j\le C_{ij}$ for all n, so one may define a single common rescaling sequence (choose $\lambda_n^{(k)}=\lambda_n^{j_k}$) and write all class-$k$ profiles in the same coordinates. Within each class, finite sums of divergence-free profiles remain divergence-free.

Step 2: compound profile reduction inside each class. For each class set

$$\Phi_n^k := \sum_{j\in J_k}\lambda_n^j\phi^j(x_n^j+\lambda_n^j y).$$

If the L^3-size of this rescaled sum satisfies

$$\Big\|\sum_{j\in J_k}\phi^j\Big\|_{L^3(\mathbb{R}^3)}<\varepsilon_\star,$$

with $\varepsilon_\star$ the perturbative small-data threshold, then standard local small-data theory for Navier–Stokes in the critical space eliminates this class from the active alternatives.

If the class is above threshold, it is retained as a single compound active core; at most one active core from that class is needed for local selection.

Step 3: local alternatives. If the retained compound core is unique and its scale-centering Jacobian is nondegenerate, we obtain the single selected core alternative. If this Jacobian is degenerate, the localized choice of active center or scale is not unique, which is the gauge-degenerate alternative in Theorem 4.6.

Step 4: strict same-point scale separation. Assume now the branch is not reduced to comparable scales and choose the innermost active scale λ_n^1 in that physical point. By Theorem 4.14, every profile with larger scale vanishes in the λ_n^1-frame in local L^3, up to constant drift terms. These constant drifts are absorbed into translation modulation by admissibility. Let S_n be the sum of all such outer profiles plus the rescaled remainder r_n. Then, by Step 2 and Theorem 4.14,

$$S_n \to 0 \quad \text{in } L_I^\infty L^3(B_{2R}).$$

Subtracting the equation for the innermost selected profile from the full-frame equation gives a perturbation equation for the selected profile whose forcing belongs to $L_I^1 H^{-1}(B_R)$ and vanishes. Thus Theorem 4.17 applies, and one of three outcomes holds:

(a) **Single-core branch:** after subtracting S_n, the branch is governed by one active component, so it reduces to the compact single-core branch of Theorem 4.20;
(b) **Scale-rigid branch:** the relative Jacobian between comparable scales becomes rigid in the sense of Theorem 4.21, so Theorem 4.21 excludes it;
(c) **Purely perturbative branch:** $S_n \to 0$ and the selected profile satisfies a vanishing forcing perturbation in $L_I^1 H^{-1}(B_R)$. No independent same-point cascade mechanism remains; after subtracting S_n, the selected component is reduced to the single-core or scale-rigidity analysis.

These are the strict same-point scale-separation alternatives, so the same-point reduction is exhausted. □

Theorem 4.19 (Separated-point decoupling)**.** *Separated active physical points are locally invisible in each other's active rescaled frames. Consequently every separated-point multibubble reduces in each selected active frame to a single-core, scale-rigid, or perturbative terminal branch, provided the terminal family in that selected active frame is admissible in the sense of Theorem 4.16.*

Proof. Fix an active point p with frame (x_n^p, λ_n^p). Let $q \neq p$ be another active point. By separation,

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

Thus Theorem 4.13 gives local L^3 convergence to zero of the q-profile in the p-frame on every compact ball. The same conclusion holds for each profile centered at any physical point different from p. Since only finitely many active profiles remain after the critical mass threshold is imposed, the sum of all profiles outside the p-frame tends to zero locally in L^3.

The pure pressure generated by a separated profile is locally invisible by the same change of variables whenever its profile pressure belongs to $L^{3/2}_{\mathrm{loc}}$ with the standard Calderon–Zygmund control. Mixed pressure terms contain at least one exterior velocity factor, hence vanish in $L^1 H^{-1}_{\mathrm{loc}}$ by the pressure part of Theorem 4.17. Localization and modulation commutators are localized to $B_{2R} \setminus B_R$, so their H^{-1}-norm is controlled by the L^3-smallness already proved in Theorems 4.13 and 4.14 and by admissibility item (vii).

Therefore, in the selected p-frame, all other physical points contribute only a perturbative forcing tending to zero in L^1H^{-1} on compact cylinders. The selected frame remains a positive finite-mass branch with the same compact-cylinder and pressure reconstruction properties. Hence the branch reduces to the single-core case, the scale-rigid case, or the perturbative terminal branch. Since p was arbitrary, the same reduction holds in every active frame. □

4.5. **Multibubble and cascade exclusion.** The following assumptions isolate the analytic inputs needed for the multibubble reduction: the single-core exclusion, the scale-rigidity reduction, and terminal active-frame admissibility. They are stated explicitly so that the following theorem uses no unstated compactness, pressure, or profile completeness conclusion.

Assumption 4.20 (Local single-core exclusion)**.** Every compact single-core branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded from the local Type II regime.

Assumption 4.21 (Local scale-rigidity exclusion)**.** Every scale-rigid local branch produced by the same-point or separated-point reductions, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit or reduces to the single-core branch in Theorem 4.20. Consequently no such branch is compatible with the local Type II condition.

Assumption 4.22 (Terminal admissibility in active frames)**.** Every terminal family produced by the same-point and separated-point reductions is admissible on each compact selected cylinder in the sense of Theorem 4.16.

Theorem 4.23 (Local multibubble and cascade exclusion)**.** *Assume the local single-core and scale-rigidity exclusions Theorems 4.20 and 4.21, and assume terminal admissibility in active frames, Theorem 4.22. Let a positive local Type II branch admit an active family satisfying the hypotheses of Theorem 4.11. Then no local multibubble, cascade, or gauge-degenerate Type II case can occur in this class.*

Proof. Let a local Type II branch be in the multibubble/cascade class of Theorem 4.6. Apply Theorem 4.11. The positive local concentration is carried by an active frame or compound profile, and one of the alternatives listed there occurs.

Case 1: compactness-failure reduction. Compactness failure yields, after changing radius by a fixed factor and rescaling as in Theorem 4.15, a bounded-cylinder concentration branch. Relabel this branch as an active core. After passing to a subsequence, the new core and every previously retained active core have one of three relations: comparable scale at the same physical point, strictly separated scale at the same physical point, or separated physical center. Comparable same-point cores are grouped into compound profiles. A perturbative compound profile is removed by small-data stability; an active compound profile is treated as one selected core in its active frame. If the scale-center selection inside such a compound core is degenerate, the branch is gauge-degenerate and belongs to the non-uniqueness alternative in Theorem 4.6, rather than to a separate core dynamics.

Case 1a: gauge degeneracy. If the scale-center selection of the selected compound or terminal frame is degenerate, then the branch is in the gauge-degenerate alternative of Theorem 4.6. By the repaired-gauge representation from Section 3, nondegeneracy is the hypothesis needed to enter the single-core branch. Persistent degeneracy therefore belongs to the scale-rigidity alternative in Theorem 4.21; if the degeneracy disappears after passing to a subsequence, the branch enters the single-core or terminal-frame alternatives below.

Case 1b: same-point comparable scales. Same-point comparable-scale active frames are grouped into the compound profile of Theorem 4.8. If the compound profile is below the critical small-data threshold, it is part of the perturbative remainder. If it is above threshold and its scale-center selection is nondegenerate, it is one selected active core and enters the single-core branch. If the corresponding scale-center selection is degenerate, it is covered by Case 1a. Thus comparable same-point profiles do not produce an additional multibubble mechanism.

Case 2: same-point cascading. Strict same-point scale cascades are treated by Theorem 4.18. In the innermost frame, all outer scales are locally invisible after constant drifts are absorbed. The branch therefore reduces to one of the three terminal alternatives

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

Case 3: separated-point interactions. For separated physical points, Theorem 4.19 gives the same three terminal alternatives as in Case 2:

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

The single-core alternative is excluded by Theorem 4.20. The scale-rigid alternative is excluded by Theorem 4.21, which includes the scale-collapse and autonomous-limit exclusions in Section 6 in this setting. The terminal perturbative alternative gives no independent multibubble obstruction: Theorem 4.17 removes all exterior profiles and remainders from the selected active equation in $L^1H^{-1}_{\mathrm{loc}}$, preserves the positive finite-mass selected branch, and leaves only the single-core or scale-rigid possibilities already excluded. Consequently every alternative in Theorem 4.6 is incompatible with the local Type II regime. □

Corollary 4.24 (Removal of the multibubble alternative). *Under the hypotheses of Theorem 4.23, every failure of the single nondegenerate compact core alternative is excluded.*

Proof. By Theorem 4.6, failure of the single nondegenerate compact core alternative means that at least one of the following occurs: the compact-cylinder suitable estimates are unbounded; positive local CKN mass splits between active cores; a strict same-point scale cascade occurs; separated physical points remain active in distinct local frames; or the localized scale or center selection is degenerate. The compactness-failure case produces an additional core by Theorem 4.15. Same-point and separated cases are reduced by Theorems 4.18 and 4.19. Theorem 4.23 excludes the remaining cases. Hence no failure of the single nondegenerate compact core alternative remains. □

5. Local Caccioppoli and Rough-Core Exclusion

5.1. **Analytic inputs and notation.** The renormalized local energy inequality from the preceding renormalized-chart analysis is used as an analytic input. The renormalized suitable structure and pressure reconstruction are not rederived. All estimates are conditional on the following local hypotheses on the compact renormalized cylinders under consideration.

Assumption 5.1 (Renormalized local energy inequality)**.** Let (V, P, a, b) be defined on a compact renormalized cylinder $Q_{2R,J} := B_{2R} \times J$, where $J \subset \mathbb{R}$ is a bounded interval. Assume

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V = a(\tau)V + a(\tau)(y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0, \tag{6}$$

with $\nu > 0$, $\nabla \cdot V = 0$, and $a, b \in L^\infty(J)$. For every nonnegative $\phi \in C_c^\infty(Q_{2R,J})$ and every $\tau_1 < \tau_2$ in J,

$$\begin{aligned}
&\int_{B_{2R}} \frac{|V(\cdot,\tau_2)|^2}{2}\phi(\cdot,\tau_2) - \int_{B_{2R}} \frac{|V(\cdot,\tau_1)|^2}{2}\phi(\cdot,\tau_1) \\
&\quad + \nu \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \phi\,|\nabla V|^2 \le \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2}(\partial_\tau \phi + \nu\Delta\phi) \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2} V \cdot \nabla\phi + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} PV \cdot \nabla \phi \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} a\frac{|V|^2}{2}(-\phi - y \cdot \nabla\phi) \\
&\quad - \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi.
\end{aligned} \tag{7}$$

The pressure is understood modulo functions of time, and the mean-normalized pressure on compact balls belongs to the local class used below.

Throughout, for a space–time set $E \subset \mathbb{R}^3 \times \mathbb{R}$, write

$$\int_E f := \int_E f(y,\tau)\,dy\,d\tau.$$

For $r > 0$ and a bounded interval $J \subset \mathbb{R}$, define

$$Q_{r,J} := B_r \times J, \qquad |Q_{r,J}| := |B_r|\,|J|.$$

For a ball $B \subset \mathbb{R}^3$, denote the spatial mean of pressure by

$$(P)_B(\tau) := \frac{1}{|B|}\int_B P(y,\tau)\,dy.$$

The local quantities are

$$\mathcal{A}_J(r) := \operatorname*{ess\,sup}_{\tau \in J} \int_{B_r} |V(y,\tau)|^2\,dy,$$

$$\mathcal{H}_J(r) := \int_J\int_{B_r} |\nabla V|^2\,dy\,d\tau,$$

$$\mathcal{C}_J(r) := \int_J\int_{B_r} |V|^3\,dy\,d\tau, \qquad \mathcal{D}_J(r) := \int_J\int_{B_r} |P - (P)_{B_r}(\tau)|^{3/2}\,dy\,d\tau.$$

Lemma 5.2 (Finite-measure conversion)**.** *For every bounded interval J and every $r > 0$,*

$$\int_J\int_{B_r} |V|^2 \le |Q_{r,J}|^{1/3}\mathcal{C}_J(r)^{2/3} \le |Q_{r,J}|^{1/3}\big(1 + \mathcal{C}_J(r)\big).$$

Proof. Hölder's inequality on $Q_{r,J}$ gives

$$\int_J\int_{B_r} |V|^2 \le |Q_{r,J}|^{1/3} \left(\int_J\int_{B_r} |V|^3\right)^{2/3}.$$

The second inequality follows from $s^{2/3} \le 1 + s$ for $s \ge 0$. $\square$

Lemma 5.3 (Comparison of pressure normalizations)**.** *Let $1 \le p < \infty$ and $0 < r \le R$. For almost every τ such that $P(\tau) \in L^p(B_R)$,*

$$\|P(\tau) - (P)_{B_r}(\tau)\|_{L^p(B_r)} \le 2\|P(\tau) - (P)_{B_R}(\tau)\|_{L^p(B_R)}.$$

Consequently,

$$\mathcal{D}_J(r) \le 2^{3/2}\mathcal{D}_J(R)$$

whenever the right-hand side is finite.

Proof. Set $c = (P)_{B_R}(\tau)$. Then

$$\|P - (P)_{B_r}\|_{L^p(B_r)} \le \|P - c\|_{L^p(B_r)} + |B_r|^{1/p}|(P)_{B_r} - c|.$$

Since

$$|(P)_{B_r} - c| \le |B_r|^{-1}\int_{B_r} |P - c| \le |B_r|^{-1/p}\|P - c\|_{L^p(B_r)},$$

the displayed pointwise estimate follows. Raising to the power $3/2$ and integrating in time gives the last assertion. $\square$

5.2. **Compact-cylinder Caccioppoli estimate.**

Theorem 5.4 (Compact-cylinder Caccioppoli estimate)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume Theorem 5.1 on $Q_{2R,J}$ and assume*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty.$$

Then there is a constant

$$C = C\big(R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \operatorname{dist}(J', \partial J)\big)$$

such that

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) \le C\Big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\Big).$$

The estimate is outer-to-inner: all right-hand quantities are measured on $B_{2R} \times J$, while the conclusion is on $B_R \times J'$.

Proof. **Step 1: localized cut-off and the two energy inequalities.** Set

$$\delta := \operatorname{dist}(J', \partial J) > 0.$$

Choose $\eta \in C_c^\infty(J)$ and $\zeta \in C_c^\infty(B_{2R})$ with

$$0 \le \eta \le 1, \qquad 0 \le \zeta \le 1, \qquad \eta \equiv 1 \text{ on } J', \qquad \zeta \equiv 1 \text{ on } B_R,$$

and

$$\|\eta'\|_{L^\infty(J)} \le C_\eta\,\delta^{-1}, \qquad \|\nabla\zeta\|_{L^\infty(B_{2R})} \le C_\zeta R^{-1}, \qquad \|\Delta\zeta\|_{L^\infty(B_{2R})} \le C_\zeta R^{-2}.$$

Set

$$\phi(y,\tau) := \eta(\tau)\,\zeta(y)^2.$$

Then $\phi \in C_c^\infty(Q_{2R,J})$, $0 \le \phi \le 1$, and

$$\phi \equiv 1 \text{ on } B_R \times J'.$$

Choose $\tau_- < \inf J'$ and $\tau_+ > \sup J'$ in J with $\eta(\tau_-) = \eta(\tau_+) = 0$. Applying (7) on (τ_-, τ_+) gives

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi\,|\nabla V|^2 \le |R_\tau| + |R_\Delta| + |R_{\text{conv}}| + |R_{\text{pres}}| + |R_{\text{mod}}|,$$

where

$$R_\tau := \int_J \int_{B_{2R}} \frac{|V|^2}{2}\,\partial_\tau \phi, \qquad R_\Delta := \nu \int_J \int_{B_{2R}} \frac{|V|^2}{2}\,\Delta\phi,$$

$$R_{\text{conv}} := \int_J \int_{B_{2R}} \big(\tfrac{|V|^2}{2} V\big)\cdot\nabla\phi, \qquad R_{\text{pres}} := \int_J \int_{B_{2R}} \big(P - (P)_{B_{2R}}(\tau)\big)V\cdot\nabla\phi,$$

$$R_{\text{mod}} := \int_J \int_{B_{2R}} a\frac{|V|^2}{2}(-\phi - y\cdot\nabla\phi) - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b\cdot\nabla\phi.$$

For almost every $\sigma \in J'$, applying (7) on (τ_-, σ) gives the same right-hand estimates and leaves the terminal term $\frac{1}{2}\int_{B_R} |V(\sigma)|^2$ on the left. Therefore the bounds below control both $\mathcal{H}_{J'}(R)$ and $\mathcal{A}_{J'}(R)$. Define

$$L := \int_J \int_{B_{2R}} |V|^2, \qquad C_3 := \mathcal{C}_J(2R), \qquad D_{3/2} := \mathcal{D}_J(2R).$$

By Theorem 5.2,

$$L \le |Q_{2R,J}|^{1/3} C_3^{2/3} \le C(R, |J|)(1 + C_3).$$

Since $\phi(y,\tau) = \eta(\tau)\zeta(y)^2$,

$$\partial_\tau\phi = \eta'(\tau)\zeta(y)^2, \qquad \nabla\phi = 2\eta(\tau)\zeta(y)\nabla\zeta(y), \qquad \Delta\phi = \eta(\tau)\Delta(\zeta(y)^2),$$

hence

$$\|\partial_\tau\phi\|_{L^\infty(Q_{2R,J})} \le C_\eta \delta^{-1}, \qquad \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \le 2C_\zeta R^{-1}, \qquad \|\Delta\phi\|_{L^\infty(Q_{2R,J})} \le C_\zeta R^{-2}.$$

Step 2: derivative terms from the cut-off.

$$|R_\tau| \le \frac{1}{2}\|\partial_\tau\phi\|_{L^\infty(Q_{2R,J})} L \le C\delta^{-1}(1 + C_3),$$

$$|R_\Delta| \le \frac{\nu}{2}\|\Delta\phi\|_{L^\infty(Q_{2R,J})} L \le C\nu R^{-2}(1 + C_3).$$

Step 3: convection term. Using $|\nabla\phi| \le CR^{-1}$,

$$|R_{\text{conv}}| \le \frac{1}{2}\|\nabla\phi\|_{L^\infty(Q_{2R,J})} \int_J \int_{B_{2R}} |V|^3 \le CR^{-1}C_3.$$

Step 4: pressure term. For each time τ,

$$\int_{B_{2R}} (P - (P)_{B_{2R}}(\tau))V\cdot\nabla\phi = \int_{B_{2R}} PV\cdot\nabla\phi,$$

with

$$\int_{B_{2R}} (P)_{B_{2R}}(\tau)V\cdot\nabla\phi = (P)_{B_{2R}}(\tau)\int_{B_{2R}} V\cdot\nabla\phi = -(P)_{B_{2R}}(\tau)\int_{B_{2R}} \phi\,\nabla\cdot V = 0$$

(using $\phi \in C_c^\infty(B_{2R})$ and $\nabla\cdot V = 0$). Thus subtracting any time-dependent pressure constant is legitimate in the pressure flux term. Hence

$$|R_{\text{pres}}| \le \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \int_J \|P - (P)_{B_{2R}}(\tau)\|_{L_x^{3/2}(B_{2R})} \|V(\tau,\cdot)\|_{L_x^3(B_{2R})}\,d\tau \le CR^{-1} D_{3/2}^{2/3} C_3^{1/3}.$$

Young's inequality gives

$$|R_{\text{pres}}| \le C(R)(C_3 + D_{3/2}).$$

Step 5: modulation terms. The local energy inequality already contains the modulation terms after the spatial integrations by parts from the repaired-gauge equation:

$$\int_{B_{2R}} (y \cdot \nabla V) \cdot V\,\phi = \frac{1}{2}\int_{B_{2R}} y \cdot \nabla(|V|^2)\phi = -\frac{1}{2}\int_{B_{2R}} |V|^2 \nabla \cdot (y\phi),$$

and

$$\int_{B_{2R}} (b(\tau) \cdot \nabla V) \cdot V\,\phi = \frac{1}{2}\int_{B_{2R}} b(\tau) \cdot \nabla(|V|^2)\phi = -\frac{1}{2}\int_{B_{2R}} |V|^2 b(\tau) \cdot \nabla \phi.$$

Thus we estimate the additional terms directly. Split

$$R_{\mathrm{mod}} = R_a + R_{ay} + R_b,$$

where

$$R_a := -\int_J\int_{B_{2R}} a\,\frac{|V|^2}{2}\phi, \quad R_{ay} := -\int_J\int_{B_{2R}} a\,\frac{|V|^2}{2}\,y \cdot \nabla\phi, \quad R_b := -\int_J\int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla\phi.$$

$$M_a := \|a\|_{L^\infty(J)}, \qquad M_b := \|b\|_{L^\infty(J)}.$$

For R_a,

$$|R_a| \le \frac{1}{2} M_a L \le C M_a(1 + C_3).$$

For the scale-gradient part, $|y| \le 2R$ on B_{2R} and $\|\nabla\phi\|_{L^\infty(Q_{2R,J})} \le CR^{-1}$, hence

$$|y \cdot \nabla\phi| \le C.$$

Therefore

$$|R_{ay}| \le C M_a L \le C M_a (1 + C_3).$$

For R_b,

$$|R_b| \le C M_b R^{-1} L \le C M_b R^{-1}(1 + C_3).$$

Step 6: absorption and conclusion. Collecting estimates from Steps 2–5 yields

$$\nu \int_{\tau_-}^{\tau_+}\int_{B_{2R}} \phi |\nabla V|^2 \le C\big(1 + C_3 + D_{3/2}\big),$$

where C depends only on

$$R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \delta.$$

Since $\phi \equiv 1$ on $B_R \times J'$,

$$\mathcal{H}_{J'}(R) \le C\big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\big).$$

Repeating the same estimates with terminal time $\sigma \in J'$ gives

$$\frac{1}{2}\int_{B_R} |V(y,\sigma)|^2\,dy \le C\big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\big)$$

for almost every $\sigma \in J'$. Taking the essential supremum over $\sigma \in J'$ and renaming C proves the theorem. □

Definition 5.5 (Local compact-cylinder control)**.** Local Caffarelli–Kohn–Nirenberg control on $Q_{r,J}$ means

$$\mathcal{C}_J(r) + \mathcal{D}_J(r) < \infty.$$

Local windowed H^1 control on $Q_{r,J}$ means

$$\mathcal{H}_J(r) < \infty.$$

Corollary 5.6 (Outer control implies inner suitable control)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume Theorem 5.1 on $Q_{2R,J}$ and $\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty$. Then*

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) < \infty.$$

More quantitatively, the bound is the one in Theorem 5.4. If an argument uses pressure means on a smaller ball, Theorem 5.3 converts those normalizations to the outer-ball normalization above.

Proof. The estimate follows from Theorem 5.4. The comparison of pressure normalizations follows from Theorem 5.3. □

Corollary 5.7 (Rough-core exclusion on compact windows)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume Theorem 5.1 on $Q_{2R,J}$. If*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty,$$

then

$$\mathcal{H}_{J'}(R) < \infty.$$

Equivalently, if $\mathcal{H}_{J'}(R) = +\infty$ for an inner cylinder $Q_{R,J'}$, then for every enclosing cylinder $Q_{2R,J}$ with $J' \Subset J$ on which the local energy inequality and bounded modulation hypotheses hold,

$$\mathcal{C}_J(2R) = +\infty \qquad \text{or} \qquad \mathcal{D}_J(2R) = +\infty.$$

Proof. The direct implication is the gradient part of Theorem 5.4. The last assertion is its contrapositive with the same outer-to-inner geometry: finite outer $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ forces finite inner $\mathcal{H}$ on $Q_{R,J'}$. □

Remark 5.8 (Meaning of rough core)*.* Here a rough core means failure of local windowed H^1 control on compact inner renormalized cylinders while the modulation coefficients are bounded. Theorem 5.7 shows that such a failure is not independent of compact-cylinder Caffarelli–Kohn–Nirenberg control: it can occur only if the velocity-pressure bounds $\mathcal{C} + \mathcal{D}$ fail on every relevant larger enclosing compact cylinder.

5.3. **Consequences for the assembly.** We shall use the following consequences in the final assembly.

(1) On every compact renormalized window satisfying the local energy inequality and bounded modulation, finite $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ implies finite $\mathcal{A} + \mathcal{H}$ on $Q_{R,J'}$ whenever $J' \Subset J$.
(2) The pressure normalization may be taken on any comparable ball, after using Theorem 5.3.
(3) Failure of local windowed H^1 control on an inner compact cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on each larger enclosing cylinder to which Theorem 5.4 applies.

6. Scale-Collapse Cost and Autonomous Limits

Notation and functional conventions. For a bounded spatial region $K \subset \mathbb{R}^3$, we use

$$(f,g)_{L^2(K)} := \int_K f \cdot g \, dx, \qquad \langle F, \phi \rangle_{H^{-1}(K), H_0^1(K)} := \int_0^T \langle F(\cdot, s), \phi(\cdot, s) \rangle_{H^{-1}(K), H_0^1(K)} \, ds$$

whenever $F \in L^1(0,T; H^{-1}(K))$ and $\phi \in L^\infty(0,T; H_0^1(K))$. Weak limits and weak convergences are taken in the distributional sense on C_c^∞-test functions.

$$\|f\|_{L_t^p L_x^q} := \left(\int_0^T \|f(\cdot, s)\|_{L^q(\mathbb{R}^3)}^p \, ds \right)^{1/p},$$

for any fixed $T > 0$, with the usual modification $p = \infty$.

6.1. Renormalized scale-collapse setting.

Position of the result. The scale-collapse analysis has two components:

1. Sections 6.2–6.4 contain the cost exclusion arguments, which do not use compactness.
2. Section 6.5 contains the autonomous compactness statement under explicitly stated compactness hypotheses.

Throughout the scale-collapse analysis, the renormalized representation established earlier is assumed: on the time interval under consideration, the fields (V, P) satisfy (8) and its distributional weak formulation. In particular, the solenoidal formulation is obtained by restricting the test fields below to divergence-free fields.

Let (u, p) be a divergence-free suitable weak solution of

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

in $\mathbb{R}^3$, with viscosity $\nu > 0$. Let $\lambda(\tau) > 0$, $X(\tau) \in \mathbb{R}^3$, and a measurable gauge function $c(t)$ be given, and define the renormalized fields by

$$u(x,t) = \lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right), \qquad p(x,t) = \lambda(\tau)^{-2} P\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right) + c(t).$$

Pressure is defined only up to an arbitrary function of time; $c(t)$ has no effect on ∇p, so it is irrelevant for all subsequent estimates. Assume that (V, P) satisfies on $(0, \infty) \times \mathbb{R}^3$

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0. \tag{8}$$

For every $\psi \in C_c^\infty((0,\infty) \times \mathbb{R}^3; \mathbb{R}^3)$ with $\nabla \cdot \psi = 0$, the pair (V, P) satisfies the weak formulation

$$\begin{aligned} 0 = &\int_0^\infty \int_{\mathbb{R}^3} \big(-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi + \nu \nabla V : \nabla \psi \big) \, dy \, d\tau \\ &+ \int_0^\infty \int_{\mathbb{R}^3} \big(-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi + P \, \nabla \cdot \psi \big) \, dy \, d\tau. \end{aligned}$$

For every $0 < T < \infty$, restricting the τ-integration to $(0, T)$ gives the same identity on $(0, T) \times \mathbb{R}^3$. Equivalently, for every $\psi \in C_c^\infty((0,\infty) \times \mathbb{R}^3; \mathbb{R}^3)$, not necessarily divergence free,

$$\begin{aligned} 0 = &\int_0^\infty \int_{\mathbb{R}^3} \big(-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi \nu \nabla V : \nabla \psi - P \, \nabla \cdot \psi \big) \, dy \, d\tau \\ &+ \int_0^\infty \int_{\mathbb{R}^3} \big(-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi \big) \, dy \, d\tau. \end{aligned}$$

Define positive and negative parts of the scalar field a by

$$a_+(\tau) := \max\{a(\tau), 0\}, \qquad a_-(\tau) := \max\{-a(\tau), 0\}.$$

Then for a.e. τ, $a = a_+ - a_-$, $a_+, a_- \geq 0$, and $a_+ a_- = 0$. Assume

$$a \in L^1_{\mathrm{loc}}([\tau_0, \infty)), \qquad b \in L^1_{\mathrm{loc}}([\tau_0, \infty); \mathbb{R}^3),$$

and the scale convention

$$\frac{d}{d\tau} \log \lambda(\tau) = a(\tau) \tag{9}$$

for a.e. $\tau \geq \tau_0$, with $\log \lambda$ absolutely continuous on compact intervals.

All integrals below are interpreted in $[0, \infty]$. When we write $\int_{\tau_0}^{\infty}$, partial integrals are taken over $[\tau_0, T]$ and monotone convergence is used as $T \to \infty$.

Fix $R > 0$. Choose $\Phi \in C_c^\infty(\mathbb{R}^3; [0, 1])$ satisfying

$$\Phi(y) = 1 \text{ if } |y| \leq R, \quad \mathrm{supp}_y \, \Phi \subset B_{2R}. \tag{10}$$

Define the localized kinetic mass

$$M_R(\tau) := \int_{\mathbb{R}^3} |V(y, \tau)|^2 \Phi(y) \, dy.$$

Since Φ is fixed in τ, M_R is measurable in τ, and only this measurability is used below.

Definition 6.1 (Nonvanishing localized core)**.** A renormalized branch has nonvanishing localized core if there exist constants $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

In applications, this lower bound is supplied by the selected-core construction; here it is stated as an explicit hypothesis.

Definition 6.2 (Scale-collapse and absolute scale-drift densities)**.** Define

$$\mathcal{C}_{\mathrm{sc}}(\tau) := a_-(\tau) M_R(\tau), \qquad \mathcal{C}_{\mathrm{abs}}(\tau) := \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \Phi(y) \, dy + |a(\tau)| M_R(\tau).$$

$\mathcal{C}_{\mathrm{sc}}$ is the negative drift weighted by localized core mass, while $\mathcal{C}_{\mathrm{abs}}$ controls both collapse drift and local dissipation. Their total costs are extended real-valued by

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau) \, d\tau, \qquad \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau) \, d\tau.$$

Definition 6.3 (Finite-cost branch)**.** A branch has finite scale-collapsing cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau) \, d\tau < \infty.$$

It has finite absolute cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau) \, d\tau < \infty.$$

6.2. **Elementary alternative.**

Lemma 6.4 (Finite-or-infinite alternative)**.** *Let $f \geq 0$ be measurable on $[\tau_0, \infty)$. Then exactly one of the following holds:*

$$\int_{\tau_0}^{\infty} f(\tau) \, d\tau < \infty \quad \text{or} \quad \int_{\tau_0}^{\infty} f(\tau) \, d\tau = \infty.$$

Proof. (1) For $T > \tau_0$, define partial costs

$$F(T) := \int_{\tau_0}^{T} f(\tau) \, d\tau \in [0, \infty].$$

(2) If $T_2 > T_1$, then by nonnegativity of f,

$$F(T_2) - F(T_1) = \int_{T_1}^{T_2} f(\tau)\, d\tau \geq 0,$$

so F is monotone nondecreasing.

(3) A monotone function on $[\tau_0, \infty)$ has an extended limit $L \in [0, \infty]$,

$$L := \lim_{T\to\infty} F(T).$$

By definition of improper integral, this limit equals $\int_{\tau_0}^{\infty} f(\tau)\, d\tau$.

(4) Therefore L is either finite or equal to $+\infty$, and these cases are mutually exclusive.

Hence exactly one item in the lemma holds. □

Lemma 6.5 (Logarithmic scale identity)**.** *Under* (9)*, for all* τ_1, τ_2 *with* $\tau_0 \leq \tau_1 < \tau_2 < \infty$*,*

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Proof. (1) Since $a \in L^1_{\mathrm{loc}}$, the map $\log \lambda$ is absolutely continuous on $[\tau_1, \tau_2]$, hence has a representative with a.e. derivative equal to a.

(2) By the fundamental theorem of calculus for absolutely continuous functions, for every $\tau \in [\tau_1, \tau_2]$,

$$\log \lambda(\tau) = \log \lambda(\tau_1) + \int_{\tau_1}^{\tau} a(s)\, ds.$$

(3) Setting $\tau = \tau_2$ yields the identity.

□

6.3. **Scale collapse versus localized core lower bounds.**

Theorem 6.6 (Collapse with nonvanishing localized core forces infinite scale-collapsing cost)**.** *Assume:*

(1) $\lambda(\tau) \to 0$ *as* $\tau \to \infty$*;*

(2) the branch has nonvanishing localized core.

Then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. (1) Choose $m_0 > 0$ and $\tau_1 \geq \tau_0$ from the nonvanishing core assumption so that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

(2) By Lemma 6.5, for any $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Taking $\tau_2 \to \infty$ and using $\lambda(\tau_2) \to 0$,

$$\lim_{\tau_2\to\infty} \int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = -\infty.$$

(3) Decompose $a = a_+ - a_-$. Then for each $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\, d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau \geq -\int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau.$$

Hence

$$\int_{\tau_1}^{\tau_2} a_-(\tau)\,d\tau \ge -\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau.$$

(4) Taking limits and using Step 2 yields

$$\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty.$$

Indeed, by Step 2,

$$-\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau \to +\infty.$$

Since the partial integrals of a_- dominate this quantity, they are unbounded. Because $a_- \ge 0$, monotone convergence gives $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$.

(5) Split the cost into early and late times:

$$\int_{\tau_0}^{\infty} a_- M_R\,d\tau = \int_{\tau_0}^{\tau_1} a_- M_R\,d\tau + \int_{\tau_1}^{\infty} a_- M_R\,d\tau.$$

The first term is finite or infinite independently of the argument; the second term satisfies

$$\int_{\tau_1}^{\infty} a_-(\tau) M_R(\tau)\,d\tau \ge m_0 \int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$$

by the bound in (1) and Step 4. Hence the total integral is infinite.

□

6.4. **Finite-cost exclusions.**

Theorem 6.7 (Scale-collapse cost exclusion)**.** *No branch with finite scale-collapsing cost (Definition 6.3) can satisfy simultaneously $\lambda(\tau) \to 0$ and nonvanishing localized core.*

Proof. Assume, for contradiction, that such a branch exists. The two hypotheses are exactly those of Theorem 6.6; therefore

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau = \infty.$$

This contradicts finite scale-collapsing cost. Therefore no such branch exists. □

Lemma 6.8 (Absolute-cost domination)**.** *For a.e. $\tau \in [\tau_0, \infty)$,*

$$\mathcal{C}_{\mathrm{abs}}(\tau) \ge \mathcal{C}_{\mathrm{sc}}(\tau).$$

Hence, pointwise and after integrating in τ,

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau \le \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\,d\tau,$$

as extended real numbers. In particular, finite absolute cost implies finite scale-collapsing cost, and equivalently

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau = \infty \quad \Longrightarrow \quad \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\,d\tau = \infty.$$

Proof. At each time τ,

$$\mathcal{C}_{\mathrm{abs}}(\tau) = \nu \int |\nabla V|^2 \Phi + (a_+ + a_-) M_R \ge (a_+ + a_-) M_R \ge a_- M_R = \mathcal{C}_{\mathrm{sc}}(\tau),$$

since $a_+, a_- \ge 0$ and $\nu \int |\nabla V|^2 \Phi \ge 0$. Integrating over $[\tau_0, \infty)$ yields the claimed implication. □

Corollary 6.9 (Absolute scale-cost exclusion)**.** *No branch with finite absolute cost in the sense of Definition 6.3 can satisfy simultaneously $\lambda(\tau) \to 0$ and nonvanishing localized core.*

Proof. Assume instead such a branch exists. By Theorem 6.6, $\int \mathcal{C}_{\mathrm{sc}} = \infty$. By Lemma 6.8, the absolute cost is pointwise above $\mathcal{C}_{\mathrm{sc}}$, so its integral must also be infinite, contradicting finite absolute cost. □

Remark 6.10. The exclusions above are purely integral and use neither compactness nor any autonomous-limit argument.

6.5. **Autonomous reduced-limit theorem from the selected-window estimates.** The preceding exclusions are integral statements and do not rely on compactness. On a fixed finite autonomous window, compactness requires not only the cost bound but also the selected-window estimates supplied by the preceding renormalized representation, pressure reconstruction, and local Caccioppoli estimates.

Analytic inputs. The preceding sections supply the following ingredients on selected compact renormalized windows:

1. the repaired-gauge renormalized equation and weak formulation for (V, P);
2. pressure reconstruction modulo spatial means, with local $L^{3/2}$ bounds on compact cylinders;
3. local windowed bounds $V \in L_t^\infty L_x^2 \cap L_t^2 H_x^1$ on compact cylinders;
4. a retained compact core lower bound preventing the selected profile from vanishing.

Together with the thick-window convergence of the modulation coefficients, these estimates imply the compactness and limit passage below.

Definition 6.11 (Thick autonomous windows)**.** A branch has *thick autonomous windows* if there exist constants $T > 0$, $c > 0$, $M_1 \in (0, \infty)$, a sequence (τ_n) with $\tau_n \to \infty$, and limits $a_\infty \in \mathbb{R}$, $b_\infty \in \mathbb{R}^3$, such that for every n:

(i)
$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau)\, d\tau \geq c;$$

(ii)
$$0 \leq M_R(\tau) \leq M_1 \quad \text{for a.e. } \tau \in [\tau_n, \tau_n + T];$$

(iii)
$$a(\tau_n + \cdot) \to a_\infty \text{ in } L^1(0, T), \qquad b(\tau_n + \cdot) \to b_\infty \text{ in } L^1(0, T; \mathbb{R}^3).$$

Remark 6.12. This notion isolates a limiting drift coefficient $a_\infty < 0$. The compactness to extract a profile limit is obtained from the selected-window compactness estimates, not from the scalar cost inequalities of Sections 6.2–6.4.

Lemma 6.13 (Thick drift implies negative autonomous limit parameter)**.** *On thick autonomous windows, the limiting autonomous coefficient is negative:*

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. (1) From (ii), for a.e. $\tau \in [\tau_n, \tau_n + T]$, $0 \leq M_R(\tau) \leq M_1$. Therefore

$$\frac{a_-(\tau) M_R(\tau)}{M_1} \leq a_-(\tau).$$

Therefore

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau)\, d\tau \geq \frac{1}{M_1 T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau)\, d\tau \geq \frac{c}{M_1}.$$

(2) Set $a_n(s) := a(\tau_n + s)$. By (iii), $a_n \to a_\infty$ in $L^1(0,T)$. The map $r \mapsto r^-$ is 1-Lipschitz, so $a_n^- \to a_\infty^-$ in $L^1(0,T)$:

$$\|a_n^- - a_\infty^-\|_{L^1(0,T)} \le \|a_n - a_\infty\|_{L^1(0,T)} \to 0.$$

Thus

$$\lim_{n\to\infty} \frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)\,d\tau = \frac{1}{T}\int_0^T (a_\infty)^-\,ds = (a_\infty)^-.$$

(3) Combine Step 1 and Step 2: $(a_\infty)^- \ge c/M_1$, i.e. $a_\infty \le -c/M_1 < 0$.

□

Assumption 6.14 (Selected-window suitable estimates)**.** Let (τ_n) be a thick autonomous-window sequence and set

$$V_n(y,s) := V(y,\tau_n+s), \qquad P_n(y,s) := P(y,\tau_n+s),$$
$$\alpha_n(s) := a(\tau_n+s), \qquad \beta_n(s) := b(\tau_n+s), s \in (0,T).$$

For every bounded Lipschitz domain $K \Subset \mathbb{R}^3$, assume there is a constant $C_K < \infty$, independent of n, such that

$$\|V_n\|_{L^\infty(0,T;L^2(K))} + \|V_n\|_{L^2(0,T;H^1(K))} + \|P_n - (P_n)_K\|_{L^{3/2}((0,T)\times K)} \le C_K,$$

where

$$(P_n)_K(s) := \frac{1}{|K|}\int_K P_n(y,s)\,dy.$$

Assume also that each V_n is divergence free and that there exist $\chi \in C_c^\infty(B_R)$, $\chi \ge 0$, $\eta > 0$, and $N \in \mathbb{N}$ such that for all $n \ge N$,

$$\int_{\mathbb{R}^3} |V_n(y,s)|^2 \chi(y)\,dy \ge \eta \quad \text{for a.e. } s \in (0,T).$$

Proposition 6.15 (Derivation of the selected-window estimates from the preceding sections)**.** *Let a repaired-gauge branch satisfy the renormalized representation and pressure reconstruction of the compact-window representation theorem, Theorem 3.35, on the windows $[\tau_n, \tau_n + T]$. Assume that on each compact cylinder $(0,T)\times K$ the local compact-cylinder Caffarelli–Kohn–Nirenberg quantities are finite uniformly in n, and that the selected core has the localized lower bound stated in Assumption 6.14. Then Assumption 6.14 holds.*

Proof. The renormalized representation theorem supplies the shifted equation, divergence-free condition, and pressure reconstruction modulo functions of time. The pressure reconstruction gives, after subtracting spatial means,

$$\sup_n \|P_n - (P_n)_K\|_{L^{3/2}((0,T)\times K)} < \infty$$

on every compact K. The local Caccioppoli estimate applied on a slightly larger compact cylinder gives the inner bounds

$$\sup_n \|V_n\|_{L^\infty(0,T;L^2(K))} + \sup_n \|V_n\|_{L^2(0,T;H^1(K))} < \infty.$$

These are precisely the local velocity and pressure estimates in Assumption 6.14. The retained selected-core lower bound is the last condition in that assumption. Hence all items in Assumption 6.14 follow from the representation, pressure, and local Caccioppoli results. □

Lemma 6.16 (Local time regularity on autonomous windows)**.** *Assume thick autonomous windows and Assumption 6.14. Let $K_0 \Subset K_1 \Subset \mathbb{R}^3$ be bounded Lipschitz domains, and choose an integer m so large that $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$. Then*

$$\{\partial_s V_n\}_{n\ge1} \quad \textit{is bounded in } L^1(0,T;H^{-m}(K_1)).$$

Proof. Fix $\zeta \in C_c^\infty(K_1)$ with $\zeta \equiv 1$ on a neighborhood of K_0. Let $\phi \in H_0^m(K_1;\mathbb{R}^3)$. Since $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$, there is C_m such that

$$\|\phi\|_{L^\infty(K_1)} + \|\nabla\phi\|_{L^\infty(K_1)} \le C_m\|\phi\|_{H^m(K_1)}.$$

The shifted equation is

$$\partial_s V_n = -(V_n\cdot\nabla)V_n - \nabla P_n + \nu\Delta V_n + \alpha_n(V_n + y\cdot\nabla V_n) + \beta_n\cdot\nabla V_n$$

in distributions. We estimate each term against ϕ.

For the convection term, integration by parts gives

$$|\langle (V_n\cdot\nabla)V_n, \phi\rangle| = \left|\int_{K_1} V_{n,i}V_{n,j}\partial_j\phi_i\,dy\right| \le C_m\|V_n(s)\|^2_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the viscous term,

$$|\langle \nu\Delta V_n, \phi\rangle| = \nu\left|\int_{K_1}\nabla V_n : \nabla\phi\,dy\right| \le \nu C_m\|\nabla V_n(s)\|_{L^2(K_1)}|K_1|^{1/2}\|\phi\|_{H^m(K_1)}.$$

For the pressure term, subtracting the spatial mean does not change the gradient:

$$|\langle \nabla P_n, \phi\rangle| = \left|\int_{K_1}(P_n - (P_n)_{K_1})\,\nabla\cdot\phi\,dy\right| \le C_m|K_1|^{1/3}\|P_n - (P_n)_{K_1}\|_{L^{3/2}(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the V_n-part of the scaling drift,

$$|\alpha_n(s)|\left|\int_{K_1} V_n\cdot\phi\,dy\right| \le C_m|\alpha_n(s)|\,|K_1|^{1/2}\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the $y\cdot\nabla V_n$-part, integrate by parts:

$$\int_{K_1}(y\cdot\nabla V_n)\cdot\phi\,dy = -\int_{K_1} V_n\cdot(y\cdot\nabla\phi + 3\phi)\,dy,$$

because ϕ has zero trace. Hence

$$|\alpha_n(s)|\left|\int_{K_1}(y\cdot\nabla V_n)\cdot\phi\,dy\right| \le C_{K_1,m}|\alpha_n(s)|\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

Similarly,

$$\left|\int_{K_1}(\beta_n\cdot\nabla V_n)\cdot\phi\,dy\right| = \left|\int_{K_1} V_n\cdot(\beta_n\cdot\nabla\phi)\,dy\right| \le C_m|\beta_n(s)|\,|K_1|^{1/2}\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

Combining the estimates and taking the supremum over $\|\phi\|_{H^m(K_1)} \le 1$, we obtain

$$\begin{aligned}\|\partial_s V_n(s)\|_{H^{-m}(K_1)} \le C_{K_1,m}\big(&\|V_n(s)\|^2_{L^2(K_1)} + \|\nabla V_n(s)\|_{L^2(K_1)} \\ &+ \|P_n(s) - (P_n)_{K_1}(s)\|_{L^{3/2}(K_1)} + (|\alpha_n(s)| + |\beta_n(s)|)\|V_n(s)\|_{L^2(K_1)}\big).\end{aligned}$$

Integrating in $s \in (0,T)$, using Assumption 6.14, and using $\alpha_n \to a_\infty$, $\beta_n \to b_\infty$ in L^1, hence uniform L^1-bounds for α_n, β_n, proves the claimed $L^1(0,T;H^{-m})$ bound. □

Lemma 6.17 (Local compactness on autonomous windows)**.** *Assume thick autonomous windows and Assumption 6.14. Then after passing to a subsequence there exist*

$$U \in L^\infty(0,T;L^2_{\mathrm{loc}}(\mathbb{R}^3)) \cap L^2(0,T;H^1_{\mathrm{loc}}(\mathbb{R}^3)), \qquad \Pi \in L^{3/2}_{\mathrm{loc}}((0,T)\times\mathbb{R}^3),$$

such that, for every compact $K \Subset \mathbb{R}^3$,

(a)

$$V_n \to U \quad \textit{strongly in } L^2(0,T;L^2(K));$$

(b)

$$V_n \rightharpoonup U \quad \textit{weakly in } L^2(0,T;H^1(K));$$

(c)

$$V_n \rightharpoonup^* U \quad \textit{weakly-* in } L^\infty(0,T;L^2(K));$$

(d)

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \textit{weakly in } L^{3/2}((0,T)\times K),$$

where the local distributions $\nabla\Pi_K$ agree on overlaps and define a pressure gradient $\nabla\Pi$ modulo functions of time.

Proof. Fix $K_0 \Subset K_1 \Subset \mathbb{R}^3$. Assumption 6.14 gives boundedness of V_n in $L^2(0,T;H^1(K_1))$, and Lemma 6.16 gives boundedness of $\partial_s V_n$ in $L^1(0,T;H^{-m}(K_1))$ for m large. Since

$$H^1(K_1) \Subset L^2(K_0) \hookrightarrow H^{-m}(K_1),$$

the Aubin–Lions–Simon compactness theorem implies, after extracting a subsequence,

$$V_n \to U \quad \text{strongly in } L^2(0,T;L^2(K_0)).$$

The local $L^\infty L^2$ and L^2H^1 bounds also give, after extraction,

$$V_n \rightharpoonup^* U \quad \text{in } L^\infty(0,T;L^2(K_0)), \qquad V_n \rightharpoonup U \quad \text{in } L^2(0,T;H^1(K_0)).$$

The pressure bound in Assumption 6.14 gives weak compactness of $P_n-(P_n)_{K_0}$ in $L^{3/2}((0,T)\times K_0)$. A diagonal extraction over balls B_j gives a single subsequence for every compact K. On overlaps, two pressure limits can differ only by a function of s, because both represent the same distributional pressure gradient in the limit equation. Thus the gradients assemble into $\nabla\Pi$, with Π defined modulo functions of time. □

Lemma 6.18 (Compatibility of local pressure limits)**.** *Let $K_1, K_2 \Subset \mathbb{R}^3$ be bounded Lipschitz domains with nonempty connected overlap $K_{12} := K_1 \cap K_2$. Suppose that, after passing to a common subsequence,*

$$P_n - (P_n)_{K_i} \rightharpoonup \Pi_i \quad \textit{in } L^{3/2}((0,T)\times K_i), \qquad i = 1,2.$$

Then there exists $c_{12} \in L^{3/2}(0,T)$ such that

$$\Pi_1(y,s) - \Pi_2(y,s) = c_{12}(s) \quad \textit{for a.e. } (y,s) \in K_{12}\times(0,T).$$

Consequently the pressure gradient $\nabla\Pi$ obtained from the local limits is globally well-defined on $(0,T)\times\mathbb{R}^3$, with the pressure itself defined modulo functions of time.

Proof. For each n,

$$\big(P_n - (P_n)_{K_1}\big) - \big(P_n - (P_n)_{K_2}\big) = (P_n)_{K_2}(s) - (P_n)_{K_1}(s)$$

on K_{12}. Hence its spatial gradient is zero in $\mathcal{D}'((0,T)\times K_{12})$. Passing to the weak limit gives

$$\nabla_y(\Pi_1 - \Pi_2) = 0 \quad \text{in } \mathcal{D}'((0,T)\times K_{12}).$$

Since K_{12} is connected, the standard distributional characterization of functions with zero spatial gradient implies that $\Pi_1 - \Pi_2 = c_{12}(s)$ for some $c_{12} \in L^{3/2}(0,T)$. The last assertion follows because adding a time-dependent scalar to pressure does not change $\nabla\Pi$. □

Theorem 6.19 (Autonomous reduced-limit theorem)**.** *Assume that a branch has thick autonomous windows and satisfies Assumption 6.14 on the same window sequence. Let (U,Π) be the limit extracted in Lemma 6.17. Then*

$$\nabla\cdot U = 0,$$

$$U \not\equiv 0$$

and (U, Π) *satisfies*

$$\partial_s U + (U \cdot \nabla)U + \nabla\Pi = \nu\Delta U + a_\infty(U + y\cdot\nabla U) + b_\infty \cdot \nabla U, \qquad \nabla\cdot U = 0, \tag{11}$$

in $\mathcal{D}'((0,T)\times\mathbb{R}^3)$*. Equivalently, for every* $\varphi \in C_c^\infty((0,T)\times\mathbb{R}^3;\mathbb{R}^3)$*,*

$$\int_0^T\int \big(-U\cdot\partial_s\varphi - (U\otimes U) : \nabla\varphi + \nu\nabla U : \nabla\varphi - \Pi\,\nabla\cdot\varphi - a_\infty(U + y\cdot\nabla U)\cdot\varphi - (b_\infty\cdot\nabla U)\cdot\varphi\big)\,dy\,ds = 0.$$

Remark 6.20. The theorem asserts only the compact finite-window autonomous limit from the selected-window estimates. A rigidity contradiction requires a separate Liouville theorem for the class containing (U, Π).

Proof. Step 1: divergence-free property of the limit.

Each V_n is divergence free by Assumption 6.14, so for any compactly supported scalar $\psi \in C_c^\infty((0,T)\times\mathbb{R}^3)$,

$$\int_{(0,T)\times\mathbb{R}^3} V_n\cdot\nabla\psi\,dy\,ds = 0.$$

Hence, for every such ψ, since $\nabla\psi \in C_c^\infty((0,T)\times\mathbb{R}^3)$ and $V_n \to U$ in $L^2(0,T;L^2_{\mathrm{loc}})$,

$$\int_{(0,T)\times\mathbb{R}^3} (V_n - U)\cdot\nabla\psi\,dy\,ds \to 0,$$

we deduce

$$\int_{(0,T)\times\mathbb{R}^3} U\cdot\nabla\psi\,dy\,ds = 0.$$

Hence $\nabla\cdot U = 0$ in $\mathcal{D}'((0,T)\times\mathbb{R}^3)$.

Step 2: nontriviality of the limit profile. Define for $s \in (0,T)$

$$f_n(s) := \int_{\mathbb{R}^3} |V_n(y,s)|^2\chi(y)\,dy, \qquad f(s) := \int_{\mathbb{R}^3} |U(y,s)|^2\chi(y)\,dy.$$

Here $K_\chi := \operatorname{supp}\chi$, so $\chi^{1/2}V_n, \chi^{1/2}U \in L^2(0,T;L^2(K_\chi))$. Then

$$f_n - f = \int_{\mathbb{R}^3} (V_n - U)\cdot(V_n + U)\chi\,dy.$$

Applying Cauchy–Schwarz in space,

$$|f_n(s) - f(s)| \le \|\chi^{1/2}(V_n(s) - U(s))\|_{L^2(\mathbb{R}^3)}\|\chi^{1/2}(V_n(s) + U(s))\|_{L^2(\mathbb{R}^3)}.$$

Integrating in s and applying Cauchy–Schwarz in time gives

$$\|f_n - f\|_{L^1(0,T)} \le \|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)}\|\chi^{1/2}(V_n + U)\|_{L^2(0,T;L^2)}.$$

Since K_χ is compact, Assumption 6.14 and the membership $U \in L^2(0,T;L^2(K_\chi))$ give

$$\sup_n \|\chi^{1/2}(V_n + U)\|_{L^2(0,T;L^2)} \le C_\chi\left(\sup_n \|V_n\|_{L^2(0,T;L^2(K_\chi))} + \|U\|_{L^2(0,T;L^2(K_\chi))}\right) < \infty.$$

Moreover

$$\|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)} \le \|\chi\|_{L^\infty}^{1/2}\|V_n - U\|_{L^2(0,T;L^2(K_\chi))} \to 0.$$

Therefore

$$\|f_n - f\|_{L^1(0,T)} \to 0.$$

Hence $f_n \to f$ in $L^1(0,T)$.

Assumption 6.14 gives for large n, $f_n(s) \ge \eta$ for a.e.s. Therefore

$$\int_0^T f(s)\,ds = \lim_{n\to\infty}\int_0^T f_n(s)\,ds \ge \eta T > 0.$$

Hence $f \not\equiv 0$, so $U \not\equiv 0$.

Step 3: shifted weak formulation for V_n. Take any test field $\varphi \in C_c^\infty((0,T) \times \mathbb{R}^3; \mathbb{R}^3)$, and set

$$K_\varphi := \overline{\{x \in \mathbb{R}^3 : \exists s \in (0,T),\ \varphi(x,s) \neq 0\}} \subset \mathbb{R}^3, \qquad K_\varphi \text{ bounded.}$$

Since (V,P) satisfies (8) in distributions, the change of variables $\tau = \tau_n + s$ gives the shifted weak formulation

$$\begin{aligned} 0 &= \int_0^T \int_{\mathbb{R}^3} \Big(-V_n \cdot \partial_s \varphi - (V_n \otimes V_n) : \nabla\varphi + \nu \nabla V_n : \nabla \varphi - P_n \nabla \cdot \varphi \Big) \, dy \, ds \\ &\quad + \int_0^T \int_{\mathbb{R}^3} \Big(-a(\tau_n + s)(V_n + y \cdot \nabla V_n) \cdot \varphi - (b(\tau_n + s) \cdot \nabla V_n) \cdot \varphi \Big) \, dy \, ds. \end{aligned} \tag{12}$$

Step 4: term-by-term passage to the limit. Write $\alpha_n(s) := a(\tau_n + s)$, $\beta_n(s) := b(\tau_n + s)$.

(i) Time derivative.

By strong convergence in $L^2(0,T; L^2(K_\varphi))$,

$$\left| \int_0^T \int (V_n - U) \cdot \partial_s \varphi \, dy \, ds \right| \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \|\partial_s \varphi\|_{L^2(0,T;L^2(K_\varphi))} \to 0.$$

Hence

$$\int V_n \cdot \partial_s \varphi \to \int U \cdot \partial_s \varphi.$$

(ii) Convection term. Decompose

$$V_n \otimes V_n - U \otimes U = (V_n - U) \otimes V_n + U \otimes (V_n - U).$$

Then

$$\begin{aligned} &\left| \int_0^T \int ((V_n \otimes V_n - U \otimes U) : \nabla \varphi) \, dy \, ds \right| \\ &\quad \le \|\nabla \varphi\|_{L^\infty_{x,t}} \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \big(\|V_n\|_{L^2(0,T;L^2(K_\varphi))} + \|U\|_{L^2(0,T;L^2(K_\varphi))} \big) \to 0, \end{aligned}$$

using boundedness of V_n in $L^2(0,T; L^2(K_\varphi))$. Therefore $\int (V_n \otimes V_n) : \nabla\varphi \to \int (U \otimes U) : \nabla \varphi$.

(iii) Viscous term. From weak convergence of V_n in $L^2(0,T; H^1_{\mathrm{loc}})$,

$$\int_0^T \int \nabla V_n : \nabla \varphi \, dy \, ds \to \int_0^T \int \nabla U : \nabla \varphi \, dy \, ds.$$

(iv) Pressure term. Let K be a bounded Lipschitz domain with $K_\varphi \Subset K$. Since φ is compactly supported in K,

$$\int_K (P_n)_K(s) \, \nabla \cdot \varphi(y,s) \, dy = (P_n)_K(s) \int_K \nabla \cdot \varphi(y,s) \, dy = 0.$$

Therefore

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi \, dy \, ds = \int_0^T \int_K (P_n - (P_n)_K) \nabla \cdot \varphi \, dy \, ds.$$

By Lemma 6.17,

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{in } L^{3/2}((0,T) \times K),$$

and $\nabla \cdot \varphi \in L^3((0,T) \times K)$. The compatibility Lemma 6.18 permits us to write the resulting pressure distribution as Π, modulo functions of time. Hence

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi \, dy \, ds \to \int_0^T \int_K \Pi_K \nabla \cdot \varphi \, dy \, ds.$$

The value of the last integral is independent of the chosen K, because replacing Π_K by $\Pi_K + c(s)$ changes the integral by $\int_0^T c(s) \int_K \nabla \cdot \varphi \, dy \, ds = 0$. The limit is therefore the pressure term $\int \Pi \nabla \cdot \varphi$.

(v) Drift term with a: part involving V_n. Define

$$Y_n^0(s) := \int_{\mathbb{R}^3} V_n(y,s) \cdot \varphi(y,s) \, dy, \qquad Y^0(s) := \int_{\mathbb{R}^3} U(y,s) \cdot \varphi(y,s) \, dy.$$

For each s, Cauchy–Schwarz gives

$$|Y_n^0(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \|\varphi(s)\|_{L^2(K_\varphi)}.$$

By Assumption 6.14, there is $C_0 < \infty$ such that $\sup_n \|Y_n^0\|_{L^\infty(0,T)} \le C_0$. Moreover,

$$Y_n^0 \to Y^0 \quad \text{in } L^2(0,T),$$

because $V_n \to U$ in $L^2(0,T; L^2(K_\varphi))$. Now

$$\int_0^T \alpha_n(s) Y_n^0(s) \, ds = \int_0^T (\alpha_n(s) - a_\infty) Y_n^0(s) \, ds + a_\infty \int_0^T Y_n^0(s) \, ds.$$

For the first term,

$$\left| \int_0^T (\alpha_n - a_\infty) Y_n^0 \, ds \right| \le \sup_{s \in (0,T)} |Y_n^0(s)| \, \|\alpha_n - a_\infty\|_{L^1(0,T)} \to 0.$$

For the second term, strong convergence in $L^2(0,T)$ implies $\int_0^T Y_n^0 \to \int_0^T Y^0$, so

$$\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi \, dy \, ds \to a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi \, dy \, ds.$$

Thus the signed contribution in (12) satisfies

$$-\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi \, dy \, ds \to -a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi \, dy \, ds.$$

(vi) Drift term with a: part involving $y \cdot \nabla V_n$. Since $\varphi \in C_c^\infty$, integrating by parts in y gives

$$\int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi = -\int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy.$$

Define

$$Y_n^1(s) := \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy, \qquad Y^1(s) := \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy.$$

For each s, Cauchy–Schwarz gives

$$|Y_n^1(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \|y \cdot \nabla \varphi(s) + 3\varphi(s)\|_{L^2(K_\varphi)}.$$

The $L^\infty(0,T; L^2(K_\varphi))$ bound in Assumption 6.14 gives

$$\sup_n \|Y_n^1\|_{L^\infty(0,T)} \le C_1 < \infty.$$

Also,

$$\|Y_n^1 - Y^1\|_{L^2(0,T)} \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \|y \cdot \nabla \varphi + 3\varphi\|_{L^\infty(0,T;L^2(K_\varphi))} \to 0.$$

Therefore

$$-\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi \, dy \, ds \to a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy \, ds.$$

Integrating by parts once more in the limit gives

$$a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla\varphi + 3\varphi)\, dy\, ds = -a_\infty \int_0^T \int_{\mathbb{R}^3} (y \cdot \nabla U) \cdot \varphi\, dy\, ds.$$

(vii) Term with b. Since $\beta_n(s)$ is independent of y,

$$\int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi = -\int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla\varphi)\, dy,$$

so

$$-\int_0^T \int (\beta_n \cdot \nabla V_n) \cdot \varphi\, dy\, ds = \int_0^T \int V_n \cdot (\beta_n \cdot \nabla\varphi)\, dy\, ds.$$

Let

$$Z_n(s) := \left(\int_{\mathbb{R}^3} V_n \cdot \partial_1\varphi\, dy,\ \int_{\mathbb{R}^3} V_n \cdot \partial_2\varphi\, dy,\ \int_{\mathbb{R}^3} V_n \cdot \partial_3\varphi\, dy \right),$$

$$Z(s) := \left(\int_{\mathbb{R}^3} U \cdot \partial_1\varphi\, dy,\ \int_{\mathbb{R}^3} U \cdot \partial_2\varphi\, dy,\ \int_{\mathbb{R}^3} U \cdot \partial_3\varphi\, dy \right).$$

Then

$$\int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla\varphi)\, dy = \beta_n(s) \cdot Z_n(s), \qquad \int_{\mathbb{R}^3} U \cdot (b_\infty \cdot \nabla\varphi)\, dy = b_\infty \cdot Z(s).$$

Consequently

$$\left| \int_0^T (\beta_n - b_\infty) \cdot Z_n\, ds \right| \le \|\beta_n - b_\infty\|_{L^1(0,T)} \sup_{s\in(0,T)} |Z_n(s)| \to 0,$$

because

$$|Z_n(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \left(\sum_{j=1}^3 \|\partial_j\varphi(s)\|^2_{L^2(K_\varphi)} \right)^{1/2}$$

and Assumption 6.14 bounds the right-hand side uniformly in n and s. Furthermore,

$$\|Z_n - Z\|_{L^2(0,T;\mathbb{R}^3)} \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \left(\sum_{j=1}^3 \|\partial_j\varphi\|^2_{L^\infty(0,T;L^2(K_\varphi))} \right)^{1/2} \to 0.$$

Therefore

$$\int_0^T b_\infty \cdot Z_n(s)\, ds \to \int_0^T b_\infty \cdot Z(s)\, ds.$$

Hence

$$-\int_0^T \int (\beta_n \cdot \nabla V_n) \cdot \varphi\, dy\, ds \to \int_0^T \int U \cdot (b_\infty \cdot \nabla\varphi)\, dy\, ds.$$

Since b_∞ is constant in y, integration by parts yields

$$\int_0^T \int U \cdot (b_\infty \cdot \nabla\varphi)\, dy\, ds = -\int_0^T \int (b_\infty \cdot \nabla U) \cdot \varphi\, dy\, ds.$$

Step 5: assemble the limit. Passing to the limit in (12) in each term (Steps 4(i)–4(vii)) gives

$$\int_0^T \int \big(-U \cdot \partial_s\varphi - (U \otimes U) : \nabla\varphi + \nu\nabla U : \nabla\varphi - \Pi\nabla\cdot\varphi - a_\infty(U + y\cdot\nabla U)\cdot\varphi - (b_\infty \cdot \nabla U)\cdot\varphi \big)\, dy\, ds = 0.$$

Since this holds for every compactly supported vector field φ, equation (11) holds in distributions on $(0,T) \times \mathbb{R}^3$. □

6.6. **Decomposition of alternatives of the scale-collapse branch.** The cost argument in Sections 6.2 and 6.4 excludes the finite-cost scale-collapse branch. The remaining scale-collapse obstruction is treated by a decomposition of alternatives, following the local estimate and autonomous-reduction argument of the preceding sections. The conclusion separates the excluded branches from the residual alternatives; it does not assert a Liouville theorem for every autonomous negative-drift equation.

Theorem 6.21 (Selected core retention)**.** *Let a represented Type II branch be obtained from the selected-core construction of the preceding sections. Then there exist $R > 0$, a cutoff $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, a constant $\eta > 0$, and a terminal time τ_1 such that, along every selected window used below,*

$$\int_{\mathbb{R}^3} |V(y,\tau)|^2 \chi(y)\, dy \geq \eta \quad \textit{for a.e. } \tau \geq \tau_1.$$

In particular, every translated sequence $V_n(y,s) = V(y,\tau_n+s)$ satisfies the localized nontriviality condition in Assumption 6.14, after passing to a tail of the sequence.

Proof. The selected-core construction chooses a compact renormalized core carrying a fixed positive amount of localized kinetic mass. Since the cutoff χ is supported strictly inside that core and equals a positive spatial weight there, the selected-core lower bound gives the stated estimate. Translating $\tau = \tau_n + s$ gives the final assertion on every finite selected window. □

Theorem 6.22 (Autonomous modulation selection)**.** *Let a represented scale-collapse branch satisfy the modulation bounds obtained in the repaired gauge construction. Suppose that a sequence of windows $[\tau_n, \tau_n+T]$ is selected in the autonomous regime. Then, after passing to a subsequence, there exist constants $a_\infty \in \mathbb{R}$ and $b_\infty \in \mathbb{R}^3$ such that*

$$a(\tau_n+\cdot) \to a_\infty \quad \textit{in } L^1(0,T), \qquad b(\tau_n+\cdot) \to b_\infty \quad \textit{in } L^1(0,T;\mathbb{R}^3).$$

Proof. The autonomous-window selection in the repaired-gauge modulation analysis gives convergence of the modulation functions to constant autonomous parameters on the fixed window length T. Passing to the corresponding subsequence gives the displayed L^1-convergences. □

Theorem 6.23 (Scale-collapse decomposition of alternatives)**.** *Let a represented Type II branch have genuine scale collapse and nonvanishing selected core. Then exactly one of the following alternatives occurs after passing to selected terminal windows:*

(i) the branch has finite scale-collapsing cost, which is excluded by Theorem 6.7;
(ii) the branch has finite absolute cost, which is excluded by Corollary 6.9;
(iii) the branch admits thick autonomous windows in the sense of Definition 6.11, satisfies the selected-window estimates of Assumption 6.14, and has autonomous modulation convergence;
(iv) the branch has a thin-drift alternative: no selected fixed-length window carries a uniform positive lower bound for the averaged weighted negative drift;
(v) the branch has an autonomous-modulation alternative: thick windows exist, but no subsequence has a constant-coefficient L^1 modulation limit in the sense needed for Theorem 6.19;
(vi) the branch has a compactness alternative: the local estimates needed for Assumption 6.14 fail on the selected windows.

Proof. The scale-collapse selection theorem decomposes terminal scale-collapse windows according to the averaged negative-drift mass

$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau)\, d\tau.$$

If the accumulated cost is finite, alternatives (i) or (ii) apply according to the cost functional under consideration. If the averaged negative-drift mass stays bounded below on a subsequence, one obtains thick windows; otherwise the branch lies in the thin-drift alternative. On thick windows, either the local estimates and core retention hold, in which case Proposition 6.15 and Theorem 6.21 give Assumption 6.14, or the branch lies in the compactness alternative. If the estimates hold but no subsequence has the required constant-coefficient modulation limit, the branch lies in the autonomous-modulation alternative. The remaining case is the thick autonomous alternative. ☐

Definition 6.24 (Stationary omega-limit class)**.** Let (U, Π) be a nonzero autonomous limit produced by Theorem 6.19. We say that it has a stationary omega-limit in the admissible class if there exists a nonzero pair

$$(W, Q), \qquad W \in L^3(\mathbb{R}^3),$$

with the local energy and pressure reconstruction properties inherited from the selected windows, such that

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty (W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \qquad \nabla \cdot W = 0,$$

in $\mathcal{D}'(\mathbb{R}^3)$.

Definition 6.25 (Rigidity-covered stationary class)**.** For a parameter pair (a_∞, b_∞), the stationary class is rigidity-covered if every admissible stationary solution in Definition 6.24 is either zero or is not a terminal Type II branch. In the classical backward self-similar normalization this is the role of the Nečas–Růžička–Šverák rigidity theorem, after the constant translation drift is removed [NRŠ96]. Other parameter regimes require their own stated Liouville or classification theorem.

Theorem 6.26 (Stationary-rigidity exclusion)**.** *Assume a thick autonomous branch produces a nonzero autonomous limit by Theorem 6.19. If that autonomous limit has a stationary omega-limit in the admissible class and the corresponding stationary class is rigidity-covered, then the branch cannot occur as a terminal Type II scale-collapse branch.*

Proof. The stationary omega-limit gives a nonzero stationary profile (W, Q) in the admissible class. The rigidity-covered hypothesis says that no such nonzero profile can represent a terminal Type II branch. This contradicts membership in the terminal scale-collapse Type II class. ☐

Theorem 6.27 (Scale-collapse alternatives)**.** *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. Then one of the following mutually exclusive outcomes occurs:*

- *(i) the finite scale-collapsing cost exclusion applies;*
- *(ii) the finite absolute cost exclusion applies;*
- *(iii) the branch has a thin-drift alternative;*
- *(iv) the branch has an autonomous-modulation alternative;*
- *(v) the branch has a compactness alternative;*
- *(vi) the branch has a nonzero autonomous reduced limit, and either stationary rigidity excludes it or the remaining obstruction is the corresponding stationary/generalized self-similar Liouville problem.*

Proof. Apply Theorem 6.23. The finite-cost alternatives are items (i) and (ii). The thin-drift, modulation, and compactness alternatives are items (iii)–(v). In the remaining thick autonomous case, Lemma 6.13 gives $a_\infty < 0$, and Theorem 6.19 produces a nonzero autonomous reduced limit. If Definitions 6.24 and 6.25 apply, then Theorem 6.26 excludes the branch. If not, the remaining obstruction is the stationary or generalized self-similar Liouville problem for the parameter pair (a_∞, b_∞). ☐

6.7. **Conclusion.** The exclusions in Sections 6.2–6.4 are proved unconditionally within the renormalized setting: no branch with finite scale-collapsing (hence finite absolute) cost can have $\lambda(\tau) \to 0$ together with a nonvanishing localized core.

Lemma 6.13 shows that any thick autonomous window admits a negative autonomous drift limit a_∞. Lemmas 6.16 and 6.17, together with Theorem 6.19, assemble the selected-window estimates from the preceding sections into a nontrivial autonomous finite-window limit (U, Π) satisfying the full distributional autonomous equation.

Section 6.6 gives the corresponding decomposition of alternatives for the scale-collapse branch. The autonomous reduction gives a contradiction only in parameter regimes where a stationary omega-limit exists and the resulting stationary class is covered by a Liouville or rigidity theorem, such as the classical Nečas–Růžička–Šverák self-similar class. Outside those regimes the conclusion is the thin-drift, modulation, compactness, or Liouville alternative, rather than an unconditional exclusion.

7. Critical Tightness and Windowed-Control Estimates

This section records the local estimates used to exclude failure of uniform critical tightness and failure of local windowed H^1 control. The statements are phrased as analytic alternatives for a renormalized Navier–Stokes solution $V(\tau)$.

7.1. **Uniform critical tightness.**

Definition 7.1 (Uniform critical tightness)**.** A renormalized orbit $V(\tau)$, $\tau \geq \tau_0$, is uniformly critically tight if

$$\forall \varepsilon > 0 \ \exists R_\varepsilon < \infty : \sup_{\tau \geq \tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3 \, dy < \varepsilon.$$

The non-tight alternative is the failure of this property.

Lemma 7.2 (Critical precompactness implies uniform tightness)**.** *If*

$$\mathcal{O} := \{V(\tau) : \tau \geq \tau_0\}$$

is contained in a compact subset of $L^3(\mathbb{R}^3)$, then V is uniformly critically tight.

Proof. Compact subsets of $L^3(\mathbb{R}^3)$ are uniformly tight. Let $K \subset L^3$ be compact and fix $\varepsilon > 0$. Set $\delta = \varepsilon^{1/3}/2$. Choose a finite δ-net $f_1, \ldots, f_N$ in L^3. For each i, choose R_i so that

$$\|f_i\|_{L^3(|y|>R_i)} < \delta.$$

Let $R = \max_i R_i$. For any $f \in K$, choose i with $\|f - f_i\|_3 < \delta$. Then

$$\|f\|_{L^3(|y|>R)} \leq \|f - f_i\|_3 + \|f_i\|_{L^3(|y|>R)} < \varepsilon^{1/3}.$$

Taking the cube gives

$$\int_{|y|>R} |f(y)|^3 \, dy < \varepsilon.$$

Applying this to the compact set containing $\mathcal{O}$ gives uniform critical tightness. □

Definition 7.3 (Compact core with vanishing critical remainder)**.** The orbit has a compact critical core with vanishing remainder if there exist a compact set $\mathcal{K}^{\mathrm{c}} \subset L^3(\mathbb{R}^3)$, a map $Q : [\tau_0, \infty) \to \mathcal{K}^{\mathrm{c}}$, and a remainder $r(\tau) \in L^3(\mathbb{R}^3)$, such that

$$V(\tau) = Q(\tau) + r(\tau), \qquad \lim_{\tau\to\infty} \|r(\tau)\|_{L^3} = 0,$$

and, for every $T > \tau_0$, the set

$$\{V(\tau) : \tau \in [\tau_0, T]\}$$

is uniformly L^3-tight.

Lemma 7.4 (Compact core plus vanishing remainder implies tightness)**.** *If Theorem 7.3 holds, then V is uniformly critically tight.*

Proof. Fix $\varepsilon > 0$ and set $\delta = \varepsilon^{1/3}$. By compactness of $\mathcal{K}^{\mathrm{c}}$, apply Theorem 7.2 to $\mathcal{K}^{\mathrm{c}}$ and choose R_1 such that

$$\sup_{q\in\mathcal{K}^{\mathrm{c}}} \|q\|_{L^3(|y|>R_1)} < \delta/3.$$

Since $\|r(\tau)\|_3 \to 0$, choose T so that

$$\|r(\tau)\|_3 < \delta/3 \qquad \text{for all } \tau \geq T.$$

Then for $\tau \geq T$,

$$\|V(\tau)\|_{L^3(|y|>R_1)} \leq \|Q(\tau)\|_{L^3(|y|>R_1)} + \|r(\tau)\|_3 < 2\delta/3.$$

By finite-initial-window tightness on $[\tau_0, T]$, choose R_2 such that

$$\sup_{\tau\in[\tau_0,T]} \int_{|y|>R_2} |V(y,\tau)|^3 \, dy < \varepsilon.$$

With $R = \max(R_1, R_2)$, the L^3-norm of the tail of $V(\tau)$ is $< \delta$ for all $\tau \geq T$, while the tail integral is $< \varepsilon$ on $[\tau_0, T]$. Cubing on the tail $\tau \geq T$ gives the required L^3-tail integral $< \varepsilon$ for all $\tau \geq \tau_0$. $\square$

Definition 7.5 (No-splitting profile condition)**.** A critical profile decomposition has no critical-mass splitting if it has:

(i) a primary compact core family $\mathcal{K}^{c} \subset L^3$;
(ii) asymptotic L^3-decoupling of core, secondary profiles, and remainder;
(iii) single-profile saturation, so every non-primary profile has zero critical L^3-mass;
(iv) a remainder whose L^3-norm vanishes on the renormalized tail;
(v) finite-initial-window tightness.

Lemma 7.6 (No splitting gives compact core with vanishing remainder)**.** *If Theorem 7.5 holds, then Theorem 7.3 holds.*

Proof. By single-profile saturation and nonnegative L^3-mass decoupling, every secondary profile has zero critical L^3-mass and therefore vanishes in L^3. The profile decomposition then reduces to a primary compact core plus a remainder whose L^3-norm tends to zero on the renormalized tail. These provide the core $Q(\tau)$, compact set $\mathcal{K}^{c}$, and remainder $r(\tau)$ required by Theorem 7.3, with finite-initial-window tightness supplied as part of Theorem 7.5. $\square$

Corollary 7.7 (No splitting gives uniform critical tightness)**.** *If Theorem 7.5 holds, then V is uniformly critically tight.*

Proof. Theorem 7.6 gives a compact core with vanishing critical remainder, and Theorem 7.4 gives uniform critical tightness. $\square$

Definition 7.8 (Finite critical profile approximation)**.** The orbit admits a finite critical profile approximation if there are finitely many profiles

$$\mathcal{B}_{NS}^{II} := \{B_1, \ldots, B_N\} \subset L^3(\mathbb{R}^3)$$

such that, for every $\delta > 0$, there is τ_δ with

$$\forall \tau \geq \tau_\delta \ \exists i(\tau) \in \{1, \ldots, N\} : \|V(\tau) - B_{i(\tau)}\|_{L^3} < \delta,$$

and the finite initial window

$$\{V(\tau) : \tau \in [\tau_0, \tau_\delta]\}$$

is uniformly L^3-tight for each $\delta > 0$.

Lemma 7.9 (Finite critical profile approximation implies tightness)**.** *If Theorem 7.8 holds, then V is uniformly critically tight.*

Proof. Fix $\varepsilon > 0$ and set $\delta = \varepsilon^{1/3}$. Since the profile family is finite and each $B_i \in L^3(\mathbb{R}^3)$, there is R_1 such that

$$\max_{1\leq i\leq N} \|B_i\|_{L^3(|y|>R_1)} < \delta/3.$$

Choose the approximation tolerance $\delta/3$. For $\tau \geq \tau_{\delta/3}$, pick $i(\tau)$ with $\|V(\tau) - B_{i(\tau)}\|_3 < \delta/3$. Then

$$\|V(\tau)\|_{L^3(|y|>R_1)} \leq \|V(\tau) - B_{i(\tau)}\|_3 + \|B_{i(\tau)}\|_{L^3(|y|>R_1)} < 2\delta/3.$$

By finite-initial-window tightness, choose R_2 so that the L^3 tail is $< \varepsilon$ on $[\tau_0, \tau_{\delta/3}]$. With $R = \max(R_1, R_2)$, the tail integral is $< \varepsilon$ for every $\tau \geq \tau_0$, so the orbit is uniformly L^3-tight. $\square$

Definition 7.10 (Compact parameter realization)**.** The orbit has a compact parameter realization if there exist a compact parameter set Θ_{II}, a continuous map

$$\Theta_{II} \ni \theta \mapsto V_\theta \in L^3(\mathbb{R}^3),$$

a curve $\theta(\tau) \in \Theta_{II}$, and a remainder $r(\tau) \to 0$ in L^3, such that

$$V(\tau) = V_{\theta(\tau)} + r(\tau),$$

and, for every $T > \tau_0$, the set

$$\{V(\tau) : \tau \in [\tau_0, T]\}$$

is uniformly L^3-tight.

Lemma 7.11 (Compact realization implies tightness)**.** *If Theorem 7.10 holds, then V is uniformly critically tight.*

Proof. The continuous image of compact Θ_{II} in L^3 is compact. Thus

$$\mathcal{K}^{\mathrm{c}} := \{V_\theta : \theta \in \Theta_{II}\}$$

is compact in L^3. The representation $V(\tau) = V_{\theta(\tau)} + r(\tau)$ with $r(\tau) \to 0$ in L^3 gives the compact-core vanishing-remainder situation, because finite-initial-window tightness is included in Theorem 7.10. Theorem 7.4 gives uniform critical tightness. □

Theorem 7.12 (Exclusion of the non-tight alternative)**.** *For a renormalized Type II Navier–Stokes solution, any one of the following conditions gives uniform critical tightness:*

(i) $\mathcal{O} \Subset L^3(\mathbb{R}^3)$*;*
(ii) *compact core with vanishing critical remainder;*
(iii) *no-splitting profile condition;*
(iv) *finite critical profile approximation;*
(v) *compact parameter realization.*

Proof. The five alternatives are handled respectively by Theorems 7.2, 7.4, 7.7, 7.9 and 7.11. In every case V is uniformly critically tight, so the failure of uniform critical tightness is unavailable. □

Corollary 7.13 (After exclusion of the non-tight alternative)**.** *Assume a finite-cost renormalized Type II solution not excluded by the compact cost-divergence criterion has a two-alternative classification whose only remaining alternatives are failure of uniform critical tightness and failure of tail windowed H^1 control. If Theorem 7.1 holds, then the only remaining alternative is failure of tail windowed H^1 control.*

Proof. The two-alternative classification gives exactly one of the two alternatives. Uniform critical tightness rules out failure of tightness, so only the windowed H^1-failure alternative remains. □

7.2. Tail windowed H^1 control.

Lemma 7.14 (Uniform local pressure control)**.** *Assume*

$$M_{3,R} := \sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(B_{4R})} < \infty$$

and

$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathbb{R}^3$$

for each $\tau \geq \tau_0$. Assume also the local tail estimate

$$\mathcal{T}_R(\tau) := \sup_{x \in B_R} \int_{|z| > 2R} \frac{|V(z,\tau)|^2}{|x - z|^5} \, dz, \qquad \sup_{\tau \geq \tau_0} \mathcal{T}_R(\tau) \leq C R^{-4} M_{3,R}^2.$$

Then there is a measurable choice of constants $c_R(\tau) \in \mathbb{R}$ such that

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_{3,R}^2.$$

Proof. The local pressure decomposition gives

$$\|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C \left(\|V(\tau)\|_{L^3(B_{4R})}^2 + R^2 \mathcal{T}_R(\tau) \right).$$

The tail hypothesis gives

$$\sup_{\tau \geq \tau_0} \mathcal{T}_R(\tau) \leq C R^{-4} M_{3,R}^2,$$

and the local critical bound gives

$$\|V(\tau)\|_{L^3(B_{4R})} \leq M_{3,R}.$$

Combining the two estimates yields

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_{3,R}^2.$$

The constants $c_R(\tau)$ can be chosen by fixing the displayed pressure decomposition, hence measurably. □

Proposition 7.15 (Renormalized Caccioppoli gives uniform windowed gradients)**.** *Assume V, P are smooth on compact renormalized cylinders and solve*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Assume local compact-cylinder critical control on the outer cylinders used in the estimate, bounded modulation

$$M_{ab} := \sup_{\tau \geq \tau_0} (|a(\tau)| + |b(\tau)|) < \infty,$$

and pressure normalized as in Theorem 7.14. Then for every $R > 0$ there exists

$$C_R = C(R, \nu, M_{3,R}, M_{ab})$$

such that

$$\sup_{T \geq \tau_0 + 1} \int_T^{T+1} \int_{B_R} |\nabla V(y, \tau)|^2 \, dy \, d\tau \leq C_R.$$

Proof. Fix $R > 0$ and $T \geq \tau_0 + 1$. Use the renormalized Caccioppoli estimate with outer radius $2R$, inner radius R, and time intervals

$$I_{2R} := (T-1, T+2), \qquad I_R := [T, T+1].$$

Thus $R_{\text{out}} - R_{\text{in}} = R$ and the inner time length is 1. Keeping the pressure term before taking absolute values gives

$$\begin{aligned} \nu \int_T^{T+1} \int_{B_R} |\nabla V|^2 \leq C_R \iint_{(T-1,T+2)\times B_{2R}} |V|^2 + C_R \iint_{(T-1,T+2)\times B_{2R}} |V|^3 \\ + C_R \mathcal{P}_{T,R} + C_R M_{ab} \iint_{(T-1,T+2)\times B_{2R}} |V|^2, \end{aligned}$$

where C_R absorbs the powers of R^{-1}, R^{-2}, and ν, and

$$\mathcal{P}_{T,R} := \left| \iint_{(T-1,T+2)\times B_{2R}} P \, V \cdot \nabla\zeta \right|.$$

The L^2-term is controlled by the local critical bound and boundedness of B_{2R}:

$$\int_{B_{2R}} |V(\tau)|^2\,dy \le |B_{2R}|^{1/3} \|V(\tau)\|^2_{L^3(B_{2R})} \le C_R M_{3,R}^2.$$

Since the outer time interval has length 3,

$$\iint_{(T-1,T+2)\times B_{2R}} |V|^2 \le C_R M_{3,R}^2,$$

and similarly

$$\iint_{(T-1,T+2)\times B_{2R}} |V|^3 \le 3M_{3,R}^3.$$

For any time-dependent constant $c(\tau)$,

$$\int_{B_{2R}} c(\tau) V\cdot\nabla\zeta\,dy = -\int_{B_{2R}} c(\tau)\zeta\,\nabla\cdot V\,dy = 0.$$

Thus P may be replaced by $P - c_{2R}(\tau)$. Holder's inequality gives

$$\mathcal{P}_{T,R} \le C_R \iint_{(T-1,T+2)\times B_{2R}} |P - c_{2R}(\tau)|\,|V|\,dy\,d\tau.$$

For almost every τ,

$$\int_{B_{2R}} |P - c_{2R}(\tau)|\,|V(\tau)|\,dy \le \|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})} \|V(\tau)\|_{L^3(B_{2R})}.$$

The pressure estimate with radius $2R$ gives

$$\|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})} \le C_R M_{3,R}^2,$$

and the local critical bound gives

$$\|V(\tau)\|_{L^3(B_{2R})} \le M_{3,R}.$$

Hence $\mathcal{P}_{T,R} \le C_R M_{3,R}^3$. Substituting these estimates into Caccioppoli yields

$$\nu \int_T^{T+1} \int_{B_R} |\nabla V|^2 \le C_R \left(M_{3,R}^2 + M_{3,R}^3 + M_{ab} M_{3,R}^2 \right),$$

uniformly in $T \ge \tau_0 + 1$. Dividing by ν and renaming the constant proves the claim. □

Corollary 7.16 (Tail windowed local H^1 bound)**.** *Under the hypotheses of Theorem 7.15, for every $R > 0$,*

$$\sup_{T\ge\tau_0+1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_R)}\,d\tau < \infty.$$

Proof. The gradient part is Theorem 7.15. The L^2-part follows from the same bounded-set Holder estimate:

$$\int_T^{T+1} \|V(\tau)\|^2_{L^2(B_R)}\,d\tau \le C_R \int_T^{T+1} \|V(\tau)\|^2_{L^3(B_{2R})}\,d\tau \le C_R M_{3,R}^2.$$

Adding the two bounds gives the asserted $H^1(B_R)$ estimate. □

Corollary 7.17 (Exclusion of failure of local windowed H^1 control)**.** *Assume the renormalized Type II solution satisfies the repaired-gauge renormalized Navier–Stokes equation on compact cylinders, local compact-cylinder critical control*

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(B_{2R})}<\infty$$

for each relevant R, bounded modulation

$$\sup_{\tau\geq\tau_0}(|a(\tau)|+|b(\tau)|)<\infty,$$

and pressure reconstruction

$$-\Delta P=\partial_i\partial_j(V_iV_j).$$

Then failure of local windowed H^1 control,

$$\exists m\geq 1:\sup_{T\geq\tau_0+1}\int_T^{T+1}\|V(\tau)\|^2_{H^1(B_m)}\,d\tau=\infty$$

is impossible.

Proof. Apply Theorem 7.16 with $R=m$. The resulting finite bound contradicts the displayed failure of local windowed H^1 control. □

7.3. **Modulation and weighted forcing inputs.**

Lemma 7.18 (Positive finite critical annulus implies bounded critical norm)**.** *If there exist $0<\eta\leq M<\infty$ such that*

$$\eta\leq\|V(\tau)\|_{L^3(\mathbb{R}^3)}\leq M\qquad\forall\tau\geq\tau_0,$$

then

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(\mathbb{R}^3)}\leq M<\infty.$$

Proof. This is obtained by taking the supremum in τ in the displayed upper bound. □

Definition 7.19 (Modulation matrix Schur data)**.** Write the repaired modulation matrix in block form

$$M(V)=\begin{pmatrix}\alpha(V) & u(V)^T\\ v(V) & A(V)\end{pmatrix}.$$

Assume

$$\alpha(V(\tau))\geq\alpha_0>0,\qquad\|A(V(\tau))^{-1}\|_{\infty\to\infty}\leq C_A,$$
$$\|u(V(\tau))\|_\infty\leq C'_u,\qquad\|v(V(\tau))\|_\infty\leq C'_v,$$

and

$$S(V(\tau)):=\alpha(V(\tau))-u(V(\tau))^TA(V(\tau))^{-1}v(V(\tau))\geq\sigma_0>0.$$

Lemma 7.20 (Uniform inverse from Schur data)**.** *Under Theorem 7.19,*

$$\sup_{\tau\geq\tau_0}\|M(V(\tau))^{-1}\|_{\infty\to\infty}\leq\sigma_0^{-1}+3\sigma_0^{-1}C'_uC_A+\sigma_0^{-1}C_AC'_v+C_A+3C_A^2C'_vC'_u\sigma_0^{-1}.$$

Proof. For each τ, write

$$M(V(\tau))=\begin{pmatrix}\alpha & u^T\\ v & A\end{pmatrix},\qquad S=\alpha-u^TA^{-1}v.$$

By the translation-block inverse bound, A^{-1} exists and is uniformly bounded. By the Schur gap, S^{-1} exists and $|S^{-1}|\leq\sigma_0^{-1}$. The block inverse formula gives

$$M^{-1}=\begin{pmatrix}S^{-1} & -S^{-1}u^TA^{-1}\\ -A^{-1}vS^{-1} & A^{-1}+A^{-1}vS^{-1}u^TA^{-1}\end{pmatrix}.$$

Using the ℓ^∞ matrix norm and the finite dimension of the mixed blocks,

$$\|S^{-1}u^T A^{-1}\|_{\infty\to\infty} \le 3\sigma_0^{-1} C_u' C_A,$$

$$\|A^{-1}vS^{-1}\|_{\infty\to\infty} \le \sigma_0^{-1} C_A C_v',$$

and

$$\|A^{-1}vS^{-1}u^T A^{-1}\|_{\infty\to\infty} \le 3C_A^2 C_v' C_u' \sigma_0^{-1}.$$

Adding the four block estimates gives the displayed uniform bound. □

Lemma 7.21 (Scale-normalization nondegeneracy gives Schur data)**.** *Assume the scale normalization constraint is satisfied, the scale derivative obeys*

$$\alpha(V) = DG_{\rm sc}(V)[Z_{\rm sc}(V)] = p\Theta_0 > 0,$$

the translation block has a uniform inverse bound, and the mixed blocks obey the weighted first-moment bound and the Schur-compatible estimate

$$\alpha(V(\tau)) - u(V(\tau))^T A(V(\tau))^{-1} v(V(\tau)) \ge \frac{\alpha_0}{2}.$$

Then the hypotheses of Theorem 7.19 hold with $\sigma_0 = \alpha_0/2$.

Proof. The scale-normalization identity gives $\alpha(V) = DG_{\rm sc}(V)[Z_{\rm sc}(V)] = p\Theta_0$, hence the scale transversality bound. The translation-block inverse hypothesis gives the uniform bound on A^{-1}. The mixed-block hypotheses give the bounds on u and v, and the displayed estimate is the Schur gap with $\sigma_0 = \alpha_0/2$. □

Definition 7.22 (Weighted scale-force components)**.** Let $0 < p < 3$ and set

$$H(V,P) = \nu\Delta V - (V\cdot\nabla)V - \nabla P.$$

Define

$$I_\Delta(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot \Delta V\,dy,$$

$$I_{\rm nl}(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot((V\cdot\nabla)V)\,dy,$$

and

$$I_P(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot\nabla P\,dy,$$

whenever these integrals are absolutely convergent or are defined by the distributional pairing used for the renormalized solution.

Lemma 7.23 (Scale-force decomposition)**.** *If*

$$\sup_\tau |I_\Delta(\tau)| \le C_\Delta, \qquad \sup_\tau |I_{\rm nl}(\tau)| \le C_{\rm nl}, \qquad \sup_\tau |I_P(\tau)| \le C_P,$$

then

$$\sup_{\tau\ge\tau_0}\left|\int |y|^{-p}|V|V\cdot H(V,P)\,dy\right| \le \nu C_\Delta + C_{\rm nl} + C_P.$$

Proof. By the definition of H,

$$\int |y|^{-p}|V|V\cdot H(V,P)\,dy = \nu I_\Delta - I_{\rm nl} - I_P.$$

The triangle inequality gives the displayed uniform bound. □

Lemma 7.24 (Weighted fourth moment controls the nonlinear scale component)**.** *Assume $V(\tau)$ is divergence-free and belongs to the approximation class needed to justify the weighted integration by parts. If*

$$\sup_{\tau\geq\tau_0}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty,$$

then

$$|I_{\mathrm{nl}}(\tau)|\leq\frac{p}{3}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy.$$

Proof. For smooth compactly supported approximants,

$$|V|V\cdot((V\cdot\nabla)V)=\frac{1}{3}V\cdot\nabla(|V|^3).$$

Since $\nabla\cdot V=0$, integration by parts gives

$$I_{\mathrm{nl}}=\frac{1}{3}\int|y|^{-p}V\cdot\nabla(|V|^3)\,dy=-\frac{1}{3}\int|V|^3V\cdot\nabla(|y|^{-p})\,dy.$$

Because

$$\nabla(|y|^{-p})=-p|y|^{-p-2}y,$$

we obtain

$$I_{\mathrm{nl}}=\frac{p}{3}\int|y|^{-p-2}(V\cdot y)|V|^3\,dy.$$

Therefore

$$|I_{\mathrm{nl}}|\leq\frac{p}{3}\int|y|^{-p-1}|V|^4\,dy.$$

Taking the supremum in τ proves the assertion. The general admissible case follows by the same approximation convention used for the weighted scale normalization. □

Corollary 7.25 (Pointwise scale-force bound)**.** *If*

$$\sup_{\tau}|I_\Delta(\tau)|<\infty,\qquad\sup_{\tau\geq\tau_0}\int|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty,\qquad\sup_{\tau}|I_P(\tau)|<\infty,$$

then

$$\sup_{\tau\geq\tau_0}\left|\int|y|^{-p}|V|V\cdot H(V,P)\,dy\right|<\infty.$$

Proof. Theorem 7.24 bounds I_{nl}. Together with the Laplacian and pressure bounds, this gives the hypotheses of Theorem 7.23. □

Lemma 7.26 (Bounded modulation from the modulation system)**.** *Assume*

$$M(V(\tau))\begin{pmatrix}a(\tau)\\ b(\tau)\end{pmatrix}=-F(V(\tau)),$$

$$\sup_{\tau\geq\tau_0}\|M(V(\tau))^{-1}\|_{\infty\to\infty}<\infty,\qquad\sup_{\tau\geq\tau_0}\|F(V(\tau))\|_\infty<\infty.$$

Then

$$\sup_{\tau\geq\tau_0}(|a(\tau)|+\|b(\tau)\|_\infty)<\infty.$$

Proof. The system gives

$$\begin{pmatrix}a(\tau)\\ b(\tau)\end{pmatrix}=-M(V(\tau))^{-1}F(V(\tau)),$$

so

$$\max\{|a(\tau)|,\|b(\tau)\|_\infty\}\leq\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty.$$

Taking the supremum in τ gives bounded modulation. □

Proposition 7.27 (Windowed H^1 exclusion from modulation bounds)**.** *Assume a renormalized Type II solution has the repaired-gauge orbit and pressure reconstruction, bounded critical L^3 norm, uniformly invertible modulation matrix, uniformly bounded modulation forcing vector, and the compact-cylinder regularity needed for Caccioppoli. Then*

$$\forall R > 0 : \qquad \sup_{T \geq \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|_{H^1(B_R)}^2 \, d\tau < \infty,$$

and failure of local windowed H^1 control is excluded for this solution.

Proof. The inverse matrix and forcing bounds imply bounded modulation by Theorem 7.26. The renormalized solution supplies the renormalized equation and pressure reconstruction. Bounded critical norm gives the local compact-cylinder critical bounds, and the regularity hypothesis allows the Caccioppoli estimate. Therefore Theorem 7.16 applies. Failure of local windowed H^1 control is the negation of the displayed tail windowed H^1 estimate on some bounded ball. $\square$

Corollary 7.28 (Two-alternative exclusion after windowed control)**.** *Assume a finite-cost renormalized Type II solution not excluded by the compact cost-divergence criterion has reached the two-alternative classification. If the hypotheses of Theorem 7.27 are supplied for that solution, then failure of local windowed H^1 control is excluded. If, in addition, uniform critical tightness is available, then no finite-cost alternative remains outside the compact cost-divergence criterion.*

Proof. Theorem 7.27 gives the positive tail windowed H^1 bound, contradicting failure of local windowed H^1 control. If uniform critical tightness is also available, then failure of tightness is excluded as well, and the two-alternative classification has no remaining finite-cost alternative outside the compact cost-divergence criterion. $\square$

Theorem 7.29 (Exclusion of local windowed H^1 failure)**.** *If every renormalized Type II solution in the class satisfies the hypotheses of Theorem 7.27, then no such solution can satisfy*

$$\exists m \geq 1 : \sup_{T \geq \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|_{H^1(B_m)}^2 \, d\tau = \infty.$$

Proof. Apply Theorem 7.27 to each renormalized solution. The displayed condition is the negation of the resulting windowed H^1-bound for $R = m$. $\square$

7.4. **Window-Integrated Modulation Forcing and Weighted Scale Components.** This subsection records sufficient analytic estimates for the modulation forcing. The estimates are stated directly for the differentiated modulation conditions and for the weighted scale component; no pointwise modulation bound is used unless it is explicitly assumed.

Let the modulation forcing operator be

$$H(V, P) := \nu \Delta V - (V \cdot \nabla) V - \nabla P.$$

For gauge functionals G_k, $0 \leq k \leq 3$, define

$$F_k(V) := DG_k(V)[H(V, P)].$$

Thus the modulation system has the form

$$M(V(\tau)) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -F(V(\tau)).$$

Lemma 7.30 (Forcing bound for the translation constraints)**.** *Let*

$$G_j(V) = \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y) \, dy, \qquad 1 \leq j \leq 3,$$

where $\psi_R \in C_c^\infty(B_{2R})$, and set

$$\Phi_j(y) := y_j \psi_R(y).$$

For

$$F_j(V) := DG_j(V)[H(V,P)] = 2\int_{\mathbb{R}^3} \Phi_j\, V \cdot H(V,P)\, dy,$$

assume, uniformly for $\tau \geq \tau_0$,

$$\|V(\tau)\|_{L^3(B_{2R})} \leq M_3, \qquad \|\nabla V(\tau)\|_{L^2(B_{2R})} \leq M_\nabla,$$

and

$$\inf_{c\in\mathbb{R}} \|P(\tau) - c\|_{L^{3/2}(B_{2R})} \leq M_P.$$

Then

$$\sup_{\tau\geq\tau_0} \max_{1\leq j\leq 3} |F_j(V(\tau))| \leq C(R,\nu,\psi_R)\left(M_\nabla^2 + M_3^2 + M_3^3 + M_P M_3\right).$$

Proof. Fix j. Since Φ_j is supported in B_{2R},

$$F_j = 2\nu I_\Delta - 2I_{\mathrm{nl}} - 2I_P,$$

where

$$I_\Delta := \int \Phi_j V \cdot \Delta V, \qquad I_{\mathrm{nl}} := \int \Phi_j V \cdot (V\cdot\nabla V), \qquad I_P := \int \Phi_j V \cdot \nabla P.$$

For the Laplacian term, using compact support of Φ_j and summing over the component index α,

$$\begin{aligned}
I_\Delta &= \sum_{\alpha=1}^{3} \int \Phi_j V_\alpha \Delta V_\alpha\, dy \\
&= -\sum_{\alpha,k} \int \partial_k(\Phi_j V_\alpha)\partial_k V_\alpha\, dy \\
&= -\int \Phi_j |\nabla V|^2\, dy - \sum_{\alpha,k}\int (\partial_k\Phi_j) V_\alpha \partial_k V_\alpha\, dy.
\end{aligned}$$

Equivalently,

$$I_\Delta = -\int \Phi_j|\nabla V|^2 - \int (\nabla\Phi_j \cdot \nabla V)\cdot V.$$

Consequently, by the Cauchy–Schwarz inequality,

$$|I_\Delta| \leq \|\Phi_j\|_{L^\infty}\|\nabla V\|^2_{L^2(B_{2R})} + \|\nabla\Phi_j\|_{L^\infty}\|V\|_{L^2(B_{2R})}\|\nabla V\|_{L^2(B_{2R})}.$$

Hölder's inequality on the bounded set B_{2R} gives

$$\|V\|_{L^2(B_{2R})} \leq |B_{2R}|^{1/6}\|V\|_{L^3(B_{2R})} \leq C_R M_3.$$

Therefore

$$|I_\Delta| \leq C_R(M_\nabla^2 + M_3 M_\nabla) \leq C_R(M_\nabla^2 + M_3^2).$$

The last inequality is Young's inequality, $ab \leq (a^2+b^2)/2$.

For the nonlinear term,

$$V\cdot(V\cdot\nabla V) = V_i V_\alpha \partial_i V_\alpha = \frac{1}{2} V_i \partial_i(|V|^2) = \frac{1}{2} V\cdot\nabla(|V|^2).$$

Since $\nabla\cdot V = 0$ and Φ_j is compactly supported,

$$I_{\mathrm{nl}} = \frac{1}{2}\int \Phi_j V\cdot\nabla(|V|^2) = -\frac{1}{2}\int |V|^2 V\cdot\nabla\Phi_j.$$

Thus

$$|I_{\rm nl}| \le \frac{1}{2}\|\nabla\Phi_j\|_{L^\infty}\|V\|^3_{L^3(B_{2R})} \le C_R M_3^3.$$

For the pressure term, replace P by $P - c(\tau)$. Since $\nabla\cdot V = 0$, the identity

$$\nabla\cdot(\Phi_j V) = V\cdot\nabla\Phi_j$$

holds distributionally. Therefore

$$I_P = \int \Phi_j V\cdot\nabla(P-c) = -\int (P-c)V\cdot\nabla\Phi_j.$$

Consequently

$$|I_P| \le \|\nabla\Phi_j\|_{L^\infty}\|P-c\|_{L^{3/2}(B_{2R})}\|V\|_{L^3(B_{2R})} \le C_R M_P M_3.$$

Taking the infimum over c, summing the three estimates in F_j, and then taking the supremum in τ proves the bound. □

Remark 7.31 (Independence of the local gradient hypothesis). The pointwise local gradient bound in Theorem 7.30 is an additional analytic input. When the translation estimate is used to prove local windowed H^1-control, this input is required to come from an independent Caccioppoli, regularity, or approximation estimate, not from the conclusion being proved.

Definition 7.32 (Weighted scale forcing)**.** Let

$$G_{\rm sc}(V) = \int_{\mathbb{R}^3} |y|^{-p}|V|^3\,dy - \Theta_0, \qquad 0 < p < 3.$$

The weighted scale component is

$$F_0(V) := DG_{\rm sc}(V)[H(V,P)] = 3\int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot H(V,P)\,dy.$$

All weighted pairings in this subsection are assumed to be defined either as absolutely convergent integrals or as specified distributional pairings represented by $L^1_{\rm loc}(d\tau)$ functions.

Lemma 7.33 (Weighted scale forcing bound)**.** *Assume*

$$\sup_{\tau\ge\tau_0}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right| < \infty.$$

Then

$$\sup_{\tau\ge\tau_0}|F_0(V(\tau))| < \infty.$$

Proof. By the derivative formula in Theorem 7.32,

$$|F_0(V(\tau))| \le 3\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|.$$

Taking the supremum in τ gives the result. □

Corollary 7.34 (Pointwise modulation forcing bound)**.** *Assume the hypotheses of Theorem 7.30 for the three translation components and the weighted scale forcing hypothesis of Theorem 7.33. Then*

$$\sup_{\tau\ge\tau_0}\|F(V(\tau))\|_\infty < \infty.$$

Proof. The vector $F(V)\in\mathbb{R}^4$ has the scale component F_0 and the three translation components F_j, $1\le j\le 3$. The scale component is bounded by Theorem 7.33, and the translation components are bounded by Theorem 7.30. Taking the maximum over the four components proves the ℓ^∞-bound. □

Corollary 7.35 (Analytic inputs for pointwise modulation forcing)**.** *Pointwise boundedness of the modulation forcing follows from the following analytic estimates:*

(i) *local L^3, local H^1, and local pressure-oscillation control on the support of the compactly supported centering constraints;*
(ii) *weighted control of the Laplacian, pressure, and fourth-moment terms in the scale component, sufficient to bound the weighted scale forcing.*

The scale-component estimate is independent of the algebraic scale transversality identity

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0.$$

Proof. The first item is the set of hypotheses used in Theorem 7.30. The second item bounds the weighted scale component through Theorem 7.33, or through the explicit decomposition in Theorem 7.52 below in the corresponding integrated setting. Theorem 7.34 then gives the modulation forcing bound. The displayed transversality identity controls the modulation matrix, not the forcing vector, so it is a separate algebraic input. □

Lemma 7.36 (Window-integrated forcing gives window-integrated modulation)**.** *Assume*

$$\sup_{\tau\ge\tau_0} \|M(V(\tau))^{-1}\|_{\infty\to\infty} \le C_M < \infty$$

and

$$\sup_{T\ge\tau_0} \int_T^{T+1} \|F(V(\tau))\|_\infty \, d\tau \le C_F < \infty.$$

Then

$$\sup_{T\ge\tau_0} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big)\, d\tau \le 2C_M C_F.$$

Proof. For almost every τ,

$$\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -M(V(\tau))^{-1}F(V(\tau)).$$

Hence

$$\max\{|a(\tau)|, \|b(\tau)\|_\infty\} \le C_M \|F(V(\tau))\|_\infty.$$

Since

$$|a(\tau)| + \|b(\tau)\|_\infty \le 2\max\{|a(\tau)|, \|b(\tau)\|_\infty\},$$

we have, for every $T \ge \tau_0$,

$$\begin{aligned} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big)\, d\tau &\le 2C_M \int_T^{T+1} \|F(V(\tau))\|_\infty \, d\tau \\ &\le 2C_M C_F. \end{aligned}$$

Taking the supremum over T proves the claim. □

Lemma 7.37 (Selection from an L^1 modulation bound)**.** *Let $I_n = [T_n, T_n + 1]$, where $T_n \to \infty$. Let*

$$d_n : I_n \to [0,\infty), \qquad q_n : I_n \to [0,\infty)$$

be measurable functions satisfying

$$\int_{I_n} d_n(\tau)\, d\tau \to 0, \qquad \sup_n \int_{I_n} q_n(\tau)\, d\tau \le Q < \infty.$$

Then, after choosing measurable representatives, there are times $\tau_n \in I_n$ such that

$$d_n(\tau_n) \to 0$$

and, if $Q > 0$,

$$q_n(\tau_n) \le 2Q \qquad \text{for every } n.$$

If $Q = 0$, the times may be chosen so that

$$q_n(\tau_n) = 0, \qquad d_n(\tau_n) \to 0.$$

Proof. Assume first that $Q > 0$, and set

$$E_n := \{\tau \in I_n : q_n(\tau) \le 2Q\}.$$

By Chebyshev's inequality,

$$|I_n \setminus E_n| \le \frac{1}{2Q} \int_{I_n} q_n(\tau)\, d\tau \le \frac{1}{2}.$$

Thus $|E_n| \ge 1/2$. Since $d_n \ge 0$,

$$\inf_{\tau \in E_n} d_n(\tau) \le \frac{1}{|E_n|} \int_{E_n} d_n(\tau)\, d\tau \le 2 \int_{I_n} d_n(\tau)\, d\tau.$$

Choose $\tau_n \in E_n$, outside the null set on which the representatives are undefined, such that

$$d_n(\tau_n) \le 2 \int_{I_n} d_n(\tau)\, d\tau + \frac{1}{n}.$$

The right-hand side tends to zero, hence $d_n(\tau_n) \to 0$, and the definition of E_n gives $q_n(\tau_n) \le 2Q$.

If $Q = 0$, then $q_n = 0$ almost everywhere on I_n. Applying the same averaging argument to d_n on the full-measure set where $q_n = 0$ gives the asserted times. □

Corollary 7.38 (Selection on unit windows)**.** *Assume*

$$q(\tau) := |a(\tau)| + \|b(\tau)\|_\infty$$

satisfies

$$\sup_{T \ge \tau_0} \int_T^{T+1} q(\tau)\, d\tau < \infty.$$

For every sequence of unit windows $I_n = [T_n, T_n + 1]$ and every nonnegative measurable density d_n satisfying

$$\int_{I_n} d_n(\tau)\, d\tau \to 0,$$

there are times $\tau_n \in I_n$ such that

$$d_n(\tau_n) \to 0, \qquad \sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty.$$

Proof. Apply Theorem 7.37 with $q_n = q|_{I_n}$ and

$$Q = \sup_n \int_{I_n} q(\tau)\, d\tau.$$

This Q is finite because of the uniform window L^1-bound. If $Q > 0$, the selected times satisfy

$$|a(\tau_n)| + \|b(\tau_n)\|_\infty = q(\tau_n) \le 2Q.$$

If $Q = 0$, the same lemma gives $q(\tau_n) = 0$. In both cases $d_n(\tau_n) \to 0$ and the modulation is uniformly bounded along the selected times. □

Lemma 7.39 (Product-form windowed modulation estimate)**.** *Assume*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty\,d\tau<\infty.$$

Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\big(|a(\tau)|+\|b(\tau)\|_\infty\big)\,d\tau<\infty.$$

Proof. The modulation system gives

$$\begin{pmatrix}a(\tau)\\ b(\tau)\end{pmatrix}=-M(V(\tau))^{-1}F(V(\tau)).$$

Therefore

$$|a(\tau)|+\|b(\tau)\|_\infty\leq 2\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty.$$

Integrating this pointwise estimate on $[T,T+1]$ and taking the supremum in T gives the result. □

Lemma 7.40 (Integrated translation forcing bound)**.** *Let G_j, Φ_j, $H(V,P)$, and $F_j(V)$ be as in Theorem 7.30. Assume that, for some $R>0$,*

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(B_{2R})}\leq M_3,$$

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|\nabla V(\tau)\|^2_{L^2(B_{2R})}\,d\tau\leq A_R,$$

and

$$\sup_{\tau\geq\tau_0}\inf_{c\in\mathbb{R}}\|P(\tau)-c\|_{L^{3/2}(B_{2R})}\leq M_P.$$

Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau\leq C(R,\nu,\psi_R)(A_R+M_3^2+M_3^3+M_PM_3).$$

Proof. Fix j. The proof of Theorem 7.30 gives, for almost every τ,

$$|F_j(V(\tau))|\leq C(R,\nu,\psi_R)\left(\|\nabla V(\tau)\|^2_{L^2(B_{2R})}+M_3\|\nabla V(\tau)\|_{L^2(B_{2R})}+M_3^3+M_PM_3\right).$$

Integrate this estimate over $I_T=[T,T+1]$. The first term contributes at most A_R. For the mixed term, Cauchy's inequality on the unit interval gives

$$\int_{I_T}M_3\|\nabla V(\tau)\|_{L^2(B_{2R})}\,d\tau\leq M_3\left(\int_{I_T}\|\nabla V(\tau)\|^2_{L^2(B_{2R})}\,d\tau\right)^{1/2}\leq M_3A_R^{1/2}.$$

Since

$$M_3A_R^{1/2}\leq\frac{1}{2}(M_3^2+A_R),$$

the mixed term is bounded by $C(A_R+M_3^2)$. The terms M_3^3 and M_PM_3 are constant on the unit window. Hence

$$\int_T^{T+1}|F_j(V(\tau))|\,d\tau\leq C(R,\nu,\psi_R)(A_R+M_3^2+M_3^3+M_PM_3).$$

Since $\max_{1\leq j\leq 3}|F_j|\leq\sum_{j=1}^3|F_j|$, taking the maximum over the three centering constraints changes only the constant. □

Remark 7.41 (Independence of the windowed gradient hypothesis). The windowed gradient input in Theorem 7.40 is an independent hypothesis whenever the integrated translation estimate is used to prove windowed H^1-control. It may come, for example, from a preliminary Caccioppoli estimate depending only on already available data, from a direct approximation or regularity estimate on the support of the centering functions, or from an independent verification obtained without using the conclusion of the windowed H^1 estimate under consideration.

Lemma 7.42 (Window-integrated scale forcing)**.** *Assume the window-integrated weighted scale-forcing bound*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|d\tau<\infty,$$

where the weighted pairing is interpreted as in Theorem 7.32. Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}|F_0(V(\tau))|\,d\tau<\infty.$$

Proof. By the identity

$$F_0(V)=3\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy$$

we have, for each $T\geq\tau_0$,

$$\int_T^{T+1}|F_0(V(\tau))|\,d\tau=3\int_T^{T+1}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|d\tau.$$

Taking the supremum over T gives the asserted L^1-bound. □

Lemma 7.43 (Full integrated modulation forcing)**.** *Assume the integrated translation forcing estimate*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau<\infty$$

and the window-integrated weighted scale-forcing bound in Theorem 7.42. Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|F(V(\tau))\|_\infty\,d\tau<\infty.$$

Proof. The vector $F(V)\in\mathbb{R}^4$ consists of F_0 and the three translation components F_j, $1\leq j\leq 3$. Since

$$\|F(V(\tau))\|_\infty\leq|F_0(V(\tau))|+\max_{1\leq j\leq 3}|F_j(V(\tau))|,$$

for almost every τ. Integrating this inequality on $[T,T+1]$ gives

$$\int_T^{T+1}\|F(V(\tau))\|_\infty\,d\tau\leq\int_T^{T+1}|F_0(V(\tau))|\,d\tau+\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau.$$

The first term is uniformly bounded in T by Theorem 7.42, and the second term is uniformly bounded by the assumed integrated translation forcing estimate. Taking the supremum over T proves the claim. □

Corollary 7.44 (Integrated modulation criterion on selected windows)**.** *Assume the uniform inverse bound for $M(V(\tau))$, the integrated translation forcing estimate, and the window-integrated weighted scale-forcing bound. Then*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\big(|a(\tau)|+\|b(\tau)\|_\infty\big)\,d\tau<\infty.$$

Consequently every sequence of unit windows with vanishing nonnegative measurable density contains selected times where this density vanishes asymptotically and the modulation remains uniformly bounded.

Proof. Theorem 7.43 gives the window-integrated bound on $\|F(V)\|_\infty$. Combining this with the uniform inverse bound in Theorem 7.36 gives the window-integrated modulation bound. The selection conclusion is Theorem 7.38. □

Definition 7.45 (Weighted scale-force components)**.** Let $w(y) = |y|^{-p}$, $0 < p < 3$. Define

$$F_0^\Delta(V) := 3\nu \int_{\mathbb{R}^3} w|V|V \cdot \Delta V \, dy,$$

$$F_0^{\mathrm{nl}}(V) := -3 \int_{\mathbb{R}^3} w|V|V \cdot (V \cdot \nabla V) \, dy,$$

and

$$F_0^P(V, P) := -3 \int_{\mathbb{R}^3} w|V|V \cdot \nabla P \, dy.$$

The three pairings are required to be compatible with the decomposition of $H(V, P)$ into diffusion, transport, and pressure.

Lemma 7.46 (Integrated scale-force decomposition)**.** *For every time at which the three pairings in Theorem 7.45 are defined,*

$$F_0(V(\tau)) = F_0^\Delta(V(\tau)) + F_0^{\mathrm{nl}}(V(\tau)) + F_0^P(V(\tau), P(\tau)).$$

Proof. Substitute

$$H(V, P) = \nu\Delta V - (V \cdot \nabla)V - \nabla P$$

into

$$F_0(V) = 3 \int w|V|V \cdot H(V, P) \, dy.$$

Then

$$\begin{aligned} F_0(V) = 3\nu \int w|V|V \cdot \Delta V \, dy - 3 \int w|V|V \cdot (V \cdot \nabla V) \, dy \\ - 3 \int w|V|V \cdot \nabla P \, dy. \end{aligned}$$

The three terms coincide with $F_0^\Delta(V)$, $F_0^{\mathrm{nl}}(V)$, and $F_0^P(V, P)$, respectively. Thus linearity of the weighted pairing gives the stated decomposition. □

Lemma 7.47 (Component bounds imply integrated scale-force control)**.** *Assume*

$$\sup_{T \ge \tau_0} \int_T^{T+1} |F_0^\Delta(V(\tau))| \, d\tau < \infty,$$

$$\sup_{T \ge \tau_0} \int_T^{T+1} |F_0^{\mathrm{nl}}(V(\tau))| \, d\tau < \infty,$$

and

$$\sup_{T \ge \tau_0} \int_T^{T+1} |F_0^P(V(\tau), P(\tau))| \, d\tau < \infty.$$

Then the window-integrated weighted scale-forcing bound in Theorem 7.42 holds.

Proof. By Theorem 7.46,

$$|F_0(V(\tau))| \le |F_0^{\Delta}(V(\tau))| + |F_0^{\mathrm{nl}}(V(\tau))| + |F_0^{P}(V(\tau), P(\tau))|$$

for almost every τ. Integrating on $[T, T+1]$, taking the supremum in T, and applying the three component bounds gives

$$\sup_{T\ge\tau_0} \int_T^{T+1} |F_0(V(\tau))| \, d\tau < \infty.$$

Since F_0 is three times the weighted pairing in Theorem 7.32, this is equivalent to the window-integrated weighted scale-forcing bound. □

Definition 7.48 (Singular-weight convective integration by parts)**.** The singular-weight convective integration-by-parts condition means that, for almost every τ,

$$-\int_{\mathbb{R}^3} w\, V\cdot\nabla(|V|^3)\, dy = \int_{\mathbb{R}^3} |V|^3 V\cdot\nabla w\, dy,$$

with no boundary contribution from $y = 0$ or from spatial infinity, either classically or by convergence of smooth compactly supported divergence-free approximants.

Lemma 7.49 (Convective scale-force bound)**.** *Assume the condition of Theorem 7.48 and*

$$\sup_{T\ge\tau_0} \int_T^{T+1} \int_{\mathbb{R}^3} |y|^{-p-1} |V(y,\tau)|^4 \, dy\, d\tau < \infty.$$

Then

$$\sup_{T\ge\tau_0} \int_T^{T+1} |F_0^{\mathrm{nl}}(V(\tau))| \, d\tau < \infty.$$

More precisely,

$$|F_0^{\mathrm{nl}}(V(\tau))| \le p \int_{\mathbb{R}^3} |y|^{-p-1} |V(y,\tau)|^4 \, dy$$

for almost every τ.

Proof. Fix such a time τ and suppress it from the notation. Since

$$\partial_i(|V|^3) = 3|V| V_j \partial_i V_j$$

for each i, and therefore

$$V\cdot\nabla(|V|^3) = 3|V| V_i V_j \partial_i V_j = 3|V|\, V\cdot((V\cdot\nabla)V),$$

we have

$$w|V| V\cdot(V\cdot\nabla V) = \frac{1}{3} w\, V\cdot\nabla(|V|^3).$$

Therefore

$$F_0^{\mathrm{nl}}(V) = -3\int w|V|V\cdot(V\cdot\nabla V) = -\int w\, V\cdot\nabla(|V|^3).$$

Using $\nabla\cdot V = 0$ and Theorem 7.48,

$$-\int w\, V\cdot\nabla(|V|^3) = \int |V|^3 V\cdot\nabla w.$$

Since

$$\nabla w(y) = -p|y|^{-p-2} y, \qquad y \ne 0,$$

we have

$$|V\cdot\nabla w| = p|y|^{-p-2}|V\cdot y| \le p|y|^{-p-1}|V|.$$

Thus

$$|F_0^{\mathrm{nl}}(V)| \le p\int |y|^{-p-1}|V|^4\, dy.$$

Integrating over $[T,T+1]$ and taking the supremum in T proves the window-integrated convective bound. □

Definition 7.50 (Annular convective regularity)**.** Annular convective regularity means that, for almost every τ, $V(\tau)$ is divergence free and, on every compact annulus $0<r<|y|<R<\infty$, the chain-rule identities

$$\partial_i(|V|^3)=3|V|V_j\partial_iV_j, \qquad V\cdot\nabla(|V|^3)=3|V|V_iV_j\partial_iV_j$$

hold in the sense of distributions, either classically or by passage to the limit from smooth divergence-free approximants on annuli.

Lemma 7.51 (Annular regularity gives singular-weight integration by parts)**.** *Assume annular convective regularity and the weighted L^4-window bound*

$$\sup_{T\ge\tau_0}\int_T^{T+1}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy\,d\tau<\infty.$$

Then the singular-weight convective integration-by-parts condition of Theorem 7.48 holds for almost every τ.

Proof. Fix a time τ for which

$$\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty$$

and annular convective regularity holds. Such times form a full-measure set on every unit window.

Let $\eta\in C^\infty([0,\infty))$ satisfy $0\le\eta\le1$, $\eta(s)=0$ for $s\le1$, and $\eta(s)=1$ for $s\ge2$. Define

$$\chi_{\varepsilon,R}(y):=\eta(|y|/\varepsilon)\eta(R/|y|).$$

Then $\chi_{\varepsilon,R}$ is supported in $\{\varepsilon<|y|<R\}$, equals one on $\{2\varepsilon<|y|<R/2\}$, and satisfies

$$|\nabla\chi_{\varepsilon,R}(y)|\le C\varepsilon^{-1}\mathbf{1}_{\{\varepsilon<|y|<2\varepsilon\}}+CR^{-1}\mathbf{1}_{\{R/2<|y|<R\}}.$$

By annular convective regularity, the following integration by parts is valid on the annulus $\{\varepsilon<|y|<R\}$. Since $\nabla\cdot V=0$ and $w\chi_{\varepsilon,R}|V|^3V$ is compactly supported in this annulus,

$$0=\int\nabla\cdot\big(w\chi_{\varepsilon,R}|V|^3V\big)\,dy.$$

Expanding the divergence gives

$$0=\int|V|^3V\cdot\nabla(w\chi_{\varepsilon,R})\,dy+\int w\chi_{\varepsilon,R}V\cdot\nabla(|V|^3)\,dy,$$

and hence

$$-\int w\chi_{\varepsilon,R}V\cdot\nabla(|V|^3)=\int|V|^3V\cdot\nabla(w\chi_{\varepsilon,R}).$$

Expanding the derivative,

$$-\int w\chi_{\varepsilon,R}V\cdot\nabla(|V|^3)=\int\chi_{\varepsilon,R}|V|^3V\cdot\nabla w+\int w|V|^3V\cdot\nabla\chi_{\varepsilon,R}.$$

On the inner cutoff region $\{\varepsilon<|y|<2\varepsilon\}$, one has $|y|\simeq\varepsilon$, and on the outer cutoff region $\{R/2<|y|<R\}$, one has $|y|\simeq R$. Therefore the derivative bound for $\chi_{\varepsilon,R}$ implies

$$\begin{aligned}\left|\int w|V|^3V\cdot\nabla\chi_{\varepsilon,R}\right|&\le C\int_{\{\varepsilon<|y|<2\varepsilon\}}|y|^{-p-1}|V|^4\,dy\\&\quad+C\int_{\{R/2<|y|<R\}}|y|^{-p-1}|V|^4\,dy.\end{aligned}$$

The integrand $|y|^{-p-1}|V|^4$ belongs to $L^1(\mathbb{R}^3)$ at the fixed time. Hence absolute continuity of the Lebesgue integral gives

$$\int_{\{\varepsilon<|y|<2\varepsilon\}} |y|^{-p-1}|V|^4\,dy \to 0 \qquad \text{as } \varepsilon \downarrow 0,$$

and integrability at infinity gives

$$\int_{\{R/2<|y|<R\}} |y|^{-p-1}|V|^4\,dy \to 0 \qquad \text{as } R \uparrow \infty.$$

Moreover

$$|\chi_{\varepsilon,R}|V|^3V\cdot\nabla w| \le p|y|^{-p-1}|V|^4 \in L^1(\mathbb{R}^3),$$

so dominated convergence gives

$$\int \chi_{\varepsilon,R}|V|^3V\cdot\nabla w \to \int |V|^3V\cdot\nabla w.$$

Passing first $\varepsilon \downarrow 0$ and then $R \uparrow \infty$ yields

$$-\int_{\mathbb{R}^3} w\,V\cdot\nabla(|V|^3)\,dy = \int_{\mathbb{R}^3} |V|^3V\cdot\nabla w\,dy,$$

with no boundary contribution at $y=0$ or at infinity. □

Corollary 7.52 (Reduced integrated scale-force criterion)**.** *Assume annular convective regularity, the weighted L^4-window bound, and the two component estimates*

$$\sup_{T\ge\tau_0}\int_T^{T+1} |F_0^{\Delta}(V(\tau))|\,d\tau < \infty,$$

$$\sup_{T\ge\tau_0}\int_T^{T+1} |F_0^{P}(V(\tau),P(\tau))|\,d\tau < \infty.$$

Then the window-integrated weighted scale-forcing bound holds.

Proof. Theorem 7.51 gives the singular-weight integration-by-parts condition. Theorem 7.49 then gives the integrated convective scale-force bound. Combining this with the diffusive and pressure component estimates, and applying Theorem 7.47, gives the window-integrated weighted scale-forcing bound. □

Corollary 7.53 (Remaining analytic alternatives for the scale component)**.** *Before the singular-weight integration by parts is justified, failure of the window-integrated weighted scale-forcing bound is reduced to failure of at least one of the following four analytic statements:*

(i) *the integrated diffusive weighted scale-force bound;*
(ii) *the singular-weight convective integration-by-parts identity;*
(iii) *the weighted L^4-window bound;*
(iv) *the integrated weighted pressure scale-force bound.*

If annular convective regularity is available, then the second item is replaced by failure of annular convective regularity itself, together with the same weighted L^4-window input.

Proof. Theorem 7.47 shows that the diffusive, convective, and pressure component bounds imply the integrated scale-force bound. Theorem 7.49 reduces the convective component to the singular-weight integration-by-parts identity and the weighted L^4-window bound. Finally, Theorem 7.51 shows that annular convective regularity and the weighted L^4-window bound imply the singular-weight integration-by-parts identity. Therefore the listed failures exhaust the remaining ways in which this scale-component estimate can fail. □

7.5. **Pointwise Type II Exclusion.**

Definition 7.54 (Pointwise exclusion estimates)**.** For a renormalized Type II solution ω, the non-tight alternative is excluded when V_ω satisfies Theorem 7.1. The windowed-control alternative is excluded when

$$\forall R > 0 : \qquad \sup_{T \geq \tau_0 + 1} \int_T^{T+1} \|V_\omega(\tau)\|^2_{H^1(B_R)} \, d\tau < \infty.$$

The universal versions require the corresponding pointwise statement for every renormalized solution in the class.

Definition 7.55 (Remaining Type II case after local estimates)**.** A renormalized Type II solution remains after the local estimates if it reaches the local two-alternative classification, is not already excluded by the compact cost-divergence criterion, and is not excluded by the Navier–Stokes cost-divergence implication in Theorem 7.58.

Theorem 7.56 (Type II exclusion after tightness and windowed control)**.** *Assume the renormalized Type II class satisfies:*

(i) *the local classification has exactly the following alternatives: the compact cost-divergence criterion, failure of uniform critical tightness, and failure of local windowed H^1 control;*
(ii) *every renormalized solution is uniformly critically tight;*
(iii) *every renormalized solution satisfies the tail windowed H^1 bound;*
(iv) *the Navier–Stokes cost-divergence implication in Theorem 7.58 is available for solutions satisfying the compact cost-divergence criterion.*

Then every renormalized Type II solution is excluded by the compact cost-divergence criterion and the Navier–Stokes cost-divergence implication. Consequently no remaining Type II case exists.

Proof. Fix a renormalized Type II solution ω. By the local classification, ω reaches exactly one of the following alternatives:

(1) the compact cost-divergence criterion;
(2) failure of uniform critical tightness;
(3) failure of windowed H^1 control, after uniform critical tightness has been established.

By the universal tightness hypothesis and Theorem 7.54, the second alternative is unavailable. By the universal windowed H^1 hypothesis and Theorem 7.54, the third alternative is unavailable. The only remaining alternative is the compact cost-divergence criterion. The Navier–Stokes cost-divergence implication then excludes this case. Since ω was arbitrary, no remaining Type II case exists. □

Corollary 7.57 (Finite-cost exclusion after pointwise tests)**.** *Under the hypotheses of Theorem 7.56, there is no finite-cost renormalized Type II solution outside the compact cost-divergence criterion.*

Proof. Theorem 7.56 shows that every renormalized Type II solution is excluded. Hence no renormalized solution remains outside the compact cost-divergence criterion, regardless of whether its Type II cost is finite. □

7.6. **Physical energy versus renormalized Type II cost.**

Assumption 7.58 (Navier–Stokes cost-divergence implication)**.** The renormalized Navier–Stokes Type II class admits a sound implication from the compact cost-divergence criterion to exclusion from the Type II class under consideration. Without this assumption, the compact criterion gives only the cost-divergence conclusion.

Lemma 7.59 (Conditional Navier–Stokes cost-divergence exclusion)**.** *Assume a renormalized Type II solution has the active supercritical scaling condition, satisfies the compact cost-divergence criterion, and satisfies Theorem 7.58. Then the solution is excluded from the Type II class under consideration.*

Proof. Theorem 7.58 is the applicability assumption that permits the compact cost-divergence criterion, or a Navier–Stokes-specific replacement theorem with the same hypotheses, to be applied to the active supercritical scaling condition. Applying that implication gives exclusion from the stated Type II class. □

Lemma 7.60 (Physical dissipation in renormalized variables)**.** *Let*

$$u(x,t) = \lambda(t)^{-1} V\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t)\right), \qquad \frac{d\tau}{dt} = \lambda(t)^{-2}.$$

Then

$$\|\nabla_x u(t)\|_{L_x^2}^2 = \lambda(t)^{-1} \|\nabla_y V(\tau(t))\|_{L_y^2}^2.$$

Consequently,

$$\int_{t_1}^{t_2} \|\nabla_x u(t)\|_{L_x^2}^2 \, dt = \int_{\tau(t_1)}^{\tau(t_2)} \lambda(\tau) \|\nabla_y V(\tau)\|_{L_y^2}^2 \, d\tau.$$

Proof. Since $u = \lambda^{-1} V$ and $y = (x - x_c)/\lambda$, $\nabla_x u = \lambda^{-2} \nabla_y V$. With $dx = \lambda^3 dy$,

$$\|\nabla_x u\|_{L_x^2}^2 = \int \lambda^{-4} |\nabla_y V|^2 \lambda^3 \, dy = \lambda^{-1} \|\nabla_y V\|_{L_y^2}^2.$$

Since $d\tau/dt = \lambda^{-2}$, $dt = \lambda^2 d\tau$. Multiplying gives the second identity. □

Definition 7.61 (Physical-renormalized dissipation cost)**.** Define

$$\mathfrak{D}_{\mathrm{phys-ren}}(\tau) := \nu \lambda(\tau) \|\nabla_y V(\tau)\|_{L_y^2}^2$$

and the localized version

$$\mathfrak{D}_{\mathrm{phys-ren},R_0}(\tau) := \nu \lambda(\tau) \int |\nabla_y V|^2 \phi_{R_0}.$$

Theorem 7.62 (Physical energy controls scale-weighted dissipation)**.** *Assume finite physical energy and the Navier–Stokes energy inequality*

$$E(t) + \nu \int_0^t \|\nabla_x u(s)\|_{L_x^2}^2 \, ds \le E(0), \qquad E(t) = \frac{1}{2} \|u(t)\|_{L_x^2}^2.$$

Then, for every $t_1 < T^*$,

$$\int_{\tau(t_1)}^{\infty} \mathfrak{D}_{\mathrm{phys-ren}}(\tau) \, d\tau \le E(t_1) \le E(0).$$

In particular,

$$\int_{\tau(t_1)}^{\infty} \mathfrak{D}_{\mathrm{phys-ren},R_0}(\tau) \, d\tau < \infty.$$

Proof. The physical energy inequality on $[t_1, t]$ gives

$$\nu \int_{t_1}^{t} \|\nabla_x u(s)\|_2^2 \, ds \le E(t_1).$$

Use Theorem 7.60 and let $t \uparrow T^*$. Monotone convergence gives the full weighted bound. The localized estimate follows from $0 \le \phi_{R_0} \le 1$. □

Definition 7.63 (Weighted positive scale-drift cost control)**.** Let

$$a_+(\tau) = \max\{a(\tau), 0\}.$$

Weighted positive scale-drift cost control means

$$\int_{\tau_1}^{\infty} \lambda(\tau)a_+(\tau) \int |V(y,\tau)|^2 \phi_{R_0}(y)\, dy\, d\tau < \infty.$$

Definition 7.64 (Scale-weighted Type II cost)**.** For

$$\widetilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla_y V|^2 \phi_{R_0} + a_+(\tau) \int |V|^2 \phi_{R_0},$$

define

$$\mathfrak{C}^{NS}_{\mathrm{II,phys}}(\tau) := \lambda(\tau)\widetilde{\mathfrak{D}}_{R_0}(\tau).$$

Theorem 7.65 (Physical energy controls the scale-weighted Type II cost)**.** *Assume finite physical energy, the Navier–Stokes energy inequality, and Theorem 7.63. Then*

$$\int_{\tau_1}^{\infty} \mathfrak{C}^{NS}_{\mathrm{II,phys}}(\tau)\, d\tau < \infty.$$

Proof. By definition,

$$\mathfrak{C}^{NS}_{\mathrm{II,phys}} = \lambda\nu \int |\nabla V|^2 \phi_{R_0} + \lambda a_+ \int |V|^2 \phi_{R_0}.$$

The first term is integrable by Theorem 7.62; the second is integrable by Theorem 7.63. □

Definition 7.66 (Scale-decoupled infinite Type II cost)**.** A renormalized Type II solution has scale-decoupled infinite Type II cost if

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty, \qquad \int_{\tau_0}^{\infty} \lambda(\tau)\widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau < \infty.$$

The physically forbidden weighted-infinite-cost alternative is

$$\int_{\tau_0}^{\infty} \mathfrak{C}^{NS}_{\mathrm{II,phys}}(\tau)\, d\tau = \infty,$$

which is impossible under Theorem 7.65.

Theorem 7.67 (Infinite-cost trichotomy)**.** *Assume finite physical energy, the Navier–Stokes energy inequality, and Theorem 7.63. Let a renormalized Type II solution satisfy*

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Then:

(i) *the compact Type II cost criterion gives a scale-collapse cost-divergence conclusion;*
(ii) *if Theorem 7.58 holds, then the solution is excluded from the Type II class under consideration;*
(iii) *relative to physical energy, exactly one of the following occurs: the physically forbidden weighted-infinite-cost alternative or the scale-decoupled infinite-cost alternative of Theorem 7.66.*

Proof. The unweighted infinite-cost conclusion is precisely the compact Type II cost divergence, so the compact criterion gives a scale-collapse cost-divergence conclusion. If Theorem 7.58 is present, Theorem 7.59 gives exclusion from the Type II class under consideration.

For the physical-energy classification, evaluate $\int \mathfrak{C}^{NS}_{\mathrm{II,phys}}$. If it is infinite, then the physically forbidden weighted-infinite-cost alternative occurs, contradicting Theorem 7.65. If it is finite, then the solution is scale-decoupled by Theorem 7.66. The two physical-energy outcomes are disjoint and exhaustive. □

Remark 7.68. Finite physical energy controls the scale-weighted renormalized cost, not the unweighted compact Type II cost. Thus infinite scale-weighted renormalized cost forces infinite physical energy dissipation, while infinite unweighted renormalized cost does not by itself force infinite physical energy without an additional scale-weight estimate. Likewise, a compact Type II cost-divergence conclusion does not automatically imply exclusion from the Type II class unless Theorem 7.58 is present.

7.7. **Navier–Stokes Cost-Divergence Exclusion.**

Definition 7.69 (Type II exclusion conclusion)**.** For a renormalized Type II solution, the exclusion conclusion means that after the active supercritical scaling condition and the compact cost-divergence criterion have both been established, the solution is excluded from the Type II class under consideration. This is stronger than cost divergence alone, which only states that the solution satisfies the compact Type II cost-divergence criterion.

Assumption 7.70 (Pointwise cost-divergence exclusion hypotheses)**.** For each renormalized Type II solution ω, assume the following hypotheses are available:

(i) class membership: ω lies in the renormalized Navier–Stokes Type II class and satisfies the Type II analytic data before the Type II cost criterion is evaluated;

(ii) cost compatibility: the active compact Type II cost criterion is the Navier–Stokes identity-cost criterion
$$\mathfrak{C}_{\mathrm{II}}^{NS} = \widetilde{\mathfrak{D}}_{R_0},$$
and its conclusion is the compact cost-divergence conclusion for ω;

(iii) stability of the active scaling condition: the active supercritical scaling condition remains in force after the repaired-gauge representation, cost comparison, and criterion evaluation are attached;

(iv) stability of the cost-divergence conclusion: every renormalized continuation preserving the active scaling condition and compact cost-divergence conclusion remains in the same compact Type II cost-divergence regime, unless it first enters one of the local alternatives already isolated above;

(v) exclusion implication: the renormalized class supplies an implication
$$\Pi_{\mathrm{NS,II}}(\omega) : \mathcal{S}_\omega \wedge \mathcal{B}_\omega \Longrightarrow \mathcal{E}_\omega.$$
Here $\mathcal{S}_\omega$ is the active supercritical scaling condition, $\mathcal{B}_\omega$ is the compact Type II cost-divergence conclusion, and $\mathcal{E}_\omega$ is exclusion of ω from the Type II class. The implication is obtained either by verifying an abstract Type II exclusion theorem for the Navier–Stokes solution or by proving a Navier–Stokes-specific replacement theorem with the same hypotheses and conclusion.

Assumption 7.71 (Uniform cost-divergence exclusion hypotheses)**.** Theorem 7.70 holds for every renormalized Type II solution in the class under consideration.

Definition 7.72 (Abstract cost-divergence exclusion theorem applicability)**.** An abstract Type II cost-divergence exclusion theorem is applicable to a renormalized Type II solution if the following analytic identifications are available:

(i) an abstract Type II exclusion theorem based on localized renormalization-cost divergence;

(ii) hypotheses supplying supercritical scaling, a Type II concentration scenario with bounded physical energy, and a localized energy monotonicity formula at the active scale $\lambda(t)$;

(iii) an identification of the compact Navier–Stokes cost
$$\mathfrak{C}_{\mathrm{II}}^{NS} = \widetilde{\mathfrak{D}}_{R_0}$$
with the theorem's renormalization-cost integral;
(iv) a domain embedding placing the repaired-gauge Navier–Stokes orbit and its scale variables into the parabolic domain expected by the theorem without changing the active scaling condition, cost-divergence conclusion, or exclusion conclusion;
(v) identification of the theorem's conclusion with Theorem 7.69.

Lemma 7.73 (Applicability of the abstract cost-divergence theorem)**.** *Assume a renormalized Type II solution has the active supercritical scaling condition, the compact Type II cost-divergence conclusion, and the abstract applicability data of Theorem 7.72. Then the solution is excluded from the Type II class under consideration.*

Proof. The abstract Type II theorem is applied with supercritical scaling, bounded-energy Type II concentration, and the localized energy monotonicity formula at the active scale. The cost identification matches the compact Navier–Stokes cost criterion with the theorem's renormalization-cost integral. The domain embedding places the repaired-gauge Navier–Stokes solution in the theorem's parabolic domain, and the conclusion is identified with Theorem 7.69. Therefore the theorem gives exclusion from the Type II class. □

Theorem 7.74 (Pointwise Navier–Stokes cost-divergence exclusion)**.** *Let ω be a renormalized Type II solution. If ω has the active supercritical scaling condition, the compact Type II cost-divergence conclusion, and the pointwise hypotheses of Theorem 7.70, then ω is excluded from the Type II class under consideration.*

Proof. Class membership and stability of the active scaling condition ensure that ω is evaluated in the renormalized Type II class and that the active supercritical scaling condition is still the relevant one. Cost compatibility ensures that the compact cost-divergence conclusion is the Navier–Stokes compact cost conclusion for ω, not an external or incomparable cost statement. Stability of the cost-divergence conclusion rules out an escape from the compact Type II cost-divergence regime to a Type II case outside the compact cost-divergence criterion inside the same class.

It remains only to justify the transition from cost divergence to exclusion. By the exclusion implication, the class supplies $\Pi_{\mathrm{NS,II}}(\omega)$. Applying this implication to the active supercritical scaling condition and compact cost-divergence conclusion gives the claimed exclusion. In the abstract theorem setting, this implication is precisely the one licensed by Theorem 7.73. □

Corollary 7.75 (Uniform Navier–Stokes cost-divergence exclusion)**.** *Assume Theorem 7.71. Then every renormalized Type II solution with active supercritical scaling condition and compact Type II cost-divergence conclusion is excluded from the Type II class under consideration.*

Proof. By Theorem 7.71, the pointwise cost-divergence exclusion hypotheses are available for every renormalized Type II solution. Apply Theorem 7.74 pointwise. □

Definition 7.76 (Missing cost-divergence exclusion hypotheses)**.** If the active supercritical scaling condition and compact cost-divergence conclusion are present but exclusion is not obtained, the first missing hypothesis lies in the ordered list:

class membership, cost compatibility, stability of the active scaling condition,
stability of the cost-divergence conclusion, exclusion implication.

These are missing hypotheses for the conditional exclusion step, not additional PDE Type II alternatives.

Lemma 7.77 (Missing exclusion hypotheses are not PDE alternatives)**.** *Assume a renormalized Type II solution already has the active supercritical scaling condition and the compact Type II cost-divergence conclusion. If one of the missing hypotheses in Theorem 7.76 occurs, then the conditional exclusion step is not classifying the solution as non-tight or as failing local windowed H^1 control. It is recording a missing hypothesis needed to pass from cost divergence to exclusion.*

Proof. The two local alternatives are failure of uniform critical tightness and failure of windowed local H^1-control. The conditional exclusion step is entered only after the active supercritical scaling condition and divergent compact cost have already been identified. Its missing hypotheses concern the passage from cost divergence to exclusion. None of these hypotheses asserts failure of tightness or failure of windowed H^1-control, so none defines a new PDE Type II alternative. □

Remark 7.78 (Cost divergence and conditional exclusion)*.* The compact Type II cost argument gives infinite localized renormalization cost under the compact PDE hypotheses. Passing from that cost divergence to exclusion from the Type II class is a separate conditional step. Without the corresponding Navier–Stokes cost-divergence exclusion hypothesis, the strongest conclusion is the cost-divergence statement itself.

8. Terminal Profile Decoupling and Exterior Regularity

This section records the terminal-profile estimates used to remove multibubble configurations, small critical profiles, and profiles carried at a positive physical distance from the selected Type II core. The statements are conditional on the analytic inputs explicitly stated below: critical profile decomposition, terminal windowed profile completeness, repaired-gauge representation, local Caccioppoli estimates, exterior regularity, and the stationary rigidity input for the compact scale-collapse limit.

8.1. **Renormalized frames and active critical profiles.** A renormalized frame consists of a center $x_c(t)$, a scale $\lambda(t) > 0$, and renormalized time

$$\frac{d\tau}{dt} = \lambda(t)^{-2}.$$

It represents the physical solution by

$$V(y,\tau) = \lambda(t)u(x_c(t) + \lambda(t)y, t).$$

When the repaired gauge is imposed, the represented velocity satisfies

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0,$$

where

$$a(\tau) = \partial_\tau \log\lambda(\tau),$$

and b is the translation modulation coefficient determined by the centering condition. In a fixed frame, every other profile either remains at bounded relative position and comparable scale, escapes to spatial infinity in the frame variables, appears at a much larger same-point scale as a perturbative ambient field in the innermost frame, or lies at a smaller scale and therefore requires selecting that smaller active frame.

Definition 8.1 (Active critical profile data)**.** Let $t_n \uparrow T^*$. Active critical profile data consist of divergence-free profiles $\phi^j \in L^3(\mathbb{R}^3)$, scales $\lambda_j^n \to 0$, centers $x_j^n \to x_j^*$, and finite truncation remainders r_n^J such that, for every finite index set J,

$$u(\cdot, t_n) = \sum_{j\in J}\Lambda_{\lambda_j^n, x_j^n}\phi^j + r_n^J, \qquad (\Lambda_{\lambda,x_0}\phi)(x) = \lambda^{-1}\phi\left(\frac{x-x_0}{\lambda}\right).$$

After grouping comparable parameters into compound profiles, the parameters are pairwise orthogonal and the critical mass decoupling

$$\|u(t_n)\|_3^3 = \sum_{j\in J}\|\phi^j\|_3^3 + \|r_n^J\|_3^3 + o_n(1)$$

holds for every finite J. A profile is active if its nonlinear evolution is not controlled by the small-data perturbative theory. The small-data threshold is denoted by $\varepsilon_{\mathrm{sd}} > 0$; the corresponding active-profile mass floor is stated and proved in Theorem 8.27.

Lemma 8.2 (Finite active profile count)**.** *Assume*

$$\sup_{t<T^*}\|u(t)\|_3 \le M < \infty.$$

Assume also that every active profile satisfies

$$\|\phi^j\|_3 \ge \varepsilon_{\mathrm{sd}}.$$

Then the number of active profiles is finite and satisfies

$$\#\mathcal{J}_{\mathrm{act}} \le \left\lfloor \frac{M^3}{\varepsilon_{\mathrm{sd}}^3}\right\rfloor.$$

Proof. Let $J \subset \mathcal{J}_{\mathrm{act}}$ be finite. By critical L^3-mass decoupling,

$$\|u(t_n)\|_3^3 = \sum_{j\in J} \|\phi^j\|_3^3 + \|r_n^J\|_3^3 + o_n(1).$$

Taking the limsup and using $\|r_n^J\|_3^3 \geq 0$,

$$M^3 \geq \sum_{j\in J} \|\phi^j\|_3^3 \geq (\#J)\varepsilon_{\mathrm{sd}}^3.$$

Thus $\#J \leq M^3/\varepsilon_{\mathrm{sd}}^3$ for every finite active set J. If the active set had more than $\lfloor M^3/\varepsilon_{\mathrm{sd}}^3 \rfloor$ elements, one could choose a finite subset with one additional element, contradicting the preceding inequality. This proves the integer bound and finiteness. □

8.2. **Same-point and separated-point reductions.**

Lemma 8.3 (Comparable same-point profiles form a compound profile)**.** *Let a finite set $\mathcal{C}$ of active profiles have the same physical center $x_j^* = x_*$. Suppose that for a representative center and scale (x_*^n, λ_*^n),*

$$\frac{\lambda_j^n}{\lambda_*^n} \to \rho_j \in (0,\infty), \qquad \frac{x_j^n - x_*^n}{\lambda_*^n} \to z_j \in \mathbb{R}^3.$$

Then, in the frame (x_*^n, λ_*^n),

$$\lambda_*^n (\lambda_j^n)^{-1} \phi^j \left(\frac{x_*^n + \lambda_*^n y - x_j^n}{\lambda_j^n} \right) \to \rho_j^{-1} \phi^j \left(\frac{y - z_j}{\rho_j} \right)$$

strongly in L^3_{loc}, and strongly in $L^3(\mathbb{R}^3)$ when the profile convergence is global. Hence the comparable class has compound profile

$$\Phi_{\mathcal{C}}(y) := \sum_{j\in\mathcal{C}} \rho_j^{-1} \phi^j \left(\frac{y - z_j}{\rho_j} \right).$$

If $\|\Phi_{\mathcal{C}}\|_3 \geq \varepsilon_{\mathrm{sd}}$, this compound profile is one active profile in the selected frame; if $\|\Phi_{\mathcal{C}}\|_3 < \varepsilon_{\mathrm{sd}}$, it is perturbative by small-data stability.

Proof. Set

$$\rho_j^n = \frac{\lambda_j^n}{\lambda_*^n}, \qquad z_j^n = \frac{x_j^n - x_*^n}{\lambda_*^n}.$$

The transformed profile is

$$T_n \phi^j(y) = (\rho_j^n)^{-1} \phi^j \left(\frac{y - z_j^n}{\rho_j^n} \right),$$

and the proposed limit is

$$T\phi^j(y) = \rho_j^{-1} \phi^j \left(\frac{y - z_j}{\rho_j} \right).$$

For $\rho \in (0,\infty)$ and $z \in \mathbb{R}^3$, the map

$$T_{\rho,z}\phi(y) = \rho^{-1} \phi \left(\frac{y - z}{\rho} \right)$$

is strongly continuous on $L^3(\mathbb{R}^3)$. Indeed, for $\phi \in C_c^\infty$, convergence of $\rho_j^n \to \rho_j$ and $z_j^n \to z_j$ gives uniform convergence on a common compact support and therefore L^3-convergence. For a general $\phi \in L^3$, choose $\psi \in C_c^\infty$ with $\|\phi - \psi\|_3 < \varepsilon$. Since each $T_{\rho,z}$ is an L^3-isometry,

$$\|T_{\rho_j^n, z_j^n}\phi - T_{\rho_j, z_j}\phi\|_3 \leq 2\varepsilon + \|T_{\rho_j^n, z_j^n}\psi - T_{\rho_j, z_j}\psi\|_3.$$

Letting $n \to \infty$ and then $\varepsilon \downarrow 0$ proves global L^3-convergence; local convergence follows immediately. Summing over the finite class gives $\Phi_{\mathcal{C}}$. The final alternative is the small-data criterion: a compound profile above the threshold is active, while one below the threshold belongs to the perturbative class. ☐

Assumption 8.4 (Strict same-point scale separation)**.** Suppose all active profiles have the same physical center x_* and split into distinct scale classes. If λ_1^n is the smallest active scale, then every outer scale $\lambda_j^n \gg \lambda_1^n$ satisfies

$$\mu_j^n := \lambda_1^n/\lambda_j^n \to 0,$$

and in the innermost frame the j-th outer profile has the form

$$W_j^n(y) = \mu_j^n \phi^j \left(\frac{x_*^n - x_j^n}{\lambda_j^n} + \mu_j^n y \right).$$

The analytic input for this case is that finite-offset regular outer profiles are constant drifts plus $O((\mu_j^n)^2)$ shear on fixed inner balls, the constant drifts are absorbed into the translation modulation coefficient, escaping-offset profiles are locally L^3-invisible, and the remaining pressure and nonlinear interactions are perturbative in the compact scale-collapse topology.

Lemma 8.5 (Separated physical centers are locally invisible)**.** *Let $p \neq q$ be two active physical concentration points,*

$$x_p^* \neq x_q^*, \qquad d_{pq} := |x_p^* - x_q^*| > 0.$$

In the p-frame define the q-profile contribution

$$W_{q\to p}^n(y) := \lambda_p^n (\lambda_q^n)^{-1} \phi^q \left(\frac{x_p^n + \lambda_p^n y - x_q^n}{\lambda_q^n} \right).$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then for every $R < \infty$,

$$\|W_{q\to p}^n\|_{L^3(B_R)} \to 0.$$

If the corresponding profile pressure $\Pi^q \in L^{3/2}(\mathbb{R}^3)$, then the pure pressure contribution

$$P_{q\to p}^n(y) = (\lambda_p^n)^2 (\lambda_q^n)^{-2} \Pi^q \left(\frac{x_p^n + \lambda_p^n y - x_q^n}{\lambda_q^n} \right)$$

satisfies

$$\|P_{q\to p}^n\|_{L^{3/2}(B_R)} \to 0.$$

Proof. By the change of variables $x = x_p^n + \lambda_p^n y$,

$$\|W_{q\to p}^n\|_{L^3(B_R)}^3 = \int_{B(x_p^n, \lambda_p^n R)} (\lambda_q^n)^{-3} \left| \phi^q \left(\frac{x - x_q^n}{\lambda_q^n} \right) \right|^3 dx.$$

Set $z = (x - x_q^n)/\lambda_q^n$. Then

$$\|W_{q\to p}^n\|_{L^3(B_R)}^3 = \int_{B(c_n, \rho_n)} |\phi^q(z)|^3 \, dz,$$

where

$$c_n = \frac{x_p^n - x_q^n}{\lambda_q^n}, \qquad \rho_n = \frac{\lambda_p^n R}{\lambda_q^n}.$$

Since $x_p^n \to x_p^*$, $x_q^n \to x_q^*$, and $\lambda_q^n \to 0$,

$$|c_n| \sim d_{pq}/\lambda_q^n \to \infty.$$

Moreover

$$\frac{\rho_n}{|c_n|} = \frac{\lambda_p^n R}{|x_p^n - x_q^n|} \to 0.$$

Thus the balls $B(c_n, \rho_n)$ escape to spatial infinity: for every $A < \infty$, they are contained in $\{|z| > A\}$ for all sufficiently large n. Since $\phi^q \in L^3$,

$$\int_{|z|>A} |\phi^q(z)|^3 \, dz \to 0 \qquad (A \to \infty),$$

and hence $\|W^n_{q\to p}\|_{L^3(B_R)} \to 0$.

For the pressure term the same change of variables gives

$$\|P^n_{q\to p}\|^{3/2}_{L^{3/2}(B_R)} = \int_{B(c_n,\rho_n)} |\Pi^q(z)|^{3/2} \, dz.$$

The same escaping-ball argument and $\Pi^q \in L^{3/2}$ imply the stated pressure convergence. □

Definition 8.6 (Multipoint decoupling conditions)**.** For each active point and selected frame, the required multipoint decoupling conditions are:

(i) local velocity decoupling in the selected frame;
(ii) pressure decoupling modulo time-dependent constants in $L^1_\tau H^{-1}_y$ on compact cylinders;
(iii) localization errors from spatial cutoffs vanish in $L^1_\tau H^{-1}_y$;
(iv) repaired-gauge modulation coefficients remain bounded after exterior ambient terms are absorbed;
(v) each active point has positive finite critical mass in its own frame;
(vi) after profiles centered at other physical points are treated as perturbative exterior terms, the selected frame satisfies the compact scale-collapse hypotheses: repaired-gauge representation, pressure reconstruction, Caccioppoli regularity, bounded modulation, and the compactness and tightness inputs used in the limiting passage.

Theorem 8.7 (Reduction of active multibubbles)**.** *Assume the Type II profile data, the active-profile mass floor, the same-point nonlinear decoupling conditions, the multipoint decoupling conditions of Theorem 8.6, and the compact scale-collapse stationary-rigidity input. Then no active multibubble Type II configuration can occur.*

Proof. By Theorem 8.2, there are only finitely many active profiles. Partition them by physical concentration point x^*_α, and within each physical point partition them by comparable scale class.

If a scale class contains several comparable profiles, Theorem 8.3 combines them into one compound profile. If the compound profile is below the small-data threshold, it is perturbative; if it is active, it is a single active profile for that frame.

If a physical point contains strict same-point scale separation, Theorem 8.4 and the same-point nonlinear decoupling conditions reduce the innermost active scale to a single repaired-gauge solution with perturbative outer interactions. The compact scale-collapse input then rules out the corresponding same-point cascade.

It remains to consider separated physical points. Choose any active point x^*_α and place the selected frame at its active scale and center. By Theorem 8.5, every profile centered at a different physical point is locally L^3-invisible in this frame. By the multipoint decoupling conditions, its pressure, cutoff, and modulation interactions are also perturbative in the compact scale-collapse topology. Therefore the selected frame contains a single active repaired-gauge solution with positive finite critical mass, bounded modulation, pressure reconstruction, Caccioppoli regularity, and the compactness and tightness inputs required by the scale-collapse argument.

The scale of that active solution tends to zero, so the scale-collapse input applies. The compactness argument extracts a nonzero stationary $L^3(\mathbb{R}^3)$ profile in the stationary-rigidity class, contradicting the stationary rigidity theorem. This excludes the chosen active point. Since the chosen active point was arbitrary, no active multibubble configuration can occur. $\square$

8.3. Local perturbation and pressure decoupling.

Lemma 8.8 (Local perturbation criterion)**.** *Let $I \subset \mathbb{R}$ be compact and $Q_R = B_R \times I$. Let U_n be a repaired-gauge solution on Q_{2R} satisfying*

$$\|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R.$$

Let S_n be a divergence-free perturbation on Q_{2R} with

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_{2R}),$$

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

$$\mathcal{R}_n := \partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R),$$

where $|a_n| + |b_n| \le M_{ab}$, and suppose the pressure cross-term satisfies

$$\nabla \mathcal{P}_{\text{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Then replacing $U_n + S_n$ by U_n changes the repaired-gauge Navier–Stokes equation on Q_R by an error tending to zero in

$$L^1_\tau H^{-1}_y(B_R).$$

Proof. The linear part is precisely $\mathcal{R}_n$, which tends to zero by hypothesis. The nonlinear difference is

$$(U_n + S_n) \cdot \nabla(U_n + S_n) - U_n \cdot \nabla U_n = \operatorname{div} F_n, \qquad F_n := U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n,$$

using $\nabla \cdot U_n = \nabla \cdot S_n = 0$ distributionally. By the assumed $L^1_\tau L^2_y$ stress convergence, this divergence tends to zero in

$$L^1_\tau H^{-1}_y(B_R),$$

because for $\varphi \in H^1_0(B_R; \mathbb{R}^3)$,

$$\left| \int_{B_R} (U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n) : \nabla\varphi \, dy \right| \le \|F_n\|_{L^2(B_R)} \times \|\nabla\varphi\|_{L^2(B_R)}.$$

The pressure contribution vanishes by the assumed cross-pressure convergence. Therefore the total perturbative error tends to zero in

$$L^1_\tau H^{-1}_y(B_R).$$

$\square$

Lemma 8.9 (Static same-point outer profiles vanish locally)**.** *Let*

$$W_n(y) = \mu_n \phi(\xi_n + \mu_n y), \qquad \mu_n \to 0, \qquad \phi \in L^3(\mathbb{R}^3).$$

Then, for every $R < \infty$,

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. Changing variables $z = \xi_n + \mu_n y$,

$$\|W_n\|_{L^3(B_R)}^3 = \int_{B(\xi_n,\mu_n R)} |\phi(z)|^3\, dz.$$

The measure of $B(\xi_n, \mu_n R)$ tends to zero. Since $|\phi|^3 \in L^1(\mathbb{R}^3)$, absolute continuity of the integral gives

$$\int_{B(\xi_n,\mu_n R)} |\phi(z)|^3\, dz \to 0,$$

uniformly in ξ_n. Hence $\|W_n\|_{L^3(B_R)} \to 0$. □

Lemma 8.10 (Pressure decoupling from kernel tails)**.** *Let Γ_{ij} be the Calderon–Zygmund kernel for*

$$P = \mathcal{R}_i\mathcal{R}_j(F_{ij}).$$

Assume

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_A) \quad \text{for every fixed } A,$$

and

$$\sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y(\mathbb{R}^3)} < \infty.$$

Then the pressure generated by F_n satisfies, for every $R < \infty$,

$$P_n - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R)$$

for suitable constants $c_{n,R}(\tau)$, and consequently

$$\nabla P_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. Fix $A > 2R$ and write

$$F_n^{\mathrm{loc}} = F_n \mathbf{1}_{B_A}, \qquad F_n^{\mathrm{far}} = F_n \mathbf{1}_{\mathbb{R}^3\setminus B_A}.$$

For the local part, Calderon–Zygmund boundedness gives

$$\|P_n^{\mathrm{loc}}\|_{L^1_\tau L^2_y(B_A)} \le C_A \|F_n\|_{L^1_\tau L^2_y(B_A)} \to 0.$$

For the far part define

$$c_{n,R}(\tau) = \int_{|z|>A} \Gamma_{ij}(-z) F_{n,ij}(z,\tau)\, dz,$$

first for compactly supported approximations and then by passage to the $L^{3/2}$ limit. For $y \in B_R$ and $|z| > A > 2R$,

$$|\Gamma_{ij}(y - z) - \Gamma_{ij}(-z)| \le C_R |z|^{-4}.$$

Hence

$$|P_n^{\mathrm{far}}(y,\tau) - c_{n,R}(\tau)| \le C_R \int_{|z|>A} |z|^{-4} |F_n(z,\tau)|\, dz.$$

By Holder's inequality with exponents $3/2$ and 3,

$$\int_{|z|>A} |z|^{-4}|F_n|\, dz \le \|F_n(\tau)\|_{L^{3/2}} \left(\int_{|z|>A} |z|^{-12}\, dz\right)^{1/3} \le CA^{-3}\|F_n(\tau)\|_{L^{3/2}}.$$

Therefore

$$\|P_n^{\mathrm{far}} - c_{n,R}\|_{L^1_\tau L^\infty_y(B_R)} \le C_{R,I} A^{-3} \sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y}.$$

Since B_R has finite measure, this also controls the far part in $L^1_\tau L^2_y(B_R)$. Taking $n \to \infty$ in the local term and then $A \to \infty$ in the far term gives

$$P_n - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

For $\varphi \in H^1_0(B_R;\mathbb{R}^3)$,

$$|\langle \nabla(P_n - c_{n,R}), \varphi\rangle| \le \left|\int_{B_R} (P_n - c_{n,R})\nabla \cdot \varphi \, dy\right| \le \|P_n - c_{n,R}\|_{L^2(B_R)} \|\nabla \varphi\|_{L^2(B_R)}.$$

Thus $\nabla P_n \to 0$ in $L^1_\tau H^{-1}_y(B_R)$. □

Definition 8.11 (Terminal nonlinear profile decoupling)**.** On every compact terminal-frame cylinder $B_R \times I$, let U_n be the retained active profile evolution and let S_n be the sum of all nonretained profiles and the critical perturbative remainder. Terminal nonlinear profile decoupling means:

(i) $\nabla \cdot S_n = 0$ distributionally;
(ii) $S_n \to 0$ in $L^\infty_\tau L^3_y(B_{2R})$;
(iii)
$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R);$$
(iv)
$$\partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R);$$
(v) pressure sources involving at least one factor of S_n satisfy Theorem 8.10, or directly satisfy
$$\nabla \mathcal{P}_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R);$$
(vi) cutoff errors introduced in isolating the selected frame vanish in $L^1_\tau H^{-1}_y(B_R)$;
(vii) the effective modulation coefficients remain bounded;
(viii) after subtracting S_n, the retained solution has positive finite critical mass, repaired-gauge representation, pressure reconstruction, Caccioppoli regularity, compactness, tightness, and the compact scale-collapse limiting inputs.

Theorem 8.12 (Same-point and multipoint decoupling from terminal decoupling)**.** *If terminal nonlinear profile decoupling holds in the sense of Theorem 8.11, then both same-point scale-decoupling and separated-point frame-decoupling hold.*

Proof. For same-point active profiles, group comparable scales by Theorem 8.3. For strict scale separation choose the innermost active scale as the selected terminal frame. By Theorem 8.9, every outer scale is locally invisible in velocity. Terminal nonlinear profile decoupling supplies the nonlinear-stress, linear-residual, pressure-source, localization, modulation, and compact scale-collapse admissibility conditions on every compact cylinder. Theorem 8.8 then shows that the outer profiles and remainder alter the selected equation only by an error tending to zero in $L^1_\tau H^{-1}_y(B_R)$. Thus the same-point configuration reduces to one retained compact scale-collapse solution with perturbative interactions.

For separated physical points, choose an active physical point x_p^* and its selected frame. For every profile centered at $x_q^* \ne x_p^*$, Theorem 8.5 gives local velocity convergence to zero in the p-frame. If a pure profile pressure is represented in L^2, the same escaping-ball calculation gives local L^2 pressure convergence and therefore H^{-1} convergence of the gradient. With only $L^{3/2}$-pressure, the mixed pressure sources are controlled by Theorem 8.10 through the L^2-strength hypotheses included in terminal nonlinear profile decoupling. The same decoupling assumption supplies localization-error convergence, bounded modulation, and compact scale-collapse admissibility. Applying Theorem 8.8 shows that separated physical-point profiles change the selected equation only by an error tending to zero in $L^1_\tau H^{-1}_{\mathrm{loc}}$. This proves separated-point decoupling. □

8.4. **Terminal profile decoupling.**

Definition 8.13 (Terminal active frame)**.** Partition active profiles by physical point $x_j^n \to x_\alpha^*$. At a fixed physical point, group comparable scales. For a comparable-scale class $\mathcal{C}$, a representative frame $(x_{\mathcal{C}}^n, \lambda_{\mathcal{C}}^n)$ has compound profile

$$\Phi_{\mathcal{C}}(y) = \sum_{j\in\mathcal{C}} \rho_j^{-1}\phi^j\left(\frac{y-z_j}{\rho_j}\right).$$

The class $\mathcal{C}$ is terminal at x_α^* if no active class at the same physical point has strictly smaller scale:

$$\lambda_{\mathcal{D}}^n/\lambda_{\mathcal{C}}^n \not\to 0 \quad \text{for every active class } \mathcal{D}.$$

The terminal frame is

$$y = \frac{x-x_{\mathcal{C}}^n}{\lambda_{\mathcal{C}}^n}, \qquad \tau = \int^t (\lambda_{\mathcal{C}}(s))^{-2}\,ds,$$

and

$$V_n(y,\tau) = \lambda_{\mathcal{C}}^n u(x_{\mathcal{C}}^n + \lambda_{\mathcal{C}}^n y, t_n + (\lambda_{\mathcal{C}}^n)^2\tau).$$

The retained solution U_n is the repaired-gauge nonlinear evolution of $\mathcal{C}$ in this frame, and $S_n := V_n - U_n$ is the nonretained remainder field.

Assumption 8.14 (Terminal windowed profile completeness)**.** For a sequence f_n and a frame (x_n, λ_n), define

$$\mathfrak{m}_R(f_n; x_n, \lambda_n) := \limsup_{n\to\infty} \|\lambda_n f_n(x_n + \lambda_n\cdot)\|_{L^3(B_R)}$$

and

$$\mathfrak{M}_{R,I}(f_n; x_n, \lambda_n) := \limsup_{n\to\infty}\sup_{\tau\in I}\left\|\lambda_n f_n(x_n+\lambda_n\cdot, t_n+\lambda_n^2\tau)\right\|_{L^3(B_R)}.$$

Terminal windowed profile completeness means that if $\mathfrak{M}_{R,I}(f_n; x_n, \lambda_n) > 0$ for a sequence left in the remainder, then after passing to a subsequence the profile decomposition contains a nonzero profile with physical center $x_n + O(\lambda_n)$ and scale comparable to λ_n. If that nonlinear profile is above the small-data threshold, it is active; if it is below the threshold, it is controlled by the small-data perturbative theory uniformly on compact windows.

Lemma 8.15 (Terminal local velocity decoupling)**.** *Let $\mathcal{C}$ be a terminal active class and S_n be the nonretained remainder field in the $\mathcal{C}$-frame. Then for every compact cylinder $B_R \times I$,*

$$S_n \to 0 \quad \text{in } L_\tau^\infty L_y^3(B_R).$$

Proof. Assume otherwise. Then there exist $R < \infty$, a compact interval I, $\delta > 0$, a subsequence, and times $\tau_n \in I$ such that

$$\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta.$$

In physical variables, after the retained terminal class has been subtracted, the remainder carries at least δ critical L^3-mass inside a ball of radius $O(\lambda_{\mathcal{C}}^n)$, centered at

$$x_{\mathcal{C}}^n + O(\lambda_{\mathcal{C}}^n),$$

at time

$$t_n + (\lambda_{\mathcal{C}}^n)^2\tau_n.$$

By terminal windowed profile completeness, this nonvanishing mass produces a profile whose physical center is x_α^* and whose scale is comparable to $\lambda_{\mathcal{C}}^n$.

If the resulting profile is active, it belongs to the same comparable-scale class as $\mathcal{C}$. But all comparable active profiles at that physical point were grouped into $\mathcal{C}$, so it cannot remain in S_n, a contradiction. If the resulting profile is below the small-data threshold, it belongs to the perturbative part of the decomposition. The small-data stability input and the choice of the terminal truncation give a contribution smaller than $\delta/2$ on the compact window, so it cannot account for $\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta$. This is again a contradiction.

Thus no such δ exists. The same argument covers same-point larger-scale profiles and separated-point profiles: any nonzero mass seen in the terminal frame would generate an omitted profile at the terminal scale or at the selected physical point, contradicting the terminal partition. $\square$

Lemma 8.16 (Uniform terminal H^1 bounds)**.** *Assume the repaired-gauge representation and the local Caccioppoli estimates hold on every compact terminal cylinder $B_{2R} \times I$. Then*

$$\sup_n \|V_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|V_n\|_{L^2_\tau H^1_y(B_{2R})} \leq C_R,$$

$$\sup_n \|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \leq C_R,$$

and, since $S_n = V_n - U_n$,

$$\sup_n \|S_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|S_n\|_{L^2_\tau H^1_y(B_{2R})} \leq C_R.$$

In particular, by Sobolev embedding on B_{2R},

$$U_n, S_n \quad \text{are bounded in} \quad L^2_\tau L^6_y(B_{2R}).$$

Proof. The first estimate is the repaired-gauge Caccioppoli estimate for V_n on the compact terminal cylinder. The retained solution U_n is itself a repaired-gauge Navier–Stokes solution with positive finite critical mass and bounded modulation on the same compact cylinder, so the same estimate applies to U_n. Since $S_n = V_n - U_n$, the displayed bound for S_n follows by the triangle inequality in the stated norms. The $L^2_\tau L^6_y$ bound follows from the Sobolev embedding $H^1(B_{2R}) \hookrightarrow L^6(B_{2R})$. $\square$

Lemma 8.17 (Nonlinear stress decoupling in terminal frames)**.** *Let*

$$F_n := U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n.$$

Then, for every compact $B_R \times I$,

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

Moreover

$$\sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y(\mathbb{R}^3)} < \infty$$

whenever U_n, S_n, V_n are uniformly bounded in global $L^\infty_\tau L^3_y$.

Proof. By Theorem 8.15,

$$\|S_n\|_{L^\infty_\tau L^3_y(B_R)} \to 0.$$

Using Holder's inequality in space and time,

$$\|U_n \otimes S_n\|_{L^1_\tau L^2_y(B_R)} \leq \|S_n\|_{L^\infty_\tau L^3_y(B_R)} \|U_n\|_{L^1_\tau L^6_y(B_R)}$$

and

$$\|U_n\|_{L^1_\tau L^6_y(B_R)} \leq |I|^{1/2} \|U_n\|_{L^2_\tau L^6_y(B_R)} \leq C_{R,I}.$$

Thus $U_n \otimes S_n \to 0$ in $L^1_\tau L^2_y(B_R)$, and the same estimate gives $S_n \otimes U_n \to 0$. Finally,

$$\|S_n \otimes S_n\|_{L^1_\tau L^2_y(B_R)} \leq \|S_n\|_{L^\infty_\tau L^3_y(B_R)} \|S_n\|_{L^1_\tau L^6_y(B_R)} \to 0,$$

because $\|S_n\|_{L^1_\tau L^6_y(B_R)} \leq C_{R,I}$. Summing the three terms proves the local convergence. The global $L^{3/2}$ bound follows from Holder's inequality and the global $L^\infty_\tau L^3_y$ bounds. $\square$

Lemma 8.18 (Terminal pressure decoupling)**.** *Let $P_{\mathrm{cross},n}$ solve*

$$-\Delta P_{\mathrm{cross},n} = \partial_i \partial_j (F_n)_{ij}.$$

Assume, in addition, that U_n, S_n, and V_n are uniformly bounded in $L^\infty_\tau L^3_y(\mathbb{R}^3)$. Then for every $R < \infty$ there are constants $c_{n,R}(\tau)$ such that

$$P_{\mathrm{cross},n} - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

and

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. Theorem 8.17 gives

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_A)$$

for every fixed A. The global $L^\infty_\tau L^3_y$ bounds and Holder's inequality give

$$\sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y(\mathbb{R}^3)} < \infty.$$

Thus the hypotheses of Theorem 8.10 hold. That lemma gives local L^2 pressure convergence modulo constants and $L^1_\tau H^{-1}_y(B_R)$ convergence of the pressure gradient. □

Lemma 8.19 (Linear residual decoupling)**.** *Let $V_n = U_n + S_n$ solve*

$$\partial_\tau V_n + (V_n \cdot \nabla)V_n + \nabla P_n = \nu \Delta V_n + a_n(V_n + y \cdot \nabla V_n) + b_n \cdot \nabla V_n.$$

Suppose U_n satisfies, in the same repaired gauge,

$$\partial_\tau U_n + (U_n \cdot \nabla)U_n + \nabla P_{U,n} = \nu \Delta U_n + a_n(U_n + y \cdot \nabla U_n) + b_n \cdot \nabla U_n + e_n,$$

with

$$e_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R)$$

on compact terminal cylinders. Then

$$\partial_\tau S_n - \nu \Delta S_n - a_n(S_n + y \cdot \nabla S_n) - b_n \cdot \nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Equivalently, the same conclusion holds with the opposite sign convention for the modulation terms.

Proof. Subtracting the U_n-equation from the V_n-equation gives

$$\partial_\tau S_n - \nu \Delta S_n = -\operatorname{div} F_n - \nabla P_{\mathrm{cross},n} - e_n + a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n.$$

Equivalently,

$$\partial_\tau S_n - \nu \Delta S_n - a_n(S_n + y \cdot \nabla S_n) - b_n \cdot \nabla S_n = -\operatorname{div} F_n - \nabla P_{\mathrm{cross},n} - e_n.$$

By Theorem 8.17,

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

hence

$$\operatorname{div} F_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

By Theorem 8.18,

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Together with $e_n \to 0$, this proves the displayed residual convergence.

For the opposite sign convention, the two residuals differ by

$$2a_n(S_n + y \cdot \nabla S_n) + 2b_n \cdot \nabla S_n.$$

By Theorem 8.15 and finite measure,

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^2_y(B_R).$$

For $\varphi \in H_0^1(B_R)$,

$$|\langle y \cdot \nabla S_n, \varphi\rangle| = \left| \int_{B_R} S_n(3\varphi + y \cdot \nabla\varphi)\, dy \right| \le C_R \|S_n\|_{L^2(B_R)} \|\varphi\|_{H^1(B_R)},$$

and

$$|\langle b_n \cdot \nabla S_n, \varphi\rangle| \le |b_n|\, \|S_n\|_{L^2(B_R)} \|\nabla\varphi\|_{L^2(B_R)}.$$

The term S_n itself is controlled by the same L^2-convergence. Since a_n and b_n are bounded, the sign-convention difference tends to zero in $L^1_\tau H^{-1}_y(B_R)$. □

Lemma 8.20 (Terminal modulation compatibility)**.** *Same-point larger-scale profiles in a terminal frame have the form*

$$W_j^n(y,\tau) = \mu_j^n U^j(\theta_j^n(\tau), \xi_j^n + \mu_j^n y), \qquad \mu_j^n \to 0.$$

When the profile is regular at the observed offset, its leading term is a spatially constant ambient velocity. Absorbing this constant into the translation coefficient replaces b_n by $b_n - b_{\mathrm{amb},n}$, and the remaining terms vanish on compact terminal cylinders. If the offset escapes, the entire profile is locally invisible and no absorption is needed. In particular,

$$\sup_n \sup_{\tau\in I} \big(|a_n(\tau)| + |b_n(\tau)|\big) < \infty.$$

Proof. The local invisibility of the nonconstant remainder follows from Theorem 8.15 and the H^{-1} product estimates in Theorem 8.19. Separated physical-point profiles are locally invisible by Theorem 8.5; their pressures and cutoff commutators vanish by Theorem 8.18 and Theorem 8.19. Therefore only the spatially constant ambient velocity changes the translation coefficient, and after this absorption the effective modulation coefficients remain uniformly bounded. □

Lemma 8.21 (Terminal compact scale-collapse admissibility)**.** *The retained terminal solution U_n has positive finite critical mass, pressure reconstruction, bounded modulation, local windowed H^1 control, critical tightness on the selected windows, and the compactness input used by the compact scale-collapse argument.*

Proof. The terminal class is active, so after the perturbative remainder and nonretained profiles have been removed,

$$\inf_{\tau\in I} \|U_n(\tau)\|_{L^3(B_R)} \ge c_{\mathrm{act}} > 0$$

on the selected compact windows, after passing to the subsequences used in the profile decomposition. The upper bound follows from the retained local compact-cylinder bound:

$$\sup_{\tau\in I} \|U_n(\tau)\|_{L^3(B_R)} \le C(M).$$

The terminal profile completeness also gives the required tightness. If U_n were not uniformly L^3-tight on the selected windows, then there would be $\varepsilon > 0$, radii $R_k \to \infty$, and times τ_n such that

$$\int_{|y|>R_k} |U_n(y,\tau_n)|^3\, dy \ge \varepsilon.$$

Applying the profile decomposition to this escaping sequence produces either an exterior-regular component, which is removed by the exterior regularity argument below, or an additional active profile at a different physical point or larger same-point scale. Such a profile belongs to S_n and is handled by terminal decoupling. Hence no non-tight retained terminal solution remains.

The repaired-gauge representation gives pressure reconstruction and bounded modulation. The local Caccioppoli estimate gives local windowed H^1 control, and the Aubin–Lions compactness layer gives the compactness input on the selected windows. □

Theorem 8.22 (Terminal nonlinear profile decoupling theorem)**.** *Assume finite critical norm, terminal windowed profile completeness, small-data perturbative stability, the exterior regularity input, repaired-gauge representation, and local Caccioppoli regularity. Then terminal nonlinear profile decoupling holds in the sense of Theorem 8.11.*

Proof. Let $\mathcal{C}$ be any terminal active compound class, let U_n be the retained repaired-gauge nonlinear solution in its terminal frame, and set

$$S_n = V_n - U_n.$$

Theorem 8.15 proves

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_{2R})$$

on every compact terminal cylinder. Divergence-free compatibility follows from the divergence-free profile data and the divergence-free Navier–Stokes evolution of each nonlinear profile.

Theorem 8.17 proves

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

The finite critical norm and the retained profile bounds give the global $L^\infty_\tau L^3_y$ control required in the pressure lemma below. Theorem 8.18 proves

$$\nabla P_{\text{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R)$$

after subtracting spatially constant pressure terms. Theorem 8.19 proves

$$\partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y\cdot\nabla S_n) + b_n\cdot\nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Theorem 8.20 gives modulation compatibility, and Theorem 8.21 gives compact scale-collapse admissibility of the retained terminal solution. These are the items in Theorem 8.11. □

Corollary 8.23 (Multibubble exclusion from terminal profile decoupling)**.** *Under the Type II profile data, terminal windowed profile completeness, small-data perturbative stability, the exterior regularity input, repaired-gauge representation, local Caccioppoli regularity, and the compact scale-collapse stationary-rigidity input, no active multibubble Type II configuration occurs.*

Proof. Theorem 8.22 gives terminal nonlinear profile decoupling. Theorem 8.12 then supplies the same-point and separated-point decoupling hypotheses used in Theorem 8.7. The active-profile mass floor follows from Theorem 8.27 and the small-data perturbative stability assumption. Applying Theorem 8.7 gives the claimed exclusion of active multibubble configurations. □

8.5. **Small critical profiles.** For $I = [0,T]$ or $I = [0,\infty)$, define the Kato critical space $X(I)$ by

$$\|u\|_{X(I)} := \sup_{t\in I}\|u(t)\|_{L^3_x} + \sup_{\substack{t\in I\\ t>0}} t^{1/8}\|u(t)\|_{L^4_x}.$$

Let $\mathbb{P}$ be the Leray projector and set

$$B(u,v)(t) := -\int_0^t e^{\nu(t-s)\Delta}\mathbb{P}\nabla\cdot(u\otimes v)(s)\,ds.$$

The critical mild formulation is

$$u(t) = e^{\nu t\Delta}u_0 + B(u,u)(t).$$

Lemma 8.24 (Critical heat and bilinear estimates)**.** *There are constants $A_\nu, B_\nu < \infty$ such that for every divergence-free $u_0 \in L^3(\mathbb{R}^3)$,*

$$\|e^{\nu t\Delta}u_0\|_{X([0,\infty))} \le A_\nu\|u_0\|_{L^3},$$

and for all $u, v \in X([0,\infty))$,

$$\|B(u,v)\|_{X([0,\infty))} \le B_\nu \|u\|_{X([0,\infty))} \|v\|_{X([0,\infty))}.$$

Proof. The $L^\infty_t L^3_x$ heat estimate follows from heat semigroup contractivity:

$$\|e^{\nu t\Delta} u_0\|_3 \le \|u_0\|_3.$$

The L^4 Kato estimate follows from the heat kernel bound

$$\|e^{\nu t\Delta} f\|_q \le C_{\nu,p,q} t^{-\frac{3}{2}(\frac{1}{p}-\frac{1}{q})} \|f\|_p,$$

with $p = 3$, $q = 4$:

$$\|e^{\nu t\Delta} u_0\|_4 \le C_\nu t^{-1/8} \|u_0\|_3.$$

Multiplying by $t^{1/8}$ and taking the supremum gives the heat estimate in X.

For the bilinear term, the kernel of $e^{\nu(t-s)\Delta}\mathbb{P}\nabla\cdot$ satisfies

$$\|e^{\nu(t-s)\Delta}\mathbb{P}\nabla \cdot F\|_q \le C_{\nu,p,q}(t-s)^{-\frac{1}{2}-\frac{3}{2}(\frac{1}{p}-\frac{1}{q})} \|F\|_p.$$

With $F = u \otimes v$, Holder's inequality gives

$$\|u(s) \otimes v(s)\|_2 \le \|u(s)\|_4 \|v(s)\|_4 \le s^{-1/4} \|u\|_X \|v\|_X.$$

Using $p = 2, q = 3$,

$$\|B(u,v)(t)\|_3 \le C\|u\|_X\|v\|_X \int_0^t (t-s)^{-3/4} s^{-1/4}\, ds \le C\|u\|_X\|v\|_X.$$

Using $p = 2, q = 4$,

$$\|B(u,v)(t)\|_4 \le C\|u\|_X\|v\|_X \int_0^t (t-s)^{-7/8} s^{-1/4}\, ds \le Ct^{-1/8}\|u\|_X\|v\|_X.$$

Multiplying by $t^{1/8}$, taking the supremum in t, and combining the two components gives the bilinear estimate. □

Theorem 8.25 (Small-data critical Navier–Stokes)**.** *There exists $\varepsilon_{\rm sd} > 0$ such that if $u_0 \in L^3(\mathbb{R}^3)$ is divergence-free and*

$$\|u_0\|_3 \le \varepsilon_{\rm sd},$$

then Navier–Stokes has a unique global mild solution $u \in X([0,\infty))$ satisfying

$$\|u\|_{X([0,\infty))} \le 2A_\nu \|u_0\|_3.$$

Moreover, for every finite interval $[0,T]$,

$$\|u - e^{\nu t\Delta} u_0\|_{X([0,T])} \le C_\nu \|u_0\|_3^2.$$

Proof. Choose

$$\varepsilon_{\rm sd} \le \frac{1}{4A_\nu B_\nu}.$$

Let

$$R = 2A_\nu \|u_0\|_3,$$

and consider the closed ball

$$\mathcal{B}_R = \{u \in X([0,\infty)) : \|u\|_X \le R\}.$$

Define

$$\Phi(u) = e^{\nu t\Delta} u_0 + B(u,u).$$

For $u \in \mathcal{B}_R$, Theorem 8.24 gives

$$\|\Phi(u)\|_X \le A_\nu \|u_0\|_3 + B_\nu R^2.$$

Since $R = 2A_\nu \|u_0\|_3$,

$$B_\nu R^2 = 4B_\nu A_\nu^2 \|u_0\|_3^2 \le A_\nu \|u_0\|_3$$

by the choice of $\varepsilon_{\mathrm{sd}}$. Hence

$$\|\Phi(u)\|_X \le R.$$

For $u, v \in \mathcal{B}_R$,

$$\Phi(u) - \Phi(v) = B(u - v, u) + B(v, u - v),$$

so

$$\|\Phi(u) - \Phi(v)\|_X \le B_\nu(\|u\|_X + \|v\|_X)\|u - v\|_X \le 2B_\nu R\|u - v\|_X.$$

The smallness condition gives

$$2B_\nu R = 4A_\nu B_\nu \|u_0\|_3 \le \frac{1}{2}.$$

Thus Φ is a contraction on $\mathcal{B}_R$, and Banach's fixed point theorem gives a unique global mild solution with the asserted bound.

The perturbative estimate follows from

$$u - e^{\nu t\Delta} u_0 = B(u, u)$$

and Theorem 8.24:

$$\|u - e^{\nu t\Delta} u_0\|_X \le B_\nu \|u\|_X^2 \le 4B_\nu A_\nu^2 \|u_0\|_3^2.$$

□

Corollary 8.26 (Small profiles are perturbative)**.** *If $\phi \in L^3(\mathbb{R}^3)$ is divergence-free and*

$$\|\phi\|_3 \le \varepsilon_{\mathrm{sd}},$$

then the nonlinear Navier–Stokes profile generated by ϕ is global in $X([0, \infty))$, obeys the estimate of Theorem 8.25, and is perturbative on every compact renormalized window. In particular, ϕ cannot be an active Type II profile.

Proof. Theorem 8.25 gives the whole-space mild solution u^ϕ with

$$\|u^\phi\|_{X([0,\infty))} \le 2A_\nu \|\phi\|_3.$$

The X-norm is invariant under Navier–Stokes scaling

$$u(x, t) \mapsto \lambda u(\lambda x, \lambda^2 t).$$

Therefore every rescaled copy of u^ϕ remains uniformly small in the same critical topology on compact windows. The perturbative estimate shows that its nonlinear contribution is controlled by the heat evolution plus an $O(\|\phi\|_3^2)$ error. Such a profile is placed in the perturbative remainder class and cannot be an active singular profile. □

Theorem 8.27 (Positive critical mass floor for active profiles)**.** *For every critical Navier–Stokes profile family satisfying the small-data stability condition, every active profile satisfies*

$$\phi^j \ \textit{active} \quad \Longrightarrow \quad \|\phi^j\|_3 > \varepsilon_{\mathrm{sd}}.$$

Proof. Let ϕ^j be a profile with

$$\|\phi^j\|_3 \le \varepsilon_{\mathrm{sd}}.$$

By Theorem 8.26, its nonlinear evolution is global and perturbative in the critical stability space. Active Type II profiles are those not removed by the perturbative small-data class. Therefore ϕ^j is not active. Taking the contrapositive gives the displayed mass floor. □

8.6. **Exterior regularity.**

Assumption 8.28 (Exterior regularity near a single Type II core)**.** Let u, p solve Navier–Stokes on $\mathbb{R}^3 \times [0, T^*)$, and let (x_*, T^*) be the selected Type II core point. For every $r > 0$, every $R < \infty$, and every integer $k \geq 0$, there are $\delta > 0$ and $C_{r,R,k} < \infty$ such that

$$\sup_{t \in (T^*-\delta, T^*)} \Big(\|u(t)\|_{C^k(\overline{B_R} \setminus B_r(x_*))} + \|p(t)\|_{C^k(\overline{B_R} \setminus B_r(x_*))} \Big) \leq C_{r,R,k}.$$

Lemma 8.29 (Divergence-free core and exterior decomposition)**.** *Fix $0 < r_0 < R_0 < \infty$. Let $\chi \in C_c^\infty(B_{2r_0}(x_*))$ satisfy*

$$\chi \equiv 1 \quad \text{on } B_{r_0}(x_*).$$

Then there are divergence-free fields u_{core} and u_{ext} such that

$$u = u_{\text{core}} + u_{\text{ext}}, \qquad u_{\text{core}} = u \text{ on } B_{r_0}(x_*), \qquad u_{\text{ext}} = 0 \text{ on } B_{r_0}(x_*),$$

and all derivatives of u_{ext} are uniformly bounded on compact exterior spacetime regions up to T^*.

Proof. Set

$$g = \nabla\chi \cdot u.$$

The support of g lies in the annulus

$$A_{r_0} = B_{2r_0}(x_*) \setminus B_{r_0}(x_*),$$

where u is smooth by Theorem 8.28. Since

$$\int_{A_{r_0}} g \, dx = \int_{\mathbb{R}^3} \nabla\chi \cdot u \, dx = -\int_{\mathbb{R}^3} \chi \, \nabla \cdot u \, dx = 0,$$

the Bogovskii operator on A_{r_0} gives a vector field

$$w = \mathcal{B}_{A_{r_0}} g$$

supported in A_{r_0}, smooth up to T^*, and satisfying

$$\nabla \cdot w = g.$$

Define

$$u_{\text{core}} := \chi u - w, \qquad u_{\text{ext}} := u - u_{\text{core}}.$$

Then

$$\nabla \cdot u_{\text{core}} = \nabla\chi \cdot u + \chi \nabla \cdot u - \nabla \cdot w = 0,$$

and therefore $\nabla \cdot u_{\text{ext}} = 0$. Since $\chi = 1$ and $w = 0$ on $B_{r_0}(x_*)$, $u_{\text{core}} = u$ and $u_{\text{ext}} = 0$ there. The uniform exterior bounds follow from Theorem 8.28 and standard boundedness of the Bogovskii construction on the fixed annulus. □

Lemma 8.30 (Exterior fields vanish in compact core frames)**.** *Let (x_n, λ_n) be a core frame with*

$$x_n \to x_*, \qquad \lambda_n \to 0.$$

For a compact frame ball B_R, define

$$V^n_{\text{ext}}(y, \tau) := \lambda_n u_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 \tau).$$

Then, for every compact $B_R \times I$,

$$V^n_{\text{ext}} \equiv 0$$

on $B_R \times I$ for all sufficiently large n. Consequently

$$V^n_{\text{ext}} \to 0 \quad \text{in every local } L^q_\tau W^{k,q}_y(B_R) \text{ norm.}$$

Proof. Since $x_n \to x_*$ and $\lambda_n \to 0$, for fixed R,

$$x_n + \lambda_n B_R \subset B_{r_0}(x_*)$$

for all sufficiently large n. On $B_{r_0}(x_*)$,

$$u_{\text{ext}} = 0.$$

Therefore $V^n_{\text{ext}} = 0$ on $B_R \times I$ for all large n, which implies convergence in every stated local norm. □

Lemma 8.31 (Exterior pressure vanishes modulo constants). *Let p_{ext} be any pressure contribution generated by source terms supported in*

$$\{|x - x_*| \geq r_0\}.$$

In a core frame set

$$P^n_{\text{ext}}(y,\tau) := \lambda_n^2 p_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 \tau).$$

Then, for every compact $B_R \times I$, there are constants $c_{n,R}(\tau)$ such that

$$P^n_{\text{ext}} - c_{n,R} \to 0 \quad \text{in } L^\infty_\tau C^1_y(B_R),$$

and hence

$$\nabla_y P^n_{\text{ext}} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. The exterior pressure source is at positive physical distance from

$$x_n + \lambda_n B_R$$

for all sufficiently large n. Therefore p_{ext} is harmonic, smooth, and uniformly C^2-bounded on the physical neighborhood swept out by the compact frame cylinder. Take

$$c_{n,R}(\tau) := \lambda_n^2 p_{\text{ext}}(x_n, t_n + \lambda_n^2 \tau).$$

By the mean value theorem,

$$\|P^n_{\text{ext}} - c_{n,R}\|_{C^1_y(B_R)} \leq C_R \lambda_n^3 \sup |\nabla_x p_{\text{ext}}| + C_R \lambda_n^4 \sup |\nabla_x^2 p_{\text{ext}}|.$$

The right-hand side tends to zero because the physical derivatives are uniformly bounded and $\lambda_n \to 0$. This proves the asserted convergence. □

Lemma 8.32 (Positive-distance profiles are exterior). *Let (x_n, λ_n) be a core frame with $x_n \to x_*$ and $\lambda_n \to 0$. Let (z_n, μ_n) be a profile sequence satisfying*

$$\liminf_{n\to\infty} |z_n - x_*| \geq r_* > 0.$$

Then this profile is invisible in every compact core frame:

$$\Lambda^{-1}_{\lambda_n,x_n} \Lambda_{\mu_n,z_n} \phi \to 0$$

locally in $L^3_y(B_R)$, and its pressure contribution vanishes modulo constants as in Theorem 8.31. Hence it is an exterior regular contribution, not part of the Type II core.

Proof. For fixed $R < \infty$, the physical frame ball

$$x_n + \lambda_n B_R$$

lies in $B_{r_*/4}(x_*)$ for all sufficiently large n, while the profile center z_n lies outside $B_{r_*/2}(x_*)$. Thus the profile is a positive physical distance from the compact core frame. The exterior velocity part is covered by Theorem 8.30 after the core/exterior decomposition, and the exterior pressure part is covered by Theorem 8.31. The conclusion follows. □

Theorem 8.33 (Exterior regular contributions are removed)**.** *Under Theorem 8.28, every component carried at a positive physical distance from the selected Type II core point is assigned to the exterior regular class and is invisible in compact Type II core frames, including its pressure contribution modulo constants.*

Proof. Use the divergence-free decomposition of Theorem 8.29. The exterior component is identically zero in every compact core frame for large n by Theorem 8.30. Its pressure contribution is a locally scaled smooth harmonic field and vanishes modulo constants by Theorem 8.31. The cutoff and Bogovskii correction terms are supported in the fixed exterior annulus A_{r_0}, so they are also invisible in compact core frames by the same argument.

Thus exterior regular mass can be removed from the Type II core without changing the local repaired-gauge equation in any compact terminal frame, except by errors converging to zero in the perturbative topology used in Theorem 8.8 and Theorem 8.11. $\square$

9. Corrected Monotonicity and Scale-Drift Cost Criteria

This section records the remaining conditional analytic inputs needed for a localized cost-divergence exclusion argument on renormalized Type II branches. The statements are written entirely in the variables of the repaired gauge:

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0.$$

All pressure terms are understood modulo functions of time.

9.1. **Hypotheses for a Conditional Cost Criterion.**

Definition 9.1 (Conditional cost hypotheses)**.** A renormalized local Type II branch satisfies the conditional cost hypotheses if the following analytic facts hold.

(i) the branch lies in the local Type II concentration alternative;
(ii) the physical Navier–Stokes energy inequality gives bounded energy along the branch;
(iii) the branch is written in the repaired-gauge variables (V, P, a, b) above, including the scale, center, pressure normalization, and modulation data;
(iv) the localized cost is identified with

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu\int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2\phi_{R_0}(y)\,dy + a_+(\tau)\int_{\mathbb{R}^3}|V(y,\tau)|^2\phi_{R_0}(y)\,dy,$$

or with an explicitly stated tail-comparable nonnegative replacement;
(v) a localized energy identity or suitable-solution local energy inequality is available for the cutoff used in the cost;
(vi) a corrected localized monotonicity inequality, in the sense of Theorem 9.6, is available.

Assumption 9.2 (Cost-divergence exclusion criterion)**.** For every renormalized local Type II branch satisfying Theorem 9.1, divergence of the corresponding nonnegative localized cost,

$$\int_{\tau_0}^{\infty} \mathcal{D}_{R_0}(\tau)\,d\tau = \infty,$$

excludes that branch from the local Type II class under consideration.

Lemma 9.3 (Concentration and energy)**.** *If a branch lies in the local Type II concentration alternative and satisfies the physical Navier–Stokes energy inequality, then it satisfies Theorem 9.1*(i) *and* (ii)*.*

Proof. The local Type II concentration alternative supplies the selected concentration branch. The physical energy inequality gives the bounded-energy part. These are precisely items (i) and (ii). □

Lemma 9.4 (Cost identification)**.** *Assume the localized cost is the Navier–Stokes cost $\tilde{\mathfrak{D}}_{R_0}$. Then divergence of*

$$\int_{\tau_0}^{\infty}\tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau$$

is the cost divergence in Theorem 9.2.

Proof. The active cost is assumed to be equal to $\tilde{\mathfrak{D}}_{R_0}$. Thus the cost appearing in the local Navier–Stokes estimates and the cost appearing in the conditional exclusion criterion are the same nonnegative functional. The divergence condition is therefore identical in the two formulations. □

Lemma 9.5 (Parabolic representation class)**.** *If the repaired-gauge representation theorem applies to a Type II branch, then the branch has the parabolic transport-diffusion structure required in Theorem 9.1.*

Proof. The repaired-gauge theorem supplies the renormalized orbit (V, P, a, b), the scale and center, the pressure reconstruction modulo functions of time, and the modulation coefficients. The displayed equation is a parabolic Navier–Stokes equation with transport, dilation, and translation drift terms. Consequently the branch has the parabolic transport-diffusion structure required by the conditional cost hypotheses. $\square$

Definition 9.6 (Corrected localized monotonicity input)**.** A renormalized branch has corrected localized monotonicity if there are a localized renormalized energy $\mathcal{E}_{R_0}$, a nonnegative cost $\mathcal{D}_{R_0}$, a constant $c_0 > 0$, and a nonnegative remainder $\mathcal{R}_{R_0}$ such that, on the tail of the branch,

$$\frac{d}{d\tau}\mathcal{E}_{R_0}(\tau) + c_0\mathcal{D}_{R_0}(\tau) + \mathcal{R}_{R_0}(\tau) \le 0$$

in distributions. If the localized energy identity has an integrable error B_{R_0}, the corrected energy is

$$\mathcal{E}^{\mathrm{corr}}_{R_0}(\tau) = \mathcal{E}_{R_0}(\tau) + \int_\tau^\infty B_{R_0}(s)\, ds,$$

and the same inequality is required for $\mathcal{E}^{\mathrm{corr}}_{R_0}$. Multiplication of a nonnegative cost by $c_0 > 0$ does not change whether its tail integral diverges.

Lemma 9.7 (Energy and Caccioppoli bounds alone are not monotonicity)**.** *The global physical energy inequality and compact-cylinder Caccioppoli estimates do not, by themselves, imply the corrected localized monotonicity input of Theorem 9.6.*

Proof. The global physical energy inequality controls total physical energy and physical dissipation, but it does not identify a localized renormalized energy at the selected Type II scale whose derivative is monotone after pressure flux, cutoff flux, transport, and modulation terms are included. The Caccioppoli estimate gives local spacetime H^1 control from the local energy inequality. It is an a priori estimate, not a corrected monotonicity formula. Therefore an additional absorption or tail-correction argument is needed. $\square$

Proposition 9.8 (Conditional cost-divergence exclusion)**.** *Let a renormalized Type II branch satisfy Theorem 9.1 and Theorem 9.2. If its nonnegative localized cost has divergent tail integral, then the branch is excluded from the local Type II class under consideration.*

Proof. Theorem 9.3 gives the local Type II concentration and bounded-energy inputs. Theorem 9.4 identifies the localized Navier–Stokes cost with the cost used in the exclusion criterion. Theorem 9.5 gives the parabolic transport-diffusion class. The corrected monotonicity input is an explicit hypothesis of Theorem 9.1. Once the cost diverges, Theorem 9.2 gives the stated exclusion. $\square$

9.2. **Fixed-Cutoff Monotonicity.** Let $\phi = \phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$, $0 \le \phi \le 1$, be the cutoff used in the localized cost. Define

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi(y)\, dy,$$

$$A_\phi(\tau) = \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi(y)\, dy, \qquad M_\phi(\tau) = \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi(y)\, dy,$$

and

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu A_\phi(\tau) + a_+(\tau) M_\phi(\tau), \qquad a_+(\tau) = \max\{a(\tau), 0\}.$$

The localized energy identity used below is

$$\begin{aligned}\frac{d}{d\tau}\mathcal{E}_\phi = &-\nu A_\phi + \frac{\nu}{2}\int |V|^2\Delta\phi + \frac{1}{2}\int |V|^2 V\cdot\nabla\phi + \int P\, V\cdot\nabla\phi \\ &-\frac{a}{2}M_\phi - \frac{a}{2}\int |V|^2 y\cdot\nabla\phi - \frac{1}{2}b\cdot\int |V|^2\nabla\phi.\end{aligned} \tag{13}$$

Definition 9.9 (Fixed-cutoff error density)**.** Set $a_-(\tau) = \max\{-a(\tau), 0\}$ and define

$$B_{R_0}(\tau) := \frac{\nu}{2}\left|\int |V|^2\Delta\phi\right| + \frac{1}{2}\left|\int |V|^2 V\cdot\nabla\phi\right| + \left|\int P\,V\cdot\nabla\phi\right| \\ + \frac{1}{2}|b(\tau)|\left|\int |V|^2\nabla\phi\right| + \frac{1}{2}a_-(\tau)M_\phi(\tau) + \frac{1}{2}|a(\tau)|\left|\int |V|^2 y\cdot\nabla\phi\right|.$$

The fixed-cutoff finite-error condition is

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\,d\tau < \infty.$$

Remark 9.10 (Pressure normalization)*.* The pressure term in B_{R_0} is independent of the representative of the pressure modulo functions of time. Replacing P by $P + c(\tau)$ changes the pressure flux by

$$c(\tau)\int V\cdot\nabla\phi = -c(\tau)\int \phi\,\nabla\cdot V = 0.$$

Theorem 9.11 (Fixed-cutoff corrected monotonicity)**.** *Assume* (13) *holds on* (τ_0,∞) *and that* $B_{R_0} \in L^1(\tau_0,\infty)$*. Define*

$$\mathcal{E}_\phi^{\mathrm{corr}}(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\,ds.$$

Then, in distributions on (τ_0,∞)*,*

$$\frac{d}{d\tau}\mathcal{E}_\phi^{\mathrm{corr}}(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le 0.$$

Proof. Add

$$\frac{1}{2}\tilde{\mathfrak{D}}_{R_0} = \frac{\nu}{2}A_\phi + \frac{1}{2}a_+M_\phi$$

to both sides of (13). The gradient term becomes

$$-\nu A_\phi + \frac{\nu}{2}A_\phi = -\frac{\nu}{2}A_\phi \le 0.$$

The core scale term satisfies

$$-\frac{a}{2}M_\phi + \frac{1}{2}a_+M_\phi = \frac{1}{2}a_-M_\phi \le B_{R_0}(\tau).$$

Every remaining term in (13) is bounded above by its corresponding absolute-value contribution in B_{R_0}. Therefore

$$\frac{d}{d\tau}\mathcal{E}_\phi(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le B_{R_0}(\tau)$$

in distributions. Since $B_{R_0} \in L^1(\tau_0,\infty)$, the correction tail is finite for every $\tau \ge \tau_0$, absolutely continuous on compact subintervals, and

$$\frac{d}{d\tau}\int_\tau^\infty B_{R_0}(s)\,ds = -B_{R_0}(\tau).$$

Adding this identity gives the claimed inequality. □

Corollary 9.12 (Fixed-cutoff monotonicity input)**.** *If the repaired-gauge representation, the cost identification, the localized energy identity, and the fixed-cutoff finite-error condition all hold, then Theorem 9.6 holds with* $\mathcal{D}_{R_0} = \tilde{\mathfrak{D}}_{R_0}$ *and* $c_0 = 1/2$*.*

Proof. The repaired-gauge representation supplies (V, P, a, b). The cost identification supplies $\tilde{\mathfrak{D}}_{R_0}$. The localized energy identity and finite-error condition allow Theorem 9.11 to be applied. □

Remark 9.13 (Finite-tail subconditions)**.** The fixed-cutoff finite-error condition is equivalent to the conjunction of tail integrability for the viscous cutoff term, the convective flux, the pressure flux, the center-drift cutoff term, the negative-scale term, and the scale-cutoff term:

$$\left|\int |V|^2\Delta\phi\right|, \qquad \left|\int |V|^2 V\cdot\nabla\phi\right|, \qquad \left|\int P\,V\cdot\nabla\phi\right|,$$
$$|b|\left|\int |V|^2\nabla\phi\right|, \qquad a_- M_\phi, \qquad |a|\left|\int |V|^2 y\cdot\nabla\phi\right|.$$

Uniform compact-window bounds do not by themselves give this global tail integrability on $[\tau_0,\infty)$.

Definition 9.14 (Fixed-cutoff cost hypotheses)**.** A renormalized Type II branch satisfies the fixed-cutoff cost hypotheses if it has the local Type II concentration alternative, bounded physical energy, repaired-gauge representation, cost identification with $\tilde{\mathfrak{D}}_{R_0}$, divergent nonnegative localized cost, localized energy identity, and the finite-tail bound $\int_{\tau_0}^{\infty} B_{R_0} < \infty$.

Theorem 9.15 (Fixed-cutoff cost criterion)**.** *Every branch satisfying the fixed-cutoff cost hypotheses is covered by Theorem 9.8.*

Proof. By Theorem 9.12, the finite-tail condition supplies the corrected monotonicity input. The remaining assumptions are precisely the other items in Theorem 9.1, together with cost divergence and Theorem 9.2. □

Definition 9.16 (Compact cost-divergence replacement)**.** A compact cost-divergence replacement is available when the compact Type II hypotheses of the preceding sections imply

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau = \infty,$$

and this divergence is the cost divergence used in Theorem 9.8. In particular, it may be supplied by the combination of classification of local alternatives, uniform critical tightness, local windowed H^1 control, and the cost identification above whenever those are the active hypotheses in the compact Type II theorem.

Theorem 9.17 (Compact cost-divergence criterion)**.** *If a renormalized branch has all fixed-cutoff cost hypotheses except that cost divergence is supplied through the compact cost-divergence replacement of Theorem 9.16, then Theorem 9.8 applies.*

Proof. The compact cost-divergence replacement supplies the missing divergent cost. Substituting this into the fixed-cutoff hypotheses gives the hypotheses of Theorem 9.15. □

9.3. **Moving-Cutoff Monotonicity.** Let $\phi \in C_c^\infty(\mathbb{R}^3)$ be radial, $0 \le \phi \le 1$, $\phi = 1$ on B_1, and $\phi = 0$ outside B_2. For $R(\tau) \ge R_0$, locally absolutely continuous and nondecreasing, set

$$\Phi(\tau,y) = \phi(y/R(\tau)), \qquad A_{R(\tau)} = \{R(\tau) \le |y| \le 2R(\tau)\},$$
$$\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\Phi(\tau,y)\,dy,$$

and

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu\int |\nabla V|^2\Phi + a_+(\tau)\int |V|^2\Phi.$$

Definition 9.18 (Moving-cutoff error density)**.** Define

$$B_{\mathrm{mov}}(\tau) := \frac{1}{2}\left|\int |V|^2\partial_\tau\Phi\right| + \frac{\nu}{2}\left|\int |V|^2\Delta\Phi\right| + \frac{1}{2}\left|\int |V|^2 V\cdot\nabla\Phi\right| + \left|\int P\,V\cdot\nabla\Phi\right|$$

$$+\frac{1}{2}|b(\tau)|\left|\int |V|^2\nabla\Phi\right| + \frac{1}{2}a_-(\tau)\int |V|^2\Phi + \frac{1}{2}|a(\tau)|\left|\int |V|^2 y\cdot\nabla\Phi\right|.$$

The moving finite-error condition is

$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\,d\tau < \infty.$$

Theorem 9.19 (Moving-cutoff corrected monotonicity)**.** *Assume the localized energy identity is valid with the time-dependent cutoff Φ, and assume $B_{\mathrm{mov}} \in L^1(\tau_0,\infty)$. Then*

$$\mathcal{E}_R^{\mathrm{corr}}(\tau) = \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\mathrm{mov}}(s)\,ds$$

satisfies

$$\frac{d}{d\tau}\mathcal{E}_R^{\mathrm{corr}}(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le 0$$

in distributions.

Proof. Use the localized energy identity with the time-dependent test function $\Phi(\tau,y)$. Compared with the fixed-cutoff identity, the only new term is

$$\frac{1}{2}\int |V|^2\partial_\tau\Phi,$$

which comes from differentiating the cutoff inside $\mathcal{E}_R$. All other terms are the fixed-cutoff terms with ϕ replaced by Φ. Add $\frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}$ to both sides. The gradient term leaves

$$-\frac{\nu}{2}\int |\nabla V|^2\Phi \le 0,$$

and the core scale term contributes

$$\frac{1}{2}a_-\int |V|^2\Phi.$$

Each remaining term is bounded above by the corresponding absolute-value term in B_{mov}. Hence

$$\frac{d}{d\tau}\mathcal{E}_R(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le B_{\mathrm{mov}}(\tau).$$

The moving finite-error condition makes the correction tail finite and absolutely continuous on compact subintervals. Differentiating that tail subtracts B_{mov}, giving the stated inequality. ☐

Definition 9.20 (Summable moving-annulus errors)**.** The moving-annulus summability condition is the existence of a nondecreasing locally absolutely continuous $R(\tau) \to \infty$ such that each of the seven quantities

$$\left|\int |V|^2\partial_\tau\Phi\right|,\quad \left|\int |V|^2\Delta\Phi\right|,\quad \left|\int |V|^2 V\cdot\nabla\Phi\right|,\quad \left|\int P\,V\cdot\nabla\Phi\right|,$$
$$|b|\left|\int |V|^2\nabla\Phi\right|,\quad a_-\int |V|^2\Phi,\quad |a|\left|\int |V|^2 y\cdot\nabla\Phi\right|$$

is integrable on $[\tau_0,\infty)$.

Lemma 9.21 (Summable annuli imply moving finite error)**.** *If Theorem 9.20 holds, then $B_{\mathrm{mov}} \in L^1(\tau_0,\infty)$.*

Proof. The density B_{mov} is a finite positive linear combination of the seven quantities listed in Theorem 9.20. If each is integrable, then their linear combination is integrable. ☐

Definition 9.22 (Summable tightness schedule)**.** A summable tightness schedule is a nondecreasing sequence $R_n \to \infty$ such that, on unit windows $I_n = [\tau_0 + n, \tau_0 + n + 1]$,

$$\sum_{n=0}^{\infty} \sup_{\tau \in I_n} \|V(\tau)\|^3_{L^3(A_{R_n})} < \infty,$$

and the same schedule is compatible with the pressure-tail normalization used for P. This is stronger than uniform critical tightness, which permits large radii on each window but does not by itself give a monotone radius schedule with a summable series.

Definition 9.23 (Moving-cost compatibility)**.** The moving localized cost is compatible with the cost-divergence criterion if

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + a_+(\tau) \int |V|^2 \phi_{R(\tau)}$$

is the localized cost appearing in Theorem 9.2, or is compared to that cost by a positive constant on the tail so that cost divergence is unchanged.

Theorem 9.24 (Moving-cutoff replacement)**.** *Assume the repaired-gauge representation, localized energy identity, moving-cost compatibility, and summable moving-annulus errors. Then the moving cutoff supplies the corrected monotonicity input needed in Theorem 9.8.*

Proof. Theorem 9.21 gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$. Applying Theorem 9.19 gives corrected monotonicity with cost $\tilde{\mathfrak{D}}_{R(\tau)}$. The moving-cost compatibility assumption identifies this cost, or a tail-comparable version of it, with the localized cost in the conditional exclusion criterion. $\square$

9.4. **Negative Scale Drift.** Set

$$M_R(\tau) = \int |V(y,\tau)|^2 \Phi(\tau, y)\, dy.$$

The negative-scale contribution in the moving cutoff argument is

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau.$$

Definition 9.25 (Negative scale-drift alternatives)**.** We say that the branch has finite negative drift if

$$\int_{\tau_0}^{\infty} a_-(\tau)\, d\tau < \infty.$$

It has bounded moving L^2 core mass if

$$\sup_{\tau \geq \tau_0} M_R(\tau) < \infty.$$

It has a moving L^2 core floor if there are $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \qquad \text{for a.e. } \tau \geq \tau_1.$$

The scale-collapse drift obstruction is the divergent alternative

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Lemma 9.26 (Finite negative drift controls the scale term)**.** *If the branch has bounded moving L^2 core mass and finite negative drift, then*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau < \infty.$$

Proof. Let $M_* = \sup_{\tau \ge \tau_0} M_R(\tau) < \infty$. Since $a_- \ge 0$,

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau \le M_* \int_{\tau_0}^{\infty} a_-(\tau)\, d\tau < \infty.$$

□

Theorem 9.27 (Finite-or-infinite scale-negative alternative)**.** *For every renormalized orbit for which a_- and M_R are measurable and nonnegative, precisely one of the following alternatives holds:*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau < \infty, \qquad \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. The integral

$$I_R = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau$$

is an extended nonnegative number. Hence either $I_R < \infty$ or $I_R = \infty$, and the two cases are mutually exclusive. □

Lemma 9.28 (Logarithmic scale identity)**.** *For the repaired-gauge convention*

$$\tau_t = \lambda(t)^{-2}, \qquad a(\tau) = \lambda(t)\lambda_t(t),$$

one has

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1)$$

whenever $\tau_0 \le \tau_1 < \tau_2 < \infty$.

Proof. The chain rule gives

$$\frac{d}{d\tau} \log \lambda = \frac{\lambda_t}{\lambda} \frac{dt}{d\tau} = \frac{\lambda_t}{\lambda} \lambda^2 = \lambda \lambda_t = a.$$

Integrating over $[\tau_1, \tau_2]$ proves the identity. □

Theorem 9.29 (Collapse with core floor forces scale-drift obstruction)**.** *Assume*

$$\lambda(\tau) \to 0 \qquad \text{as } \tau \to \infty,$$

and assume the moving L^2 core floor of Theorem 9.25. Then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. By Theorem 9.28,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Since $\lambda(\tau) \to 0$, the right-hand side tends to $-\infty$ as $\tau_2 \to \infty$. If $\int_{\tau_1}^{\infty} a_-(\tau)\, d\tau < \infty$, then

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\, d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau \ge -\int_{\tau_1}^{\infty} a_-(\tau)\, d\tau > -\infty,$$

which contradicts the preceding limit. Thus $\int_{\tau_1}^{\infty} a_-(\tau)\, d\tau = \infty$.

By the moving L^2 core floor, $M_R(\tau) \ge m_0$ for a.e. $\tau \ge \tau_1$. Therefore

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau \ge m_0 \int_{\tau_1}^{\infty} a_-(\tau)\, d\tau = \infty.$$

□

Corollary 9.30 (Scale-drift alternatives)**.** *In the moving-cutoff setup, the negative-scale term has the following exhaustive alternatives. If bounded moving L^2 core mass and finite negative drift hold, then the negative-scale term is integrable. If the weighted negative-drift integral is infinite, the branch lies in the scale-collapse drift obstruction. If genuine collapse and a moving L^2 core floor also hold, then the obstruction is forced.*

Proof. The first case is Theorem 9.26. The exhaustive split is Theorem 9.27. The final assertion is Theorem 9.29. □

9.5. **Scale-Drift Cost Criteria.**

Definition 9.31 (Scale-collapse drift cost)**.** For a renormalized branch define

$$\mathfrak{C}_{\rm sc}(\tau) = a_-(\tau)M_R(\tau), \qquad M_R(\tau) = \int |V(y,\tau)|^2\phi_{R(\tau)}(y)\,dy,$$

and

$$\mathcal{C}_{\rm sc}[\tau_0,\infty) = \int_{\tau_0}^{\infty} \mathfrak{C}_{\rm sc}(\tau)\,d\tau.$$

Thus the scale-collapse drift obstruction is precisely $\mathcal{C}_{\rm sc}[\tau_0,\infty) = \infty$. The density is nonnegative and local in the renormalized core.

Definition 9.32 (Scale-collapse cost compatibility)**.** The scale-collapse drift cost is compatible with the cost-divergence criterion when:

(i) $\mathfrak{C}_{\rm sc}$ is measurable, nonnegative, and locally integrable on finite τ-intervals for renormalized branches;
(ii) divergence of $\mathcal{C}_{\rm sc}[\tau_0,\infty)$ is included among the hypotheses in Theorem 9.2;
(iii) the resulting conclusion is the same local Type II exclusion conclusion used for the fixed localized cost.

Theorem 9.33 (Scale-collapse drift cost criterion)**.** *If the scale-collapse drift obstruction holds and the cost $\mathfrak{C}_{\rm sc}$ is compatible with the cost-divergence criterion, then the divergent-cost hypothesis of Theorem 9.8 is satisfied for the scale-collapse branch.*

Proof. By Theorem 9.31, the obstruction is precisely

$$\mathcal{C}_{\rm sc}[\tau_0,\infty) = \infty.$$

By Theorem 9.32, this divergence is a nonnegative cost-divergence hypothesis with the same local conclusion. □

Corollary 9.34 (Scale-collapse exclusion by drift cost)**.** *Assume a renormalized local Type II branch has scale-collapse drift obstruction, compatible scale-collapse drift cost, and the remaining conditional cost hypotheses. If Theorem 9.2 holds, then the branch is excluded from the local Type II class under consideration.*

Proof. Theorem 9.33 supplies the divergent localized cost. Applying Theorem 9.8 gives the exclusion. □

Definition 9.35 (Absolute scale-drift cost)**.** Define

$$\mathfrak{C}_{\rm abs}(\tau) = \nu \int |\nabla V|^2\phi_{R(\tau)} + |a(\tau)|M_R(\tau).$$

The absolute scale-drift cost is compatible with the cost-divergence criterion if divergence of $\int_{\tau_0}^{\infty} \mathfrak{C}_{\rm abs}(\tau)\,d\tau$ is included as a Type II cost-divergence condition with the same local conclusion.

Lemma 9.36 (Scale-collapse drift forces absolute-cost divergence)**.** *If the scale-collapse drift obstruction holds, then*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{abs}}(\tau)\, d\tau = \infty.$$

Proof. Since

$$\mathfrak{C}_{\text{abs}}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + |a(\tau)| M_R(\tau) \geq a_-(\tau) M_R(\tau),$$

divergence of $\int a_- M_R$ forces divergence of $\int \mathfrak{C}_{\text{abs}}$. □

Corollary 9.37 (Absolute scale-cost criterion)**.** *If the scale-collapse drift obstruction holds and the absolute scale-drift cost is compatible with the cost-divergence criterion, then the divergent-cost hypothesis of Theorem 9.8 is satisfied.*

Proof. Theorem 9.36 gives divergence of the absolute scale-drift cost, and compatibility makes this divergence a Type II cost-divergence condition. □

Theorem 9.38 (Moving cutoff with scale-collapse alternative)**.** *Assume the moving-cutoff setup and a renormalized local Type II branch.*

(i) *If the negative-scale term is integrable and the remaining moving-annulus errors are summable, then the moving-cutoff corrected monotonicity estimate is available.*
(ii) *If the scale-collapse drift obstruction holds and either the scale-collapse drift cost or the absolute scale-drift cost is compatible with the cost-divergence criterion, then the divergent-cost hypothesis is satisfied directly.*
(iii) *If, in addition, the remaining hypotheses of Theorem 9.1 and Theorem 9.2 hold, then the branch is excluded.*

Proof. The first branch is Theorem 9.24. The second branch is Theorem 9.33 or Theorem 9.37, depending on which compatible cost is used. The third branch is Theorem 9.8. □

9.6. **Absolutely Continuous Repaired-Gauge Charts and Renormalized Equations.** The following statements isolate the parts of the repaired-gauge representation that follow from an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path. They do not prove extraction of the chart from arbitrary data and do not prove solvability of the gauge constraints.

Definition 9.39 (Absolutely continuous concentration chart)**.** Let u, p solve the Navier–Stokes equations on $\mathbb{R}^3 \times (t_0, T^*)$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0.$$

An absolutely continuous concentration chart consists of absolutely continuous functions

$$x_c : (t_0, T^*) \to \mathbb{R}^3, \qquad \lambda : (t_0, T^*) \to (0, \infty),$$

and a strictly increasing absolutely continuous time ρ such that

$$\rho_t = \lambda^{-2} \quad \text{a.e.},$$

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x,t) = \lambda(t)^{-1} U(y, \rho(t)),$$

and

$$p(x,t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t)$$

for some scalar function $c(t)$. The chart is assumed to have the local integrability required for the distributional chain rule on compact sets in the y-variables. Define

$$A(\rho(t)) := \lambda(t)\lambda_t(t), \qquad B(\rho(t)) := \lambda(t) x_c'(t),$$

or equivalently

$$A = \partial_\rho \log \lambda, \qquad B = \lambda^{-1}\partial_\rho x_c,$$

where λ and x_c are regarded as functions of ρ.

Lemma 9.40 (Initial renormalized equation)**.** *Assume Definition 9.39. Then U, Q satisfy*

$$\partial_\rho U + (U\cdot\nabla)U + \nabla Q = \nu\Delta U + A(\rho)(U + y\cdot\nabla U) + B(\rho)\cdot\nabla U, \qquad \nabla\cdot U = 0$$

distributionally on compact y-sets.

Proof. For smooth functions the computation is pointwise. The distributional case is obtained by testing against compactly supported functions and using the assumed chart chain rule. The spatial derivatives are

$$\nabla_x u = \lambda^{-2}\nabla_y U, \qquad \Delta_x u = \lambda^{-3}\Delta_y U, \qquad (u\cdot\nabla_x)u = \lambda^{-3}(U\cdot\nabla_y)U.$$

Since

$$y_t = -\frac{x_c'}{\lambda} - \frac{\lambda_t}{\lambda}y, \qquad \rho_t = \lambda^{-2},$$

one has

$$\partial_t u = \lambda^{-3}\partial_\rho U - \lambda^{-2}\lambda_t(U + y\cdot\nabla U) - \lambda^{-2}x_c'\cdot\nabla U.$$

Also

$$\nabla_x p = \lambda^{-3}\nabla_y Q.$$

Substitution into the physical Navier–Stokes equation and multiplication by λ^3 gives

$$\partial_\rho U - \lambda\lambda_t(U + y\cdot\nabla U) - \lambda x_c'\cdot\nabla U + (U\cdot\nabla)U + \nabla Q = \nu\Delta U.$$

Moving the two chart-velocity terms to the right gives the displayed equation. Finally,

$$\nabla_x\cdot u = \lambda^{-2}\nabla_y\cdot U,$$

so $\nabla_y\cdot U = 0$. $\square$

Definition 9.41 (Absolutely continuous repaired-gauge path)**.** An absolutely continuous repaired-gauge path for the initial chart consists, on every compact final time interval, of absolutely continuous functions

$$\mu(\tau) > 0, \qquad q(\tau)\in\mathbb{R}^3, \qquad \rho = \rho(\tau),$$

such that

$$\rho_\tau = \mu(\tau)^2 \quad \text{a.e.},$$
$$V(Y,\tau) = \mu(\tau)U(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

and

$$P(Y,\tau) = \mu(\tau)^2 Q(\mu(\tau)Y + q(\tau), \rho(\tau)).$$

The scale and centering constraints defining the repaired gauge are assumed to hold along this path. This hypothesis assumes the path has already been selected; it does not prove solvability of the gauge constraints.

Lemma 9.42 (Physical chart after gauge repair)**.** *Assume Definitions 9.39 and 9.41. Let $t = t(\rho(\tau))$, and define*

$$\Lambda(\tau) := \lambda(t(\rho(\tau)))\mu(\tau), \qquad X(\tau) := x_c(t(\rho(\tau))) + \lambda(t(\rho(\tau)))q(\tau).$$

Then, on every compact final time interval, Λ and X are absolutely continuous, $\Lambda > 0$, and

$$\frac{d\tau}{dt} = \Lambda(\tau)^{-2}.$$

Moreover,

$$x = X(\tau) + \Lambda(\tau)Y$$

is the final physical chart and

$$u(x,t) = \Lambda(\tau)^{-1}V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2}P(Y,\tau) + c(t).$$

Proof. The intermediate time ρ is strictly increasing because $\rho_t = \lambda^{-2} > 0$ a.e.; hence it has an absolutely continuous inverse on compact subintervals of its range. The compositions $\lambda(t(\rho(\tau)))$ and $x_c(t(\rho(\tau)))$ are therefore absolutely continuous on compact final windows. Products and sums with the absolutely continuous functions μ, q are absolutely continuous, so Λ and X are absolutely continuous.

The identity $\rho_\tau = \mu^2$ implies

$$\frac{d\tau}{d\rho} = \mu^{-2}.$$

Since $d\rho/dt = \lambda^{-2}$,

$$\frac{d\tau}{dt} = \frac{d\tau}{d\rho}\frac{d\rho}{dt} = \mu^{-2}\lambda^{-2} = (\lambda\mu)^{-2} = \Lambda^{-2}.$$

If $x = X + \Lambda Y$, then

$$x = x_c + \lambda q + \lambda\mu Y,$$

and hence the intermediate coordinate is

$$y = \frac{x - x_c}{\lambda} = q + \mu Y.$$

Therefore

$$u(x,t) = \lambda^{-1}U(q + \mu Y, \rho) = \lambda^{-1}\mu^{-1}V(Y,\tau) = \Lambda^{-1}V(Y,\tau),$$

and similarly

$$p(x,t) = \lambda^{-2}Q(q + \mu Y, \rho) + c(t) = \lambda^{-2}\mu^{-2}P(Y,\tau) + c(t) = \Lambda^{-2}P(Y,\tau) + c(t).$$

□

Lemma 9.43 (Modulation coefficients in repaired gauge)**.** *Define*

$$a(\tau) := \partial_\tau \log \Lambda(\tau), \qquad b(\tau) := \Lambda(\tau)^{-1}X_\tau(\tau),$$

where derivatives are understood a.e. on compact final windows. Under the hypotheses of Lemma 9.42,

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau)B(\rho(\tau)) + \mu(\tau)A(\rho(\tau))q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

In particular $a, b \in L^1_{\mathrm{loc}}$ on every compact final time interval.

Proof. Since $\Lambda = \lambda(\rho(\tau))\mu(\tau)$,

$$\partial_\tau \log \Lambda = \rho_\tau \partial_\rho \log \lambda + \partial_\tau \log \mu = \mu^2 A + \frac{\mu_\tau}{\mu}.$$

This proves the formula for a. Next

$$X = x_c(\rho(\tau)) + \lambda(\rho(\tau))q(\tau).$$

Using

$$\partial_\rho x_c = \lambda B, \qquad \partial_\rho \lambda = \lambda A, \qquad \rho_\tau = \mu^2,$$

we get

$$X_\tau = \mu^2\lambda B + \mu^2\lambda A q + \lambda q_\tau.$$

Division by $\Lambda = \lambda\mu$ gives
$$\Lambda^{-1} X_\tau = \mu B + \mu A q + \frac{q_\tau}{\mu}.$$

Local integrability follows from absolute continuity and positivity of μ. On every compact final interval, μ is bounded above and below away from zero, q is bounded, and $\mu_\tau, q_\tau \in L^1$. The A-terms are locally integrable because
$$\int |A(\rho(\tau))| \mu(\tau)^2 \, d\tau = \int |A(\rho)| \, d\rho,$$
and the extra factors μ, μ^{-1}, and q are bounded on the same compact window. Likewise, since μ is bounded away from zero,
$$\int |B(\rho(\tau))| \mu(\tau) \, d\tau \le C \int |B(\rho(\tau))| \mu(\tau)^2 \, d\tau = C \int |B(\rho)| \, d\rho,$$
so the B-term in b is locally integrable. □

Lemma 9.44 (Renormalized equation in repaired gauge)**.** *The final variables V, P satisfy*
$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0$$
distributionally on compact Y-sets.

Proof. It suffices to compute in the smooth case and then pass to distributions by testing against compactly supported functions. Set
$$z = \mu Y + q.$$
Since $V(Y, \tau) = \mu U(z, \rho(\tau))$ and $\rho_\tau = \mu^2$, the spatial derivatives are
$$\nabla_Y V = \mu^2 \nabla_z U, \qquad \Delta_Y V = \mu^3 \Delta_z U,$$
$$(V \cdot \nabla_Y) V = \mu^3 (U \cdot \nabla_z) U, \qquad \nabla_Y P = \mu^3 \nabla_z Q.$$
Differentiation in τ gives
$$V_\tau = \mu_\tau U + \mu\mu^2 U_\rho + \mu(\mu_\tau Y + q_\tau) \cdot \nabla_z U.$$
Using the initial renormalized equation,
$$U_\rho = \nu \Delta_z U - (U \cdot \nabla_z) U - \nabla_z Q + A(U + z \cdot \nabla_z U) + B \cdot \nabla_z U,$$
we obtain
$$\begin{aligned} &V_\tau + (V \cdot \nabla_Y) V + \nabla_Y P - \nu \Delta_Y V \\ &\quad = \mu_\tau U + \mu^3 A(U + z \cdot \nabla_z U) + \mu^3 B \cdot \nabla_z U + \mu(\mu_\tau Y + q_\tau) \cdot \nabla_z U. \end{aligned}$$
Now
$$U = \mu^{-1} V, \qquad \nabla_z U = \mu^{-2} \nabla_Y V, \qquad z = \mu Y + q.$$
Therefore
$$\mu_\tau U + \mu^3 A U = \left(\frac{\mu_\tau}{\mu} + \mu^2 A \right) V,$$
$$\mu^3 A \, z \cdot \nabla_z U + \mu(\mu_\tau Y + q_\tau) \cdot \nabla_z U = \left(\mu^2 A + \frac{\mu_\tau}{\mu} \right) Y \cdot \nabla_Y V + \left(\mu A q + \frac{q_\tau}{\mu} \right) \cdot \nabla_Y V,$$
$$\mu^3 B \cdot \nabla_z U = \mu B \cdot \nabla_Y V.$$
Combining these identities, the coefficient multiplying
$$V + Y \cdot \nabla_Y V$$
is
$$\frac{\mu_\tau}{\mu} + \mu^2 A,$$

and the coefficient multiplying $\nabla_Y V$ is

$$\mu B + \mu A q + \frac{q_\tau}{\mu}.$$

By Lemma 9.43 these coefficients are a and b. Hence the displayed repaired-gauge equation holds in the distributional sense on compact Y-sets. Incompressibility follows from

$$\nabla_Y \cdot V = \mu^2 \nabla_z \cdot U = 0.$$

□

Lemma 9.45 (Pressure reconstruction under the charts)**.** *Assume the physical pressure satisfies*

$$-\Delta_x p = \partial_i \partial_j (u_i u_j)$$

in distributions. Then the intermediate pressure satisfies

$$-\Delta_y Q = \partial_i \partial_j (U_i U_j)$$

modulo functions of ρ, and the final pressure satisfies

$$-\Delta_Y P = \partial_i \partial_j (V_i V_j)$$

modulo functions of τ.

Proof. The additive function $c(t)$ in

$$p = \lambda^{-2} Q + c(t)$$

has zero spatial derivatives. Since

$$\Delta_x p = \lambda^{-4} \Delta_y Q, \qquad \partial_{x_i} \partial_{x_j} (u_i u_j) = \lambda^{-4} \partial_{y_i} \partial_{y_j} (U_i U_j),$$

the intermediate pressure equation follows. For the repaired variables,

$$P(Y) = \mu^2 Q(\mu Y + q), \qquad V(Y) = \mu U(\mu Y + q).$$

Therefore

$$-\Delta_Y P = \mu^4 (-\Delta_z Q)(z) = \mu^4 \partial_{z_i} \partial_{z_j} (U_i U_j)(z) = \partial_{Y_i} \partial_{Y_j} (V_i V_j)(Y).$$

Adding functions of time to Q or P does not change the identity. □

Lemma 9.46 (Scaling of local L^r norms)**.** *For every $1 \le r < \infty$ for which the norms are finite,*

$$\|V(\tau)\|_{L^r_Y}^r = \Lambda(\tau)^{r-3} \|u(t(\tau))\|_{L^r_x}^r.$$

In particular,

$$\|V(\tau)\|_{L^3_Y} = \|u(t(\tau))\|_{L^3_x}.$$

Thus membership in $L^3(\mathbb{R}^3)$, finiteness, infiniteness, and positivity are invariant under the final chart.

Proof. Using

$$u(X + \Lambda Y, t) = \Lambda^{-1} V(Y, \tau), \qquad dx = \Lambda^3 \, dY,$$

we obtain

$$\|u(t)\|_{L^r_x}^r = \int |\Lambda^{-1} V(Y, \tau)|^r \Lambda^3 \, dY = \Lambda^{3-r} \|V(\tau)\|_{L^r_Y}^r.$$

Rearranging gives the stated formula. The case $r = 3$ is scale invariant. □

Lemma 9.47 (Regularity of scale, center, and modulation)**.** *If λ, x_c, μ, q are absolutely continuous and $\mu > 0$, then on compact final time intervals the final scale Λ, center X, and coefficients a, b are absolutely continuous or locally integrable as specified above: $\Lambda, X \in W^{1,1}_{\mathrm{loc}}$ and*

$$a, b \in L^1_{\mathrm{loc}}.$$

Proof. This follows from Lemmas 9.42 and 9.43. Positivity of μ and compactness of the time window give a positive lower bound for μ, so the divisions by μ in the formulas for a, b are legitimate. ☐

Theorem 9.48 (Consequences of absolutely continuous repaired-gauge charts)**.** *Assume an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path in the sense of Definitions 9.39 and 9.41. Then the following analytic conclusions hold:*

(i) *the initial and repaired renormalized Navier–Stokes equations hold distributionally on compact renormalized sets;*
(ii) *the final physical chart satisfies*

$$u(x,t) = \Lambda(\tau)^{-1} V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2} P(Y,\tau) + c(t), \qquad \frac{d\tau}{dt} = \Lambda^{-2};$$

(iii) *pressure reconstruction holds modulo functions of time;*
(iv) *the final modulation coefficients are the functions a, b in Lemma 9.43;*
(v) *the critical L^3 norm is invariant under the final chart;*
(vi) *the final scale, center, and modulation coefficients have the local absolute-continuity and integrability stated in Lemma 9.47.*

Proof. Lemma 9.40 gives the initial renormalized equation. Lemmas 9.42 and 9.44 give the final chart and the final repaired-gauge equation with the stated coefficients. Lemma 9.45 gives pressure reconstruction modulo functions of time. Lemma 9.46 gives critical norm invariance, and Lemma 9.47 gives the regularity inheritance. ☐

Corollary 9.49 (Remaining representation hypotheses)**.** *Once the absolutely continuous concentration chart and absolutely continuous repaired-gauge path are available, pressure reconstruction, modulation coefficient realization, local $W^{1,1}$ regularity of the chart, and critical L^3-invariance are analytic consequences rather than separate hypotheses. The remaining representation hypotheses are:*

(i) *extraction of an absolutely continuous concentration chart from the selected Type II branch;*
(ii) *solvability of the repaired scale and centering equations on the admissible profile class;*
(iii) *selection of the repaired scale and center as an absolutely continuous path satisfying $\rho_\tau = \mu^2$;*
(iv) *terminal admissibility of the final scale and time, including $\tau \to \infty$ and preservation of the Type II core scale.*

Proof. This is Theorem 9.48. The enumerated items are the complementary analytic hypotheses that must be available before that theorem can be applied: one must produce the initial concentration chart, solve the repaired gauge equations, select the resulting scale and center absolutely continuously, and verify that the resulting final chart is terminally admissible. ☐

9.7. **Nontrivial Critical Core and Vanishing Critical L^3 Norm.**

Definition 9.50 (Tail nontrivial critical core)**.** A repaired-gauge Type II branch (V, P, a, b) has a tail nontrivial critical core if there exist $R_* < \infty$, $\eta_* > 0$, and $\tau_* \geq \tau_0$ such that

$$\|V(\tau)\|_{L^3(B_{R_*})} \geq \eta_* \qquad \text{for all } \tau \geq \tau_*.$$

Equivalently, the selected terminal core remains nonzero in the critical topology on the renormalized tail.

Definition 9.51 (Subsequential nontrivial critical core)**.** A branch has subsequential nontriviality if there exist $R_* < \infty$, $\eta_* > 0$, and a sequence $\tau_n \to \infty$ such that

$$\|V(\tau_n)\|_{L^3(B_{R_*})} \geq \eta_* \qquad \text{for all } n.$$

Definition 9.52 (Core persistence)**.** Core persistence is the analytic implication that subsequential nontriviality of a selected terminal Type II core implies tail nontriviality:

$$\text{subsequential nontriviality} \Longrightarrow \text{tail nontriviality}.$$

Failure of this implication is a failure of persistence of the selected core, not a new Type II alternative.

Definition 9.53 (Vanishing critical L^3 alternative)**.** The vanishing critical L^3 alternative for a repaired-gauge branch is

$$\sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty, \qquad \liminf_{\tau \to \infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} = 0.$$

Theorem 9.54 (Concentration extraction gives subsequential nontriviality)**.** *Assume a Type II branch has been obtained from a concentration extraction and placed in a repaired-gauge representation. Assume also that the extracted branch is selected as a terminal Type II core, rather than accounted for by scattering, exterior regularity, or loss of the extracted profile in the limiting procedure. Then the branch satisfies the subsequential nontriviality condition of Definition 9.51.*

Proof. The concentration extraction supplies a nonzero profile in the Navier–Stokes profile decomposition. The repaired-gauge representation transports that profile to the renormalized trajectory $V(\tau)$ by Navier–Stokes critical symmetries and admissible chart maps. These symmetries preserve the scale-critical L^3 norm.

Because the extracted profile is nonzero in L^3_{loc} after choosing the terminal coordinate frame, there are $R_* < \infty$, $\eta_* > 0$, and the corresponding extraction times $\tau_n \to \infty$ such that

$$\|V(\tau_n)\|_{L^3(B_{R_*})} \geq \eta_*.$$

This is subsequential nontriviality. □

Remark 9.55 (Persistence is an additional analytic hypothesis). Theorem 9.54 does not by itself rule out

$$\liminf_{\tau \to \infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} = 0.$$

That stronger conclusion requires the persistence hypothesis of Definition 9.52; a subsequence extraction theorem is not a uniform-in-time lower bound.

Theorem 9.56 (Vanishing critical L^3 norm excluded by persistent nontriviality)**.** *For a repaired-gauge Type II branch, core persistence together with subsequential nontriviality excludes the vanishing critical L^3 alternative of Definition 9.53. Equivalently, if the vanishing critical L^3 alternative occurs, then either subsequential nontriviality fails or core persistence fails.*

Proof. By core persistence and subsequential nontriviality, the branch has a tail nontrivial critical core. Hence there are $R_* < \infty$, $\eta_* > 0$, and τ_* such that

$$\|V(\tau)\|_{L^3(B_{R_*})} \geq \eta_* \qquad \text{for all } \tau \geq \tau_*.$$

Since $B_{R_*} \subset \mathbb{R}^3$, this gives the global lower bound

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} \geq \eta_* \qquad \text{for all } \tau \geq \tau_*.$$

Hence

$$\liminf_{\tau \to \infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \geq \eta_* > 0.$$

This contradicts the vanishing critical L^3 alternative, which requires the same liminf to be zero. Therefore a selected nontrivial core satisfying persistence cannot have vanishing critical L^3 norm. ☐

Corollary 9.57 (Critical L^3 alternatives after exclusion of vanishing norm)**.** *On selected Type II cores satisfying core persistence and subsequential nontriviality, the vanishing critical L^3 alternative is excluded from the ordered critical L^3 alternatives. The remaining possibilities are failure of membership in $L^3(\mathbb{R}^3)$ along the terminal tail, divergence of the critical L^3 norm, or a positive finite critical L^3 annulus.*

Proof. Apply Theorem 9.56 to exclude the vanishing critical L^3 alternative of Definition 9.53. The other alternatives are the remaining critical L^3 outcomes. ☐

Corollary 9.58 (Terminal boundedness after exclusion of vanishing norm)**.** *In the terminal boundedness alternatives of Theorem 9.82, if core persistence and subsequential nontriviality hold, then vanishing critical L^3 norm is not a terminal-sequence obstruction. The remaining obstructions are failure of sampling from the repaired-gauge trajectory, failure of membership in $L^3(\mathbb{R}^3)$ along the terminal tail, and divergence of the critical L^3 norm.*

Proof. Theorem 9.82 imports the critical L^3 alternatives. Corollary 9.57 excludes the vanishing critical L^3 alternative on selected nontrivial cores. The listed obstructions are the remaining terminal alternatives. ☐

Remark 9.59 (Remaining critical L^3 boundary). The preceding results prove only the lower-bound half of critical normalization, modulo subsequential nontriviality and persistence. The remaining analytic hypotheses are persistence of the selected core, membership in $L^3(\mathbb{R}^3)$ along the terminal tail, classification of the infinite critical L^3-norm alternative, and an upper finite critical L^3 bound on the selected repaired-gauge branch. Once these are supplied, the positive finite critical annulus follows.

9.8. **Local Caccioppoli Estimate for Absolutely Continuous Suitable Branches.**

Definition 9.60 (Suitable repaired-gauge Type II branch)**.** A suitable repaired-gauge Type II branch is a repaired-gauge Type II branch for which:

(i) an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path are available;
(ii) the conclusions of Theorem 9.48 hold, including pressure reconstruction and final modulation coefficients;
(iii) the physical branch (u, p) is a local suitable weak solution on the terminal interval:

$$u \in L^\infty_t L^2_{\text{loc}} \cap L^2_t H^1_{\text{loc}}, \qquad p \in L^{3/2}_{\text{loc}}, \qquad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold distributionally, and the physical local energy inequality holds for every nonnegative smooth compactly supported test function.

Lemma 9.61 (Local energy inequality under absolutely continuous Type II charts)**.** *Let (u, p) be physically suitable on a compact physical cylinder, and let (V, P) be obtained from an absolutely continuous final chart*

$$u(x, t) = \Lambda(s)^{-1} V(Y, s), \qquad p(x, t) = \Lambda(s)^{-2} P(Y, s) + c(t), \qquad Y = \frac{x - X(s)}{\Lambda(s)},$$

with $s = \tau$, $dt = \Lambda(s)^2\, ds$, $\Lambda > 0$, and

$$X, \Lambda \in W^{1,1}_{\text{loc}}.$$

Then (V,P) satisfies the renormalized local energy inequality on compact (Y,s)-cylinders:

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|V(s_2)|^2}{2}\phi(s_2)\,dY + \nu\int_{s_1}^{s_2}\int_{\mathbb{R}^3} |\nabla V|^2\phi\,dY\,ds \\
&\le \int_{\mathbb{R}^3} \frac{|V(s_1)|^2}{2}\phi(s_1)\,dY + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \frac{|V|^2}{2}(\partial_s\phi + \nu\Delta\phi)\,dY\,ds \\
&\quad + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P\right) V\cdot\nabla\phi\,dY\,ds \\
&\quad + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} a(s)\frac{|V|^2}{2}(-\phi - Y\cdot\nabla\phi)\,dY\,ds \\
&\quad - \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \frac{|V|^2}{2} b(s)\cdot\nabla\phi\,dY\,ds,
\end{aligned}$$

for every nonnegative $\phi \in C^\infty([s_1,s_2]\times\mathbb{R}^3)$ that is compactly supported in the spatial variable. Here a,b are the final modulation coefficients.

Proof. First suppose X,Λ are C^1. Use the physical test function

$$\psi(x,t) = \Lambda(s)^{-1}\phi\left(\frac{x-X(s)}{\Lambda(s)}, s\right), \qquad t = t(s),$$

where $dt = \Lambda^2 ds$. The spatial support of ϕ is compact, and Λ is bounded above and below on the selected compact time interval, so ψ is an admissible compactly supported test function in the physical local energy inequality.

Let

$$Y = \frac{x - X(s)}{\Lambda(s)}.$$

At fixed x,

$$\frac{dY}{ds} = -\frac{X_s}{\Lambda} - \frac{\Lambda_s}{\Lambda}Y.$$

Consequently,

$$\nabla_x\psi = \Lambda^{-2}\nabla_Y\phi, \qquad \Delta_x\psi = \Lambda^{-3}\Delta_Y\phi,$$

and, since $ds/dt = \Lambda^{-2}$,

$$\partial_t\psi = \Lambda^{-3}\left[\partial_s\phi - \frac{\Lambda_s}{\Lambda}(\phi + Y\cdot\nabla\phi) - \frac{X_s}{\Lambda}\cdot\nabla\phi\right].$$

Changing variables

$$x = X(s) + \Lambda(s)Y, \qquad dt = \Lambda^2 ds,$$

and using

$$u = \Lambda^{-1}V, \qquad p = \Lambda^{-2}P + c(t),$$

each term in the physical local energy inequality has the following renormalized form. The endpoint and dissipation terms transform as

$$\int \frac{|u(t_i)|^2}{2}\psi(t_i)\,dx = \int \frac{|V(s_i)|^2}{2}\phi(s_i)\,dY,$$

and

$$\nu\int_{t_1}^{t_2}\int |\nabla_x u|^2\psi\,dx\,dt = \nu\int_{s_1}^{s_2}\int |\nabla_Y V|^2\phi\,dY\,ds.$$

The test-function derivative term becomes

$$\int_{t_1}^{t_2}\int \frac{|u|^2}{2}(\partial_t\psi+\nu\Delta_x\psi)\,dx\,dt$$
$$=\int_{s_1}^{s_2}\int \frac{|V|^2}{2}\left[\partial_s\phi+\nu\Delta_Y\phi-a(\phi+Y\cdot\nabla\phi)-b\cdot\nabla\phi\right]dY\,ds,$$

where

$$a=\frac{\Lambda_s}{\Lambda},\qquad b=\frac{X_s}{\Lambda}.$$

The transport and pressure term becomes

$$\int_{t_1}^{t_2}\int\left(\frac{|u|^2}{2}+p\right)u\cdot\nabla_x\psi\,dx\,dt$$
$$=\int_{s_1}^{s_2}\int_{\mathbb{R}^3}\left(\frac{|V|^2}{2}+P\right)V\cdot\nabla_Y\phi\,dY\,ds+\int_{s_1}^{s_2}\Lambda(s)^2c(t(s))\left(\int_{\mathbb{R}^3}V\cdot\nabla_Y\phi\,dY\right)ds.$$

The pressure constant drops out because

$$\int_{\mathbb{R}^3}V\cdot\nabla_Y\phi\,dY=-\int_{\mathbb{R}^3}\phi\,\nabla_Y\cdot V\,dY=0.$$

Combining the transformed terms gives the displayed renormalized local energy inequality.

For $X,\Lambda\in W^{1,1}_{\mathrm{loc}}$, the same identity follows at the level of test functions. On the selected compact time interval, Λ has positive lower and finite upper bounds. The pulled-back test

$$\psi(x,t)=\Lambda(s)^{-1}\phi((x-X(s))/\Lambda(s),s)$$

is compactly supported, nonnegative, bounded, has bounded spatial derivatives on the compact cylinder, and has time derivative in $L^1_tL^\infty_x$ because $X',\Lambda'\in L^1$ and ϕ is smooth compactly supported. The physical local energy inequality, initially stated for C_c^∞ tests, extends to this class by standard mollification of the test function: the suitability classes give the local integrability needed for every term in the inequality, and the mollified tests keep support inside a slightly larger compact cylinder. Applying the extended inequality to ψ and changing variables gives the displayed renormalized inequality. The only derivatives of the chart that appear are the L^1-functions X',Λ', hence the modulation terms are locally integrable against local $|V|^2$. □

Lemma 9.62 (Pressure and modulation for suitable repaired-gauge branches)**.** *For a suitable repaired-gauge branch, the pressure identity*

$$-\Delta P=\partial_i\partial_j(V_iV_j)$$

holds modulo functions of τ, and the final coefficients a,b in the renormalized equation are measurable and locally integrable on compact renormalized windows.

Proof. This is the pressure and modulation part of Theorem 9.48. Pressure pullback comes from the physical pressure equation and the chart/gauge scaling. The final coefficients are forced by the absolutely continuous scale-translation gauge transform:

$$a=\mu^2A+\frac{\mu_\tau}{\mu},\qquad b=\mu B+\mu Aq+\frac{q_\tau}{\mu}.$$

Since the chart and gauge parameters are absolutely continuous on compact windows, these coefficients are locally integrable. □

Theorem 9.63 (Caccioppoli estimate for suitable absolutely continuous repaired-gauge branches)**.** *Every suitable repaired-gauge Type II branch satisfies the local renormalized Caccioppoli estimate on each compact repaired-gauge cylinder. The constants may depend on the cylinder, the cutoff, and local upper and lower bounds for the chart. No local Caffarelli–Kohn–Nirenberg smallness, global windowed H^1, critical L^3-norm, tightness, bounded modulation, finite energy at infinity, or whole-space compactness conclusion is asserted.*

Proof. By the suitable repaired-gauge hypotheses and Theorem 9.48, the branch has an absolutely continuous final chart, an admissible absolutely continuous repaired-gauge path, pressure reconstruction, and final modulation coefficients. Lemma 9.61 pulls the physical local energy inequality to the repaired variables. Lemma 9.62 supplies the pressure and coefficient identities needed to interpret the renormalized inequality in the same PDE class used by the local windowed estimates.

Choose a standard compactly supported Caccioppoli cutoff

$$\phi(Y,\tau) = \eta(\tau)\zeta(Y)^2.$$

Let $I = (\tau_1, \tau_2)$, let $I' \Subset I$, and let $0 < r < R$. Take $0 \le \eta \le 1$, $\eta \in C_c^\infty(I)$, with $\eta \equiv 1$ on I', and take $0 \le \zeta \le 1$, $\zeta \in C_c^\infty(B_R)$, with $\zeta \equiv 1$ on B_r. Applying Lemma 9.61 on (τ_1, σ) and then taking the essential supremum over $\sigma \in I'$ controls the endpoint term. Applying the same inequality on the full interval I controls the dissipation term. Adding the two estimates gives

$$\begin{aligned}
&\operatorname*{ess\,sup}_{\tau\in I'} \int_{B_r} \frac{|V(\tau)|^2}{2}\, dY + \nu \int_{I'}\int_{B_r} |\nabla V|^2\, dY\, d\tau \\
&\le C \int_I \int_{B_R} \frac{|V|^2}{2} \left(|\eta'|\zeta^2 + \nu\eta|\Delta(\zeta^2)|\right)\, dY\, d\tau \\
&\quad + C \int_I \int_{B_R} \left(\frac{|V|^2}{2} + |P|\right) |V|\, \eta|\nabla(\zeta^2)|\, dY\, d\tau \\
&\quad + C \int_I |a(\tau)| \int_{B_R} \frac{|V|^2}{2}\eta \left(\zeta^2 + |Y|\,|\nabla(\zeta^2)|\right)\, dY\, d\tau \\
&\quad + C \int_I |b(\tau)| \int_{B_R} \frac{|V|^2}{2}\eta|\nabla(\zeta^2)|\, dY\, d\tau.
\end{aligned}$$

Here C is an absolute constant. Since the cutoff derivatives are bounded by constants depending only on I, I', r, R, this is the local Caccioppoli estimate on the compact renormalized cylinder.

No global estimate is used in this step. On the compact cylinder under consideration, the absolutely continuous chart has

$$0 < \Lambda_- \le \Lambda(\tau) \le \Lambda_+ < \infty.$$

Pulling back physical suitability therefore gives, for every compact spatial set K,

$$V \in L^\infty_\tau L^2_Y(K) \cap L^2_\tau H^1_Y(K), \qquad P \in L^{3/2}_{\tau,Y}(K).$$

Local Sobolev interpolation on the finite cylinder gives

$$V \in L^{10/3}_{\tau,Y}(K),$$

hence $V \in L^3_{\tau,Y}(K)$. Thus $PV \cdot \nabla\phi$ is integrable by Hölder's inequality. The representation identities give

$$a, b \in L^1_{\mathrm{loc}}(d\tau),$$

and

$$\tau \mapsto \int_K |V(Y,\tau)|^2 \, dY$$

is locally essentially bounded; therefore the scale and translation modulation terms in the pulled-back local energy inequality are integrable on the same compact cylinder. This proves the local Caccioppoli estimate, but not a global windowed H^1 or critical L^3-norm bound. □

Corollary 9.64 (No separate Caccioppoli obstruction for suitable absolutely continuous branches)**.** *On suitable repaired-gauge branches, failure of local Caccioppoli regularity, failure of pressure reconstruction, and failure of local gauge regularity are not separate obstructions to the local H^1 analysis. The remaining hypotheses for the uniform windowed H^1 estimates are the separate bounded critical L^3 norm and modulation bounds required by those estimates.*

Proof. Theorem 9.63 gives local Caccioppoli regularity. Theorem 9.48 gives pressure reconstruction and the final modulation coefficients from the final chart and absolutely continuous gauge path. Thus the listed failures cannot occur separately on suitable repaired-gauge branches. The uniform windowed H^1 estimate may still require bounded critical norm and bounded modulation beyond the local Caccioppoli statement. □

9.9. **Caccioppoli Regularity Criterion.**

Definition 9.65 (Suitable physical hypotheses)**.** Let u, p be a local suitable weak solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (0, T^*)$. Thus

$$u \in L^\infty_t L^2_{\mathrm{loc}} \cap L^2_t H^1_{\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$

the equations hold distributionally, and for every nonnegative $\psi \in C_c^\infty(\mathbb{R}^3 \times (0, T^*))$ and a.e. $0 < t_1 < t_2 < T^*$,

$$\begin{aligned} &\int_{\mathbb{R}^3} \frac{|u(t_2)|^2}{2} \psi(t_2) \, dx + \nu \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \psi \, dx \, dt \\ &\le \int_{\mathbb{R}^3} \frac{|u(t_1)|^2}{2} \psi(t_1) \, dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \frac{|u|^2}{2} (\partial_t \psi + \nu \Delta \psi) \, dx \, dt \\ &\quad + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \psi \, dx \, dt. \end{aligned}$$

Lemma 9.66 (Renormalized local energy inequality)**.** *Assume the suitable physical hypotheses of Theorem 9.65. Let*

$$u(x,t) = \lambda(t)^{-1} V\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t) \right), \qquad \frac{d\tau}{dt} = \lambda(t)^{-2},$$

with

$$P(y,\tau) = \lambda(t)^2 p(x_c(t) + \lambda(t) y, t),$$

where $\lambda > 0$, x_c, and τ are C^1 on compact windows. Assume that V, P satisfy the repaired-gauge equation

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Then, for every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times (\tau_0, \infty))$ and a.e. $\tau_1 < \tau_2$ in the corresponding renormalized time interval,

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|V(\tau_2)|^2}{2}\phi(\tau_2)\,dy + \nu \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} |\nabla V|^2 \phi\,dy\,d\tau \\
&\le \int_{\mathbb{R}^3} \frac{|V(\tau_1)|^2}{2}\phi(\tau_1)\,dy + \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} \frac{|V|^2}{2}(\partial_\tau\phi + \nu\Delta\phi)\,dy\,d\tau \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P\right) V\cdot\nabla\phi\,dy\,d\tau \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} a(\tau)\frac{|V|^2}{2}\big(-\phi - y\cdot\nabla\phi\big)\,dy\,d\tau \\
&\quad - \int_{\tau_1}^{\tau_2}\int_{\mathbb{R}^3} \frac{|V|^2}{2}\,b(\tau)\cdot\nabla\phi\,dy\,d\tau.
\end{aligned}$$

Proof. For smooth solutions, multiply the repaired-gauge equation by $V\phi$ and integrate by parts. Put $e = |V|^2/2$. Since $\nabla\cdot V = 0$,

$$V\cdot(V\cdot\nabla V) = V\cdot\nabla e = \nabla\cdot(eV), \qquad V\cdot\nabla P = \nabla\cdot(PV),$$

and

$$V\cdot(y\cdot\nabla V) = y\cdot\nabla e, \qquad V\cdot(b\cdot\nabla V) = b\cdot\nabla e.$$

The scale contribution is

$$\int a(2e + y\cdot\nabla e)\phi\,dy = \int ae(-\phi - y\cdot\nabla\phi)\,dy,$$

because

$$\int y\cdot\nabla e\,\phi\,dy = -\int e\,\nabla\cdot(y\phi)\,dy = -\int e(3\phi + y\cdot\nabla\phi)\,dy.$$

The translation contribution is

$$\int b\cdot\nabla e\,\phi\,dy = -\int e\,b\cdot\nabla\phi\,dy,$$

and the viscous term gives

$$\int \nu\Delta V\cdot V\phi\,dy = -\nu\int|\nabla V|^2\phi\,dy + \nu\int e\Delta\phi\,dy.$$

Combining these identities gives the displayed equality in the smooth case.

For suitable weak solutions, apply the physical local energy inequality with

$$\psi(x,t) = \phi\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t)\right).$$

Use the change of variables

$$x = x_c(t) + \lambda(t)y, \qquad dt = \lambda(t)^2\,d\tau.$$

The C^1 regularity of λ and x_c justifies the chain rule for the test function, and the Jacobian factors are those of the Navier–Stokes critical scaling. Passing to the variables y, τ gives the same formula with equality replaced by $\le$. □

Lemma 9.67 (Caccioppoli estimate from the renormalized local energy inequality)**.** *Assume Theorem 9.66 on a compact renormalized cylinder $B_{2R}\times I$, bounded modulation on I, and local bounds*

$$V \in L^\infty(I; L^3(B_{2R})), \qquad P \in L^{3/2}(I\times B_{2R}).$$

Then the compact-cylinder renormalized Caccioppoli estimate used in Theorem 7.15 is justified.

Proof. Choose a standard nonnegative cutoff

$$\phi(y,\tau) = \eta(\tau)\zeta(y)^2,$$

where $\zeta \equiv 1$ on the inner ball and ζ is supported in the outer ball. Insert this test function in Theorem 9.66. The left-hand side contains

$$\nu \int |\nabla V|^2 \eta \zeta^2.$$

Every term on the right-hand side is finite under the stated hypotheses. The term

$$\int |V|^2 |\partial_\tau \phi + \Delta \phi|$$

is controlled by $V \in L^\infty(I; L^3(B_{2R}))$ and finite measure. The convective term

$$\int |V|^3 |\nabla \phi|$$

is controlled by the same local critical bound. For the pressure term, choose a time-dependent constant $c(\tau)$. Since

$$\int c(\tau) V \cdot \nabla \phi \, dy = -\int c(\tau) \phi \, \nabla \cdot V \, dy = 0,$$

Hölder's inequality controls

$$\int |P - c(\tau)| \, |V| \, |\nabla \phi|$$

by $P - c(\tau) \in L^{3/2}(I \times B_{2R})$ and $V \in L^\infty(I; L^3(B_{2R}))$. The modulation terms are bounded by

$$\|a\|_{L^\infty} + \|b\|_{L^\infty}$$

times the local L^2-mass, and the local L^2-mass is controlled on bounded balls by local L^3-mass. These estimates give the renormalized Caccioppoli estimate on compact cylinders. □

Proposition 9.68 (Caccioppoli regularity criterion)**.** *Assume the suitable physical hypotheses, the repaired-gauge representation, pressure reconstruction modulo functions of time, and C^1 scale and center functions on compact renormalized windows. Then the compact-cylinder Caccioppoli regularity hypothesis used in the local windowed H^1 estimate holds.*

Proof. The repaired-gauge representation writes the suitable weak solution in the variables V, P, a, b. Theorem 9.66 gives the renormalized local energy inequality, and Theorem 9.67 shows that this inequality justifies the renormalized Caccioppoli estimate on every compact cylinder. □

Corollary 9.69 (Caccioppoli estimates give windowed control under boundedness hypotheses)**.** *Assume the suitable physical hypotheses, the repaired-gauge representation, pressure reconstruction, a bounded critical L^3 norm on compact cylinders, and bounded modulation. Then local windowed H^1 control follows from Theorem 7.15. Thus failure of the Caccioppoli regularity hypothesis is not an independent rough-core alternative under these hypotheses.*

Proof. Theorem 9.68 supplies the Caccioppoli regularity hypothesis. The bounded critical norm, pressure reconstruction, and bounded modulation are the remaining hypotheses used in Theorem 7.15. Therefore the local windowed H^1 estimate holds. □

9.10. **Terminal Profile Compactness Hypotheses.**

Assumption 9.70 (Critical Navier–Stokes profile decomposition)**.** Let $u_{0,n}$ be a bounded sequence in $L^3_\sigma(\mathbb{R}^3)$, or in a stronger admissible critical profile space for which the standard Navier–Stokes profile theorem and perturbation theorem are available. After passing to a subsequence, there are profiles ϕ^j, scales $\lambda_{j,n} > 0$, centers $x_{j,n} \in \mathbb{R}^3$, and remainders r_n^J such that

$$u_{0,n} = \sum_{j=1}^{J} \Lambda_{\lambda_{j,n},x_{j,n}}\phi^j + r_n^J, \qquad (\Lambda_{\lambda,x_0} f)(x) = \lambda^{-1} f\left(\frac{x - x_0}{\lambda}\right),$$

with pairwise orthogonality of the parameters and critical mass decoupling

$$\|u_{0,n}\|_{L^3}^3 = \sum_{j=1}^{J} \|\phi^j\|_{L^3}^3 + \|r_n^J\|_{L^3}^3 + o_n(1).$$

Moreover, if the corresponding nonlinear profiles are evolved and the small critical profiles are removed by the small-data theory, then

$$\lim_{J\to\infty} \limsup_{n\to\infty} \|e^{t\Delta} r_n^J\|_{X_{\mathrm{Kato}}} = 0,$$

and the nonlinear remainder is perturbatively small on every compact profile-time window. Finally, if after removal of all extracted profiles a terminal coordinate frame still contains nonzero local critical mass, then, after passing to a further subsequence, the profile theorem produces an additional nonzero profile associated to that coordinate frame.

Definition 9.71 (Bounded critical terminal sequences)**.** The bounded critical terminal-sequence hypothesis is the assertion that every terminal active sequence used in the terminal profile analysis consists of divergence-free data $u_{0,n}$ satisfying

$$\sup_n \|u_{0,n}\|_{L^3(\mathbb{R}^3)} < \infty,$$

or satisfying the corresponding uniform bound in a stronger admissible critical profile space to which Theorem 9.70 applies.

Theorem 9.72 (Kato small-data critical stability)**.** *There is $\varepsilon_{\mathrm{sd}} > 0$ such that, if $f \in L^3_\sigma(\mathbb{R}^3)$ and $\|f\|_{L^3} \le \varepsilon_{\mathrm{sd}}$, then the Navier–Stokes equations have a global mild solution in a standard critical Kato class X_{Kato},*

$$U \in C([0,\infty); L^3_\sigma(\mathbb{R}^3)) \cap X_{\mathrm{Kato}}, \qquad \|U\|_{X_{\mathrm{Kato}}} \le C\|f\|_{L^3}.$$

The solution map is Lipschitz on this small ball, and perturbations whose linear heat evolutions are small in X_{Kato} remain perturbative in the same topology.

Proof. This is the classical Kato fixed-point theorem in the critical L^3 space. Writing the mild formulation as

$$U(t) = e^{t\Delta} f - \int_0^t e^{(t-s)\Delta}\mathbb{P}\nabla \cdot (U \otimes U)(s)\, ds,$$

the heat estimate and bilinear estimate in X_{Kato} give

$$\|B(U,V)\|_{X_{\mathrm{Kato}}} \le C\|U\|_{X_{\mathrm{Kato}}}\|V\|_{X_{\mathrm{Kato}}}.$$

For $\|f\|_{L^3} \le \varepsilon_{\mathrm{sd}}$, the map is a contraction in the ball $\|U\|_{X_{\mathrm{Kato}}} \le 2C\|f\|_{L^3}$. The same estimate applied to the difference of two mild solutions gives Lipschitz dependence. If a perturbing datum has small heat evolution in X_{Kato}, the Duhamel difference inequality is absorbed by the left-hand side after the same smallness choice. □

Definition 9.73 (Small-data critical stability)**.** Small-data critical stability means that every divergence-free datum whose critical size is below $\varepsilon_{\mathrm{sd}}$, and every perturbation whose linear heat flow is small in X_{Kato}, generates only a perturbative Navier–Stokes evolution on compact profile-time windows.

Corollary 9.74 (Kato theorem gives small-data stability)**.** *Theorem 9.72 implies the small-data critical stability condition of Theorem 9.73.*

Proof. The theorem gives global critical control for data below $\varepsilon_{\mathrm{sd}}$ and Lipschitz stability under small heat-flow perturbations. Restricting the resulting global estimate to any compact profile-time window gives the stated stability condition. □

Definition 9.75 (Terminal critical profile hypotheses)**.** The terminal critical profile hypotheses consist of the bounded critical terminal-sequence hypothesis Theorem 9.71, the small-data stability condition Theorem 9.73, and the critical Navier–Stokes profile theorem Theorem 9.70.

Theorem 9.76 (Terminal critical local compactness)**.** *Assume the terminal critical profile hypotheses. Then the terminal profile analysis is complete in the following sense: after passing to subsequences, all non-scattering critical profiles appear as active profiles, and the remaining terminal component is perturbatively small on compact profile-time windows.*

Proof. Fix a terminal active sequence. By Theorem 9.71 and Theorem 9.70, a critical profile decomposition is available after passing to a subsequence:

$$u_{0,n} = \sum_{j=1}^{J} \Lambda_{\lambda_{j,n},x_{j,n}}\phi^j + r_n^J.$$

The mass decoupling identity shows that only finitely many profiles can have critical norm at least $\varepsilon_{\mathrm{sd}}$. Profiles with smaller critical norm generate perturbative evolutions by Theorem 9.74. The finitely many remaining profiles are the non-scattering active profiles.

It remains to exclude hidden terminal mass. Suppose that, after removing all active profiles and all small-data perturbative profiles, a terminal coordinate frame still sees nonzero local critical mass. The final clause of Theorem 9.70 then produces an additional nonzero profile associated to that frame. If its critical norm is below $\varepsilon_{\mathrm{sd}}$, the Kato stability estimate makes its nonlinear contribution perturbative, contradicting the assumed nonzero terminal critical mass. If its critical norm is at least $\varepsilon_{\mathrm{sd}}$, then it is another active profile, contradicting the claim that all active profiles had already been removed. Hence no hidden terminal critical mass remains, and the residual component is perturbatively small on compact profile-time windows. □

Corollary 9.77 (Profile completeness from terminal critical profile hypotheses)**.** *The terminal profile completeness hypothesis in the final argument may be replaced by the terminal critical profile hypotheses of Theorem 9.75.*

Proof. Theorem 9.76 gives the asserted terminal profile completeness. □

Remark 9.78 (Boundary of the terminal compactness hypothesis)**.** The preceding compactness conclusion applies only to terminal sequences with a uniform bound in L^3_σ, or in a stronger admissible critical profile space. If a terminal sequence is not a compact-cylinder sequence in such a space, then the failure is a critical-norm blowup defect or a failure of the coordinate-frame construction, not a failure of the critical profile theorem.

Definition 9.79 (Terminal sequences sampled from the renormalized orbit)**.** A terminal active sequence is sampled from the renormalized orbit if it is of the form

$$u_{0,n} = \Lambda_{\lambda_n,x_n}V(\tau_n), \qquad \tau_n \to \infty, \qquad \lambda_n > 0, \qquad x_n \in \mathbb{R}^3,$$

or by the equivalent inverse critical change of variables, where

$$(\Lambda_{\lambda,x_0} f)(x) = \lambda^{-1} f\left(\frac{x - x_0}{\lambda}\right).$$

Lemma 9.80 (Critical symmetry preserves the L^3 norm). *For every $f \in L^3(\mathbb{R}^3)$, $\lambda > 0$, and $x_0 \in \mathbb{R}^3$,*

$$\|\Lambda_{\lambda,x_0} f\|_{L^3} = \|f\|_{L^3}.$$

Proof. By the change of variables $x = x_0 + \lambda y$,

$$\|\Lambda_{\lambda,x_0} f\|_{L^3}^3 = \int_{\mathbb{R}^3} \lambda^{-3} \left| f\left(\frac{x - x_0}{\lambda}\right)\right|^3 dx = \int_{\mathbb{R}^3} |f(y)|^3 \, dy.$$

Taking cube roots gives the claim. □

Theorem 9.81 (Bounded terminal sequences from the critical annulus). *Assume the renormalized orbit satisfies*

$$0 < \inf_\tau \|V(\tau)\|_{L^3} \leq \sup_\tau \|V(\tau)\|_{L^3} < \infty,$$

on the terminal interval under consideration. If every terminal active sequence is sampled from the renormalized orbit, then every terminal active sequence is bounded in $L^3_\sigma(\mathbb{R}^3)$.

Proof. Let $u_{0,n} = \Lambda_{\lambda_n,x_n} V(\tau_n)$. By Theorem 9.80,

$$\|u_{0,n}\|_{L^3} = \|V(\tau_n)\|_{L^3}.$$

The assumed finite upper bound on the terminal interval gives $\sup_n \|u_{0,n}\|_{L^3} < \infty$. If the positive lower bound is also used, the sequence is nontrivial in the critical norm. □

Theorem 9.82 (Terminal boundedness alternatives). *For a candidate terminal active sequence in the Type II analysis, the following alternatives exhaust the possible boundedness outcomes, after passing to a subsequence when necessary:*

(i) *the sequence is not sampled from the renormalized orbit;*
(ii) *the terminal L^3 norm of V is not defined in the asserted critical class;*
(iii) $\|V(\tau_n)\|_{L^3} \to \infty$ *along a terminal sampling subsequence;*
(iv) $\|V(\tau_n)\|_{L^3} \to 0$ *along a terminal sampling subsequence;*
(v) *after excluding the preceding alternatives, the terminal active sequence satisfies a positive finite critical annulus and is bounded in $L^3_\sigma(\mathbb{R}^3)$.*

Proof. If the terminal sequence is not sampled from the renormalized orbit, then (i) holds. If the critical norm is not defined in the asserted class, then (ii) holds. If the critical norm is defined but has no positive finite terminal annulus, then either its limsup is infinite, which gives (iii) after passing to a subsequence, or its liminf is zero, which gives (iv) after passing to a subsequence. Otherwise the positive finite annulus condition of Theorem 9.81 holds, and (v) follows. □

Corollary 9.83 (Boundedness from orbit sampling). *In the presence of terminal orbit sampling and a positive finite terminal L^3 annulus, the bounded critical terminal-sequence hypothesis in Theorem 9.71 follows from Theorem 9.81.*

Proof. The theorem gives the required uniform L^3 bound for every terminal active sequence. □

Definition 9.84 (Terminal coordinate-frame construction). The terminal coordinate-frame construction means that every terminal active frame used in the analysis is obtained from the repaired-gauge orbit as follows. Choose sampling times $\tau_n \to \infty$ and critical symmetry parameters (λ_n, x_n); form the data by applying the critical coordinate-frame map to $V(\tau_n)$; group comparable profiles based at the same physical point before selecting the terminal active frame. No external data are introduced.

Theorem 9.85 (Terminal coordinate-frame construction from profile selection). *Assume terminal active frames are selected by first partitioning active profiles by physical concentration point, then grouping comparable profiles at the same point into compound profiles, and finally choosing a minimal-scale active class at that point. Then the terminal coordinate-frame construction of Theorem 9.84 holds for the selected terminal frames.*

Proof. By the stated selection rule, each terminal active frame is attached to a compound active class after all comparable profiles at the same physical point have been grouped. The scale and center of the terminal frame are the scale and center of this compound class. The terminal data are then obtained by applying the Navier–Stokes critical change of variables to the renormalized branch along the chosen sampling sequence $\tau_n \to \infty$. Thus the construction uses only the repaired-gauge orbit and the Navier–Stokes critical symmetries, and it satisfies all clauses of Theorem 9.84. □

Theorem 9.86 (Terminal coordinate frames are sampled from the orbit). *Assume the terminal coordinate-frame construction. Then every terminal active sequence belongs to the class described in Theorem 9.79.*

Proof. By definition of the terminal coordinate-frame construction, each terminal active frame is obtained by choosing times τ_n, selecting scale and center parameters (λ_n, x_n), and applying the Navier–Stokes critical change of variables to the renormalized solution $V(\tau_n)$. Hence the associated terminal data have the form

$$u_{0,n} = \Lambda_{\lambda_n,x_n} V(\tau_n),$$

or the equivalent inverse formulation, which is the orbit-sampling condition. □

Theorem 9.87 (Repaired-gauge representation gives terminal orbit sampling). *Assume a repaired-gauge representation of the Type II branch and the terminal coordinate-frame construction. Then every terminal active sequence is sampled from the renormalized orbit.*

Proof. The repaired-gauge representation supplies the renormalized orbit $V(\tau)$ and the critical coordinate maps between physical variables and renormalized variables. The terminal coordinate-frame construction supplies the terminal sampling times and scale-center parameters. Combining these hypotheses gives

$$u_{0,n} = \Lambda_{\lambda_n,x_n} V(\tau_n),$$

which is the assertion of Theorem 9.79. □

Corollary 9.88 (Failure of terminal orbit sampling). *If a sequence reaches the terminal profile stage but is not sampled from the renormalized orbit, then either the repaired-gauge representation is not available for that sequence or the terminal coordinate-frame construction has not been applied as specified.*

Proof. This is the contrapositive of Theorem 9.87. □

Corollary 9.89 (Terminal compactness from explicit PDE hypotheses). *Assume the repaired-gauge representation, the terminal coordinate-frame construction, a positive finite terminal L^3 annulus for V, the Kato small-data stability theorem, and the critical Navier–Stokes profile decomposition. Then the terminal profile completeness hypothesis in the final argument holds.*

Proof. Theorem 9.87 gives terminal orbit sampling. Together with the positive finite L^3 annulus, Theorem 9.81 gives bounded critical terminal sequences. The Kato theorem gives small-data stability by Theorem 9.74. Therefore the terminal critical profile hypotheses of Theorem 9.75 hold, and Theorem 9.77 gives terminal profile completeness. □

9.11. **Terminal Conditional Assembly.**

Definition 9.90 (Final conditional hypotheses)**.** The final conditional hypotheses for this section are:

(i) every renormalized local Type II branch enters one of the local alternatives used in Theorem 10.13;
(ii) compact single-core branches are excluded by Theorem 10.5;
(iii) multibubble, cascade, gauge-degenerate, and scale-rigid terminal branches are excluded under the hypotheses of Theorem 10.6;
(iv) finite-cost scale-collapse branches are excluded by Theorem 10.8, while scale-collapse branches with divergent negative scale-drift cost are covered by Theorem 9.33 or Theorem 9.37;
(v) failure of uniform critical tightness is excluded, equivalently uniform critical tightness holds for every renormalized branch in the class;
(vi) failure of local windowed H^1 control is excluded, equivalently the required local windowed H^1 control holds for every renormalized branch in the class;
(vii) the cost-divergence exclusion criterion Theorem 9.2 holds for the localized costs used in the preceding items.

Theorem 9.91 (Final conditional Type II exclusion)**.** *Under the final conditional hypotheses of Theorem 9.90, no renormalized local Type II branch in the stated class remains.*

Proof. Let a renormalized local Type II branch be given. By item (i), it lies in one of the alternatives in Theorem 10.13. Items (ii) and (iii) exclude the compact single-core, multibubble, cascade, gauge-degenerate, and scale-rigid alternatives. Item (iv) excludes finite-cost scale collapse and also excludes the divergent negative-drift scale-collapse branch whenever the corresponding cost is one of the costs covered by Theorem 9.2. Items (v) and (vi) exclude failure of uniform critical tightness and failure of local windowed H^1 control. These cases exhaust the alternatives, so no branch remains. ☐

Definition 9.92 (Expanded terminal hypotheses)**.** The following analytic hypotheses imply the final conditional hypotheses:

(i) the local alternative decomposition for renormalized Type II branches;
(ii) the compact single-core, multibubble/cascade, finite-cost scale-collapse, and scale-rigidity exclusions from the preceding sections;
(iii) a critical-tightness estimate giving uniform critical tightness;
(iv) an exhaustive terminal state-space partition and a local finite critical-mass statement for retained active terminal strata;
(v) small-data stability in L^3 and a critical Navier–Stokes profile decomposition with nonlinear stability;
(vi) removal of the scattering branch and of physically exterior regular profiles;
(vii) the repaired-gauge representation theorem;
(viii) the local Caccioppoli/windowed regularity input;
(ix) the scale-collapse rigidity input used by the multibubble and scale-collapse reductions;
(x) the cost-divergence exclusion criterion Theorem 9.2.

Lemma 9.93 (Terminal hypotheses imply critical tightness)**.** *The expanded terminal hypotheses imply uniform critical tightness.*

Proof. The repaired-gauge representation turns each renormalized orbit into the terminal sequence used in the profile analysis. The terminal state-space partition, finite critical-mass statement, small-data L^3 stability, and critical profile decomposition give the required terminal profile completeness. The scattering branch and exterior regular profiles are removed by hypothesis. Together with the multibubble and scale-collapse exclusions, this leaves only possible loss of critical tightness; the critical-tightness estimate excludes that loss. ☐

Lemma 9.94 (Terminal hypotheses imply windowed H^1 control)**.** *The expanded terminal hypotheses imply local windowed H^1 control.*

Proof. The windowed H^1 estimate requires a repaired-gauge branch, local pressure and modulation control, compact-cylinder critical bounds, and the Caccioppoli/windowed regularity input. These are among the expanded terminal hypotheses. Hence the required local windowed H^1 control holds, and the failure of local windowed H^1 control is unavailable. ☐

Lemma 9.95 (Expanded terminal hypotheses imply final hypotheses)**.** *The expanded terminal hypotheses imply the final conditional hypotheses of Theorem 9.90.*

Proof. Items (i), (ii), and (x) of Theorem 9.92 give the local alternative decomposition, the compact, multibubble, scale-collapse, and scale-rigidity exclusions, and the cost-divergence criterion. By Theorem 9.93 they also give critical tightness, and by Theorem 9.94 they give the local windowed H^1 control. These are the hypotheses listed in Theorem 9.90. ☐

Theorem 9.96 (Expanded terminal Type II exclusion)**.** *If the expanded terminal hypotheses hold, then no renormalized local Type II branch in the stated class remains.*

Proof. Theorem 9.95 gives the final conditional hypotheses. Applying Theorem 9.91 gives the exclusion. ☐

Remark 9.97 (Branchwise reading)*.* Compact single-core branches are removed by the compact criterion; scale-collapse branches are removed by the scale-collapse rigidity input, or by Theorem 9.33 when the scale-collapse drift cost is compatible with the cost-divergence criterion; multibubble and non-tight cases are removed by the terminal profile and critical-tightness inputs; failures of local windowed H^1 control are removed by the local Caccioppoli estimate; and every divergent compatible localized cost is handled by Theorem 9.8.

10. Assembly of the Local Type II Exclusion

10.1. **Local Entry.** Let (u, p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

on $\mathbb{R}^3 \times (T - \delta, T)$. For $z_0 = (x_0, T)$ and $r > 0$, set

$$Q_r(z_0) = B_r(x_0) \times (T - r^2, T),$$
$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 \, dx \, dt,$$

and

$$D(z_0, r) = r^{-2} \int_{T-r^2}^{T} \int_{B_r(x_0)} |p(x,t) - (p)_{B_r(x_0)}(t)|^{3/2} \, dx \, dt.$$

The pressure is always taken modulo functions of time, and the spatial mean normalization is the one used in the Caffarelli–Kohn–Nirenberg criterion.

Definition 10.1 (Singular set and local Type II point)**.** Let $\Sigma(T)$ be the set of points x_0 for which u is not locally bounded in any backward cylinder $Q_r(x_0, T)$. A point $x_0 \in \Sigma(T)$ is in the local Type II alternative if it carries a positive local Caffarelli–Kohn–Nirenberg concentration sequence and no local Type I scale-invariant bound holds at (x_0, T).

Definition 10.2 (Positive local Type II concentration sequence)**.** A sequence $r_n \downarrow 0$ is a positive local Type II concentration sequence at (x_0, T) if there exists $\eta > 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n,$$

and (x_0, T) is in the local Type II alternative. The associated parabolic rescalings are

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Theorem 10.3 (Local concentration and Type I/Type II dichotomy)**.** *If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \textit{for all } n.$$

At the concentration point exactly one of the following alternatives holds: there is a local Type I scale-invariant bound, or (x_0, T) is in the local Type II alternative.

Proof. This is the local concentration dichotomy used in the final assembly. □

10.2. **Local Theorems Used in the Assembly.** The local results needed for the final assembly are restated in the following form.

Theorem 10.4 (Repaired-gauge representation)**.** *Let a positive local Type II concentration sequence contain a nondegenerate selected core. Then, after passing to selected windows, there are local scale-center variables $\lambda(\tau) > 0$, $x_c(\tau)$, renormalized fields*

$$V(y, \tau) = \lambda(\tau) u(x_c(\tau) + \lambda(\tau) y, t(\tau)), \qquad P(y, \tau) = \lambda(\tau)^2 p(x_c(\tau) + \lambda(\tau) y, t(\tau)) + \pi(\tau),$$

and modulation coefficients a, b such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

On each compact renormalized window one has the local suitable energy inequality, local pressure reconstruction modulo functions of time, stability under subsequences, and the compact-window bounds proved in Section 3. Failure of the nondegenerate scale-center selection is part of the multibubble, cascade, or scale-rigidity alternatives.

Proof. Theorem 3.35 is precisely the compact-window representation theorem. □

Theorem 10.5 (Compact single-core exclusion)**.** *No compact single-core vanishing-dissipation branch can be a local Type II sequence. More explicitly, let a represented selected-window sequence have:*

(i) *positive local CKN concentration on a fixed compact cylinder;*
(ii) *uniformly bounded compact-cylinder suitable estimates $A + E + C + D$;*
(iii) *nondegenerate local scale-center representation;*
(iv) *bounded modulation coefficients;*
(v) *local pressure reconstruction and pressure stability modulo spatial means;*
(vi) *vanishing localized dissipation on some retained compact cylinder.*

Then the sequence is incompatible with the local Type II condition.

Proof. Theorem 2.18 gives the stated exclusion. □

Theorem 10.6 (Multibubble and cascade exclusion)**.** *Assume the terminal admissibility and scale-rigidity hypotheses stated in the multibubble/cascade analysis. Then no local multibubble, cascade, or gauge-degenerate Type II branch can occur in the class produced by the local active-core extraction. Equivalently, every failure of the single nondegenerate compact-core alternative is excluded by the local multibubble/cascade analysis under those stated hypotheses.*

Proof. Theorem 4.23, together with its removal corollary Theorem 4.24, gives the stated exclusion. □

Theorem 10.7 (Rough-core reduction)**.** *On any compact renormalized window satisfying the local energy inequality and bounded modulation hypotheses, finite outer velocity-pressure CKN control $\mathcal{C} + \mathcal{D}$ implies finite inner windowed H^1 control. Hence failure of local windowed H^1 control on a retained inner cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on every relevant enclosing cylinder.*

Proof. Theorem 5.7 gives the stated reduction. □

Theorem 10.8 (Finite-cost scale-collapse exclusion)**.** *No renormalized branch with finite scale-collapsing cost, and hence no branch with finite absolute scale cost, can have genuine scale collapse together with a nonvanishing selected core.*

Proof. Theorems 6.7 and 6.9 give the stated exclusion. □

Theorem 10.9 (Scale-collapse decomposition of alternatives)**.** *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. After passing to selected terminal windows, exactly one of the alternatives isolated in Section 6 occurs:*

(i) *finite scale-collapsing cost;*
(ii) *finite absolute scale cost;*
(iii) *thin-drift alternative;*
(iv) *autonomous-modulation alternative;*
(v) *compactness alternative;*
(vi) *a nonzero autonomous reduced limit in the scale-rigid state.*

In the local Type II class of alternatives used by the multibubble/cascade and scale-collapse analyses, alternatives (iii)–(vi) *do not constitute additional compact single-core mechanisms: the thin-drift, modulation, and compactness alternatives lie outside the compact finite-cost scale-collapse branch, and the scale-rigid state is covered by the scale-rigidity hypothesis in the multibubble/cascade theorem.*

Proof. The assertion follows by combining the scale-collapse decomposition of alternatives with the scale-rigidity alternative used in the multibubble/cascade reduction. □

Definition 10.10 (Virial-dissipative terminal state)**.** Let $G_\nu(y) = e^{-|y|^2/(4\nu)}$. A scale-rigid terminal state is called velocity-virial dissipative if its selected terminal representative (V, P) satisfies, after the usual pressure normalization, the autonomous self-similar equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu\Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \nabla \cdot V = 0,$$

and the Gaussian identity below is justified by cutoff approximation on the selected renormalized windows. It is called swirl-virial dissipative if, in addition, the representative is axisymmetric, $\Gamma = RV_\theta$ is the scale-invariant circulation, and the corresponding weighted circulation identity below is justified.

Theorem 10.11 (Virial exclusion for dissipative terminal states)**.** *Let a scale-rigid terminal state satisfy the hypotheses of Theorem 10.10 and be bounded for all negative renormalized times in the sense needed for the Gaussian integrals below.*

(a) *Suppose the velocity virial defect is absorbable: for some $\kappa > 0$ and a.e. τ,*

$$-\frac{1}{2\nu}\int_{\mathbb{R}^3} |V|^2(V\cdot y)G_\nu\, dy - \frac{1}{\nu}\int_{\mathbb{R}^3} P(V\cdot y)G_\nu\, dy \le (1-\kappa)\int_{\mathbb{R}^3} |V|^2 G_\nu\, dy + 2\nu(1-\kappa)\int_{\mathbb{R}^3} |\nabla V|^2 G_\nu\, dy.$$

Then

$$\int_{\mathbb{R}^3} |V(y,\tau)|^2 G_\nu(y)\, dy = 0 \qquad \text{for every } \tau.$$

Consequently no retained terminal core with positive Gaussian velocity mass can belong to this branch.

(b) *Suppose the terminal state is axisymmetric and the circulation defect is absorbable: for some $\kappa > 0$ and a.e. τ,*

$$-\frac{1}{2\nu}\int_{\mathbb{R}^3} \Gamma^2(V\cdot y)G_\nu\, dy \le (1-\kappa)\int_{\mathbb{R}^3} \Gamma^2 G_\nu\, dy + 2\nu(1-\kappa)\int_{\mathbb{R}^3} |\nabla\Gamma|^2 G_\nu\, dy.$$

Then

$$\int_{\mathbb{R}^3} \Gamma(y,\tau)^2 G_\nu(y)\, dy = 0 \qquad \text{for every } \tau.$$

Thus every retained axisymmetric terminal core whose remaining obstruction is a positive Gaussian circulation mass is excluded by the weighted circulation identity.

Proof. For the velocity statement set

$$\mathcal{E}(\tau) = \int_{\mathbb{R}^3} |V(y,\tau)|^2 G_\nu(y)\, dy.$$

The Gaussian virial identity gives

$$\partial_\tau\mathcal{E} = -2\nu\int_{\mathbb{R}^3} |\nabla V|^2 G_\nu\, dy - \int_{\mathbb{R}^3} |V|^2 G_\nu\, dy - \frac{1}{2\nu}\int_{\mathbb{R}^3} |V|^2(V\cdot y)G_\nu\, dy - \frac{1}{\nu}\int_{\mathbb{R}^3} P(V\cdot y)G_\nu\, dy.$$

The absorbability hypothesis therefore implies

$$\partial_\tau\mathcal{E} + \kappa\mathcal{E} + 2\nu\kappa\int_{\mathbb{R}^3} |\nabla V|^2 G_\nu\, dy \le 0.$$

In particular $\partial_\tau\mathcal{E} + \kappa\mathcal{E} \le 0$. Integrating from $s < \tau$ to τ gives

$$\mathcal{E}(\tau) \le e^{-\kappa(\tau-s)}\mathcal{E}(s).$$

The ancient boundedness of the terminal state makes $\sup_{s<\tau}\mathcal{E}(s) < \infty$. Sending $s \to -\infty$ yields $\mathcal{E}(\tau) = 0$.

For the axisymmetric statement set

$$\mathcal{G}(\tau) = \int_{\mathbb{R}^3} \Gamma(y,\tau)^2 G_\nu(y)\,dy.$$

The weighted circulation identity is

$$\partial_\tau \mathcal{G} + \mathcal{G} + 2\nu \int_{\mathbb{R}^3} |\nabla\Gamma|^2 G_\nu\,dy = -\frac{1}{2\nu}\int_{\mathbb{R}^3} \Gamma^2 (V\cdot y) G_\nu\,dy.$$

The assumed absorption gives

$$\partial_\tau \mathcal{G} + \kappa\mathcal{G} + 2\nu\kappa \int_{\mathbb{R}^3} |\nabla\Gamma|^2 G_\nu\,dy \le 0.$$

The same backward Gronwall argument gives $\mathcal{G}(\tau) = 0$ for every τ. □

Remark 10.12 (Controlled swirl alone). If the axisymmetric representative satisfies $\Gamma \in L^\infty$, the same weighted circulation identity gives the a priori estimate

$$\mathcal{G}(\tau)^{1/2} \lesssim \|V\|_{L^\infty}\|\Gamma\|_{L^\infty}$$

for bounded ancient terminal states. This estimate enters the local decomposition of alternatives but does not by itself exclude a terminal state. The exclusion requires the stated absorption, vanishing conclusion, or an equivalent condition which makes the virial inequality dissipative.

10.3. **Assembly of Alternatives.**

Theorem 10.13 (Local Type II decomposition into local alternatives)**.** *Every positive local Type II concentration sequence in Theorem 10.2, after passing to subsequences and selected windows, enters one of the following alternatives:*

(i) *a compact single-core vanishing-dissipation branch;*
(ii) *a multibubble, cascade, or gauge-degenerate branch;*
(iii) *loss of local windowed H^1 control on a retained compact cylinder;*
(iv) *a finite-cost scale-collapse branch with nonvanishing selected core;*
(v) *a scale-rigid state covered by the scale-rigidity hypothesis in Theorem 10.6.*

In the last alternative, any terminal subbranch satisfying Theorem 10.11 is eliminated before the remaining scale-rigid state is passed to the scale-rigidity hypothesis.

Proof. This decomposition is obtained by combining Theorems 10.4, 10.6, 10.7 and 10.9. The representation theorem gives the nondegenerate single-core chart whenever a unique retained core is selected. Failure of that selection, splitting of positive CKN mass, compactness failure, or scale cascade is the multibubble/cascade side. On a represented retained core, rough-core loss is equivalent by Theorem 10.7 to failure of compact CKN control on an enclosing cylinder, hence returns to the multibubble/cascade side. Genuine scale collapse is divided by Theorem 10.9 into finite-cost scale collapse or one of the scale-rigid terminal states used in Theorem 10.6. These cases exhaust the local alternatives. □

Proposition 10.14 (Elimination of local Type II states)**.** *No positive local Type II concentration sequence can lie in any of the alternatives* (i)–(v) *of Theorem 10.13.*

Proof. Alternative (i) is excluded by the compact single-core criterion, Theorem 10.5.

Alternative (ii) is excluded by the local multibubble and cascade theorem, Theorem 10.6, under its terminal admissibility and scale-rigidity hypotheses.

Alternative (iii) is not an independent terminal state. By Theorem 10.7, rough-core loss of local H^1 control forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on enclosing compact cylinders. Such failure yields an active compact-cylinder concentration branch and therefore belongs to the multibubble/cascade alternative, already excluded by Theorem 10.6.

Alternative (iv) is excluded by the finite-cost scale-collapse theorem, Theorem 10.8.

Alternative (v) is the scale-rigid terminal state in Theorem 10.6. If this terminal state is velocity-virial dissipative, Theorem 10.11 forces its Gaussian velocity mass to vanish, which contradicts retention of a positive terminal core. If it is axisymmetric and swirl-virial dissipative with positive retained circulation mass, the same theorem forces $\Gamma \equiv 0$, contradicting that positive circulation mass. The remaining scale-rigid terminal states are covered by the scale-rigidity hypothesis in Theorem 10.6. Under that hypothesis, the state either has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition, or it reduces to the compact single-core branch, which is excluded by Theorem 10.5. Hence it cannot be a local Type II branch. □

10.4. **Assembly Theorem.**

Theorem 10.15 (Local Type II exclusion)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$. Then no point of $\Sigma(T)$ lies in the local Type II alternative, within the class of local alternatives covered by Theorem 10.13.*

Proof. Assume, for contradiction, that $x_0 \in \Sigma(T)$ lies in the local Type II alternative. By Theorem 10.3, there is a positive local Type II concentration sequence $r_n \downarrow 0$ at (x_0, T). By the decomposition into local alternatives, Theorem 10.13, this sequence lies, after passing to selected windows, in one of the five alternatives listed there. Each alternative is impossible by Theorem 10.14. This contradiction proves that no such x_0 exists in the covered local class of alternatives. □

Corollary 10.16 (Final local dichotomy after Type I separation)**.** *Every singular point at time T in the covered class of local alternatives belongs to the local Type I alternative. Equivalently, once the Type I case is covered by the Type I part of the theory, the local Type II class of alternatives contains no remaining singular branch.*

Proof. Let $x_0 \in \Sigma(T)$. By Theorem 10.3, a concentration point is either in the local Type I alternative or in the local Type II alternative. Theorem 10.15 excludes the latter. □

10.5. **Conclusion.** The proof uses only local PDE statements. The local concentration theorem gives positive local CKN concentration and the Type I/Type II dichotomy. The repaired-gauge theorem gives the local renormalized representation for nondegenerate selected cores. The single-core theorem excludes compact single-core vanishing-dissipation branches. The multibubble/cascade theorem excludes splitting, cascades, gauge degeneracy, and the scale-rigid terminal state under its stated local hypotheses. The virial identities eliminate the velocity-virial dissipative terminal states and the axisymmetric swirl-virial dissipative terminal states before the remaining scale-rigid alternatives are covered by that hypothesis. The rough-core estimate reduces loss of windowed H^1 control to failure of compact CKN control. The scale-collapse theorem excludes finite-cost scale collapse and decomposes the remaining scale-collapse alternatives locally. Thus the local alternatives are exhausted without adding any further classification theorem.

11. PDE Form of the Type II Classification Inputs

This section records, in PDE notation, the Type II classification inputs used in the local Type II analysis. The statements are conditional classification statements for admissible Type II branches. They do not assert the existence of such branches and do not add any regularity theorem beyond the analytic hypotheses explicitly stated below.

11.1. **Admissible Type II branches and exhaustion.**

Definition 11.1 (Admissible local Type II class)**.** Let $\mathcal{U}_{\mathrm{II}}^{NS}$ be the class of admissible terminal branches (u, p, T^*) such that:

(i) (u, p) is a three-dimensional incompressible Navier–Stokes solution on a terminal interval $[0, T^*)$ in the analytic class under consideration;
(ii) $T^* < \infty$ is the terminal time of the branch;
(iii) the branch is not in the local Type I class associated with the scale-invariant Type I criterion;
(iv) the branch is a Type II branch rather than a continuation, Type I, boundary, or non-Navier–Stokes-domain alternative.

Membership in $\mathcal{U}_{\mathrm{II}}^{NS}$ specifies the class under analysis; no nonemptiness assertion is made.

Definition 11.2 (Type II analytic data)**.** A branch $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$ has the Type II analytic data if the following three analytic assertions hold:

(i) a concentration profile or blow-up germ modulo the Navier–Stokes symmetry group;
(ii) a supercritical scale alternative, meaning that the branch enters the Type II scale alternative rather than the Type I scale alternative;
(iii) local profile data in the Navier–Stokes profile class used for the subsequent compactness and representation arguments.

The analytic data are exhaustive if every admissible Type II branch either has these three properties or falls into one of the ordered failures in Theorem 11.3.

Definition 11.3 (Ordered exhaustion failures)**.** For an admissible Type II branch, the ordered failures of the Type II analytic data are:

(i) concentration extraction fails;
(ii) concentration extraction succeeds but the supercritical scale alternative does not hold;
(iii) concentration extraction and the scale alternative hold but the local profile data are insufficient for the subsequent compactness and representation arguments;
(iv) the admissible Type II branch lies outside the concentration, scale, and profile alternatives under consideration.

The first applicable failure in this order is the exhaustion alternative.

Theorem 11.4 (Type II branch exhaustion)**.** *Assume the Type II analytic data of Theorem 11.2 are exhaustive on $\mathcal{U}_{\mathrm{II}}^{NS}$. Then every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ has the concentration profile, the supercritical scale alternative, and the local profile data.*

Proof. Fix $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$. By the concentration-extraction component of the analytic data, the branch has a concentration profile or blow-up germ. By the scale component, the same branch lies in the supercritical Type II scale alternative. By the profile component, the extracted profile belongs to the local Navier–Stokes profile class used in the subsequent arguments. Since ω was arbitrary, the three assertions hold for every admissible Type II branch. □

Corollary 11.5 (Ordered exhaustion classification)**.** *For each $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$, exactly one ordered alternative occurs: the Type II analytic data hold, or one of the four failures in Theorem 11.3* *occurs. The Type II analytic data are exhaustive on $\mathcal{U}_{\mathrm{II}}^{NS}$ if and only if the first alternative holds for every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$.*

Proof. Test the four analytic requirements in the stated order. If concentration extraction, the scale alternative, and the profile condition all hold, the branch has the Type II analytic data. If a test fails, the first failed test gives the corresponding ordered failure. The alternatives are disjoint by the first failure convention and exhaustive by construction. The final equivalence is the definition of exhaustion on $\mathcal{U}_{\mathrm{II}}^{NS}$. ☐

Theorem 11.6 (Entry into the Type II alternatives)**.** *Assume the Type II analytic data are exhaustive. Then each admissible Type II branch enters the subsequent analysis in exactly one of the following ways:*

(i) *it reaches the compact Type II cost-divergence hypotheses after the representation, critical-mass, tightness, and windowed-H^1 tests;*
(ii) *it falls into an ordered representation failure, critical-mass failure, loss of critical tightness, or loss of local windowed H^1 control.*

Proof. By Theorem 11.4, every admissible Type II branch has the concentration, scale, and profile data. The representation step then either gives a repaired-gauge representation or one of the ordered representation failures in Theorem 11.12. If a repaired-gauge orbit exists, the critical L^3 test gives either the positive finite annulus or one of the ordered critical-mass alternatives in Theorem 11.22. If the positive finite annulus holds, the remaining compact cost-divergence inputs are critical tightness and local windowed H^1 control. Their failures are exactly the two local failures in Theorem 11.30. If both hold, the compact Type II cost-divergence criterion applies. ☐

11.2. **Repaired-gauge representation from Type II analytic data.**

Definition 11.7 (Raw Type II chart)**.** A raw Type II chart consists of

$$(u, p, T^*, x_c, \lambda, \tau, V, P, \mathcal{I}), \qquad \mathcal{I} = (t_0, T^*),$$

where u, p solve

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu\Delta u, \qquad \nabla \cdot u = 0,$$

$\lambda(t) > 0$, $x_c(t) \in \mathbb{R}^3$, $x_c, \lambda \in AC_{\mathrm{loc}}$, $\lambda(t) \to 0$ as $t \uparrow T^*$, and

$$\frac{d\tau}{dt} = \lambda(t)^{-2}, \qquad \tau(t) \to \infty \quad \text{as } t \uparrow T^*.$$

The renormalized variables are

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x,t) = \lambda(t)^{-1} V(y, \tau(t)), \qquad p(x,t) = \lambda(t)^{-2} P(y, \tau(t)),$$

with enough regularity to justify the distributional chain rule on compact y-sets.

Lemma 11.8 (Raw renormalized equation)**.** *For a raw Type II chart, define*

$$a_{\mathrm{raw}}(\tau(t)) = \lambda(t)\lambda_t(t), \qquad b_{\mathrm{raw}}(\tau(t)) = \lambda(t)x_c'(t).$$

Then V, P satisfy, distributionally on compact y-sets,

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu\Delta V + a_{\mathrm{raw}}(\tau)(V + y \cdot \nabla V) + b_{\mathrm{raw}}(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Proof. Write

$$u(x,t) = \lambda(t)^{-1} V(y, \tau(t)), \qquad y = \frac{x - x_c(t)}{\lambda(t)}, \qquad \tau_t = \lambda^{-2}.$$

Then
$$\nabla_x u = \lambda^{-2}\nabla_y V, \qquad \Delta_x u = \lambda^{-3}\Delta_y V, \qquad (u\cdot\nabla_x)u = \lambda^{-3}(V\cdot\nabla_y)V.$$
Since
$$y_t = -\frac{x_c'(t)}{\lambda(t)} - \frac{\lambda_t(t)}{\lambda(t)}y,$$
one has
$$\partial_t u = \lambda^{-3}\partial_\tau V - \lambda_t\lambda^{-2}(V + y\cdot\nabla V) - \lambda^{-2}x_c'(t)\cdot\nabla V.$$
Moreover $\nabla_x p = \lambda^{-3}\nabla_y P$. Substitution in Navier–Stokes and multiplication by λ^3 give
$$\partial_\tau V - \lambda\lambda_t(V + y\cdot\nabla V) - \lambda x_c'(t)\cdot\nabla V + (V\cdot\nabla)V + \nabla P = \nu\Delta V.$$
Moving the modulation terms to the right-hand side gives the equation with $a_{\mathrm{raw}} = \lambda\lambda_t$ and $b_{\mathrm{raw}} = \lambda x_c'$. Finally, $\nabla_x\cdot u = \lambda^{-2}\nabla_y\cdot V$, so incompressibility of u implies $\nabla_y\cdot V = 0$. ☐

Definition 11.9 (Repaired-gauge realization)**.** Let $0 < p < 3$, $\Theta_0 > 0$, and let ψ_R be a smooth compactly supported centering cutoff. A raw Type II chart admits a repaired-gauge realization if, after an admissible time-dependent scale and translation correction, the profile satisfies
$$G_{\mathrm{sc}}(V) = \int_{\mathbb{R}^3} |y|^{-p}|V(y)|^3\,dy - \Theta_0 = 0,$$
and
$$G_j(V) = \int_{\mathbb{R}^3} y_j|V(y)|^2\psi_R(y)\,dy = 0, \qquad j = 1,2,3,$$
for a.e. renormalized time. The corrected profile belongs to the admissible gauge class $\mathcal{A}_p$, has the regularity needed to differentiate these functionals, and all time-dependent symmetry corrections have been absorbed into final modulation coefficients a, b.

Definition 11.10 (Repaired-gauge representation)**.** A Type II branch has a repaired-gauge representation if it has:

(i) a raw Type II chart;
(ii) a repaired-gauge realization;
(iii) pressure reconstruction modulo functions of time;
(iv) final modulation coefficients a, b appearing in the repaired-gauge equation.

The representation consists of the tuple
$$(V, P, a, b, G_{\mathrm{sc}}, G_1, G_2, G_3, \tau_0).$$

Theorem 11.11 (Representation implication)**.** *Assume a Type II profile branch has a raw Type II chart, a repaired-gauge realization, pressure reconstruction modulo functions of time, and final modulation coefficients. Then the branch has a repaired-gauge representation in the sense of Theorem 11.10. In particular, (V, P, a, b) satisfies the PDE class required by the compact Type II cost-divergence criterion.*

Proof. The raw chart gives V, P and, by Theorem 11.8, the renormalized Navier–Stokes equation with raw modulation coefficients. The repaired-gauge realization places the profile on the scale and centering gauge surface and absorbs the time-dependent symmetry correction into updated coefficients. The pressure reconstruction identifies the pressure associated with V, modulo the usual functions of time. The modulation data identify the final coefficients a, b. These are the components of Theorem 11.10. ☐

Definition 11.12 (Ordered representation failures)**.** If a repaired-gauge representation is not obtained, the ordered failures are:

(i) no raw orbit is supplied by the concentration and profile data;

(ii) the raw orbit exists but the repaired gauge cannot be realized;
(iii) pressure reconstruction modulo functions of time is unavailable;
(iv) the final modulation coefficients are unavailable.

The first applicable failure is the representation alternative.

Corollary 11.13 (Ordered representation classification)**.** *Every Type II profile branch in the admissible class has exactly one ordered alternative: a repaired-gauge representation, or one of the four failures in Theorem 11.12.*

Proof. Verify the four representation inputs in order. If all are present, Theorem 11.11 gives the repaired-gauge representation. If not, the first missing input gives the corresponding ordered failure. This is exhaustive and disjoint by first-failure ordering. □

11.3. **Minimal chart and gauge realization.**

Definition 11.14 (Minimal representation data)**.** The minimal data for a Type II branch satisfying the analytic data consist of:

(i) an absolutely continuous concentration chart
$$(u, p, T^*, x_c, \lambda, \rho, U, Q, \mathcal{I}),$$
where $\mathcal{I} = (t_0, T^*)$, $\lambda(t) > 0$, $\lambda(t) \to 0$, $\rho_t = \lambda^{-2}$, and
$$u(x,t) = \lambda(t)^{-1} U(y, \rho(t)), \qquad p(x,t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t), \qquad y = \frac{x - x_c(t)}{\lambda(t)};$$

(ii) an absolutely continuous repaired-gauge correction consisting of $\rho = \rho(\tau)$, $\mu(\tau) > 0$, and $q(\tau) \in \mathbb{R}^3$, with
$$\rho_\tau = \mu(\tau)^2,$$
$$V(Y,\tau) = \mu(\tau) U(\mu(\tau) Y + q(\tau), \rho(\tau)), \qquad P(Y,\tau) = \mu(\tau)^2 Q(\mu(\tau) Y + q(\tau), \rho(\tau)),$$
and
$$G_{\mathrm{sc}}(V(\tau)) = 0, \qquad G_j(V(\tau)) = 0, \quad j = 1,2,3.$$
The condition $\rho_\tau = \mu^2$ preserves the viscosity coefficient after the time-dependent scale correction.

Lemma 11.15 (Chart data imply the raw orbit)**.** *Under the chart data in Theorem 11.14, U, Q satisfy*
$$\partial_\rho U + (U \cdot \nabla) U + \nabla Q = \nu \Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \qquad \nabla \cdot U = 0,$$
where
$$A(\rho(t)) = \lambda(t) \lambda_t(t), \qquad B(\rho(t)) = \lambda(t) x_c'(t).$$

Proof. The chart data are the raw orbit data with raw time ρ. The additive pressure function $c(t)$ has zero spatial gradient. The chain-rule computation is the one in Theorem 11.8, with τ replaced by ρ. Thus spatial derivatives scale as $\nabla_x u = \lambda^{-2} \nabla_y U$, $\Delta_x u = \lambda^{-3} \Delta_y U$, the convection term scales as $\lambda^{-3}(U \cdot \nabla) U$, and $\rho_t = \lambda^{-2}$. The terms generated by λ_t and x_c' give $A = \lambda \lambda_t$ and $B = \lambda x_c'$. Incompressibility pulls back as $\nabla_x \cdot u = \lambda^{-2} \nabla_y \cdot U$. □

Lemma 11.16 (Pressure pullback)**.** *Under the chart data,*
$$-\Delta Q = \partial_i \partial_j (U_i U_j)$$
in distributions, modulo functions of ρ. After any admissible repaired-gauge correction, the transformed pressure satisfies
$$-\Delta P = \partial_i \partial_j (V_i V_j)$$
in distributions, modulo functions of τ.

Proof. Taking divergence of the physical Navier–Stokes equation gives

$$-\Delta p = \partial_i \partial_j (u_i u_j),$$

with pressure determined up to a function of time. Pulling this identity back through $p = \lambda^{-2} Q + c(t)$, $u = \lambda^{-1} U$, and $y = (x - x_c)/\lambda$ gives the pressure equation for Q. If

$$V(Y) = \mu U(\mu Y + q), \qquad P(Y) = \mu^2 Q(\mu Y + q),$$

then

$$-\Delta_Y P = \mu^4 (-\Delta Q)(\mu Y + q) = \mu^4 \partial_i \partial_j (U_i U_j)(\mu Y + q) = \partial_i \partial_j (V_i V_j).$$

Adding a function of ρ or τ to the pressure does not affect the identity. □

Lemma 11.17 (Forced modulation coefficients)**.** *Assume the minimal data of Theorem 11.14. Then (V, P) satisfies*

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V,$$

where

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau) B(\rho(\tau)) + \mu(\tau) A(\rho(\tau)) q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

Proof. Let

$$z = \mu(\tau) Y + q(\tau), \qquad V(Y, \tau) = \mu(\tau) U(z, \rho(\tau)), \qquad \rho_\tau = \mu^2.$$

The raw equation in (z, ρ) is

$$U_\rho + (U \cdot \nabla_z) U + \nabla_z Q = \nu \Delta_z U + A(U + z \cdot \nabla_z U) + B \cdot \nabla_z U.$$

The identities

$$\nabla_Y V = \mu^2 \nabla_z U, \qquad \Delta_Y V = \mu^3 \Delta_z U, \qquad (V \cdot \nabla_Y) V = \mu^3 (U \cdot \nabla_z) U, \qquad \nabla_Y P = \mu^3 \nabla_z Q$$

together with differentiation in τ give

$$V_\tau + (V \cdot \nabla_Y) V + \nabla_Y P - \nu \Delta_Y V = \left(\frac{\mu_\tau}{\mu} + \mu^2 A \right) (V + Y \cdot \nabla_Y V) + \left(\mu B + \mu A q + \frac{q_\tau}{\mu} \right) \cdot \nabla_Y V.$$

This is the displayed equation. Since μ, q, ρ are absolutely continuous and A, B are the raw chart coefficients, a, b are measurable and locally integrable on every finite renormalized window. □

Theorem 11.18 (Minimal representation theorem)**.** *If a Type II branch satisfying the analytic data has the minimal data of Theorem 11.14, then it has a repaired-gauge representation in the sense of Theorem 11.10.*

Proof. Theorem 11.15 gives the raw orbit. The gauge correction places the profile on the repaired scale and centering surface, hence gives the repaired-gauge realization. Theorem 11.16 gives pressure reconstruction modulo functions of time. Theorem 11.17 gives the final modulation coefficients. These are the four inputs in Theorem 11.10. □

Definition 11.19 (Minimal representation failures)**.** Outside the admissible repaired-gauge representation class, the ordered failures are:

(i) no absolutely continuous concentration chart is available;
(ii) the chart exists but the repaired scale and centering gauges cannot be solved admissibly and absolutely continuously.

Pressure-pullback and modulation-coefficient failures are not independent alternatives when the chart and the absolutely continuous gauge solve exist.

Corollary 11.20 (Ordered minimal representation classification)**.** *Every Type II branch satisfying the analytic data has exactly one ordered alternative: a repaired-gauge representation, failure of the raw chart, or failure of the repaired-gauge solve.*

Proof. If the chart is absent, the first failure occurs. If the chart exists but the admissible absolutely continuous gauge solve is absent, the second failure occurs. If both are present, Theorem 11.18 gives the repaired-gauge representation. The alternatives are disjoint by their order and exhaustive for the representation row. □

11.4. **Critical L^3 annulus.**

Definition 11.21 (Critical mass and positive finite annulus)**.** For a repaired-gauge renormalized Type II branch (V, P, a, b) on $[\tau_0, \infty)$, define

$$N_3(\tau) = \|V(\tau)\|_{L^3(\mathbb{R}^3)}.$$

The orbit has positive finite critical mass if there are constants $0 < \eta \le M < \infty$ such that

$$\eta \le N_3(\tau) \le M, \qquad \tau \ge \tau_0.$$

This condition replaces the exact normalization $\|V(\tau)\|_{L^3} = 1$; no amplitude rescaling is performed.

Definition 11.22 (Critical-mass alternatives)**.** For a repaired-gauge orbit, the ordered critical-mass alternatives are:

(i) N_3 is not a well-defined extended measurable critical norm on the renormalized branch;
(ii) $N_3(\tau) = \infty$ for some τ, or $\sup_{\tau \ge \tau_0} N_3(\tau) = \infty$;
(iii) $\sup_{\tau \ge \tau_0} N_3(\tau) < \infty$ and $\liminf_{\tau \to \infty} N_3(\tau) = 0$;
(iv) after increasing the initial time at most once, the positive finite critical annulus of Theorem 11.21 holds.

The first applicable case is the corresponding alternative.

Theorem 11.23 (Nonzero-mass compact Type II cost divergence)**.** *Let (V, P, a, b) be a repaired-gauge renormalized Type II branch. Assume:*

(i) *the positive finite critical annulus holds:*

$$0 < \eta \le \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M < \infty;$$

(ii) *uniform critical tightness holds:*

$$\forall \varepsilon > 0 \ \exists R_\varepsilon : \quad \sup_{\tau \ge \tau_0} \int_{|y| > R_\varepsilon} |V(y, \tau)|^3 \, dy < \varepsilon;$$

(iii) *local windowed H^1 control holds:*

$$\forall m \ge 1 : \quad \sup_{n \in \mathbb{N}} \int_{\tau_0 + n}^{\tau_0 + n + 1} \|V(\tau)\|^2_{H^1(B_m)} \, d\tau < \infty.$$

Then finite total localized renormalization cost is impossible:

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) \, d\tau = \infty.$$

Proof. The proof is the compact Type II cost-divergence proof with the exact normalization $\|V(\tau)\|_{L^3} = 1$ replaced by the lower bound $N_3(\tau) \ge \eta$. Assume for contradiction that the total localized renormalization cost is finite. The unit-window selection and compactness argument gives times $\sigma_j \to \infty$ such that, after passing to a subsequence,

$$V(\sigma_j) \to V_\infty \quad \text{strongly in } L^3_{\text{loc}},$$

the selected sequence has vanishing local dissipation on the exhaustion used by the compactness argument, and consequently

$$\nabla V_\infty = 0$$

distributionally on every fixed ball. These conclusions use only tightness, local windowed H^1 control, and finite total cost.

The low-dissipation limit is spatially constant on each fixed ball. The constants are compatible as the balls increase, so $V_\infty \equiv c$ on $\mathbb{R}^3$. The upper critical-mass bound gives, for every R,

$$|c|^3|B_R| = \lim_{j\to\infty} \int_{B_R} |V(y,\sigma_j)|^3\,dy \le M^3.$$

Letting $R \to \infty$ forces $c = 0$. Thus

$$V(\sigma_j) \to 0 \quad \text{strongly in } L^3_{\mathrm{loc}}.$$

Choose R so that the critical L^3-tail outside B_R is $< \eta^3/4$ uniformly in τ. Strong convergence on B_R gives

$$\int_{B_R} |V(y,\sigma_j)|^3\,dy < \eta^3/4$$

for j large. Therefore

$$\|V(\sigma_j)\|^3_{L^3(\mathbb{R}^3)} < \eta^3/2,$$

contradicting $N_3(\sigma_j) \ge \eta$. Hence the total localized renormalization cost is infinite. □

Theorem 11.24 (Critical L^3-annulus classification)**.** *For a repaired-gauge Type II branch, exactly one of the ordered alternatives in Theorem 11.22 occurs. In the fourth case the compact Type II cost-divergence criterion remains valid with the positive finite critical annulus in place of exact unit L^3-normalization.*

Proof. If N_3 is not a well-defined measurable extended norm, the first alternative holds. Otherwise N_3 is an extended nonnegative measurable function. If it is infinite at some time, or has infinite supremum on the tail, the second alternative holds. If the supremum is finite but $\liminf_{\tau\to\infty} N_3(\tau) = 0$, the third alternative holds. In the remaining case,

$$0 < \liminf_{\tau\to\infty} N_3(\tau) \quad \text{and} \quad \sup_{\tau\ge\tau_0} N_3(\tau) < \infty.$$

After increasing the initial time, there exist $0 < \eta \le M < \infty$ such that

$$\eta \le N_3(\tau) \le M$$

for all remaining τ. Relabeling this tail initial time as τ_0 gives the fourth alternative. The cost-divergence statement is Theorem 11.23. □

11.5. **Localized cost and the Type II scale-exclusion condition.**

Definition 11.25 (Localized Type II cost)**.** For a repaired-gauge Type II branch, let

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\,dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\,dy,$$

where $a_+ = \max(a,0)$, ϕ_{R_0} is the compact Type II cutoff, and $\nu > 0$. A Type II scale-exclusion cost is a measurable nonnegative function $\mathfrak{C}_{\mathrm{II}}$ that is locally integrable on finite τ-intervals and whose divergence on $[\tau_0,\infty)$ gives the Type II scale-exclusion condition.

Definition 11.26 (Cost comparison)**.** A cost comparison for a repaired-gauge Type II branch consists of $R_0, \phi_{R_0}, \mathfrak{C}_{\mathrm{II}}, c_0, \tau_1$, with $c_0 > 0$ and $\tau_1 \geq \tau_0$, such that

$$\mathfrak{C}_{\mathrm{II}}(\tau) \geq c_0\, \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \text{for a.e. } \tau \geq \tau_1.$$

The canonical Navier–Stokes choice is

$$\mathfrak{C}_{\mathrm{II}}(\tau) = \tilde{\mathfrak{D}}_{R_0}(\tau), \qquad c_0 = 1, \qquad \tau_1 = \tau_0.$$

Lemma 11.27 (Identity-cost comparison)**.** *If $\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ-intervals, then the canonical choice $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$ defines a valid Type II scale-exclusion cost and satisfies the cost comparison with $c_0 = 1$ and $\tau_1 = \tau_0$.*

Proof. The assumed measurability, nonnegativity, and local finite-interval integrability give the cost properties in Theorem 11.25. The identity

$$\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$$

gives the comparison in Theorem 11.26 with $c_0 = 1$ and $\tau_1 = \tau_0$. □

Theorem 11.28 (Cost comparison theorem)**.** *Let (V, P, a, b) be a repaired-gauge Type II branch. Assume:*

(i) *$\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ-intervals;*
(ii)
$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty;$$
(iii) *either the canonical cost $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$ is used, or a cost comparison as in Theorem 11.26 is available.*

Then the Type II scale-exclusion condition associated with $\mathfrak{C}_{\mathrm{II}}$ holds.

Proof. In the canonical case, Theorem 11.27 supplies the cost comparison. In the general case, the comparison is assumed. Hence there are $c_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$\mathfrak{C}_{\mathrm{II}}(\tau) \geq c_0 \tilde{\mathfrak{D}}_{R_0}(\tau)$$

for a.e. $\tau \geq \tau_1$. Since $\tilde{\mathfrak{D}}_{R_0} \geq 0$ and is locally integrable on finite intervals, divergence on $[\tau_0, \infty)$ implies

$$\int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Therefore

$$\int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau \geq c_0 \int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Since $\mathfrak{C}_{\mathrm{II}}$ is nonnegative and locally integrable on finite intervals,

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau = \int_{\tau_0}^{\tau_1} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau + \int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau = \infty.$$

By definition of the Type II scale-exclusion cost, this divergence is the Type II scale-exclusion condition. □

Corollary 11.29 (Compact cost divergence implies scale exclusion)**.** *Assume the hypotheses of Theorem 11.23 and use the canonical Navier–Stokes cost $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$. Then the Type II scale-exclusion condition holds. If, in addition, the supercritical scale alternative is present and the corresponding Navier–Stokes exclusion hypothesis is available, then the Type II exclusion conclusion holds.*

Proof. Theorem 11.23 gives

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau = \infty.$$

By Theorem 11.27, the canonical cost satisfies the cost comparison. Theorem 11.28 converts the infinite PDE cost into the Type II scale-exclusion condition. Combining this condition with the supercritical scale alternative gives the Type II exclusion conclusion only under the stated Navier–Stokes exclusion hypothesis. □

11.6. **Two-alternative finite-cost classification.**

Definition 11.30 (Remaining compact cost-divergence tests)**.** For a repaired-gauge Type II orbit, define the critical-tightness condition by

$$\forall \varepsilon > 0\ \exists R_\varepsilon : \quad \sup_{\tau \geq \tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3\,dy < \varepsilon.$$

Failure of critical tightness is the condition

$$\exists \varepsilon_0 > 0\ \forall R > 0\ \exists \tau_R \geq \tau_0 : \quad \int_{|y|>R} |V(y,\tau_R)|^3\,dy \geq \varepsilon_0.$$

Define local windowed H^1 control by

$$\forall m \geq 1 : \quad \sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2\,d\tau < \infty.$$

Failure of local windowed H^1 control is the condition

$$\exists m \geq 1 : \quad \sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2\,d\tau = \infty.$$

Definition 11.31 (Compact Type II data)**.** A Type II branch has compact Type II data if it has:

(i) the Type II analytic data of Theorem 11.2;
(ii) the repaired-gauge representation of Theorem 11.10;
(iii) the positive finite critical annulus of Theorem 11.21;
(iv) the locally integrable nonnegative cost and cost comparison of Theorems 11.25 and 11.26.

This condition does not include critical tightness or windowed H^1 control; those are the two remaining compact cost-divergence tests.

Lemma 11.32 (Compact positive case is excluded)**.** *Assume an admissible Type II branch has compact Type II data and satisfies both compact cost-divergence tests in Theorem 11.30. If the Navier–Stokes exclusion hypothesis for the Type II scale-exclusion condition is available, then the branch satisfies the Type II exclusion conclusion.*

Proof. The compact Type II data give the supercritical scale alternative, a repaired-gauge orbit (V, P, a, b), the positive finite critical annulus, and the cost comparison. Together with critical tightness and local windowed H^1 control, the hypotheses of Theorem 11.23 hold. Thus

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\,d\tau = \infty.$$

Theorem 11.28 converts this divergence into the Type II scale-exclusion condition. Combining this condition with the supercritical scale alternative gives the Type II exclusion conclusion under the stated Navier–Stokes exclusion hypothesis. □

Theorem 11.33 (Compact exclusion or two-alternative classification)**.** *Assume every admissible Type II branch has compact Type II data and that the Navier–Stokes exclusion hypothesis is available. Then each admissible Type II branch has exactly one ordered alternative:*

(i) *the compact Type II exclusion conclusion;*
(ii) *failure of critical tightness;*
(iii) *failure of local windowed H^1 control after critical tightness has been verified.*

Proof. Fix an admissible Type II branch. Since it has compact Type II data, only the two compact cost-divergence tests remain. First evaluate critical tightness. If tightness fails, the failure-of-tightness alternative holds. If tightness holds, evaluate local windowed H^1 control. If the windowed H^1 condition fails, the windowed-control alternative holds. If it holds, then Theorem 11.32 gives the compact Type II exclusion conclusion. The alternatives are exhaustive and disjoint by this ordered evaluation. □

Definition 11.34 (Finite-cost branch not excluded by compact cost divergence)**.** An admissible Type II branch is finite-cost if

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau < \infty.$$

In the canonical Navier–Stokes cost this is equivalent to finite localized PDE cost, because $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$. It is not excluded by compact cost divergence if the compact Type II exclusion conclusion does not hold for that branch.

Corollary 11.35 (Two-alternative finite-cost classification)**.** *Under the hypotheses of Theorem 11.33, every finite-cost admissible Type II branch not excluded by compact cost divergence is either non-tight or fails local windowed H^1 control.*

Proof. Apply Theorem 11.33. Since the branch is not excluded by compact cost divergence, the first alternative is unavailable. Therefore the ordered alternative is failure of critical tightness or failure of local windowed H^1 control. The finite-cost assumption records that the resulting branch remains finite-cost. □

Remark 11.36 (Content of the classification inputs). The preceding statements put the Type II classification inputs in PDE form. Exhaustion failure, missing repaired-gauge representation, ambiguous critical L^3 normalization, and cost-comparison failure are all replaced by ordered alternatives. Once those alternatives are excluded or assumed absent, the only finite-cost Type II alternatives not excluded by the compact cost-divergence criterion are failure of critical tightness and failure of local windowed H^1 control.

References

[Aub63] Jean-Pierre Aubin. Un théorème de compacité. *Comptes Rendus de l'Académie des Sciences de Paris*, 256:5042–5044, 1963.

[BL83] Haïm Brezis and Elliott H. Lieb. A relation between pointwise convergence of functions and convergence of functionals. *Proceedings of the American Mathematical Society*, 88(3):486–490, 1983.

[CF88] Peter Constantin and Ciprian Foias. *Navier–Stokes Equations*. University of Chicago Press, Chicago, 1988.

[CKN82] Luis Caffarelli, Robert Kohn, and Louis Nirenberg. Partial regularity of suitable weak solutions of the navier–stokes equations. *Communications on Pure and Applied Mathematics*, 35(6):771–831, 1982.

[CZ52] Alberto P. Calderon and Antoni Zygmund. On the existence of certain singular integrals. *Acta Mathematica*, 88:85–139, 1952.

[ESS03] Luis Escauriaza, Gregory Seregin, and Vladimir Sverak. $L_{3,\infty}$-solutions of navier–stokes equations and backward uniqueness. *Russian Mathematical Surveys*, 58(2):211–250, 2003.

[Gal01] Isabelle Gallagher. Profile decomposition for solutions of the navier–stokes equations. *Bulletin de la Société Mathématique de France*, 129(2):285–316, 2001.

[Kat84] Tosio Kato. Strong L^p-solutions of the navier–stokes equation in $\mathbb{R}^m$, with applications to weak solutions. *Mathematische Zeitschrift*, 187:471–480, 1984.

[Lad69] Olga A. Ladyzhenskaya. *The Mathematical Theory of Viscous Incompressible Flow*. Gordon and Breach, New York, 1969.

[Ler34] Jean Leray. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Mathematica*, 63:193–248, 1934.

[Lin98] Fanghua Lin. A new proof of the caffarelli-kohn-nirenberg theorem. *Communications on Pure and Applied Mathematics*, 51(3):241–257, 1998.

[LM72] Jacques-Louis Lions and Enrico Magenes. *Non-Homogeneous Boundary Value Problems and Applications. Vol. I*. Springer, Berlin, 1972.

[LR02] Pierre Gilles Lemarié-Rieusset. *Recent Developments in the Navier–Stokes Problem*. Chapman and Hall/CRC, Boca Raton, FL, 2002.

[NRŠ96] Jindřich Nečas, Michael Růžička, and Vladimir Šverák. On leray's self-similar solutions of the navier–stokes equations. *Acta Mathematica*, 176(2):283–294, 1996.

[Sch76] Vladimir Scheffer. Partial regularity of solutions to the navier-stokes equations. *Pacific Journal of Mathematics*, 66(2):535–552, 1976.

[Sim87] Jacques Simon. Compact sets in the space $L^p(0, T; B)$. *Annali di Matematica Pura ed Applicata*, 146:65–96, 1987.

[Ste70] Elias M. Stein. *Singular Integrals and Differentiability Properties of Functions*. Princeton University Press, Princeton, NJ, 1970.

[Tem77] Roger Temam. *Navier–Stokes Equations: Theory and Numerical Analysis*. North-Holland, Amsterdam, 1977.

Email address: `guillem@fragile.tech`